MECHANICS APPLIED TO AEROSPACE ENGINEERING - PART I

Topic 5: Relative Motion - Problem Sets

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Source: sources/cuatrimestre-1/03-engineering-mechanics/problemas/Problems.pdf (2026 Edition)

Statements and solutions: Eighteen problems of the official collection that need a moving reference frame. Each card gives the statement (with the PDF page) and a worked solution in four phases: hypotheses and data, frames with $[{}_0R_1]$, $\boldsymbol{\omega}_{10}$ and $\boldsymbol{\alpha}_{10}$, step-by-step derivation with the relative-motion formulas (Notes Eqs. 3.26–3.27) or the inertia forces (Notes Eqs. 4.62–4.63), and interpretation with unit and limit checks. Problems 7, 25, 27, 33, 34, 36, 37, 40, 41 and 47 have an official key entry, compared in each card; the other results are derived here and checked symbolically.

PROBLEM 07

Skater Towed by a Vehicle: Rotating Frame

Original statement · Problems.pdf, p. 8

Rotating Frame + Inertia Forces

A skater B is being towed by a vehicle A on a perfectly smooth horizontal plane, where we define an inertial reference frame Oxy. The vehicle is moving along a circumference of radius a and center at O with constant angular velocity ω. The skater is attached to the vehicle by a bar without mass and constant length $l = a$. Initially, the vehicle is located at $(a, 0)$ and the skater is at rest at $(2a, 0)$. Defining a rotating reference frame Ox_1y_1 with origin at O and axis Ox_1 along OA, and calling ψ the angle between the axis Ox_1 and the bar AB, determine:

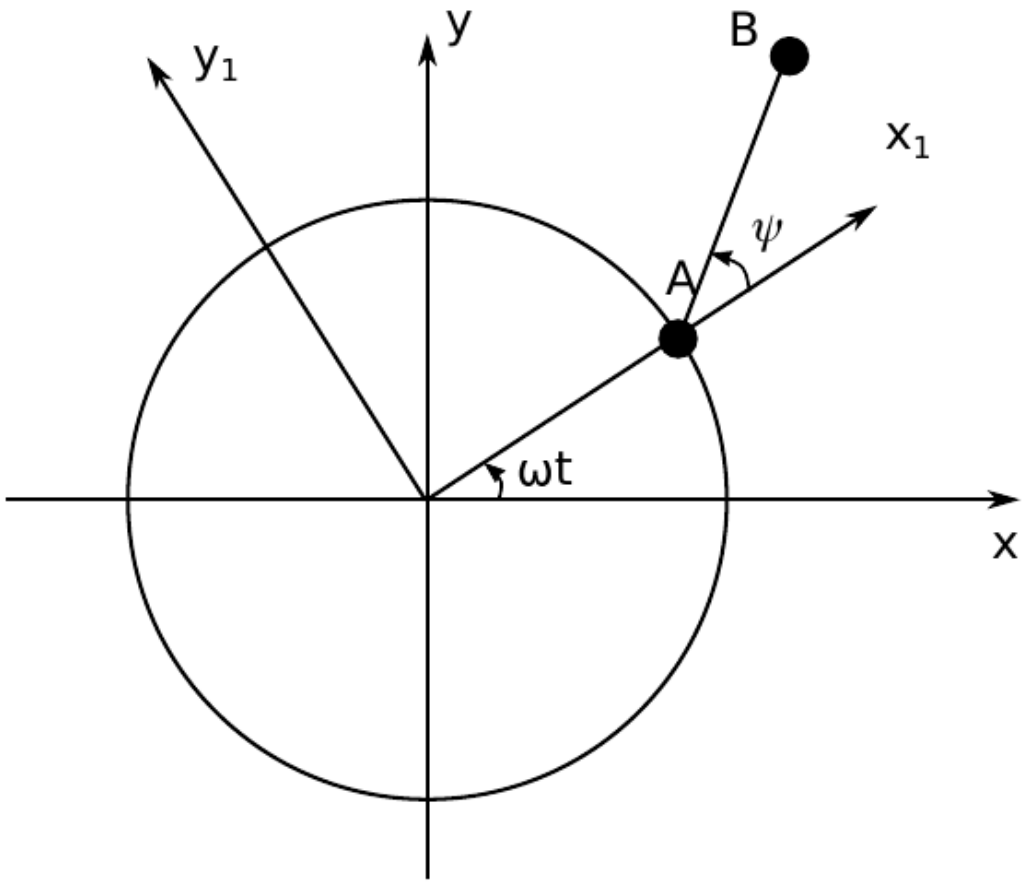


Figure: Inertial axes Oxy, rotating axes Ox1y1 turned by ωt, vehicle A on the circle of radius a, bar AB at angle ψ from Ox1, skater B (Problems.pdf, p. 8).

(a) The relative acceleration of the skater with respect to Ox_1y_1 as a function of ψ and its time derivatives.

(b) The centripetal and Coriolis acceleration terms also as a function of ψ and its time derivatives.

(c) Reduce the evolution of ψ to integrals of the form $\int [\dots]\, d\psi = \int [\dots]\, dt$ (consider that $\dot{\psi}^2 = \int 2\dot{\psi}\ddot{\psi}\, dt$).

(d) Determine the tension on the bar.

(e) If the rigid bar is substituted by a string which cannot bear compression, determine if/when the string will become loose.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** the vehicle A moves on the circle of radius a centred at O with constant angular rate ω. The skater B (mass m) is joined to A by a massless rigid bar of length $l = a$ and slides on a smooth horizontal plane.
- **Why a rotating frame.** In the frame Ox_1y_1 that turns with the line OA, the vehicle is at rest at $(a, 0)$, so the skater moves on a fixed circle of radius a around A: one coordinate ψ describes the motion and the constraint is fixed in that frame. The price is that Ox_1y_1 is non-inertial, so we must add the inertia forces of Notes Eq. 4.62 to Newton's law (Notes Eq. 4.63).
- **Hypotheses:**
  1. Oxy is inertial; the plane is smooth (its reaction is vertical and cancels the weight, so neither enters the horizontal dynamics).
  2. The bar is massless, so the force it exerts on B is directed along the bar: $-T\,\mathbf{u}_{AB}$, with $T > 0$ meaning tension.
  3. ω is constant.
- **Degrees of freedom:** B moves in the plane (2 coordinates) with one constraint $|AB| = a$: **1 DOF**, the angle ψ.
- **Initial conditions:** A at $(a, 0)$ and B at rest at $(2a, 0)$, so $\psi(0) = 0$ and $\mathbf{v}_0^B(0) = \mathbf{0}$ (this gives $\dot{\psi}(0)$ in part (c)).

Symbol	Meaning	SI unit
m	mass of the skater B	kg
a	radius of the vehicle's circle and bar length	m
ω	angular rate of the vehicle (constant)	rad/s
ψ	angle from Ox_1 to the bar AB	rad
T	tension of the bar (positive when pulling B toward A)	N

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Frames.** $S_0 = Oxy = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ (inertial) and $S_1 = Ox_1y_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ with $\mathbf{i}_1$ along OA. The origin of S_1 is O (it plays the role of the generic origin "A" of Notes Ch. 3, which here is the fixed point O; the letter A is kept for the vehicle as in the statement) and the particle is $P \equiv B$.
- **Attitude.** The vehicle starts on Ox and turns counter-clockwise, so the angle from $\mathbf{i}_0$ to $\mathbf{i}_1$ is ωt. The columns of $[{}_0R_1]$ are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Checks: $\det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1$; the columns are unit vectors ($\cos^2\omega t + \sin^2\omega t = 1$) and mutually orthogonal ($-\cos\omega t \sin\omega t + \sin\omega t \cos\omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.

- **Motion of S_1.** Simple rotation about $\mathbf{k}_0 = \mathbf{k}_1$ (Slide 16):

$$\boldsymbol{\omega}_{10} = \omega\,\mathbf{k}_1, \qquad \boldsymbol{\alpha}_{10} = \left.\frac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \dot{\omega}\,\mathbf{k}_1 = \mathbf{0}, \qquad \mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$$

- **Bar basis.** $\mathbf{u}_{AB} = \cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1$ (from A to B) and $\mathbf{u}_\perp = \mathbf{k}_1 \times \mathbf{u}_{AB} = -\sin\psi\,\mathbf{i}_1 + \cos\psi\,\mathbf{j}_1$. Their matrix $\begin{pmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{pmatrix}$ (completed with $\mathbf{k}_1$) is again a simple rotation, with $\det = 1$ and $RR^T = I$.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Relative acceleration $\mathbf{a}_1^B$

> **Pedagogical Justification:** the relative quantities are derivatives in S_1, where $\mathbf{i}_1, \mathbf{j}_1$ are constant, so only the components are differentiated.

Relative position (Notes Eq. 3.16 with $\mathbf{r}_0^O = \mathbf{0}$): $\mathbf{r}_1^B = \mathbf{OA} + \mathbf{AB}$,

$$\mathbf{r}_1^B = a\,\mathbf{i}_1 + a\,(\cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1) = a(1+\cos\psi)\,\mathbf{i}_1 + a\sin\psi\,\mathbf{j}_1$$

Relative velocity, with the chain rule $\frac{d}{dt}\cos\psi = -\sin\psi\,\dot\psi$ and $\frac{d}{dt}\sin\psi = \cos\psi\,\dot\psi$:

$$\mathbf{v}_1^B = \left.\frac{d\mathbf{r}_1^B}{dt}\right|_1 = -a\dot\psi\sin\psi\,\mathbf{i}_1 + a\dot\psi\cos\psi\,\mathbf{j}_1 = a\dot\psi\,\mathbf{u}_\perp$$

Relative acceleration, with the product rule $\frac{d}{dt}(\dot\psi\sin\psi) = \ddot\psi\sin\psi + \dot\psi^2\cos\psi$ and $\frac{d}{dt}(\dot\psi\cos\psi) = \ddot\psi\cos\psi - \dot\psi^2\sin\psi$:

$$\boxed{\mathbf{a}_1^B = -a\left(\ddot\psi\sin\psi + \dot\psi^2\cos\psi\right)\mathbf{i}_1 + a\left(\ddot\psi\cos\psi - \dot\psi^2\sin\psi\right)\mathbf{j}_1 = a\ddot\psi\,\mathbf{u}_\perp - a\dot\psi^2\,\mathbf{u}_{AB}}$$

(the last form follows by grouping the $\ddot\psi$ and $\dot\psi^2$ terms: circular motion of radius a about A, tangential plus normal part).

(b) Centripetal and Coriolis terms

> **Pedagogical Justification:** they are the two non-zero rotation terms of Notes Eq. 3.27 (Slide 25); the Euler term vanishes because $\boldsymbol{\alpha}_{10} = \mathbf{0}$ and the origin term because $\mathbf{a}_0^O = \mathbf{0}$.

Cross products used: $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$.

Centripetal term.

$$\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B = \omega\mathbf{k}_1 \times [a(1+\cos\psi)\mathbf{i}_1 + a\sin\psi\,\mathbf{j}_1] = a\omega(1+\cos\psi)\,\mathbf{j}_1 - a\omega\sin\psi\,\mathbf{i}_1$$

$$\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B\right) = \omega\mathbf{k}_1 \times [-a\omega\sin\psi\,\mathbf{i}_1 + a\omega(1+\cos\psi)\,\mathbf{j}_1] = -a\omega^2\sin\psi\,\mathbf{j}_1 - a\omega^2(1+\cos\psi)\,\mathbf{i}_1$$

$$\boxed{\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B\right) = -a\omega^2\left[(1+\cos\psi)\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1\right] = -\omega^2\,\mathbf{r}_1^B}$$

It equals $-\omega^2\mathbf{r}_\perp$ with $\mathbf{r}_\perp = \mathbf{r}_1^B$ because $\mathbf{r}_1^B$ lies in the plane normal to $\boldsymbol{\omega}_{10}$ (Slide 30).

Coriolis term.

$$2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^B = 2\omega\mathbf{k}_1\times\left(-a\dot\psi\sin\psi\,\mathbf{i}_1 + a\dot\psi\cos\psi\,\mathbf{j}_1\right) = -2a\omega\dot\psi\sin\psi\,\mathbf{j}_1 - 2a\omega\dot\psi\cos\psi\,\mathbf{i}_1$$

$$\boxed{2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^B = -2a\omega\dot\psi\left(\cos\psi\,\mathbf{i}_1+\sin\psi\,\mathbf{j}_1\right) = -2a\omega\dot\psi\,\mathbf{u}_{AB}}$$

Both agree with the official key (Problems.pdf, p. 72), where the centripetal term is called "centrifugal acceleration".

Absolute acceleration (Notes Eq. 3.27):

$$\mathbf{a}_0^B = \underbrace{\mathbf{a}_0^O}_{\mathbf{0}} + \mathbf{a}_1^B + \underbrace{\boldsymbol{\alpha}_{10}\times\mathbf{r}_1^B}_{\mathbf{0}} - \omega^2\mathbf{r}_1^B - 2a\omega\dot\psi\,\mathbf{u}_{AB}$$

(c) Equation of motion and reduction to quadratures

> **Pedagogical Justification:** in S_1 Newton's law holds with the inertia forces (Notes Eqs. 4.62–4.63): $m\mathbf{a}_1^B = \mathbf{F}+\mathbf{F}_i$. Here $\mathbf{F} = -T\,\mathbf{u}_{AB}$ (horizontal part), $\mathbf{F}_A=\mathbf{0}$, $\mathbf{F}_E=\mathbf{0}$, and

$$\mathbf{F}_C = -m\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B\right) = ma\omega^2\left[(1+\cos\psi)\,\mathbf{i}_1+\sin\psi\,\mathbf{j}_1\right],\qquad \mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10}\times\mathbf{v}_1^B = 2ma\omega\dot\psi\,\mathbf{u}_{AB}$$

The unknown T disappears if we project on $\mathbf{u}_\perp$, the only direction along which the bar exerts no force.

Projections on $\mathbf{u}_\perp = -\sin\psi\,\mathbf{i}_1+\cos\psi\,\mathbf{j}_1$:

- $m\mathbf{a}_1^B\cdot\mathbf{u}_\perp = ma\ddot\psi$ (from the grouped form of part (a), since $\mathbf{u}_{AB}\cdot\mathbf{u}_\perp=0$);
- $-T\,\mathbf{u}_{AB}\cdot\mathbf{u}_\perp = 0$ and $\mathbf{F}_{COR}\cdot\mathbf{u}_\perp = 0$;
- $\mathbf{F}_C\cdot\mathbf{u}_\perp = ma\omega^2\left[-(1+\cos\psi)\sin\psi+\sin\psi\cos\psi\right] = -ma\omega^2\sin\psi$.

$$ma\ddot\psi = -ma\omega^2\sin\psi \quad\Longrightarrow\quad \boxed{\ddot\psi+\omega^2\sin\psi=0}$$

In S_1 the skater behaves as a pendulum of length a hanging from A: about A the centrifugal force $m\omega^2(\mathbf{OA}+\mathbf{AB})$ has the same moment as the uniform field $ma\omega^2\,\mathbf{i}_1$ (because $\mathbf{AB}\times\mathbf{AB}=\mathbf{0}$), so the "effective gravity" has strength $a\omega^2$ and points along $+\mathbf{i}_1$.

First integral (hint of the statement). Multiply by $2\dot\psi$. By the chain rule $\frac{d}{dt}\left(\dot\psi^2\right)=2\dot\psi\ddot\psi$ and $\frac{d}{dt}\cos\psi=-\sin\psi\,\dot\psi$, so

$$2\dot\psi\ddot\psi=-2\omega^2\sin\psi\,\dot\psi \quad\Longleftrightarrow\quad \frac{d}{dt}\left(\dot\psi^2\right)=2\omega^2\frac{d}{dt}\left(\cos\psi\right)$$

Integrate from 0 to t and apply Barrow's rule $[F]_0^t=F(t)-F(0)$:

$$\dot\psi^2(t)-\dot\psi^2(0)=\int_0^t 2\dot\psi\ddot\psi\,dt'=2\omega^2\left[\cos\psi\right]_{\psi(0)}^{\psi(t)}=2\omega^2\left(\cos\psi-1\right)$$

Initial rate. By Notes Eq. 3.26, $\mathbf{v}_0^B=\mathbf{v}_0^O+\mathbf{v}_1^B+\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B$. At $t=0$: $\psi=0$, $\mathbf{r}_1^B=2a\,\mathbf{i}_1$, $\mathbf{v}_1^B=a\dot\psi(0)\,\mathbf{j}_1$, $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B=2a\omega\,\mathbf{j}_1$, and $\mathbf{v}_0^B=\mathbf{0}$:

$$\mathbf{0}=\Big[a\dot\psi(0)+2a\omega\Big]\,\mathbf{j}_1 \quad\Longrightarrow\quad \dot\psi(0)=-2\omega$$

Hence, with $1+\cos\psi=2\cos^2(\psi/2)$,

$$\dot\psi^2=4\omega^2+2\omega^2(\cos\psi-1)=2\omega^2(1+\cos\psi)=4\omega^2\cos^2\frac{\psi}{2}$$

For $-\pi<\psi<\pi$ the factor $\cos(\psi/2)>0$, so $\dot\psi$ cannot vanish nor change sign there; $\dot\psi(0)=-2\omega<0$ selects

$$\boxed{\dot\psi=-2\omega\cos\frac{\psi}{2},\qquad -\pi<\psi\le 0}$$

which agrees with the official key (Problems.pdf, p. 72).

Energy check in S_1 (Slides 29–30). The origin is fixed and $\boldsymbol{\omega}_{10}$ is constant, so $\mathbf{F}_C$ has the potential $V_1=-\tfrac12 m\omega^2 r_\perp^2$, $\mathbf{F}_{COR}$ does no work, and the bar force does no work in S_1 ($-T\mathbf{u}_{AB}\cdot a\dot\psi\mathbf{u}_\perp=0$). With $r_\perp^2=a^2\left[(1+\cos\psi)^2+\sin^2\psi\right]=2a^2(1+\cos\psi)$:

$$E_1^B = \frac{1}{2}ma^2\dot\psi^2 - ma^2\omega^2(1+\cos\psi) = \text{const} = \frac{1}{2}ma^2(4\omega^2) - 2ma^2\omega^2 = 0$$

which gives again $\dot\psi^2 = 2\omega^2(1+\cos\psi)$.

Quadrature. Separate variables in $\frac{d\psi}{dt} = -2\omega\cos\frac{\psi}{2}$:

$$\boxed{\int_0^{\psi}\frac{d\psi'}{2\cos(\psi'/2)} = -\int_0^t \omega\,dt'}$$

The left integral is elementary. Substitute $w = \psi'/2$, $d\psi' = 2\,dw$, limits $0 \to \psi/2$; the antiderivative of $\sec w$ is $\ln|\sec w + \tan w|$ (its derivative is $\frac{\sec w\tan w + \sec^2 w}{\sec w + \tan w} = \sec w$). By Barrow's rule:

$$\int_0^{\psi/2}\frac{dw}{\cos w} = \left[\ln|\sec w + \tan w|\right]_0^{\psi/2} = \ln\left|\sec\frac{\psi}{2} + \tan\frac{\psi}{2}\right| - \ln 1 = -\omega t$$

so $\psi(t)$ is given implicitly by $\sec\frac{\psi}{2} + \tan\frac{\psi}{2} = e^{-\omega t}$; as $t\to\infty$, $\psi \to -\pi$, i.e. B tends to the centre O ($\mathbf{r}_1^B \to \mathbf{0}$) without ever reaching it.

(d) Tension on the bar

> **Pedagogical Justification:** project $m\mathbf{a}_1^B = -T\mathbf{u}_{AB} + \mathbf{F}_C + \mathbf{F}_{COR}$ on $\mathbf{u}_{AB}$, the direction in which the bar force acts.

Projections on $\mathbf{u}_{AB} = \cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1$:

- $m\mathbf{a}_1^B\cdot\mathbf{u}_{AB} = -ma\dot\psi^2$;
- $\mathbf{F}_C\cdot\mathbf{u}_{AB} = ma\omega^2\left[(1+\cos\psi)\cos\psi + \sin^2\psi\right] = ma\omega^2(1+\cos\psi)$;
- $\mathbf{F}_{COR}\cdot\mathbf{u}_{AB} = 2ma\omega\dot\psi$.

$$-ma\dot\psi^2 = -T + ma\omega^2(1+\cos\psi) + 2ma\omega\dot\psi \quad\Longrightarrow\quad T = ma\left[\dot\psi^2 + 2\omega\dot\psi + \omega^2(1+\cos\psi)\right]$$

Substitute the first integral with $c \equiv \cos(\psi/2)$: $\dot\psi^2 = 4\omega^2c^2$, $\dot\psi = -2\omega c$, $1+\cos\psi = 2c^2$:

$$T = ma\omega^2\left(4c^2 - 4c + 2c^2\right) = ma\omega^2\left(6c^2 - 4c\right)$$

$$\boxed{T = 2ma\omega^2 \cos\frac{\psi}{2}\left(3\cos\frac{\psi}{2} - 2\right)}$$

which agrees with the official key (Problems.pdf, p. 72). At $t = 0$: $T(0) = 2ma\omega^2 > 0$.

(e) Does a string become loose?

> **Pedagogical Justification:** a string can only pull, so it stays taut while $T \geq 0$; it goes slack at the first instant where T would become negative.

As ψ decreases from 0 toward $-\pi$, $c = \cos(\psi/2)$ decreases monotonically from 1 toward 0. The factor c stays positive, and $3c - 2$ changes sign at $c = 2/3$:

$$\cos\frac{\psi}{2} = \frac{2}{3} \quad\Longrightarrow\quad \boxed{\psi_{\text{loose}} = -2\arccos\frac{2}{3} \approx -1.682\ \text{rad} \approx -96.38^\circ}$$

(official key: $\psi = -2\arccos(2/3)$). **When:** at $\psi/2 = -\arccos(2/3)$, $\sec(\psi/2) = 3/2$ and $\tan(\psi/2) = -\sqrt{1 - 4/9}/(2/3) = -\sqrt{5}/2$, so from part (c)

$$\omega t_{\text{loose}} = -\ln\left(\frac{3}{2} - \frac{\sqrt{5}}{2}\right) = \ln\frac{2}{3 - \sqrt{5}} = \ln\frac{3 + \sqrt{5}}{2} \approx 0.9624, \qquad t_{\text{loose}} \approx \frac{0.962}{\omega}$$

So yes: the string goes slack after about $0.96/\omega$ seconds, when the bar has turned 96.4° clockwise from Ox_1.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Reading in S_1.** The rotating observer sees B as a pendulum of length a hung from A in an effective field of strength $a\omega^2$ pointing away from O (the centrifugal force), whose "lowest" point is $\psi = 0$. The skater starts there with relative speed $2a\omega$, which is exactly the separatrix energy ($E_1^B = 0$): it creeps toward the "inverted" position $\psi = -\pi$ (where B would sit at O) in infinite time.

- **Why T drops.** The Coriolis force $2ma\omega\dot{\psi}\,\mathbf{u}_{AB}$ points toward A when $\dot{\psi}<0$, and the centrifugal pull along the bar, $ma\omega^2(1+\cos\psi)$, shrinks as B swings in; when the required centripetal force $ma\dot{\psi}^2$ is supplied by these inertia forces alone the bar is unloaded ($T=0$).
- **Limit $\omega\to 0$:** $\dot{\psi}\to 0, T\to 0$ (nothing moves).
- **Dimensions:** $[\mathbf{a}_1^B]=[a\ddot{\psi}]=[a\dot{\psi}^2]=\mathrm{m\,s^{-2}}$; $[\omega^2 a]=[\omega\dot{\psi}a]=\mathrm{m\,s^{-2}}$; $[T]=[ma\omega^2]=\mathrm{kg\,m\,s^{-2}}=\mathrm{N}$; $[\omega t]=1$.
- **Verification (sympy, `verify_P07.py`):** the inertial double derivative of $[{}_0R_1]\{\mathbf{r}_1^B\}$ equals $\mathbf{a}_1^B-\omega^2\mathbf{r}_1^B-2a\omega\dot{\psi}\mathbf{u}_{AB}$; the EOM, $\dot{\psi}(0)=-2\omega$, $dE_1^B/dt=0$ along the EOM, the tension and ψ_{loose} are reproduced; an RK4 integration with $\omega=1$ reaches $\cos(\psi/2)=2/3$ at $t=0.9625$ s, matching $\ln\frac{3+\sqrt{5}}{2}=0.9624$.
- **Cross-check with Topic 1.** The Topic 1 solution of this problem (inertial approach) gives the same EOM, tension and loosening angle; no discrepancy with the official key.

Source: Problems.pdf (PDF page 8); key on PDF page 72

PROBLEM 08

Kinematics, Two Trains on Tracks

Original statement · Problems.pdf, p. 8

Relative Motion

Two trains, A and B, are moving along parallel tracks. The separation between the tracks is a. The absolute velocity of each train is v_A and v_B, respectively. What is the relative velocity of each train with respect to the other?

Solve the same problem considering that train B is now moving along a circular track of radius R. The minimum distance between the circular track and the straight track of A is a.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** Two trains A and B move on separate tracks with distance of closest separation a.
- Case 1: Two parallel straight tracks with constant speeds v_A and v_B.
- Case 2: Train A on a straight track, Train B on a circular track of radius R.
- **Parameters & Units:** Separation a $[\mathrm{m}]$, radius R $[\mathrm{m}]$, speeds v_A, v_B $[\mathrm{m/s}]$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Cartesian Frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0\}$.

PHASE 3

Step-by-Step Mathematical Deduction

Case 1: Parallel Tracks

- Train A on line $y = 0$: $\mathbf{r}_A(t) = v_A t\,\mathbf{i}_0$.
- Train B on line $y = a$: $\mathbf{r}_B(t) = (x_{B0} + v_B t)\mathbf{i}_0 + a\mathbf{j}_0$.
- **Relative Velocity:**

$$\mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A = (v_B - v_A)\mathbf{i}_0$$

$$\mathbf{v}_{A/B} = -\mathbf{v}_{B/A} = (v_A - v_B)\mathbf{i}_0$$

Case 2: Train B on a Circular Track of Radius R

- Straight track of A along the x-axis: $\mathbf{r}_A = v_A t\,\mathbf{i}_0$.
- Circular track center at $(0, R + a)$ so that the closest distance to A's track is $(R + a) - R = a$.
- Position of train B traveling at angular speed $\omega_B = \frac{v_B}{R}$ with angle $\theta_B(t) = \omega_B t$:

$$\mathbf{r}_B(t) = R\sin(\omega_B t)\mathbf{i}_0 + [R + a - R\cos(\omega_B t)]\mathbf{j}_0$$

- Velocity of train B:

$$\mathbf{v}_B = \frac{d\mathbf{r}_B}{dt} = R\omega_B\cos(\omega_B t)\mathbf{i}_0 + R\omega_B\sin(\omega_B t)\mathbf{j}_0 = v_B\cos\left(\frac{v_B t}{R}\right)\mathbf{i}_0 + v_B\sin\left(\frac{v_B t}{R}\right)\mathbf{j}_0$$

- **Relative Velocity $\mathbf{v}_{B/A}$:**

$$\mathbf{v}_{B/A} = \mathbf{v}_B - \mathbf{v}_A = \left[v_B\cos\left(\frac{v_B t}{R}\right) - v_A\right]\mathbf{i}_0 + v_B\sin\left(\frac{v_B t}{R}\right)\mathbf{j}_0$$

- **Magnitude of Relative Velocity:**

$$v_{\text{rel}} = \sqrt{\left[v_B \cos\left(\frac{v_B t}{R}\right) - v_A\right]^2 + v_B^2 \sin^2\left(\frac{v_B t}{R}\right)} = \sqrt{v_B^2 + v_A^2 - 2 v_A v_B \cos\left(\frac{v_B t}{R}\right)}$$

PHASE 4

Physical Interpretation, Limiting Cases and Dimensional Check

- **Dimensional check:** $[\mathbf{v}_{B/A}] = \mathrm{m/s}$; in Case 2 the argument of the trigonometric functions is $\dfrac{v_B t}{R} = \dfrac{(\mathrm{m/s})\,\mathrm{s}}{\mathrm{m}}$, which is dimensionless (rad), and $[v_B^2 + v_A^2 - 2 v_A v_B \cos(\cdot)] = \mathrm{m}^2/\mathrm{s}^2$, so v_{rel} is in m/s.
- **Role of the separation a:** it does not appear in any relative velocity, because $\mathbf{v}_{B/A}$ is the time derivative of $\mathbf{r}_B - \mathbf{r}_A$ and a only enters the constant offset of that vector.
- **Limiting cases of Case 2 (sympy):**
- $t = 0$: $v_{\text{rel}} = \sqrt{v_A^2 + v_B^2 - 2 v_A v_B} = |v_B - v_A|$, because both velocities are parallel to $\mathbf{i}_0$ at that instant.
- $R \to \infty$ (the circular track straightens): $\cos(v_B t/R) \to 1$ and $v_{\text{rel}} \to |v_B - v_A|$, which is the Case 1 result.
- $v_A = v_B = v$: $v_{\text{rel}} = v\sqrt{2 - 2\cos(vt/R)} = 2v\left|\sin\dfrac{vt}{2R}\right|$, which starts at 0 and never exceeds $2v$.
- $v_A = 0$: $v_{\text{rel}} = v_B$ for all t (relative to a stationary train, only the speed of B matters).
- **Symbolic check:** $\|\mathbf{v}_B - \mathbf{v}_A\|^2$ computed from the components equals $v_B^2 + v_A^2 - 2 v_A v_B \cos(v_B t/R)$ with zero residual. Both cases are written in the single inertial basis $\mathcal{B}_0$, so no change-of-basis matrix is needed.

Source: Problems.pdf

PROBLEM 09

Artificial Gravity in a Rotating Cylinder

Original statement · Problems.pdf, p. 8

Centrifugal + Coriolis Forces

In "2001: A Space Odyssey" an idea to obtain artificial gravity in a spaceship was presented. It consists of a cylinder rotating at constant angular velocity around its axis. The astronauts would live in the inner surface of the cylinder and the centrifugal force would act as artificial gravity.

(a) If the cylinder has radius R, determine the angular velocity ω required to obtain a given level of artificial gravity g. Consider $R = 10$ m and $g = 9.81$ m/s^2 for a particular case.

(b) An astronaut runs inside the surface of the cylinder with a constant speed u with respect to the cylinder. Determine the artificial gravity that she feels (neglect the height of the astronaut h with respect to the radius of the cylinder R, $h \ll R$).

(c) What is the condition for her to feel no artificial gravity at all?

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** a cylindrical hull of radius R spins at constant rate ω about its axis in deep space. An astronaut P of mass m stands on, or runs along, the inner surface.
- **Why the hull frame.** The astronaut lives in the hull: the floor, her running speed u and what she "feels" are all defined with respect to the cylinder. In the frame S_1 attached to the hull the floor is fixed, and the only price is to add the inertia forces of Notes Eq. 4.62. The centrifugal force is exactly the "artificial gravity" of the statement; the Coriolis force is what changes when she runs.
- **Hypotheses:**
  1. No external gravity (deep space); the only actual force on the astronaut is the total force $\mathbf{N}$ of the floor on her feet (normal reaction plus any friction; part (b) shows that, for uniform running, $\mathbf{N}$ comes out purely radial).

2. $h \ll R$: she is a point particle on the surface, at distance R from the axis.
3. ω constant; the axis point O is fixed in an inertial frame S_0.

- **Artificial gravity felt.** A body feels as "weight" the force it has to receive from the floor. We define the artificial gravity as $g_{\text{art}} = |\mathbf{N}|/m$, the reading per unit mass of a scale under her feet, directed opposite to $\mathbf{N}$ (i.e. away from the axis, "down" for her).
- **Degrees of freedom:** on the cylinder surface, 2 coordinates: azimuth φ in S_1 and axial position z. Running at constant speed u means $u_\theta = R\dot{\varphi}$ and $u_z = \dot{z}$ constant with $u_\theta^2 + u_z^2 = u^2$.

Symbol	Meaning	SI unit
R	radius of the cylinder (10 m in the numerical case)	m
ω	spin rate of the cylinder	rad/s
g	target artificial gravity (9.81 m/s²)	m/s²
u	running speed relative to the cylinder	m/s
$u_\theta,\ u_z$	circumferential (positive in the sense of rotation) and axial components of the running velocity	m/s
m	mass of the astronaut	kg

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Frames.** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ inertial, with O on the axis and $\mathbf{k}_0$ along it; $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ fixed to the hull, $\mathbf{k}_1 = \mathbf{k}_0$. The origin of S_1 is the fixed point O.

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

(the columns are unit vectors and mutually orthogonal).

- **Motion of S_1: $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$** ($\omega$ constant and $\mathbf{k}_1$ fixed), $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$.
- **Local cylindrical basis of the astronaut (in S_1).** With φ her azimuth measured in S_1: $\mathbf{e}_r = \cos\varphi\,\mathbf{i}_1 + \sin\varphi\,\mathbf{j}_1$ (from the axis to the floor), $\mathbf{e}_\theta = -\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1$ (sense of rotation), $\mathbf{k}_1$. The matrix with these columns is a simple rotation by φ: $\det = 1$, $RR^T = I$. Derivatives in S_1 (chain rule): $\left.\frac{d\mathbf{e}_r}{dt}\right|_1 = \dot{\varphi}\,\mathbf{e}_\theta$, $\left.\frac{d\mathbf{e}_\theta}{dt}\right|_1 = -\dot{\varphi}\,\mathbf{e}_r$.
- **Inertia forces** (Notes Eq. 4.62): $\mathbf{F}_A = \mathbf{0}$, $\mathbf{F}_E = \mathbf{0}$; $\mathbf{F}_C = m\omega^2\mathbf{r}_\perp$ (Slide 30); $\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$ (Slide 31).

PHASE 3

Step-by-Step Mathematical Deduction

(a) Spin rate for a given artificial gravity

> **Pedagogical Justification:** an astronaut at rest on the floor has $\mathbf{v}_1^P = \mathbf{0}$ and $\mathbf{a}_1^P = \mathbf{0}$, so Notes Eq. 4.63 becomes a static balance in S_1 between the floor reaction and the centrifugal force (the Coriolis force vanishes because $\mathbf{v}_1^P = \mathbf{0}$).

Position $\mathbf{r}_1^P = R\,\mathbf{e}_r$ (plus an axial part along $\mathbf{k}_1$ that does not contribute to $\mathbf{r}_\perp$), so $\mathbf{r}_\perp = R\,\mathbf{e}_r$ and

$$\mathbf{F}_C = -m\,\omega\mathbf{k}_1 \times (\omega\mathbf{k}_1 \times R\,\mathbf{e}_r) = -m\,\omega\mathbf{k}_1 \times (R\omega\,\mathbf{e}_\theta) = m\omega^2 R\,\mathbf{e}_r$$

using $\mathbf{k}_1 \times \mathbf{e}_r = \mathbf{e}_\theta$ and $\mathbf{k}_1 \times \mathbf{e}_\theta = -\mathbf{e}_r$. Then

$$m\,\mathbf{0} = \mathbf{N} + m\omega^2 R\,\mathbf{e}_r \quad\Longrightarrow\quad \mathbf{N} = -m\omega^2 R\,\mathbf{e}_r, \qquad g_{\text{art}} = \frac{|\mathbf{N}|}{m} = \omega^2 R$$

The floor pushes her toward the axis, and she feels a "gravity" $\omega^2 R$ pointing away from the axis. Setting $g_{\text{art}} = g$:

$$\boxed{\omega = \sqrt{\frac{g}{R}}}$$

Numbers ($R = 10$ m, $g = 9.81$ m/s^2):

$$\omega = \sqrt{\frac{9.81\ \text{m/s}^2}{10\ \text{m}}} = \sqrt{0.981}\ \text{s}^{-1} \approx 0.9905\ \text{rad/s}, \qquad T_{\text{rev}} = \frac{2\pi}{\omega} \approx 6.34\ \text{s}, \qquad \frac{60\,\omega}{2\pi} \approx 9.46\ \text{rpm}$$

and the floor moves at $\omega R \approx 9.90$ m/s in S_0.

(b) Artificial gravity felt while running

> **Pedagogical Justification:** now $\mathbf{v}_1^P \neq \mathbf{0}$, so the Coriolis force appears, and her relative path may be curved, so $\mathbf{a}_1^P \neq \mathbf{0}$. Both are computed in S_1 and inserted in Notes Eq. 4.63 to obtain $\mathbf{N}$. The statement does not fix the running direction, so we take a general constant relative velocity with circumferential part u_θ and axial part u_z (running "around" the hull is $u_z = 0$, $u_\theta = \pm u$).

Relative kinematics. $\mathbf{r}_1^P = R\,\mathbf{e}_r + z\,\mathbf{k}_1$ with $R\dot{\varphi} = u_\theta$, $\dot{z} = u_z$ (both constant). Differentiating in S_1 with the product and chain rules:

$$\mathbf{v}_1^P = R\dot{\varphi}\,\mathbf{e}_\theta + \dot{z}\,\mathbf{k}_1 = u_\theta\,\mathbf{e}_\theta + u_z\,\mathbf{k}_1$$

$$\mathbf{a}_1^P = \dot{u}_\theta\,\mathbf{e}_\theta + u_\theta\,(-\dot{\varphi}\,\mathbf{e}_r) + \dot{u}_z\,\mathbf{k}_1 = -\frac{u_\theta^2}{R}\,\mathbf{e}_r$$

(the relative path is a helix of radius R; in S_1 she needs a centripetal acceleration u_θ^2/R).

Inertia forces. $\mathbf{r}_\perp = R\,\mathbf{e}_r$ as before, so $\mathbf{F}_C = m\omega^2 R\,\mathbf{e}_r$. For the Coriolis force, $\mathbf{k}_1 \times \mathbf{e}_\theta = -\mathbf{e}_r$ and $\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$:

$$\mathbf{F}_{COR} = -2m\,\boldsymbol{\omega}\mathbf{k}_1 \times (u_\theta\,\mathbf{e}_\theta + u_z\,\mathbf{k}_1) = 2m\omega u_\theta\,\mathbf{e}_r$$

Newton's law in S_1 (Notes Eq. 4.63), $m\mathbf{a}_1^P = \mathbf{N} + \mathbf{F}_C + \mathbf{F}_{COR}$:

$$-m\frac{u_\theta^2}{R}\,\mathbf{e}_r = \mathbf{N} + m\omega^2 R\,\mathbf{e}_r + 2m\omega u_\theta\,\mathbf{e}_r$$

$$\mathbf{N} = -m\left(\omega^2 R + 2\omega u_\theta + \frac{u_\theta^2}{R}\right)\mathbf{e}_r = -\frac{m}{R}\left(\omega R + u_\theta\right)^2\mathbf{e}_r$$

The reaction is purely radial, so no friction is needed to keep the motion uniform. The artificial gravity she feels is

$$\boxed{g_{\text{art}} = \frac{(\omega R + u_\theta)^2}{R} = g\left(1 + \frac{u_\theta}{\omega R}\right)^2 = g + 2\omega u_\theta + \frac{u_\theta^2}{R}}$$

using $\omega^2 R = g$ from part (a). Particular cases:

- running around the hull **in the sense of rotation** ($u_\theta = +u$): $g_{\text{art}} = (\omega R + u)^2/R > g$ (she feels heavier);
- running **against the rotation** ($u_\theta = -u$): $g_{\text{art}} = (\omega R - u)^2/R < g$ for $u < 2\omega R$ (she feels lighter);
- running **along the axis** ($u_\theta = 0$, $u_z = u$): $g_{\text{art}} = g$ (the Coriolis force vanishes because $\mathbf{v}_1^P \parallel \boldsymbol{\omega}_{10}$).

Numbers ($R = 10$ m, $\omega = 0.9905$ rad/s): at $u = 2$ m/s, $g_{\text{art}} = 14.17$ m/s^2 ($1.44\,g$) with the rotation and 6.25 m/s^2 ($0.64\,g$) against it; at $u = 5$ m/s, 22.2 m/s^2 ($2.26\,g$) and 2.41 m/s^2 ($0.25\,g$).

(c) Condition for no artificial gravity

> **Pedagogical Justification:** $g_{\text{art}} = (\omega R + u_\theta)^2/R$ is a perfect square, so it vanishes only when its base vanishes.

$$g_{\text{art}} = 0 \iff \omega R + u_\theta = 0 \iff \boxed{u_\theta = -\omega R, \quad u_z = 0}$$

She must run around the hull, against the rotation, at the speed of the floor ωR (≈ 9.90 m/s for the numerical case, a sprinter's speed). Then, by Notes Eq. 3.26, her absolute velocity is $\mathbf{v}_0^P = \mathbf{v}_1^P + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = -\omega R\,\mathbf{e}_\theta + \omega R\,\mathbf{e}_\theta = \mathbf{0}$: she is at rest in inertial space, floating, while the floor slides under her feet. With an axial component ($u_z \neq 0$) and $u = \omega R$ the condition cannot be met, since then $|u_\theta| < \omega R$.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Inertial reading (cross-check).** In S_0 the astronaut moves on a circle (or helix) of radius R with tangential speed $\omega R + u_\theta$, so the floor must supply the centripetal force $m(\omega R + u_\theta)^2/R$: the same result. The rotating observer splits it into centrifugal $m\omega^2 R$, Coriolis $2m\omega u_\theta$ and relative centripetal mu_θ^2/R parts.
- **Limits:** $u_\theta \to 0$ gives $g_{\text{art}} = g$; for $|u_\theta| \ll \omega R$, $g_{\text{art}} \approx g \pm 2\omega u$, a correction due to the Coriolis force alone; a larger hull ($R \to \infty$ at fixed g) needs a smaller ω and makes the Coriolis correction $2\omega u = 2u\sqrt{g/R}$ smaller, which is why large habitats are more

comfortable.

- **Dimensions:** $[\sqrt{g/R}] = \sqrt{\mathrm{m\,s^{-2}/m}} = \mathrm{s^{-1}}$; $[\omega^2 R] = [2\omega u_\theta] = [u_\theta^2/R] = \mathrm{m\,s^{-2}}$; $[N] = \mathrm{kg\,m\,s^{-2}} = \mathrm{N}$.
- **Verification (sympy, `verify_P09.py`):** $\mathbf{a}_1^P$, $\mathbf{F}_C$, $\mathbf{F}_{COR}$ and $\mathbf{N}$ are recomputed; the inertial double derivative of $[{}_0R_1]\{\mathbf{r}_1^P\}$ equals the Notes Eq. 3.27 decomposition and gives the same $\mathbf{N}$; the numbers above are reproduced.
- **Cross-check with Topic 1.** The Topic 1 solution (intrinsic acceleration in S_0, circumferential running only) gives $g_{\text{art}} = (\omega R \pm u)^2/R$ and $u = \omega R$ against the rotation: identical. No official key exists for this problem.

Source: Problems.pdf (PDF page 8); no official key

PROBLEM 20

Heavy Particle on a Rotating Circumference

Original statement · Problems.pdf, p. 10

Energy in a Rotating Frame

A heavy particle is moving along a smooth circumference of radius R and center O. One of the diameters of the circumference is along the local vertical, and the circumference is rotating with constant angular velocity ω about it. The position of the particle with respect to the circumference is described by the angle θ between the particle and the lowest point of the circumference.

(a) Discuss the conservation of mechanical energy in the moving reference frame and compute its value.

(b) Compute the reaction of the circumference in the direction normal to its plane, as a function of θ.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** a particle P of mass m slides on a smooth circular wire of radius R and centre O; the wire turns with constant angular velocity ω about its vertical diameter. The angle θ is measured from the lowest point.
- **Why the moving frame.** In the frame S_1 attached to the wire the constraint is a fixed circle, the motion has one coordinate θ, and the reaction has two clear components (in the plane of the wire, normal to it). The question itself asks for the energy "in the moving reference frame", which requires the inertia forces of Notes Eq. 4.62 and their potentials (Slides 29–31).
- **Hypotheses:**
  1. S_0 fixed to the ground is inertial; O is fixed; uniform gravity $-g\,\mathbf{k}$.
  2. The wire is smooth: the reaction has no component along the tangent of the wire.
  3. ω is constant.

- **Degrees of freedom:** the particle is constrained to a curve fixed in S_1: **1 DOF**, θ.
- **Unknown reaction:** $\mathbf{N} = N_r\,\mathbf{e}_r + N_b\,\mathbf{j}_1$, with N_r in the plane of the wire (along the radius) and N_b normal to that plane.

Symbol	Meaning	SI unit
m	mass of the particle	kg
R	radius of the circumference	m
ω	spin rate of the circumference (constant)	rad/s
g	gravitational acceleration	m/s^2
θ	angle from the lowest point	rad
$N_r,\ N_b$	reaction components (radial in-plane, normal to the plane)	N
$\theta_0,\ \dot\theta_0$	initial angle and angular rate in S_1	rad, rad/s

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Frames.** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ inertial, $\mathbf{k}_0$ upward along the rotation axis. $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ attached to the wire: $\mathbf{k}_1 = \mathbf{k}_0$, $\mathbf{i}_1$ horizontal in the plane of the wire (on the side where $\theta > 0$), $\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{i}_1$ normal to the plane of the wire. With the wire plane at azimuth ωt:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- **Motion of S_1:** $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$.
- **Wire basis.** $\mathbf{e}_r = \sin\theta\,\mathbf{i}_1 - \cos\theta\,\mathbf{k}_1$ (from O to P), $\mathbf{e}_\theta = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{k}_1$ (tangent, increasing θ). Since $\mathbf{e}_r \times \mathbf{e}_\theta = (-\cos\theta)(\cos\theta)(\mathbf{k}_1 \times \mathbf{i}_1) + (\sin\theta)(\sin\theta)(\mathbf{i}_1 \times \mathbf{k}_1) = -\mathbf{j}_1$, the right-handed triad is $\{\mathbf{e}_r, \mathbf{e}_\theta, -\mathbf{j}_1\}$:

$$[{}_1R_w] = \begin{pmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \\ -\cos\theta & \sin\theta & 0 \end{pmatrix}, \qquad \det[{}_1R_w] = \sin\theta(0\cdot 0 + \sin\theta) - \cos\theta(0\cdot 0 - \cos\theta) = 1$$

and the columns are orthonormal ($\sin^2\theta + \cos^2\theta = 1$, $\sin\theta\cos\theta - \cos\theta\sin\theta = 0$), so $[{}_1R_w][{}_1R_w]^T = I$.

- **Derivatives in S_1** (chain rule): $\left.\frac{d\mathbf{e}_r}{dt}\right|_1 = \dot\theta\,(\cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{k}_1) = \dot\theta\,\mathbf{e}_\theta$ and $\left.\frac{d\mathbf{e}_\theta}{dt}\right|_1 = \dot\theta\,(-\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{k}_1) = -\dot\theta\,\mathbf{e}_r$.

PHASE 3

Step-by-Step Mathematical Deduction

Relative kinematics and forces in S_1

$$\mathbf{r}_1^P = R\,\mathbf{e}_r, \qquad \mathbf{v}_1^P = R\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a}_1^P = R\ddot\theta\,\mathbf{e}_\theta + R\dot\theta\left(-\dot\theta\,\mathbf{e}_r\right) = R\ddot\theta\,\mathbf{e}_\theta - R\dot\theta^2\,\mathbf{e}_r$$

(product rule on $R\dot\theta\,\mathbf{e}_\theta$). Forces seen from S_1 (Notes Eqs. 4.62–4.63):

- weight $-mg\,\mathbf{k}_1$ and reaction $\mathbf{N} = N_r\,\mathbf{e}_r + N_b\,\mathbf{j}_1$ (actual forces);
- $\mathbf{F}_A = \mathbf{0}$ (O fixed) and $\mathbf{F}_E = \mathbf{0}$ ($\boldsymbol{\alpha}_{10} = \mathbf{0}$);
- centrifugal force: $\mathbf{r}_\perp = R\sin\theta\,\mathbf{i}_1$ (the part of $\mathbf{r}_1^P$ normal to $\mathbf{k}_1$), so (Slide 30)

$$\mathbf{F}_C = -m\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P\right) = m\omega^2 R\sin\theta\,\mathbf{i}_1$$

Component check: $\omega\mathbf{k}_1\times\left(R\sin\theta\,\mathbf{i}_1 - R\cos\theta\,\mathbf{k}_1\right) = R\omega\sin\theta\,\mathbf{j}_1$, then $\omega\mathbf{k}_1\times R\omega\sin\theta\,\mathbf{j}_1 = -R\omega^2\sin\theta\,\mathbf{i}_1$, and the minus sign of $\mathbf{F}_C$ gives $+m\omega^2 R\sin\theta\,\mathbf{i}_1$;

- Coriolis force, with $\mathbf{k}_1\times\mathbf{i}_1 = \mathbf{j}_1$ and $\mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$:

$$\mathbf{F}_{COR} = -2m\,\boldsymbol{\omega}\mathbf{k}_1\times R\dot\theta\left(\cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{k}_1\right) = -2m\omega R\dot\theta\cos\theta\,\mathbf{j}_1$$

Equation of motion (projection on $\mathbf{e}_\theta$, where $\mathbf{N}$ has no component). Using $\mathbf{k}_1\cdot\mathbf{e}_\theta = \sin\theta$, $\mathbf{i}_1\cdot\mathbf{e}_\theta = \cos\theta$, $\mathbf{j}_1\cdot\mathbf{e}_\theta = 0$:

$$mR\ddot\theta = -mg\sin\theta + m\omega^2 R\sin\theta\cos\theta \quad\Longrightarrow\quad \ddot\theta = -\frac{g}{R}\sin\theta + \omega^2\sin\theta\cos\theta$$

(a) Mechanical energy in the moving frame

- **Justification:** E_1^P is conserved when every force that does work in S_1 derives from a potential (Slide 29). We check the forces one by one using their power in S_1, $\mathbf{F}\cdot\mathbf{v}_1^P$.
- **Reaction:** $\mathbf{N}\cdot\mathbf{v}_1^P = R\dot\theta\left(N_r\,\mathbf{e}_r + N_b\,\mathbf{j}_1\right)\cdot\mathbf{e}_\theta = 0$: the wire is smooth and **fixed in** S_1, so it does no work in S_1.
- **Coriolis force:** $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$ always (Slide 31); here explicitly $\mathbf{j}_1\cdot\mathbf{e}_\theta = 0$.
- **Euler force and origin force:** both zero.

- **Weight:** conservative, $V_g = mgz_1 = -mgR\cos\theta$ (zero at the height of O).
- **Centrifugal force:** $\boldsymbol{\omega}_{10}$ is constant, so it has the potential (Slide 30) $V_1 = -\frac{1}{2}m\omega^2 r_\perp^2 = -\frac{1}{2}m\omega^2 R^2\sin^2\theta$ (zero on the axis).

Therefore E_1^P **is conserved**:

$$\boxed{E_1^P = \frac{1}{2}mR^2\dot\theta^2 - mgR\cos\theta - \frac{1}{2}m\omega^2R^2\sin^2\theta = \frac{1}{2}mR^2\dot\theta_0^2 - mgR\cos\theta_0 - \frac{1}{2}m\omega^2R^2\sin^2\theta_0}$$

its value being fixed by the initial conditions $\theta(0)=\theta_0$, $\dot\theta(0)=\dot\theta_0$ (the statement gives none, so the constant is left in terms of them).

Check by differentiation (chain rule on each term, then the EOM):

$$\frac{dE_1^P}{dt} = mR^2\dot\theta\ddot\theta + mgR\sin\theta\,\dot\theta - m\omega^2R^2\sin\theta\cos\theta\,\dot\theta = mR^2\dot\theta\left[\ddot\theta + \frac{g}{R}\sin\theta - \omega^2\sin\theta\cos\theta\right] = 0$$

The energy in S_0 is not conserved. In S_0 the wire moves, and its reaction does work: with $\mathbf{v}_0^P = \mathbf{v}_1^P + \boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = R\dot\theta\,\mathbf{e}_\theta + \omega R\sin\theta\,\mathbf{j}_1$ (Notes Eq. 3.26), the power of $\mathbf{N}$ in S_0 is $\mathbf{N}\cdot\mathbf{v}_0^P = N_b\,\omega R\sin\theta \neq 0$. This is why the moving frame is the natural one for an energy integral here.

(b) Reaction normal to the plane of the circumference

> **Pedagogical Justification:** project Notes Eq. 4.63 on $\mathbf{j}_1$, normal to the plane. The relative motion stays in the plane, so $\mathbf{a}_1^P\cdot\mathbf{j}_1 = 0$; the weight and the centrifugal force lie in the plane; only N_b and the Coriolis force remain.

$$m\,\mathbf{a}_1^P\cdot\mathbf{j}_1 = 0 = N_b - 2m\omega R\dot\theta\cos\theta \quad\Longrightarrow\quad N_b = 2m\omega R\dot\theta\cos\theta$$

To express it as a function of θ, solve the energy integral of part (a) for $R\dot\theta$:

$$R^2\dot\theta^2 = \frac{2E_1^P}{m} + 2gR\cos\theta + \omega^2R^2\sin^2\theta \quad\Longrightarrow\quad R\dot\theta = \pm\sqrt{\frac{2E_1^P}{m} + 2gR\cos\theta + \omega^2R^2\sin^2\theta}$$

$$\boxed{N_b(\theta) = \pm\, 2m\omega\cos\theta\,\sqrt{\frac{2E_1^P}{m} + 2gR\cos\theta + \omega^2R^2\sin^2\theta}}$$

with the sign of $\dot\theta$ (positive when θ increases) and E_1^P from part (a).

In-plane component (for completeness). Projecting on $\mathbf{e}_r$ ($\mathbf{k}_1\cdot\mathbf{e}_r = -\cos\theta$, $\mathbf{i}_1\cdot\mathbf{e}_r = \sin\theta$):

$$-mR\dot\theta^2 = N_r + mg\cos\theta + m\omega^2R\sin^2\theta \quad\Longrightarrow\quad N_r = -mR\dot\theta^2 - mg\cos\theta - m\omega^2R\sin^2\theta$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Origin of N_b.** The normal-to-plane reaction is entirely due to the Coriolis force: the wire must push the particle sideways so that it keeps turning with the plane while its distance to the axis changes. It vanishes when $\dot\theta = 0$ and when $\cos\theta = 0$ (the particle crosses the horizontal diameter, where $\mathbf{v}_1^P \parallel \boldsymbol{\omega}_{10}$).
- **Relative equilibria** (extension, from $\ddot\theta = 0$ with $\dot\theta = 0$): $\sin\theta\,\left(\omega^2\cos\theta - g/R\right) = 0$, i.e. $\theta = 0$, $\theta = \pi$, and, if $\omega^2R > g$, $\cos\theta = g/(\omega^2R)$; at all of them $N_b = 0$.
- **Limit $\omega \to 0$:** $N_b \to 0$ and $E_1^P \to \tfrac{1}{2}mR^2\dot\theta^2 - mgR\cos\theta$, the energy of the simple pendulum.
- **Dimensions:** each term of E_1^P is $\mathrm{kg\,m^2\,s^{-2}} = \mathrm{J}$; $[N_b] = [m\omega R\dot\theta] = \mathrm{kg\,s^{-1}\,m\,s^{-1}} = \mathrm{N}$; inside the root, $[2E_1^P/m] = [gR] = [\omega^2R^2] = \mathrm{m^2\,s^{-2}}$.
- **Verification (sympy, `verify_P20.py`):** the inertial double derivative of $[{}_0R_1]\{\mathbf{r}_1^P\}$ equals the Notes Eq. 3.27 decomposition; solving $m\mathbf{a}_0^P = -mg\mathbf{k} + \mathbf{N}$ in S_0 returns the same EOM, N_b and N_r; $dE_1^P/dt = 0$ along the EOM, while dE_0^P/dt equals the power $N_b\,\omega R\sin\theta$ of the reaction in S_0.
- **Cross-check with Topic 3.** The Topic 3 solution gives the same E' (there called the effective energy) and $N_\phi = 2m\omega R\dot\theta\cos\theta$. No official key exists for this problem.

Source: Problems.pdf (PDF page 10); no official key

PROBLEM 21

Apparent Weight on a Spinning Planet

Original statement · Problems.pdf, p. 10

Centrifugal Force

On a spinning spherical planet of radius R and gravitational acceleration g_p, the bodies weigh twice as much on the poles as on the equator. Determine the duration of the day on that planet.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** a spherical planet of radius R spins with constant angular velocity Ω about its polar axis. At its surface the gravitational attraction is g_p per unit mass, directed toward the centre. A body of mass m rests on the ground.
- **Why the planet frame.** "Weight" is what a scale fixed to the ground reads: it is defined in the frame of the planet, where the body is at rest. In that frame the dynamics reduce to a static balance once the inertia forces (Notes Eq. 4.62) are added; the centrifugal force is what makes the weight depend on latitude.
- **Hypotheses:**
  1. The frame S_0 with origin at the centre O of the planet and axes fixed with respect to the stars is inertial (O unaccelerated).
  2. Ω is constant; the planet is a rigid sphere and g_p is the same everywhere on its surface (spherical mass distribution).
  3. The apparent weight is the magnitude of the contact force $\mathbf{N}$ the ground (or a scale) exerts on the body at rest.
- **Degrees of freedom:** the body is at rest in S_1: no motion to determine; the unknown is Ω (and then the day $T_{\text{day}} = 2\pi/\Omega$).

Symbol	Meaning	SI unit
R	radius of the planet	m
g_p	gravitational acceleration at the surface	m/s^2
Ω	spin rate of the planet	rad/s
λ	latitude of the body	rad

Symbol	Meaning	SI unit
m	mass of the body	kg
T_{day}	duration of the day (one rotation, sidereal)	s

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Frames.** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ inertial, $\mathbf{k}_0$ along the polar axis; $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ fixed to the planet, $\mathbf{k}_1 = \mathbf{k}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & -\sin\Omega t & 0 \\ \sin\Omega t & \cos\Omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\Omega t + \sin^2\Omega t = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- **Motion of S_1:** $\boldsymbol{\omega}_{10} = \Omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$.
- **Position of the body.** Choose $\mathbf{i}_1$ in the meridian plane of the body. At latitude λ: $\mathbf{r}_1^P = R\,\mathbf{e}_r$ with $\mathbf{e}_r = \cos\lambda\,\mathbf{i}_1 + \sin\lambda\,\mathbf{k}_1$; its part normal to the axis is $\mathbf{r}_\perp = R\cos\lambda\,\mathbf{i}_1$.

PHASE 3

Step-by-Step Mathematical Deduction

1. Static balance in the planet frame

> **Pedagogical Justification:** the body is at rest in S_1, so $\mathbf{v}_1^P = \mathbf{0}$ and $\mathbf{a}_1^P = \mathbf{0}$. In Notes Eq. 4.63, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$, the Coriolis force vanishes ($\mathbf{v}_1^P = \mathbf{0}$), $\mathbf{F}_A = \mathbf{0}$ and $\mathbf{F}_E = \mathbf{0}$; only the centrifugal force remains.

Actual forces: attraction $-mg_p\,\mathbf{e}_r$ and contact force $\mathbf{N}$. Centrifugal force (Slide 30):

$$\mathbf{F}_C = -m\,\Omega\mathbf{k}_1 \times (\Omega\mathbf{k}_1 \times R\,\mathbf{e}_r) = -m\,\Omega\mathbf{k}_1 \times (R\Omega\cos\lambda\,\mathbf{j}_1) = m\Omega^2 R\cos\lambda\,\mathbf{i}_1$$

(using $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$, $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$). Then

$$\mathbf{0} = -mg_p\,\mathbf{e}_r + \mathbf{N} + m\Omega^2 R\cos\lambda\,\mathbf{i}_1 \quad\Longrightarrow\quad \mathbf{N} = mg_p\,(\cos\lambda\,\mathbf{i}_1 + \sin\lambda\,\mathbf{k}_1) - m\Omega^2 R\cos\lambda\,\mathbf{i}_1$$

2. Weight at the pole and at the equator

- **Pole** ($\lambda = \pi/2$): $\cos\lambda = 0$, $\mathbf{r}_\perp = \mathbf{0}$, no centrifugal force:

$$\mathbf{N}_{\text{pole}} = m g_p \, \mathbf{k}_1, \qquad W_{\text{pole}} = m g_p$$

- **Equator** ($\lambda = 0$): $\mathbf{e}_r = \mathbf{i}_1$ and the centrifugal force is radial and outward:

$$\mathbf{N}_{\text{eq}} = m\left(g_p - \Omega^2 R\right)\mathbf{i}_1, \qquad W_{\text{eq}} = m\left(g_p - \Omega^2 R\right)$$

A body can rest on the equator only if $N_{\text{eq}} \geq 0$ (the ground can push but not pull), i.e. $\Omega^2 R \leq g_p$.

3. Condition $W_{\text{pole}} = 2W_{\text{eq}}$ and length of the day

$$m g_p = 2m\left(g_p - \Omega^2 R\right) \quad\Longrightarrow\quad 2\Omega^2 R = 2g_p - g_p = g_p \quad\Longrightarrow\quad \Omega = \sqrt{\frac{g_p}{2R}}$$

(this satisfies $\Omega^2 R = g_p/2 \leq g_p$. The other possibility, $m g_p = 2m(\Omega^2 R - g_p)$, would mean $\Omega^2 R > g_p$: equatorial bodies would fly off, so it is rejected.) The day is the time of one rotation:

$$\boxed{T_{\text{day}} = \frac{2\pi}{\Omega} = 2\pi\sqrt{\frac{2R}{g_p}}}$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Meaning.** The centrifugal force at the equator cancels half of the attraction, so $\Omega^2 R = g_p/2$: the planet spins at $1/\sqrt{2} \approx 0.71$ of the break-up rate $\sqrt{g_p/R}$ at which equatorial bodies would float ($W_{\text{eq}} = 0$).
- **Any latitude** (extension, same balance): $|\mathbf{N}|^2 = m^2\left[g_p^2 - \left(2g_p\Omega^2 R - \Omega^4 R^2\right)\cos^2\lambda\right]$, which with $\Omega^2 R = g_p/2$ gives $W(\lambda) = m g_p\sqrt{1 - \tfrac{3}{4}\cos^2\lambda}$: from $m g_p/2$ at the equator to $m g_p$ at the poles (e.g. $m g_p\sqrt{10}/4 \approx 0.79\, m g_p$ at 45°). Away from the poles and the equator $\mathbf{N}$ is not radial: the plumb line does not point to the centre.

- **Numbers** (illustration only, Earth-like $R = 6.371 \times 10^6$ m, $g_p = 9.81$ m/s^2): $T_{\text{day}} \approx 7.16 \times 10^3$ s ≈ 1.99 h.
- **Limit:** a slower spin ($\Omega \to 0$) gives $W_{\text{eq}} \to W_{\text{pole}}$; the ratio 2 requires a fast rotation.
- **Dimensions:** $[\Omega^2 R] = \mathrm{s}^{-2}\,\mathrm{m} = [g_p]$; $[T_{\text{day}}] = \sqrt{\mathrm{m}/(\mathrm{m\,s^{-2}})} = \mathrm{s}$.
- **Verification (sympy, `verify_P21.py`):** the inertial acceleration of a point at rest on the planet, obtained by differentiating $[{}_0R_1]\{\mathbf{r}_1^P\}$ twice, equals $\boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P)$; the balance gives $W_{\text{pole}} = m g_p$, $W_{\text{eq}} = m|g_p - \Omega^2 R|$, $\Omega = \sqrt{g_p/(2R)}$ and $T_{\text{day}} = 2\pi\sqrt{2R/g_p}$.
- **Cross-check with Topic 2.** The Topic 2 solution gives the same Ω and T_{day}. No official key exists for this problem.

Source: Problems.pdf (PDF page 10); no official key

PROBLEM 22

Particle on a Curve

Original statement · Problems.pdf, p. 10

1D Curve Constraint

A heavy point particle P of mass m is constrained to move without friction along the curve:

$$y = a\left(1 - \cos\frac{x}{a}\right) \tag{4}$$

in the range $-\pi a \le x \le \pi a$, where a is a positive constant. Initially, the particle is located at $x = 0$ with velocity v_0 along the positive x axis.

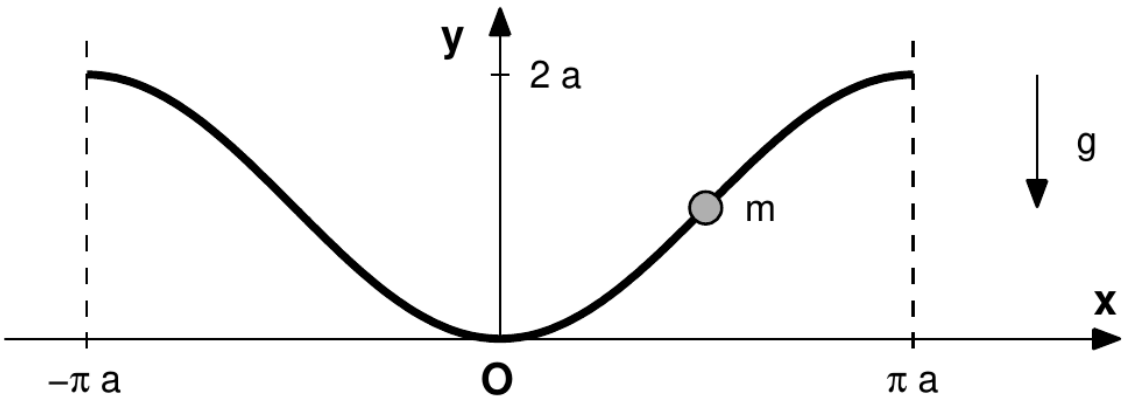


Figure: Heavy particle m on the curve y = a(1 - cos(x/a)) between x = -pi a and x = pi a, gravity g downward (Problems.pdf, p. 10).

(a) Reduce the problem to integrals.

(b) If $v_0 = \sqrt{ga}$, determine the minimum and maximum value of x along the motion.

(c) What should be the minimum value of v_0 in order to reach $x = \pi a$?

Extended Section (Moving Reference Frame):
Let's assume now that the curve containing the point particle rotates around the OY axis with constant angular velocity $\boldsymbol{\omega} = \Omega\mathbf{j}$.

(d) Define a convenient moving reference frame with origin in O, $OX_1Y_1Z_1$. Determine the inertia force that acts on the particle in this moving reference frame as a function of m, Ω, x_1 and $\dot{x}_1$, where x_1 represents the position coordinate of P along OX_1.

(e) Reduce the problem to quadratures and determine the value of the reaction force.

(f) Find out a transcendental equation whose solutions are the equilibrium positions of the point particle in $OX_1Y_1Z_1$.

(g) If Ω were a function of time, would it be possible to use the energy equation in the same way as before? Justify your answer.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle of mass m is constrained to move without friction along the smooth planar curve:

$$y = a\left(1 - \cos\frac{x}{a}\right), \qquad x \in [-\pi a, \pi a]$$

in a vertical plane under uniform downward gravity $\mathbf{g} = -g\mathbf{j}$.

- **CDOFs:** 1 degree of freedom, parametrized by Cartesian abscissa $x(t)$.
- **Initial State ($t = 0$):** $x(0) = 0 \implies y(0) = a(1 - \cos 0) = 0$, with initial velocity $\mathbf{v}_0 = v_0\mathbf{i}$ ($v_0 > 0$).
- **Parameters:** $m > 0$ [kg], $a > 0$ [m], $g > 0$ [m/s^2], $v_0 > 0$ [m/s].

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial Frame:** $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$.
- **Curve Differential Elements:**

$$\frac{dy}{dx} = \frac{d}{dx}\left[a\left(1 - \cos\frac{x}{a}\right)\right] = \sin\left(\frac{x}{a}\right)$$

Differential arc-length:

$$ds^2 = dx^2 + dy^2 = dx^2 + \sin^2\left(\frac{x}{a}\right)dx^2 = \left[1 + \sin^2\left(\frac{x}{a}\right)\right]dx^2$$

$$ds = \sqrt{1 + \sin^2\left(\frac{x}{a}\right)}\,dx$$

- **Velocity:**

$$v = \frac{ds}{dt} = \sqrt{1 + \sin^2\left(\frac{x}{a}\right)}\,\dot{x}$$

PHASE 3

Step-by-Step Mathematical Deduction

Part (a): Reduction of Motion to Integrals

Pedagogical Justification: The frictionless constraint does zero work ($W_N = 0$). Gravity is conservative with potential energy $V(y) = mgy = mga(1 - \cos(x/a))$. Total mechanical energy is constant.

$$E = \frac{1}{2}mv^2 + mgy = \frac{1}{2}mv_0^2 + mg(0) = \frac{1}{2}mv_0^2$$

$$\frac{1}{2}m\left[1+\sin^2\left(\frac{x}{a}\right)\right]\dot{x}^2+mga\left(1-\cos\frac{x}{a}\right)=\frac{1}{2}mv_0^2$$

Multiply by $2/m$:

$$\left[1+\sin^2\left(\frac{x}{a}\right)\right]\dot{x}^2=v_0^2-2ga\left(1-\cos\frac{x}{a}\right)$$

Isolate $\dot{x}=\frac{dx}{dt}$:

$$\frac{dx}{dt}=\pm\sqrt{\frac{v_0^2-2ga(1-\cos(x/a))}{1+\sin^2(x/a)}}$$

Separation of variables yields the quadrature:

$$\int_0^{x(t)}\sqrt{\frac{1+\sin^2(x'/a)}{v_0^2-2ga(1-\cos(x'/a))}}\,dx'=\pm t$$

Part (b): Turning Points for $v_0=\sqrt{ga}$

> **Pedagogical Justification:** Turning points occur where the velocity vanishes ($v=0\iff\dot{x}=0$).

$$v_0^2-2ga\left(1-\cos\frac{x}{a}\right)=0$$

Substitute $v_0^2=ga$:

$$ga-2ga\left(1-\cos\frac{x}{a}\right)=0$$

Divide by ga:

$$1 - 2\left(1 - \cos\frac{x}{a}\right) = 0 \implies 2\cos\frac{x}{a} = 1 \implies \cos\frac{x}{a} = \frac{1}{2}$$

In the domain $x/a \in [-\pi, \pi]$:

$$\frac{x}{a} = \pm\frac{\pi}{3} \implies x_{min} = -\frac{\pi}{3}a, \qquad x_{max} = +\frac{\pi}{3}a$$

Part (c): Minimum Velocity v_0 to Reach $x = \pi a$

> **Pedagogical Justification:** At the crest $x = \pi a$, the height is $y(\pi a) = a(1 - \cos\pi) = a(1 - (-1)) = 2a$. For the particle to reach this crest, its kinetic energy at the peak must be non-negative ($T(\pi a) \geq 0$).

$$E \geq V(\pi a) \implies \frac{1}{2}mv_0^2 \geq mg(2a) = 2mga$$

$$v_0^2 \geq 4ga \implies v_{0,min} = \sqrt{4ga} = 2\sqrt{ga}$$

Part (d): Moving Reference Frame with Rotation $\boldsymbol{\omega} = \Omega\mathbf{j}$

- Frame $\mathcal{F}_1 = \{O; \mathbf{i}_1, \mathbf{j}, \mathbf{k}_1\}$ rotating with $\boldsymbol{\Omega} = \Omega\mathbf{j}$.
- Particle position: $\mathbf{r}' = x_1\mathbf{i}_1 + y(x_1)\mathbf{j}$.
- Relative velocity: $\mathbf{v}' = \dot{x}_1\mathbf{i}_1 + \sin(x_1/a)\dot{x}_1\mathbf{j}$.
- Inertia forces in $\mathcal{F}_1$:

1. Centrifugal Force:

$$\mathbf{F}_{cf} = -m\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}') = m\Omega^2 x_1\mathbf{i}_1$$

2. Coriolis Force:

$$\mathbf{F}_{Cor} = -2m\boldsymbol{\Omega} \times \mathbf{v}' = -2m(\Omega\mathbf{j}) \times (\dot{x}_1\mathbf{i}_1 + \dot{y}_1\mathbf{j}) = 2m\Omega\dot{x}_1\,\mathbf{k}_1$$

Part (e): Quadratures and Reaction Force in the Moving Frame

- Effective potential in $\mathcal{F}_1$:

$$V_{eff}(x_1) = mgy(x_1) - \frac{1}{2}m\Omega^2 x_1^2 = mga\left(1 - \cos\frac{x_1}{a}\right) - \frac{1}{2}m\Omega^2 x_1^2$$

- Energy conservation in $\mathcal{F}_1$:

$$E' = \frac{1}{2}m\left[1 + \sin^2(x_1/a)\right]\dot{x}_1^2 + mga\left(1 - \cos\frac{x_1}{a}\right) - \frac{1}{2}m\Omega^2 x_1^2 = \text{constant}$$

- Reaction force: $\mathbf{N} = N_n\mathbf{n} + N_z\mathbf{k}_1$.

Along the binormal $\mathbf{k}_1$: $N_z = -F_{Cor,z} = -2m\Omega\dot{x}_1$.

Part (f): Transcendental Equilibrium Equation

Equilibrium occurs where $\frac{dV_{eff}}{dx_1} = 0$:

$$\frac{dV_{eff}}{dx_1} = mg\sin\left(\frac{x_1}{a}\right) - m\Omega^2 x_1 = 0 \implies g\sin\left(\frac{x_1}{a}\right) - \Omega^2 x_1 = 0$$

Part (g): Case $\Omega = \Omega(t)$

If Ω varies with time, the centrifugal potential becomes explicitly time-dependent ($\frac{\partial V_{cf}}{\partial t} = -m\Omega\dot{\Omega}x_1^2 \neq 0$) and an Euler inertia force $\mathbf{F}_{Euler} = -m\dot{\boldsymbol{\Omega}} \times \mathbf{r}'$ appears, which does non-zero work. Thus, mechanical energy in $\mathcal{F}_1$ is **not** conserved.

PHASE 4

Physical Interpretation & Dimensional Verification

- The potential well has width $2\pi a$ and barrier height $2ga$. Rotation creates a centrifugal destabilizing potential $-\frac{1}{2}m\Omega^2 x_1^2$ so that the origin loses stability for $\Omega^2 > g/a$. Off-centre equilibria satisfy $\sin u = \lambda u$ with $u = x_1/a$ and $\lambda = \Omega^2 a/g$; since $\sin u/u$ decreases monotonically from 1 to 0 on $(0, \pi]$, they exist (one pair $\pm u^*$ with $0 < u^* < \pi$) only for $\lambda < 1$, i.e. $\Omega^2 < g/a$, and there they are unstable ($V''_{eff} = (mg/a)(\cos u^* - \lambda) < 0$).
- **Erratum:** The first version stated that off-centre equilibria appear for $\Omega^2 > g/a$; they exist only for $\Omega^2 < g/a$ (subcritical case) and are unstable. Numeric check: a scan of $\sin u = \lambda u$ on $(0, \pi]$ finds one root for $\lambda = 0.5$ ($u^* = 1.8955$) and for 0.9, and none for $\lambda = 1, 1.1, 2$.

Source: Problems.pdf

PROBLEM 25

Ijon Tichy's Space Walk

Original statement · Problems.pdf, p. 13

Orbiting Frame + Inertia Forces

A spacecraft S is orbiting a celestial body F. The trajectory of the spacecraft with respect to an inertial reference frame Fxy is given by

$$r = \frac{p}{1 + e\cos\nu},$$

where p and e are known constants and $\nu(t)$ is a known function of time. The space explorer Ijon Tichy needs to perform some extra-vehicular activities during his Seventh Voyage (from *The Star Diaries* by Stanislaw Lem). He attaches himself with a massless cord of length L to the spacecraft and goes for a space walk. The cord remains in tension at all times. The astronaut is very light compared to the spacecraft, so his motion does not perturb the motion of the spacecraft. Furthermore, we assume that we can consider him as a point particle of mass m. We define a moving reference frame Sx_1y_1 as depicted in the figure.

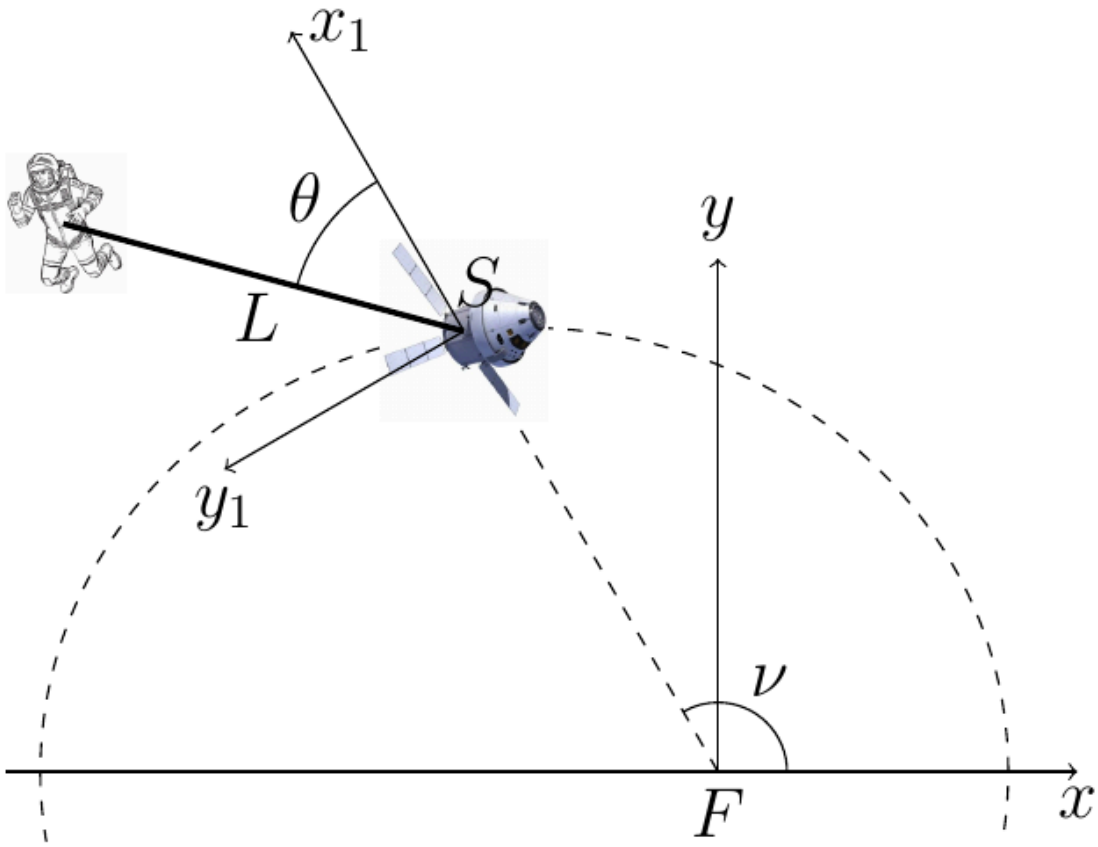


Figure: Spacecraft S on its orbit around F, true anomaly ν, moving axes Sx1 (radial) and Sy1, astronaut on a cord of length L at angle θ from Sx1 (Problems.pdf, p. 13).

(a) Determine the acceleration of the spacecraft with respect to Fxy as a function of time.

(b) Determine the tangent vector to the trajectory of S as a function of time.

(c) Determine the relative velocity and relative acceleration of Ijon Tichy with respect to Sx_1y_1 as a function of θ and its time derivatives.

(d) Determine the fictitious forces acting on Ijon Tichy if we analyze the problem with respect to the frame Sx_1y_1.

The length L is very small compared to the distance between F and S. This allows us to take an approximation: in the following, assume that the gravitational acceleration on the astronaut and on the spacecraft are exactly the same vector (magnitude and direction) as a function of time. (In

actual problems the small differences in gravity at the position of the astronaut and the spacecraft can give rise to small but important effects, and this approximation is usually not taken for that reason.)

(e) Write down one ordinary differential equation that once solved (with given initial conditions) would yield the angle θ as a function of time (you do not need to solve it).

(f) Assuming that you solved the previous equation, and therefore knew $\theta(t)$, determine the value of the tension on the cord as a function of time.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** a spacecraft S moves on the conic $r = p/(1 + e\cos\nu)$ around the body F, with the true anomaly $\nu(t)$ a known function of time. The astronaut P (mass m) is tied to S by a massless cord of length L that is always taut; θ is the angle from the axis Sx_1 to the cord SP (figure).
- **Hypotheses:** Fxy is inertial. The motion of S is prescribed (the astronaut does not perturb it). The motion is planar, in the orbital plane Fxy. The cord is massless and taut, so it pulls P towards S with a tension $T > 0$. From part (e) on, the gravitational acceleration at P equals the one at S.
- **Why a moving frame:** the cord fixes the distance $SP = L$, so in a frame attached to S the astronaut moves on a circle of radius L and one angle, θ, describes it. The price is that Sx_1y_1 accelerates and rotates, so inertia forces must be added (Notes Eqs. 4.62–4.63).
- **Degrees of freedom:** the astronaut moves in the plane (2 coordinates) with one constraint, $|\mathbf{SP}| = L$, so CDOF $= 2 - 1 = 1$, with generalized coordinate θ. The spacecraft adds no unknown, because $\nu(t)$ is given.

Symbol	Meaning	SI unit
p	semi-latus rectum of the orbit of S	m
e	eccentricity of the orbit	dimensionless
$\nu(t)$	true anomaly of S (known)	rad
$r(t)$	distance FS	m
L	cord length	m

Symbol	Meaning	SI unit
m	mass of the astronaut	kg
$\theta(t)$	angle from Sx_1 to the cord	rad
$T(t)$	cord tension	N
μ_F	gravitational parameter GM_F of F (used in the official key only)	$\mathrm{m}^3/\mathrm{s}^2$

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial frame:** $S_0 \equiv Fxy = \{F; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, with $\mathbf{k} = \mathbf{i} \times \mathbf{j}$ normal to the orbital plane.
- **Moving frame:** $S_1 \equiv Sx_1y_1 = \{S; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$. From the figure, Sx_1 lies on the line FS and points away from F, and Sy_1 is obtained from Sx_1 by a $+90^\circ$ turn about $\mathbf{k}$ (it points in the direction of increasing ν). Hence $\mathcal{B}_1$ is the polar basis of the orbit:

$$\mathbf{i}_1 = \cos\nu\,\mathbf{i} + \sin\nu\,\mathbf{j}, \qquad \mathbf{j}_1 = -\sin\nu\,\mathbf{i} + \cos\nu\,\mathbf{j}, \qquad \mathbf{k}_1 = \mathbf{k}$$

- **Change-of-basis matrix** (columns are $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ written in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\nu & -\sin\nu & 0 \\ \sin\nu & \cos\nu & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\nu + \sin^2\nu = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

The product $[{}_0R_1][{}_0R_1]^T$ has diagonal entries $\cos^2\nu + \sin^2\nu = 1$ and off-diagonal entries $\cos\nu\sin\nu - \sin\nu\cos\nu = 0$.

- **Angular velocity and acceleration of S_1:** S_1 performs a simple rotation of angle $\nu(t)$ about $\mathbf{k}$ (theory note, Sec. 5.5; Slide 16), so

$$\boldsymbol{\omega}_{10} = \dot\nu\,\mathbf{k}, \qquad \boldsymbol{\alpha}_{10} = \left.\frac{d(\dot\nu\,\mathbf{k})}{dt}\right|_0 = \ddot\nu\,\mathbf{k}$$

- **Origin of S_1:** $\mathbf{r}_0^S = r\,\mathbf{i}_1$, with $r = p/(1 + e\cos\nu)$; $\mathbf{v}_0^S$ and $\mathbf{a}_0^S$ are computed in part (a).
- **Relative position of the astronaut** (figure: θ is measured from $\mathbf{i}_1$ towards $\mathbf{j}_1$). Define the cord unit vectors

$$\mathbf{e}_c = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1, \qquad \mathbf{e}_\theta = -\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1, \qquad \mathbf{r}_1^P = L\,\mathbf{e}_c$$

$\{\mathbf{e}_c, \mathbf{e}_\theta, \mathbf{k}\}$ is again a rotation of $\mathcal{B}_1$ by θ about $\mathbf{k}$, with determinant $\cos^2\theta + \sin^2\theta = 1$.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Acceleration of the spacecraft with respect to Fxy

> **Pedagogical Justification:** $\mathbf{r}_0^S = r\,\mathbf{i}_1$ is written in the rotating basis $\mathcal{B}_1$, so its derivative in S_0 follows from the Coriolis theorem (Notes Eq. 3.8, Slide 17): derivative of the components in S_1 plus $\boldsymbol{\omega}_{10} \times (\cdot)$.

Velocity. In S_1 only the component r changes, and $\dot{\nu}\,\mathbf{k} \times r\,\mathbf{i}_1 = r\dot{\nu}\,(\mathbf{k} \times \mathbf{i}_1) = r\dot{\nu}\,\mathbf{j}_1$:

$$\mathbf{v}_0^S = \left.\frac{d(r\,\mathbf{i}_1)}{dt}\right|_1 + \boldsymbol{\omega}_{10} \times r\,\mathbf{i}_1 = \dot{r}\,\mathbf{i}_1 + r\dot{\nu}\,\mathbf{j}_1$$

Acceleration. Apply the theorem again. The derivative of the components in S_1 uses the product rule on $r\dot{\nu}$, $\frac{d}{dt}(r\dot{\nu}) = \dot{r}\dot{\nu} + r\ddot{\nu}$:

$$\left.\frac{d\mathbf{v}_0^S}{dt}\right|_1 = \ddot{r}\,\mathbf{i}_1 + (\dot{r}\dot{\nu} + r\ddot{\nu})\,\mathbf{j}_1$$

The rotation term, with $\mathbf{k} \times \mathbf{i}_1 = \mathbf{j}_1$ and $\mathbf{k} \times \mathbf{j}_1 = -\mathbf{i}_1$:

$$\boldsymbol{\omega}_{10} \times \mathbf{v}_0^S = \dot{\nu}\,\mathbf{k} \times (\dot{r}\,\mathbf{i}_1 + r\dot{\nu}\,\mathbf{j}_1) = \dot{r}\dot{\nu}\,\mathbf{j}_1 - r\dot{\nu}^2\,\mathbf{i}_1$$

Adding both:

$$\mathbf{a}_0^S = \left(\ddot{r} - r\dot{\nu}^2\right)\mathbf{i}_1 + (2\dot{r}\dot{\nu} + r\ddot{\nu})\,\mathbf{j}_1$$

Derivatives of $r(\nu(t))$. Write $r = p\,D^{-1}$ with $D = 1 + e\cos\nu$. By the chain rule, $\dfrac{dD}{dt} = \dfrac{dD}{d\nu}\dot\nu = -e\sin\nu\,\dot\nu$, so

$$\dot r = \frac{dr}{dD}\frac{dD}{dt} = -p\,D^{-2}\left(-e\sin\nu\,\dot\nu\right) = \frac{p\,e\,\dot\nu\sin\nu}{(1+e\cos\nu)^2}$$

Differentiate again with the product rule on $N = pe\,\dot\nu\sin\nu$ and D^{-2}: $\dot N = pe\left(\ddot\nu\sin\nu + \dot\nu^2\cos\nu\right)$ (chain rule $\frac{d}{dt}\sin\nu = \cos\nu\,\dot\nu$) and $\frac{d}{dt}D^{-2} = -2D^{-3}\dot D = 2e\sin\nu\,\dot\nu\,D^{-3}$:

$$\ddot r = \frac{pe\left(\ddot\nu\sin\nu + \dot\nu^2\cos\nu\right)}{(1+e\cos\nu)^2} + \frac{2pe^2\dot\nu^2\sin^2\nu}{(1+e\cos\nu)^3}$$

Since $\nu(t)$ is known, $r, \dot r, \ddot r, \dot\nu, \ddot\nu$ are known functions of time, and so is $\mathbf{a}_0^S$. In $\mathcal{B}_0$, $\mathbf{a}_0^S = \left[(\ddot r - r\dot\nu^2)\cos\nu - (2\dot r\dot\nu + r\ddot\nu)\sin\nu\right]\mathbf{i} + \left[(\ddot r - r\dot\nu^2)\sin\nu + (2\dot r\dot\nu + r\ddot\nu)\cos\nu\right]\mathbf{j}$.

(b) Tangent vector to the trajectory of S

> **Pedagogical Justification:** the unit tangent is the velocity divided by the speed, $\mathbf{u}_t = \mathbf{v}_0^S/|\mathbf{v}_0^S|$ (intrinsic frame of Topic 1). Taking $\dot\nu > 0$ (prograde motion, as in the figure) the common factor $r\dot\nu$ can be extracted.

From (a), $\dot r = r\dot\nu\,\dfrac{e\sin\nu}{1+e\cos\nu}$ (because $\dfrac{p}{(1+e\cos\nu)^2} = \dfrac{r}{1+e\cos\nu}$). Then

$$\mathbf{v}_0^S = \frac{r\dot\nu}{1+e\cos\nu}\left[e\sin\nu\,\mathbf{i}_1 + (1+e\cos\nu)\,\mathbf{j}_1\right]$$

The norm of the bracket is $\sqrt{e^2\sin^2\nu + 1 + 2e\cos\nu + e^2\cos^2\nu} = \sqrt{1+2e\cos\nu+e^2}$, hence

$$\mathbf{u}_t(t) = \frac{e\sin\nu\,\mathbf{i}_1 + (1+e\cos\nu)\,\mathbf{j}_1}{\sqrt{1+2e\cos\nu+e^2}}, \qquad |\mathbf{v}_0^S| = \frac{r\dot\nu\sqrt{1+2e\cos\nu+e^2}}{1+e\cos\nu}$$

with $\nu = \nu(t)$. In $\mathcal{B}_0$, substituting $\mathbf{i}_1$ and $\mathbf{j}_1$, the $\mathbf{i}$ component of the bracket is $e\sin\nu\cos\nu - (1+e\cos\nu)\sin\nu = -\sin\nu$ and the $\mathbf{j}$ component is $e\sin^2\nu + (1+e\cos\nu)\cos\nu = e + \cos\nu$, so $\mathbf{u}_t = \left[-\sin\nu\,\mathbf{i} + (e+\cos\nu)\,\mathbf{j}\right]/\sqrt{1+2e\cos\nu+e^2}$. For a circular orbit ($e = 0$) $\mathbf{u}_t = \mathbf{j}_1$, perpendicular to the radius, as expected.

(c) Relative velocity and acceleration of the astronaut in Sx_1y_1

> **Pedagogical Justification:** in S_1 the basis $\mathcal{B}_1$ is constant, so only the components of $\mathbf{r}_1^P = L\cos\theta\,\mathbf{i}_1 + L\sin\theta\,\mathbf{j}_1$ are differentiated (Notes Eq. 3.7); L is constant.

Chain rule: $\frac{d}{dt}\cos\theta = -\sin\theta\,\dot\theta$, $\frac{d}{dt}\sin\theta = \cos\theta\,\dot\theta$:

$$\mathbf{v}_1^P = \left.\frac{d\mathbf{r}_1^P}{dt}\right|_1 = -L\dot\theta\sin\theta\,\mathbf{i}_1 + L\dot\theta\cos\theta\,\mathbf{j}_1 = L\dot\theta\,\mathbf{e}_\theta$$

Differentiate again; product rule on $\dot\theta\sin\theta$: $\frac{d}{dt}(\dot\theta\sin\theta) = \ddot\theta\sin\theta + \dot\theta^2\cos\theta$, and on $\dot\theta\cos\theta$: $\ddot\theta\cos\theta - \dot\theta^2\sin\theta$:

$$\mathbf{a}_1^P = -L\left(\ddot\theta\sin\theta + \dot\theta^2\cos\theta\right)\mathbf{i}_1 + L\left(\ddot\theta\cos\theta - \dot\theta^2\sin\theta\right)\mathbf{j}_1 = L\ddot\theta\,\mathbf{e}_\theta - L\dot\theta^2\,\mathbf{e}_c$$

This is uniform-radius circular motion: a tangential part $L\ddot\theta$ and a centripetal part $L\dot\theta^2$ towards S.

(d) Fictitious forces in Sx_1y_1

> **Pedagogical Justification:** in the non-inertial frame S_1, Newton's law reads $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63), with the four inertia forces of Notes Eq. 4.62 (Slide 28). Each one is computed with the data of Phase 2.

1. Force due to the acceleration of the origin ($\mathbf{F}_A = -m\mathbf{a}_0^S$), with $\mathbf{a}_0^S$ from (a):

$$\mathbf{F}_A = -m\left(\ddot r - r\dot\nu^2\right)\mathbf{i}_1 - m\left(2\dot r\dot\nu + r\ddot\nu\right)\mathbf{j}_1$$

2. Euler force ($\mathbf{F}_E = -m\boldsymbol{\alpha}_{10}\times\mathbf{r}_1^P$). With $\mathbf{k}\times\mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}\times\mathbf{j}_1 = -\mathbf{i}_1$:

$$\boldsymbol{\alpha}_{10}\times\mathbf{r}_1^P = \ddot\nu\,\mathbf{k}\times(L\cos\theta\,\mathbf{i}_1 + L\sin\theta\,\mathbf{j}_1) = L\ddot\nu\,(\cos\theta\,\mathbf{j}_1 - \sin\theta\,\mathbf{i}_1) \;\Rightarrow\; \mathbf{F}_E = mL\ddot\nu\,(\sin\theta\,\mathbf{i}_1 - \cos\theta\,\mathbf{j}_1) = -mL\ddot\nu\,\mathbf{e}_\theta$$

3. Centrifugal force ($\mathbf{F}_C = -m\boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P)$). First $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = L\dot\nu\,\mathbf{e}_\theta$ (same computation as above with $\dot\nu$), then $\dot\nu\,\mathbf{k}\times L\dot\nu\,\mathbf{e}_\theta = L\dot\nu^2\,(-\sin\theta\,\mathbf{j}_1 - \cos\theta\,\mathbf{i}_1) = -L\dot\nu^2\,\mathbf{e}_c$:

$$\mathbf{F}_C = mL\dot{\nu}^2\,\mathbf{e}_c = mL\dot{\nu}^2\,(\cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1)$$

in agreement with $\mathbf{F}_C = m\omega_{10}^2\mathbf{r}_\perp$ (Slide 30), since $\mathbf{r}_1^P$ is perpendicular to $\mathbf{k}$.

4. Coriolis force ($\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$). With $\mathbf{k} \times \mathbf{e}_\theta = -\mathbf{e}_c$:

$$\mathbf{F}_{COR} = -2m\,\dot{\nu}\,\mathbf{k} \times L\dot{\theta}\,\mathbf{e}_\theta = 2mL\dot{\nu}\dot{\theta}\,\mathbf{e}_c = 2mL\dot{\nu}\dot{\theta}\,(\cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1)$$

The total inertia force is $\mathbf{F}_i = \mathbf{F}_A + \mathbf{F}_E + \mathbf{F}_C + \mathbf{F}_{COR}$. Because $\nu(t)$ is in general not uniform ($\ddot{\nu} \neq 0$ on an elliptic orbit), the Euler force is present.

(e) Differential equation for $\theta(t)$

> **Pedagogical Justification:** the real forces on the astronaut are the cord tension $\mathbf{T} = -T\,\mathbf{e}_c$ (towards S) and gravity $m\mathbf{g}_P$. Projecting Notes Eq. 4.63 on $\mathbf{e}_\theta$ removes the unknown tension, because $\mathbf{e}_c \cdot \mathbf{e}_\theta = 0$.

Gravity hypothesis. S is in free flight around F (its conic orbit is a Keplerian orbit, Topic 4), so its acceleration is the gravitational acceleration at S: $\mathbf{a}_0^S = \mathbf{g}_S$. The statement assumes $\mathbf{g}_P = \mathbf{g}_S$. Hence the weight cancels the force due to the acceleration of the origin exactly:

$$m\mathbf{g}_P + \mathbf{F}_A = m\mathbf{a}_0^S - m\mathbf{a}_0^S = \mathbf{0}$$

Newton's law in S_1:

$$m\left(L\ddot{\theta}\,\mathbf{e}_\theta - L\dot{\theta}^2\,\mathbf{e}_c\right) = -T\,\mathbf{e}_c - mL\ddot{\nu}\,\mathbf{e}_\theta + mL\dot{\nu}^2\,\mathbf{e}_c + 2mL\dot{\nu}\dot{\theta}\,\mathbf{e}_c$$

Component along $\mathbf{e}_\theta$: $mL\ddot{\theta} = -mL\ddot{\nu}$, that is

$$\ddot{\theta} + \ddot{\nu}(t) = 0$$

This is the requested ODE; $\ddot{\nu}(t)$ is a known function. (Without the approximation, the same projection gives $L(\ddot{\theta} + \ddot{\nu}) = (\mathbf{g}_P - \mathbf{a}_0^S) \cdot \mathbf{e}_\theta$, the tidal term that the statement neglects.)

Its integral (not requested, but immediate). Integrate from 0 to t; the antiderivative of $\ddot\theta + \ddot\nu$ is $\dot\theta + \dot\nu$, and Barrow's rule gives

$$\int_0^t \left(\ddot\theta + \ddot\nu\right) dt' = \Big[\dot\theta + \dot\nu\Big]_0^t = \dot\theta(t) + \dot\nu(t) - \dot\theta(0) - \dot\nu(0) = 0 \;\Rightarrow\; \dot\theta + \dot\nu = \Omega_c \equiv \dot\theta(0) + \dot\nu(0)$$

Integrating once more, $[\theta + \nu]_0^t = \Omega_c t$, so $\theta(t) = \theta(0) + \nu(0) - \nu(t) + \Omega_c t$.

(f) Tension on the cord

Component along $\mathbf{e}_c$ of the equation of (e):

$$-mL\dot\theta^2 = -T + mL\dot\nu^2 + 2mL\dot\nu\dot\theta \;\Rightarrow\; T = mL\left(\dot\theta^2 + 2\dot\theta\dot\nu + \dot\nu^2\right) = mL\left(\dot\theta + \dot\nu\right)^2$$

With $\theta(t)$ known (and $\nu(t)$ given), $T(t) = mL\left(\dot\theta(t) + \dot\nu(t)\right)^2$; by the integral of (e) it is in fact constant, $T = mL\,\Omega_c^2$. The cord is taut ($T > 0$) as long as $\Omega_c \neq 0$.

General form (as in the official key). Keeping gravity and $\mathbf{a}_0^S$ separate, with $\mathbf{g}_P = \mathbf{g}_S = -\dfrac{\mu_F}{r^2}\mathbf{i}_1$, the $\mathbf{e}_c$ component reads $-mL\dot\theta^2 = -T + m(\mathbf{g}_S - \mathbf{a}_0^S)\cdot\mathbf{e}_c + mL\dot\nu^2 + 2mL\dot\nu\dot\theta$. With $\mathbf{i}_1\cdot\mathbf{e}_c = \cos\theta$, $\mathbf{j}_1\cdot\mathbf{e}_c = \sin\theta$:

$$T = mL\dot\theta^2 - \frac{\mu_F m\cos\theta}{r^2} + mL\dot\nu^2 + 2mL\dot\theta\dot\nu + m\left(r\dot\nu^2 - \ddot r\right)\cos\theta - m\left(2\dot r\dot\nu + r\ddot\nu\right)\sin\theta$$

For the Keplerian orbit of S, $\ddot r - r\dot\nu^2 = -\mu_F/r^2$ and $2\dot r\dot\nu + r\ddot\nu = \frac{1}{r}\frac{d}{dt}\left(r^2\dot\nu\right) = 0$ (Topic 4: conservation of $h = r^2\dot\nu$ and $p = h^2/\mu_F$), so the last three terms cancel each other and this expression reduces to $T = mL(\dot\theta + \dot\nu)^2$.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Meaning of $\dot\theta + \dot\nu = \text{const}$.** $\theta + \nu$ is the angle of the cord measured from the inertial axis Fx. With equal gravity on S and P, both fall together and the astronaut sees no net gravity: relative to a non-rotating frame attached to S he simply turns around S at the constant rate Ω_c, and the cord supplies the centripetal force $mL\Omega_c^2$. In the rotating frame Sx_1y_1 this appears as the balance between the Euler force (which drives $\ddot\theta = -\ddot\nu$), and the centrifugal and Coriolis forces (which, together with $mL\dot\theta^2$, add up to the tension).

- **Limiting cases.** Circular orbit ($e = 0$): $\dot{\nu}$ is constant, $\ddot{\nu} = 0$, the Euler force vanishes and $\dot{\theta}$ is constant. If the astronaut does not rotate in inertial space ($\Omega_c = 0$), the tension vanishes and the taut-cord hypothesis is lost: he would drift freely at constant relative position in a non-rotating frame.
- **Official key (Problems.pdf, PDF pages 73–74):** it gives $\mathbf{a}_0^S$, $\dot{r}$, $\ddot{r}$, the Euler force $mL\ddot{\nu}(\sin\theta\,\mathbf{i}_1 - \cos\theta\,\mathbf{j}_1)$ and the general tension formula written above; all of them coincide with the results of this solution, and the general formula reduces to $mL(\dot{\theta}+\dot{\nu})^2$ for the Keplerian orbit.
- **Dimensions:** $[\dot{r}] = \mathrm{m}\cdot\mathrm{s}^{-1}$ (p in m times $\dot{\nu}$ in s^{-1}); $[\ddot{r}] = [r\dot{\nu}^2] = [r\ddot{\nu}] = \mathrm{m}\cdot\mathrm{s}^{-2}$; $[mL\ddot{\nu}] = [mL\dot{\nu}^2] = [mL\dot{\nu}\dot{\theta}] = \mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2} = \mathrm{N}$; $[\mu_F m/r^2] = \mathrm{m}^3\mathrm{s}^{-2}\cdot\mathrm{kg}/\mathrm{m}^2 = \mathrm{N}$; $\mathbf{u}_t$ is dimensionless.
- **Symbolic check (sympy, `verify_P25.py`):** $\mathbf{a}_0^S$ from differentiating $r(\nu(t))\,\mathbf{i}_1$ twice in $\mathcal{B}_0$, the formulas for $\dot{r}$ and $\ddot{r}$, the unit tangent, $\mathbf{v}_1^P$, $\mathbf{a}_1^P$, the decomposition of Notes Eq. 3.27, the four inertia forces, $\ddot{\theta} = -\ddot{\nu}$, $T = mL(\dot{\theta}+\dot{\nu})^2$, the equality of the general formula with the key, and $\mathbf{a}_0^S = -\mu_F/r^2\,\mathbf{i}_1$ for $\dot{\nu} = h/r^2$, $p = h^2/\mu_F$: all verified.

Source: Problems.pdf (PDF page 13); key on PDF pages 73-74

PROBLEM 26

Heavy Particle on a Rotating Half-Circumference (Moving-Frame Method)

Original statement · Problems.pdf, p. 14

Rotating Wire + Inertia Forces

A heavy point particle P of mass m is constrained to move without friction along the half-circumference wire of radius a shown in the figure. The wire containing the point particle rotates around the OY axis with constant angular velocity $\boldsymbol{\omega} = \Omega\mathbf{j}$. Initially, the particle is located at $x = 0$ with velocity v_0 along the positive x axis.

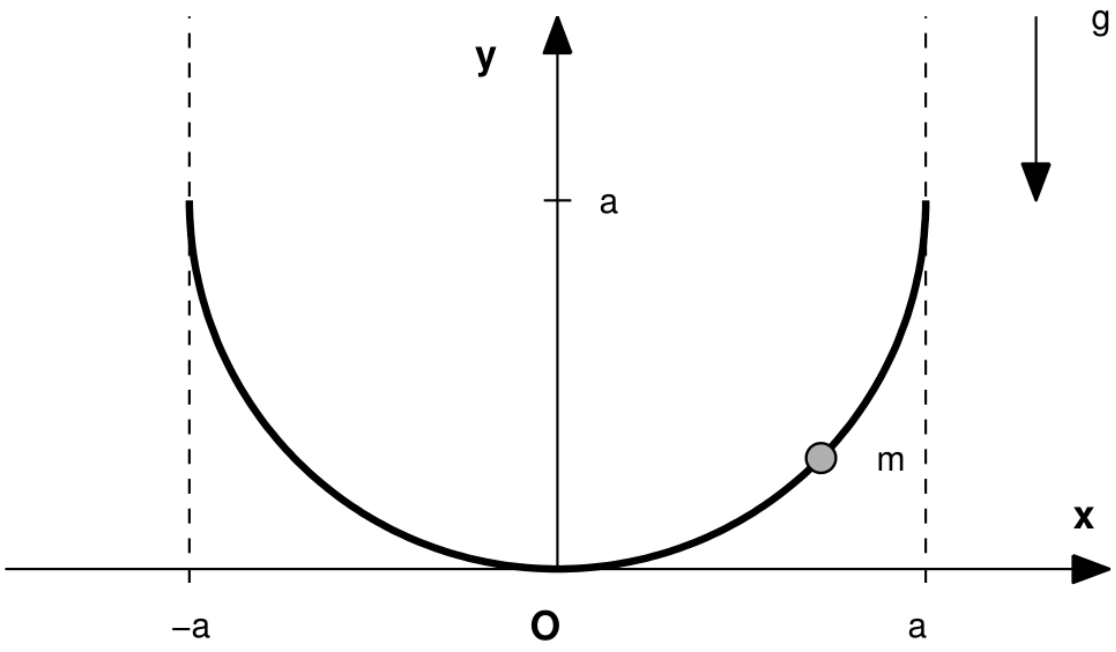


Figure: Half-circumference wire of radius a with lowest point O, ends at x = ±a, rotating about the vertical axis OY; particle of mass m on the wire (Problems.pdf, p. 14).

(a) Determine the number of degrees of freedom of the problem.

(b) Define a convenient moving reference frame with origin in O, $OX_1Y_1Z_1$. Express the inertia force that acts on the particle in this moving reference frame.

(c) Express also the components of the weight of the particle with respect to $OX_1Y_1Z_1$.

(d) Determine if there are any equilibrium points.

(e) Reduce the problem to integrals (i.e. quadratures) and determine the value of the reaction force.

(f) Is it possible to use the energy equation? If Ω were a function of time, would it be possible to use the energy equation? Justify your answer.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** a heavy particle P (mass m) slides without friction on a half-circumference wire of radius a. In the plane of the wire the lowest point is O, the centre is $C = (0, a)$ and the ends are $(\pm a, a)$ (figure). The wire turns about the vertical axis OY with constant angular velocity $\boldsymbol{\omega} = \Omega\,\mathbf{j}$.
- **Hypotheses:** the frame $OXYZ$ with OY vertical upwards is inertial; gravity is $-g\,\mathbf{j}$; the wire is smooth, so its reaction has no component along the wire; the rotation is imposed (whatever torque is needed is supplied). This problem was solved in Topic 3; here it is solved again with the relative-motion method of Topic 5 (moving frame, Notes Eq. 3.27 and inertia forces, Notes Eqs. 4.62–4.63).
- **Why a moving frame:** in a frame that turns with the wire the constraint is a fixed circle, so the motion of P is a one-dimensional motion on a fixed curve. The cost is to add the centrifugal and Coriolis forces; there is no Euler force and no force from the acceleration of the origin because Ω is constant and O is fixed.
- **Initial conditions ($t = 0$):** P at O ($x = 0$) with velocity v_0 along $+x$.

Symbol	Meaning	SI unit
m	mass of P	kg
a	radius of the wire	m
Ω	angular velocity of the wire (constant)	rad/s
g	gravitational acceleration	m/s^2
v_0	initial speed at O	m/s
θ	angle of CP from the downward vertical	rad

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial frame:** $S_0 = OXYZ = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, with $\mathbf{j}$ upwards.
- **Moving frame** (requested in (b)): $S_1 = OX_1Y_1Z_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ attached to the wire, with OX_1 in the plane of the wire (the x axis of the figure), $OY_1 \equiv OY$ and OZ_1 normal to the plane of the wire. Taking S_1 coincident with S_0 at $t = 0$, it is a simple rotation of angle Ωt about $\mathbf{j}$ (theory note, Sec. 4.1):

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i} - \sin\Omega t\,\mathbf{k}, \qquad \mathbf{j}_1 = \mathbf{j}, \qquad \mathbf{k}_1 = \sin\Omega t\,\mathbf{i} + \cos\Omega t\,\mathbf{k}$$

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & 0 & \sin\Omega t \\ 0 & 1 & 0 \\ -\sin\Omega t & 0 & \cos\Omega t \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\Omega t + \sin^2\Omega t = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

(the columns are unit vectors and $\mathbf{i}_1 \cdot \mathbf{k}_1 = \cos\Omega t \sin\Omega t - \sin\Omega t \cos\Omega t = 0$; also $\mathbf{i}_1 \times \mathbf{j}_1 = \mathbf{k}_1$).

- **Motion of S_1:** $\mathbf{r}_0^O = \mathbf{0}$, $\mathbf{v}_0^O = \mathbf{0}$, $\mathbf{a}_0^O = \mathbf{0}$ (fixed origin); $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}_1$, $\boldsymbol{\alpha}_{10} = \dot{\Omega}\,\mathbf{j}_1 = \mathbf{0}$. Check: $\left.\frac{d\mathbf{i}_1}{dt}\right|_0 = -\Omega\sin\Omega t\,\mathbf{i} - \Omega\cos\Omega t\,\mathbf{k} = -\Omega\,\mathbf{k}_1$ (chain rule), and $\Omega\,\mathbf{j} \times \mathbf{i}_1 = -\Omega\,\mathbf{k}_1$.
- **Position of P in S_1:** with θ the angle of CP measured from the downward vertical, positive towards $+X_1$, and $\theta \in [-\pi/2, \pi/2]$ on the wire:

$$\mathbf{r}_1^P = a\sin\theta\,\mathbf{i}_1 + a\,(1 - \cos\theta)\,\mathbf{j}_1$$

- **Wire basis (intrinsic frame of the circle):** tangent $\mathbf{e}_t = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1$, principal normal towards C, $\mathbf{e}_n = -\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1$, binormal $\mathbf{e}_t \times \mathbf{e}_n = (\cos^2\theta + \sin^2\theta)\,\mathbf{k}_1 = \mathbf{k}_1$. The matrix with columns $\mathbf{e}_t, \mathbf{e}_n, \mathbf{k}_1$ in $\mathcal{B}_1$ is a rotation by θ about $\mathbf{k}_1$, with determinant $\cos^2\theta + \sin^2\theta = 1$.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Degrees of freedom

Pedagogical Justification: CDOF = (coordinates of the free particle) − (independent holonomic constraints).

P has 3 coordinates. In S_1 the wire imposes two fixed equations, $z_1 = 0$ (plane of the wire) and $x_1^2 + (y_1 - a)^2 = a^2$ (circle). In S_0 the same constraints are rheonomic (they turn with Ωt), but the count is the same: CDOF $= 3 - 2 = 1$, with generalized coordinate θ. The rotation Ωt is prescribed, so it is not a degree of freedom.

(b) Inertia force in $OX_1Y_1Z_1$

> **Pedagogical Justification:** Notes Eq. 4.62 (Slide 28) gives $\mathbf{F}_i = -m\mathbf{a}_0^O - m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P - m\boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) - 2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$. The first two terms vanish ($\mathbf{a}_0^O = \mathbf{0}, \boldsymbol{\alpha}_{10} = \mathbf{0}$).

Relative velocity. In S_1 only the components change; chain rule $\frac{d}{dt}\sin\theta = \cos\theta\,\dot\theta$, $\frac{d}{dt}(1 - \cos\theta) = \sin\theta\,\dot\theta$:

$$\mathbf{v}_1^P = a\dot\theta\cos\theta\,\mathbf{i}_1 + a\dot\theta\sin\theta\,\mathbf{j}_1 = a\dot\theta\,\mathbf{e}_t$$

Centrifugal force. With $\mathbf{j}_1 \times \mathbf{i}_1 = -\mathbf{k}_1$ and $\mathbf{j}_1 \times \mathbf{j}_1 = \mathbf{0}$, $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \Omega\,\mathbf{j}_1 \times a\sin\theta\,\mathbf{i}_1 = -\Omega a\sin\theta\,\mathbf{k}_1$; then, with $\mathbf{j}_1 \times \mathbf{k}_1 = \mathbf{i}_1$, $\boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) = -\Omega^2 a\sin\theta\,\mathbf{i}_1$:

$$\mathbf{F}_C = m\Omega^2 a\sin\theta\,\mathbf{i}_1$$

It points horizontally away from the axis and equals $m\Omega^2\mathbf{r}_\perp$ (Slide 30), since $\mathbf{r}_\perp = a\sin\theta\,\mathbf{i}_1$ is the part of $\mathbf{r}_1^P$ perpendicular to $\mathbf{j}_1$.

Coriolis force. $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = \Omega\,\mathbf{j}_1 \times \Big(a\dot\theta\cos\theta\,\mathbf{i}_1 + a\dot\theta\sin\theta\,\mathbf{j}_1\Big) = -\Omega a\dot\theta\cos\theta\,\mathbf{k}_1$, so

$$\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = 2m\Omega a\dot\theta\cos\theta\,\mathbf{k}_1$$

Total inertia force:

$$\mathbf{F}_i = m\Omega^2 a\sin\theta\,\mathbf{i}_1 + 2m\Omega a\dot\theta\cos\theta\,\mathbf{k}_1$$

(c) Weight in $OX_1Y_1Z_1$

The axis of rotation is the vertical, so $\mathbf{j}_1 = \mathbf{j}$ at all times and

$$\mathbf{W} = -mg\,\mathbf{j}_1 \qquad (W_{X_1} = 0, \quad W_{Y_1} = -mg, \quad W_{Z_1} = 0)$$

In the wire basis: $\mathbf{W}\cdot\mathbf{e}_t = -mg\sin\theta$, $\mathbf{W}\cdot\mathbf{e}_n = -mg\cos\theta$, $\mathbf{W}\cdot\mathbf{k}_1 = 0$.

(d) Equilibrium points

> **Pedagogical Justification:** Newton's law in S_1, $m\mathbf{a}_1^P = \mathbf{W}+\mathbf{N}+\mathbf{F}_i$ (Notes Eq. 4.63), projected on the tangent $\mathbf{e}_t$, eliminates the smooth reaction ($\mathbf{N}\cdot\mathbf{e}_t = 0$) and the Coriolis force ($\mathbf{k}_1\cdot\mathbf{e}_t = 0$).

Relative acceleration. Product rule on $a\dot\theta\,\mathbf{e}_t$ in S_1, with $\frac{d\mathbf{e}_t}{dt}\big|_1 = \dot\theta\,(-\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1) = \dot\theta\,\mathbf{e}_n$:

$$\mathbf{a}_1^P = a\ddot\theta\,\mathbf{e}_t + a\dot\theta^2\,\mathbf{e}_n$$

Tangential equation. $\mathbf{F}_C\cdot\mathbf{e}_t = m\Omega^2 a\sin\theta\cos\theta$, so

$$ma\ddot\theta = -mg\sin\theta + m\Omega^2 a\sin\theta\cos\theta \;\Rightarrow\; \ddot\theta = \sin\theta\left(\Omega^2\cos\theta - \frac{g}{a}\right)$$

Equilibria (relative rest, $\ddot\theta = \dot\theta = 0$):

1. $\sin\theta = 0$: $\theta = 0$, the lowest point O. It exists for every Ω.
2. $\cos\theta_* = \dfrac{g}{a\Omega^2}$: two symmetric points $\theta = \pm\theta_*$, which exist only if $\Omega^2 > g/a$ (then $0 < \theta_* < \pi/2$, inside the wire).

Stability with the potential of (f), $V_1(\theta) = mga(1-\cos\theta) - \frac{1}{2}m\Omega^2a^2\sin^2\theta$ (the tangential equation is $ma^2\ddot\theta = -dV_1/d\theta$). Its derivatives are $V_1' = ma\sin\theta\left(g - \Omega^2 a\cos\theta\right)$ and $V_1'' = mga\cos\theta - m\Omega^2a^2\cos 2\theta$:

- $V_1''(0) = ma\left(g - \Omega^2 a\right)$: $\theta = 0$ is stable if $\Omega^2 < g/a$ and unstable if $\Omega^2 > g/a$.
- $V_1''(\pm\theta_*) = mga\cos\theta_* - m\Omega^2a^2\left(2\cos^2\theta_* - 1\right) = m\Omega^2a^2\left(1-\cos^2\theta_*\right) = m\Omega^2a^2\sin^2\theta_* > 0$ (using $ga = \Omega^2a^2\cos\theta_*$): the off-axis equilibria are stable whenever they exist.

(e) Quadratures and reaction force

> **Pedagogical Justification:** multiplying the tangential equation by $\dot\theta$ makes both sides exact derivatives; the result is the energy integral of (f). The remaining two projections of Newton's law (on $\mathbf{e}_n$ and $\mathbf{k}_1$) give the reaction.

First integral. Multiply $ma^2\ddot\theta = -mga\sin\theta + m\Omega^2a^2\sin\theta\cos\theta$ by $\dot\theta$. By the chain rule, $\frac{d}{dt}\left(\frac{1}{2}\dot\theta^2\right) = \dot\theta\ddot\theta$, $\frac{d}{dt}\cos\theta = -\sin\theta\,\dot\theta$ and $\frac{d}{dt}\left(\frac{1}{2}\sin^2\theta\right) = \sin\theta\cos\theta\,\dot\theta$, so

$$\frac{d}{dt}\left[\tfrac{1}{2}ma^2\dot\theta^2 - mga\cos\theta - \tfrac{1}{2}m\Omega^2a^2\sin^2\theta\right] = 0$$

Integrate from 0 to t and apply Barrow's rule, with $\theta(0)=0$ and $a\dot\theta(0)=v_0$ (at O, $\mathbf{r}_1^P=\mathbf{0}$, so $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P=\mathbf{0}$ and the given absolute velocity equals the relative one, $v_0\,\mathbf{i}_1 = a\dot\theta(0)\,\mathbf{e}_t(0)$):

$$\left[\tfrac{1}{2}ma^2\dot\theta^2 - mga\cos\theta - \tfrac{1}{2}m\Omega^2a^2\sin^2\theta\right]_0^t = 0 \;\Rightarrow\; \dot\theta^2 = \frac{v_0^2}{a^2} - \frac{2g}{a}\left(1-\cos\theta\right) + \Omega^2\sin^2\theta \equiv \Phi(\theta)$$

Quadrature. Separate variables ($d\theta = \sqrt{\Phi}\,dt$, positive root since $\dot\theta(0)=v_0/a>0$, valid until the first turning point $\Phi=0$) and integrate from 0 to t:

$$t = \int_0^{\theta(t)} \frac{d\theta'}{\sqrt{\dfrac{v_0^2}{a^2} - \dfrac{2g}{a}\left(1-\cos\theta'\right) + \Omega^2\sin^2\theta'}}$$

Reaction force. $\mathbf{N} = N_n\,\mathbf{e}_n + N_z\,\mathbf{k}_1$ (no tangential component). Projection on $\mathbf{e}_n$, with $\mathbf{F}_C\cdot\mathbf{e}_n = -m\Omega^2a\sin^2\theta$:

$$ma\dot\theta^2 = N_n - mg\cos\theta - m\Omega^2a\sin^2\theta \;\Rightarrow\; N_n = ma\dot\theta^2 + mg\cos\theta + m\Omega^2a\sin^2\theta$$

Substituting $\dot\theta^2=\Phi(\theta)$:

$$N_n(\theta) = \frac{mv_0^2}{a} + mg\left(3\cos\theta - 2\right) + 2m\Omega^2a\sin^2\theta$$

Projection on $\mathbf{k}_1$ ($\mathbf{a}_1^P\cdot\mathbf{k}_1=0$, $\mathbf{W}\cdot\mathbf{k}_1=0$):

$$0 = N_z + 2m\Omega a\dot\theta\cos\theta \;\Rightarrow\; N_z = -2m\Omega a\dot\theta\cos\theta = \mp 2m\Omega a\cos\theta\sqrt{\Phi(\theta)}$$

(upper sign while $\dot\theta>0$). The total reaction is $\mathbf{N} = N_n\left(-\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1\right) + N_z\,\mathbf{k}_1$.

Staying on the wire. The particle reaches the end $\theta=\pi/2$ only if $\Phi(\pi/2) = v_0^2/a^2 - 2g/a + \Omega^2 \geq 0$; otherwise it oscillates between the two roots of $\Phi(\theta)=0$.

(f) Energy equation

> **Pedagogical Justification:** E_1^P is conserved when every force that does work in S_1 derives from a time-independent potential (Slide 29; theory note, Sec. 12.6).

Constant Ω: yes. In S_1: the wire is fixed and smooth, so $\mathbf{N}\cdot\mathbf{v}_1^P = 0$; the Coriolis force never does work, $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$ (Slide 31); the weight has the potential $mgy_1 = mga(1-\cos\theta)$; the centrifugal force has the potential $-\frac{1}{2}m\Omega^2 r_\perp^2 = -\frac{1}{2}m\Omega^2 a^2\sin^2\theta$ (Slide 30), time-independent because Ω is constant. Hence

$$E_1^P = \tfrac{1}{2}ma^2\dot\theta^2 + mga\,(1-\cos\theta) - \tfrac{1}{2}m\Omega^2a^2\sin^2\theta = \tfrac{1}{2}mv_0^2$$

which is the first integral used in (e). (In S_0 the mechanical energy is not conserved: the reaction N_z is perpendicular to the moving wire but not to the absolute velocity, so the motor that keeps Ω constant does work.)

Variable $\Omega(t)$: no energy integral. Two things change. The Euler force appears, $\mathbf{F}_E = -m\dot\Omega\,\mathbf{j}_1\times a\sin\theta\,\mathbf{i}_1 = m\dot\Omega a\sin\theta\,\mathbf{k}_1$; it is normal to the plane of the wire, so $\mathbf{F}_E\cdot\mathbf{v}_1^P = 0$ and it does no work in S_1 (it is balanced by N_z). But the centrifugal potential now depends explicitly on time, $V_C = -\frac{1}{2}m\Omega(t)^2a^2\sin^2\theta$, and differentiating E_1^P along the motion (tangential equation with $\Omega(t)$) gives

$$\frac{dE_1^P}{dt} = \frac{\partial V_C}{\partial t} = -m\Omega\dot\Omega a^2\sin^2\theta \neq 0$$

so E_1^P is not a constant of motion and the problem cannot be reduced to a quadrature with the energy equation. The work-energy relation $dE_1^P/dt = \partial V_C/\partial t$ is still valid, but it is not a first integral.

> **Correction to the Topic 3 solution**
> The Topic 3 version of this problem placed the origin at the centre of the circle (the statement puts it at the lowest point O), gave only one off-axis equilibrium (there are two, $\pm\theta_*$), gave only the out-of-plane reaction (the in-plane component N_n is also part of the answer) and said that for variable Ω "Euler forces do work". The Euler force $m\dot\Omega a\sin\theta\,\mathbf{k}_1$ is perpendicular to $\mathbf{v}_1^P$ and does no work; the energy integral is lost because the centrifugal potential depends explicitly on time. The quadrature $\dot\theta^2 = \Phi(\theta)$, the equilibrium $\cos\theta_* = g/(a\Omega^2)$ and $N_z = -2m\Omega a\dot\theta\cos\theta$ of the Topic 3 version agree with this solution.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Bifurcation.** For $\Omega^2 < g/a$ gravity dominates and the bottom is the only (stable) equilibrium. At $\Omega_c = \sqrt{g/a}$ the bottom becomes unstable and two stable equilibria $\pm\theta_*$ appear, where the tangential components of weight and centrifugal force balance: $mg\sin\theta_* = m\Omega^2 a \sin\theta_*\cos\theta_*$.
- **Limits.** $\Omega \to 0$: $\Phi = v_0^2/a^2 - 2g(1-\cos\theta)/a$ and $N_n = mv_0^2/a + mg(3\cos\theta - 2)$, the simple-pendulum results; $N_z \to 0$. $\Omega \to \infty$: $\cos\theta_* \to 0$, the equilibria move to the ends of the wire.
- **Coriolis and the reaction.** N_z is exactly the force the wire must exert to cancel the Coriolis force; it changes sign with $\dot\theta$, pushing the particle forward or backward with the rotation.
- **Dimensions:** $[\Omega^2] = [g/a] = \mathrm{s}^{-2}$, so every term of Φ is in s^{-2} and $[t] = \mathrm{rad}/\mathrm{s}^{-1} = \mathrm{s}$; $[mv_0^2/a] = [mg] = [m\Omega^2 a] = \mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2} = \mathrm{N}$; $[m\Omega a\dot\theta] = \mathrm{N}$; $[E_1^P] = [ma^2\dot\theta^2] = \mathrm{J}$.
- **Symbolic check (sympy, `verify_P26.py`):** $[{}_0R_1]$ and $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}$; Notes Eq. 3.27 against the double derivative of $\mathbf{r}_1^P$ in $\mathcal{B}_0$; $\mathbf{F}_C$, $\mathbf{F}_{COR}$; the equation of motion and both reaction components from $m\mathbf{a}_0^P = \mathbf{W} + \mathbf{N}$ in S_0; $dE_1^P/dt = 0$; $\Phi(\theta)$ and $N_n(\theta)$; the stability derivatives; and, for $\Omega(t)$, the Euler force, its zero power and $dE_1^P/dt = -m\Omega\dot\Omega a^2\sin^2\theta$. A numerical integration ($g = 9.81$, $a = 1$, $\Omega = 4$, $v_0 = 2$, SI) keeps $\dot\theta^2 - \Phi(\theta)$ below 3×10^{-10}.

Source: Problems.pdf

PROBLEM 27

Bead on a Rotating Sine-Shaped Wire with a Spring

Original statement · Problems.pdf, p. 15

Rotating Wire + Spring

A heavy bead P of mass m slides without friction along a thin wire with shape given by the following equation:

$$y_1(x_1) = \sin x_1.$$

Moreover, the bead is joined to a straight rail located at $y_1 = 2$ by a spring of stiffness k. The connection of the spring with the rail can move freely along the rail, so that the spring is always vertical. The whole setup rotates with respect to the y_1 axis with a constant angular velocity ω. A sketch of the setup and the definition of the right-handed, rotating $Ox_1y_1z_1$ reference frame is given below.

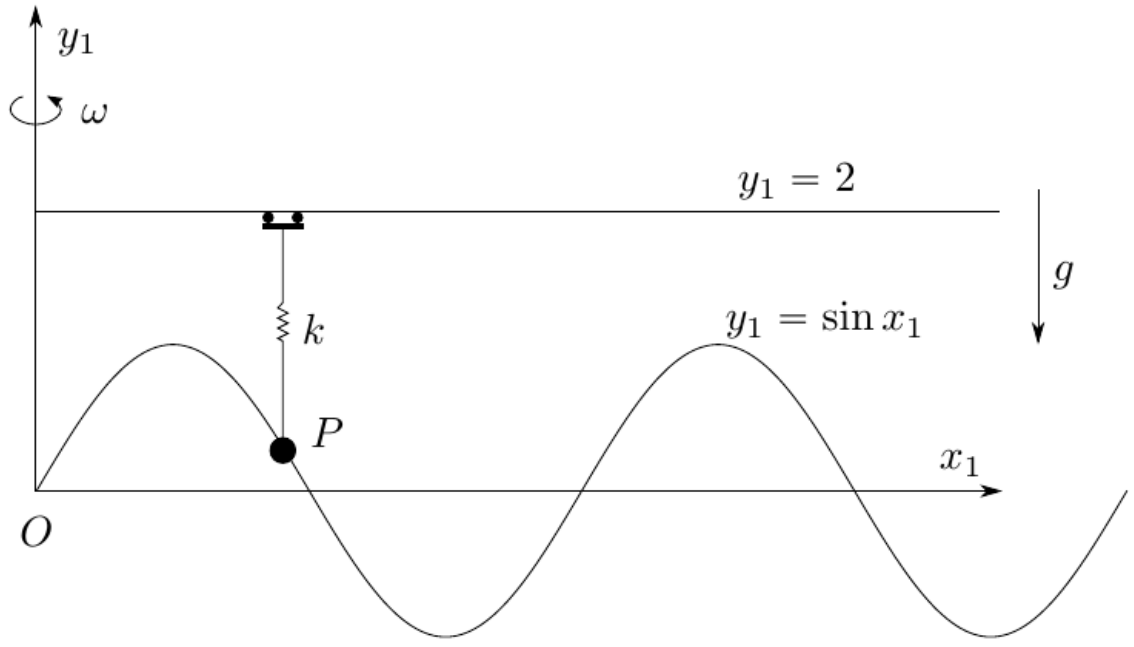


Figure: Sine-shaped wire y1 = sin x1 rotating about the vertical axis Oy1 with angular velocity ω; bead P joined by a vertical spring k to the rail y1 = 2 (Problems.pdf, p. 15).

(a) Write down the general expression for the components of the relative velocity and acceleration of P in the reference frame $Ox_1y_1z_1$ as functions of the axial position of the bead, x_1, and its time derivatives $\dot{x}_1$ and $\ddot{x}_1$.

(b) Identify all forces acting on the bead and then write down the three components of Newton's second law in the $Ox_1y_1z_1$ reference frame. From these equations, obtain an explicit expression of the components of the reaction force on the wire as a function of x_1, $\dot{x}_1$ and the parameters of the problem only, and a second order differential equation for x_1 that does not depend on the reaction force.

(c) Is it possible to use the equation of conservation of mechanical energy of the relative motion of P? If so, write it down in terms of x_1, $\dot{x}_1$ and the parameters of the problem. What is the relation of this equation with the differential equation for x_1 in the previous point?

(d) For $\omega = 0$ and $x_1 \in [0, 2\pi]$, discuss the existence of equilibrium positions as a function of the parameter $\lambda = mg/k$. Are these equilibrium positions stable?

(e) Consider now that $\omega \neq 0$, and $k = 0$. We drop the bead P at $x_1 = \pi$ with no initial velocity. Sketch the shape of the potential and discuss qualitatively the motion of P if the parameter $\beta = \omega^2/g$ is small or large. Write down the system of equations that would allow to solve for the critical value of β that divides between the two behaviors.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** a heavy bead P (mass m) slides on the smooth plane wire $y_1 = \sin x_1$. A vertical spring of stiffness k joins it to the rail $y_1 = 2$; the upper end slides freely on the rail, so the spring stays vertical. Wire, rail and spring rotate about the vertical axis Oy_1 with constant angular velocity ω.
- **Hypotheses:** the non-rotating frame with origin O and vertical axis Oy_1 is inertial; gravity is $-g\,\mathbf{j}_1$; the wire is smooth (no reaction along the tangent). The statement does not give the natural length of the spring; as in the official key it is taken as zero, so the spring force is k times its length $2 - \sin x_1$. Lengths in the wire equation are measured in metres (the equation $y_1 = \sin x_1$ hides a unit length of 1 m).

- **Why the rotating frame:** in $Ox_1y_1z_1$ the wire, the rail and the spring are fixed, so the bead moves on a fixed curve with one coordinate x_1; the rotation enters only through the centrifugal and Coriolis forces (Notes Eq. 4.62).
- **Degrees of freedom:** 3 coordinates minus 2 constraints ($z_1 = 0$ and $y_1 = \sin x_1$): CDOF $= 1$, generalized coordinate x_1.

Symbol	Meaning	SI unit
m	mass of the bead	kg
k	spring stiffness	N/m
ω	angular velocity of the setup (constant)	rad/s
g	gravitational acceleration	m/s^2
x_1	axial position of the bead	m
$\lambda = mg/k$	weight-to-stiffness ratio	m
$\beta = \omega^2/g$	centrifugal-to-gravity ratio	m^{-1}

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial frame:** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ with $\mathbf{j}_0$ vertical upwards.
- **Rotating frame:** $S_1 = Ox_1y_1z_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, attached to the wire, coincident with S_0 at $t = 0$; it is a simple rotation of angle ωt about $\mathbf{j}_0$:

$$\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 - \sin\omega t\,\mathbf{k}_0, \qquad \mathbf{j}_1 = \mathbf{j}_0, \qquad \mathbf{k}_1 = \sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{k}_0$$

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & 0 & \sin\omega t \\ 0 & 1 & 0 \\ -\sin\omega t & 0 & \cos\omega t \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- **Motion of S_1:** $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$; $\boldsymbol{\omega}_{10} = \omega\,\mathbf{j}_1$ (constant), $\boldsymbol{\alpha}_{10} = \mathbf{0}$.
- **Position of the bead:** $\mathbf{r}_1^P = x_1\,\mathbf{i}_1 + \sin x_1\,\mathbf{j}_1$. The tangent to the wire is $\mathbf{t} = \mathbf{i}_1 + \cos x_1\,\mathbf{j}_1$ (not normalised, $|\mathbf{t}|^2 = 1 + \cos^2 x_1$); the in-plane normal is $\mathbf{n} = -\cos x_1\,\mathbf{i}_1 + \mathbf{j}_1$, and $\mathbf{k}_1$ is normal to the plane of the wire.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Relative velocity and acceleration

Pedagogical Justification: $\mathcal{B}_1$ is constant in S_1, so the relative velocity and acceleration are the time derivatives of the components of $\mathbf{r}_1^P$ (Notes Eq. 3.7).

Chain rule: $\frac{d}{dt}\sin x_1(t) = \cos x_1\,\dot{x}_1$. Product and chain rule: $\frac{d}{dt}\left(\cos x_1\,\dot{x}_1\right) = -\sin x_1\,\dot{x}_1^2 + \cos x_1\,\ddot{x}_1$. Hence

$$\mathbf{v}_1^P = \dot{x}_1\,\mathbf{i}_1 + \cos x_1\,\dot{x}_1\,\mathbf{j}_1 + 0\,\mathbf{k}_1$$

$$\mathbf{a}_1^P = \ddot{x}_1\,\mathbf{i}_1 + \left(\cos x_1\,\ddot{x}_1 - \sin x_1\,\dot{x}_1^2\right)\mathbf{j}_1 + 0\,\mathbf{k}_1$$

(b) Forces, Newton's law in $Ox_1y_1z_1$, reaction and equation of motion

Pedagogical Justification: in the non-inertial frame S_1, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63). Of the four inertia forces (Notes Eq. 4.62, Slide 28), only the centrifugal and the Coriolis forces survive, because $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$.

Real forces.

1. Weight: $\mathbf{W} = -mg\,\mathbf{j}_1$.
2. Spring: its length is $2 - \sin x_1 > 0$ and it pulls the bead up towards the rail: $\mathbf{F}_s = k\,(2 - \sin x_1)\,\mathbf{j}_1$.
3. Wire reaction: $\mathbf{N} = N_x\,\mathbf{i}_1 + N_y\,\mathbf{j}_1 + N_z\,\mathbf{k}_1$ with $\mathbf{N}\cdot\mathbf{t} = 0$, that is $N_x + N_y\cos x_1 = 0$ (smooth wire).

Inertia forces. With $\mathbf{j}_1\times\mathbf{i}_1 = -\mathbf{k}_1, \mathbf{j}_1\times\mathbf{j}_1 = \mathbf{0}, \mathbf{j}_1\times\mathbf{k}_1 = \mathbf{i}_1$:

$$\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = \omega\,\mathbf{j}_1\times(x_1\,\mathbf{i}_1 + \sin x_1\,\mathbf{j}_1) = -\omega x_1\,\mathbf{k}_1, \qquad \boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P\right) = -\omega^2 x_1\,\mathbf{i}_1$$

$$\mathbf{F}_C = m\omega^2 x_1\,\mathbf{i}_1, \qquad \mathbf{F}_{COR} = -2m\,\omega\,\mathbf{j}_1\times(\dot{x}_1\,\mathbf{i}_1 + \cos x_1\,\dot{x}_1\,\mathbf{j}_1) = 2m\omega\dot{x}_1\,\mathbf{k}_1$$

Three components of Newton's law in S_1:

$$\mathbf{i}_1: \quad m\ddot{x}_1 = N_x + m\omega^2 x_1$$

$$\mathbf{j}_1: \quad m\left(\cos x_1\,\ddot{x}_1 - \sin x_1\,\dot{x}_1^2\right) = N_y - mg + k\,(2 - \sin x_1)$$

$$\mathbf{k}_1: \quad 0 = N_z + 2m\omega\dot{x}_1$$

Out-of-plane reaction. From the $\mathbf{k}_1$ equation, $N_z = -2m\omega\dot{x}_1$: the wire cancels the Coriolis force.

In-plane reaction. Substitute $N_x = -N_y\cos x_1$ in the $\mathbf{i}_1$ equation, $m\ddot{x}_1 = -N_y\cos x_1 + m\omega^2 x_1$. Multiply it by $\cos x_1$ and substitute $m\cos x_1\,\ddot{x}_1$ in the $\mathbf{j}_1$ equation:

$$-N_y\cos^2 x_1 + m\omega^2 x_1\cos x_1 - m\sin x_1\,\dot{x}_1^2 = N_y - mg + k\,(2 - \sin x_1)$$

Collect N_y:

$$N_y = \frac{m\left(\omega^2 x_1\cos x_1 - \dot{x}_1^2\sin x_1\right) + mg - k\,(2 - \sin x_1)}{1 + \cos^2 x_1}, \qquad N_x = -N_y\cos x_1$$

so the reaction is

$$\mathbf{N} = \frac{m\left(\omega^2 x_1\cos x_1 - \dot{x}_1^2\sin x_1\right) + mg - k\,(2 - \sin x_1)}{1 + \cos^2 x_1}\,(-\cos x_1\,\mathbf{i}_1 + \mathbf{j}_1) - 2m\omega\dot{x}_1\,\mathbf{k}_1$$

Equation of motion. Project Newton's law on the tangent $\mathbf{t} = \mathbf{i}_1 + \cos x_1\,\mathbf{j}_1$: add the $\mathbf{i}_1$ equation and $\cos x_1$ times the $\mathbf{j}_1$ equation; $N_x + N_y\cos x_1 = 0$ removes the reaction:

$$m\ddot{x}_1 + m\cos x_1\left(\cos x_1\,\ddot{x}_1 - \sin x_1\,\dot{x}_1^2\right) = m\omega^2 x_1 + \cos x_1\left[-mg + k\,(2 - \sin x_1)\right]$$

$$m\left(1+\cos^2 x_1\right)\ddot{x}_1 - m\sin x_1\cos x_1\,\dot{x}_1^2 = m\omega^2 x_1 - mg\cos x_1 + k\cos x_1\,(2-\sin x_1)$$

(c) Conservation of the mechanical energy of the relative motion

> **Pedagogical Justification:** E_1^P is conserved if all the forces that do work in S_1 derive from a time-independent potential (Slide 29). Here $\boldsymbol{\omega}_{10}$ is constant and the wire is fixed in S_1.

Yes, it can be used. Check of each force: $\mathbf{N}\cdot\mathbf{v}_1^P = \dot{x}_1\,(N_x + N_y\cos x_1) = 0$; $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$ (Slide 31); the weight has potential $mgy_1 = mg\sin x_1$; the spring has potential $\frac{1}{2}k\,(2-y_1)^2 = \frac{1}{2}k\,(2-\sin x_1)^2$ (its free end slides without friction, so only its length matters); the centrifugal force has potential $-\frac{1}{2}m\omega^2 r_\perp^2 = -\frac{1}{2}m\omega^2 x_1^2$ (Slide 30). With $T_1^P = \frac{1}{2}m|\mathbf{v}_1^P|^2 = \frac{1}{2}m\dot{x}_1^2\left(1+\cos^2 x_1\right)$:

$$E_1^P = \tfrac{1}{2}m\dot{x}_1^2\left(1+\cos^2 x_1\right) + mg\sin x_1 + \tfrac{1}{2}k\,(2-\sin x_1)^2 - \tfrac{1}{2}m\omega^2 x_1^2 = \text{const}$$

Expanding $\frac{1}{2}k(2-\sin x_1)^2 = 2k - 2k\sin x_1 + \frac{1}{2}k\sin^2 x_1$ and dropping the constant $2k$ gives the form of the official key, $\frac{1}{2}m\dot{x}_1^2(1+\cos^2 x_1) + \sin x_1\,(mg-2k) + \frac{1}{2}k\sin^2 x_1 - \frac{1}{2}m\omega^2 x_1^2 = C$.

Relation with the equation of motion. Differentiate E_1^P in time. Product and chain rules: $\frac{d}{dt}\left[\frac{1}{2}m\dot{x}_1^2(1+\cos^2 x_1)\right] = m\dot{x}_1\ddot{x}_1(1+\cos^2 x_1) - m\dot{x}_1^3\sin x_1\cos x_1$, $\frac{d}{dt}(mg\sin x_1) = mg\cos x_1\,\dot{x}_1$, $\frac{d}{dt}\left[\frac{1}{2}k(2-\sin x_1)^2\right] = -k(2-\sin x_1)\cos x_1\,\dot{x}_1$, $\frac{d}{dt}\left(-\frac{1}{2}m\omega^2 x_1^2\right) = -m\omega^2 x_1\dot{x}_1$. Therefore

$$\frac{dE_1^P}{dt} = \dot{x}_1\left[m\left(1+\cos^2 x_1\right)\ddot{x}_1 - m\sin x_1\cos x_1\,\dot{x}_1^2 - m\omega^2 x_1 + mg\cos x_1 - k\cos x_1\,(2-\sin x_1)\right]$$

The bracket is exactly the equation of motion of (b), so $dE_1^P/dt = 0$ along every motion: the energy equation is a first integral of the equation of motion (obtained by multiplying it by $\dot{x}_1$ and integrating once), and the equation of motion is recovered by differentiating it (for $\dot{x}_1 \neq 0$).

(d) Equilibria for $\omega = 0$, $x_1 \in [0, 2\pi]$

> **Pedagogical Justification:** with $\omega = 0$ the equation of motion is $m(1+\cos^2 x_1)\ddot{x}_1 - m\sin x_1\cos x_1\dot{x}_1^2 = -dV/dx_1$, with $V(x_1) = mg\sin x_1 + \frac{1}{2}k(2-\sin x_1)^2$. Equilibria are the critical points of V; minima are stable, maxima unstable.

First derivative (chain rule):

$$V'(x_1) = mg\cos x_1 - k\,(2 - \sin x_1)\cos x_1 = k\cos x_1\,(\lambda - 2 + \sin x_1)$$

Second derivative (product rule):

$$V''(x_1) = -k\sin x_1\,(\lambda - 2 + \sin x_1) + k\cos^2 x_1$$

Equilibria.

1. $\cos x_1 = 0$: $x_1 = \pi/2$ (crest of the wire) and $x_1 = 3\pi/2$ (trough). They exist for every λ.
2. $\sin x_1 = 2 - \lambda$: it has solutions only if $-1 \le 2 - \lambda \le 1$, that is $1 \le \lambda \le 3$. For $1 < \lambda < 2$ they are $x_1 = \arcsin(2-\lambda)$ and $\pi - \arcsin(2-\lambda)$ (on both sides of the crest); for $\lambda = 2$, $x_1 = 0, \pi, 2\pi$; for $2 < \lambda < 3$, $x_1 = \pi + \arcsin(\lambda - 2)$ and $2\pi - \arcsin(\lambda - 2)$ (on both sides of the trough). At $\lambda = 1$ and $\lambda = 3$ they merge with $\pi/2$ and $3\pi/2$.

Stability.

- $x_1 = \pi/2$: $V''(\pi/2) = -k(\lambda - 1) = k(1 - \lambda)$: stable for $\lambda < 1$ (a stiff spring holds the bead on the crest), unstable for $\lambda > 1$.
- $x_1 = 3\pi/2$: $V''(3\pi/2) = +k(\lambda - 3)$: stable for $\lambda > 3$ (a heavy bead sits in the trough), unstable for $\lambda < 3$.
- $\sin x_1 = 2 - \lambda$ ($1 < \lambda < 3$): the bracket vanishes, so $V'' = k\cos^2 x_1 > 0$: stable.
- Boundary cases: for $\lambda = 1$ near $\pi/2$ ($u = x_1 - \pi/2$) and for $\lambda = 3$ near $3\pi/2$ ($u = x_1 - 3\pi/2$), the Taylor expansion gives $V' \approx \frac{1}{2}ku^3$, so V has a quartic minimum: stable, but with zero linear frequency.

Discrepancy with the official solution

The official key (Problems.pdf, PDF page 74) states the results in terms of a parameter α: "$\pi/2$ stable if $\alpha > 1$; $3\pi/2$ stable if $\alpha < 1/3$; $\sin x_1 = 2 - 1/\alpha$, which exists and is stable if $1/3 < \alpha < 1$". These coincide with the results above only if $\alpha = 1/\lambda = k/(mg)$, the reciprocal of the parameter $\lambda = mg/k$ defined in the statement. Check: at $\pi/2$, $V''(\pi/2) = k(1-\lambda) > 0 \iff \lambda < 1 \iff 1/\lambda > 1$. The key also writes the in-plane reaction direction as $(-\cos x\,\mathbf{i}_1 + \mathbf{j}_1)$, a typo for $(-\cos x_1\,\mathbf{i}_1 + \mathbf{j}_1)$; its reaction formulas and energy integral otherwise agree with (b) and (c).

(e) $\omega \neq 0$, $k = 0$: bead dropped at $x_1 = \pi$

> **Pedagogical Justification:** the energy integral of (c) with $k = 0$ is $\frac{1}{2}m\dot{x}_1^2(1 + \cos^2 x_1) + V(x_1) = V(\pi)$, so the bead can only be where $V(x_1) \leq V(\pi)$; the motion is read from the shape of V (turning points where $V = V(\pi)$).

Dimensionless potential. With $k = 0$, $V = mg\sin x_1 - \frac{1}{2}m\omega^2 x_1^2$. Define $\tilde{V} = 2V/(mg)$:

$$\tilde{V}(x_1) = 2\sin x_1 - \beta x_1^2, \qquad \beta = \frac{\omega^2}{g}, \qquad \tilde{V}(\pi) = -\beta\pi^2$$

and the energy equation becomes $\dfrac{\dot{x}_1^2}{g}\left(1 + \cos^2 x_1\right) = \tilde{V}(\pi) - \tilde{V}(x_1)$.

Shape of the potential (sketch). $\tilde{V}$ is a sine wave of amplitude 2 riding on the downward parabola $-\beta x_1^2$: wells near the troughs $x_1 \approx 3\pi/2, 7\pi/2, \ldots$, humps near the crests $x_1 \approx 5\pi/2, 9\pi/2, \ldots$, and $\tilde{V} \to -\infty$ as $|x_1| \to \infty$. The humps flatten and disappear as β grows, because $\tilde{V}' = 2\cos x_1 - 2\beta x_1$ can vanish only where $|\beta x_1| \leq 1$. For example, $\beta = 0.02\ \mathrm{m}^{-1}$: $\tilde{V}(\pi) = -0.197$ m, $\tilde{V}(3\pi/2) = -2.44$ m, $\tilde{V}(5\pi/2) = +0.77$ m; $\beta = 0.1\ \mathrm{m}^{-1}$: $\tilde{V}(\pi) = -0.99$ m, $\tilde{V}(5\pi/2) = -4.17$ m.

Start of the motion. $\tilde{V}'(\pi) = 2\cos\pi - 2\beta\pi = -2(1 + \beta\pi) < 0$, so the generalized force $-V'$ is positive: the bead starts moving towards larger x_1 (down the wire into the trough near $3\pi/2$), and $x_1 = \pi$ is a turning point.

Small β (slow rotation). The hump near $5\pi/2$ is higher than the starting level, $\tilde{V} > -\beta\pi^2$ there, so the bead stops at the turning point x_t with $\tilde{V}(x_t) = \tilde{V}(\pi)$ and returns: it oscillates periodically between $x_1 = \pi$ and x_t (in the example $\beta = 0.02\ \mathrm{m}^{-1}$, $0.77 > -0.197$).

Large β (fast rotation). The centrifugal potential dominates, no hump to the right of π reaches the starting level, $\tilde{V}(x_1) < \tilde{V}(\pi)$ for all $x_1 > \pi$, and the bead moves outwards without bound ($x_1 \to \infty$) (in the example $\beta = 0.1\ \mathrm{m}^{-1}$, $-4.17 < -0.99$).

Critical value. The two behaviours are separated when the first hump to the right of π has its maximum exactly at the starting level: at that x_c the potential is stationary and equal to $\tilde{V}(\pi)$. The system of equations for (x_c, β_c) is

$$\begin{cases} \tilde{V}(x_c) = 2\sin x_c - \beta_c x_c^2 = -\beta_c\pi^2 \\ \tilde{V}'(x_c) = 2\cos x_c - 2\beta_c x_c = 0 \end{cases}$$

with the condition $\tilde{V}''(x_c) = -2\sin x_c - 2\beta_c < 0$ (a maximum). Solving it numerically (extension: the statement asks only for the system) gives $x_c \approx 2.401\pi \approx 7.544$ m and $\beta_c \approx 0.0405\ \mathrm{m}^{-1}$. Hence the bead oscillates for $\beta < \beta_c$ and escapes for $\beta > \beta_c$.

Discrepancy with the official solution

The official key writes the same system of equations for x_c and β_c, but states "$\beta > \beta_c$, the bead oscillates; $\beta < \beta_c$, the bead moves towards infinity". This is reversed. Proof: the centrifugal term $-\beta x_1^2$ is the only one that pushes the bead outwards, so a larger β lowers the humps; in the extreme $\beta \to \infty$, $\tilde{V}' = 2\cos x_1 - 2\beta x_1 < 0$ for all $x_1 \geq \pi$ (no hump at all), while for $\beta \to 0$ the hump at $5\pi/2$ has $\tilde{V} \to 2 > \tilde{V}(\pi) \to 0$. Numerical integration of the equation of motion (`verify_P27.py`) gives a bounded oscillation $x_1 \in [\pi, 2.31\pi]$ for $\beta = 0.95\beta_c$ and escape for $\beta = 1.05\beta_c$.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Role of each force.** The spring pulls the bead towards the rail, which on this wire favours the crest $\pi/2$; the weight favours the trough $3\pi/2$; the ratio $\lambda = mg/k$ decides, with the intermediate equilibria $\sin x_1 = 2 - \lambda$ where the vertical forces mg and $k(2 - \sin x_1)$ balance exactly. The centrifugal force pushes the bead away from the axis and, for fast rotation, throws it out; the Coriolis force only loads the wire (N_z) and never changes the energy.
- **Limits.** $\omega \to 0$: $N_z \to 0$ and the equation of motion is that of a bead on a fixed wire. $k \to 0$ in (d): $\lambda \to \infty$, only $3\pi/2$ is stable (bead in the trough). $k \to \infty$: $\lambda \to 0$, only $\pi/2$ is stable.
- **Dimensions** (with the implicit unit length of the wire): $[m\omega^2 x_1] = [mg] = [k(2 - \sin x_1)] = \mathrm{N}$; $[m\omega\dot{x}_1] = \mathrm{kg \cdot s^{-1} \cdot m \cdot s^{-1}} = \mathrm{N}$; every term of E_1^P is in J ($[k(2 - \sin x_1)^2] = \mathrm{N/m \cdot m^2}$); $[\lambda] = \mathrm{m}$ is compared with lengths ($2 - \sin x_1$); $[\beta x_1^2] = \mathrm{m^{-1} \cdot m^2} = \mathrm{m} = [2\sin x_1]$.
- **Symbolic and numerical check** (`verify_P27.py`)**:** $[{}_0R_1]$; $\mathbf{v}_1^P$, $\mathbf{a}_1^P$; Notes Eq. 3.27 against differentiation in $\mathcal{B}_0$; $\mathbf{F}_C$, $\mathbf{F}_{COR}$; N_x, N_y, N_z and the equation of motion solved from the three components (and again from $m\mathbf{a}_0^P = \mathbf{F}$ in S_0); $dE_1^P/dt = \dot{x}_1 \times$(equation of motion); V', V'' and the boundary cases; the critical point (x_c, β_c) and the two behaviours by numerical integration: all verified.

Source: Problems.pdf (PDF page 15); key on PDF page 74

PROBLEM 28

Spherical Pendulum with a Vertically Oscillating Suspension Point

Original statement · Problems.pdf, p. 16

Translating Frame + Inertia Force

A point particle P of mass m is attached with a massless cord of length L to the point A, as shown in the figure. The point A is oscillating vertically with respect to the fixed reference frame $Oxyz$ according to the law $z = a\sin\omega t$, where a and ω are known constants. The position of the particle P with respect to the moving reference frame $Ax'y'z'$ is completely specified by the angles φ and θ shown in the figure.

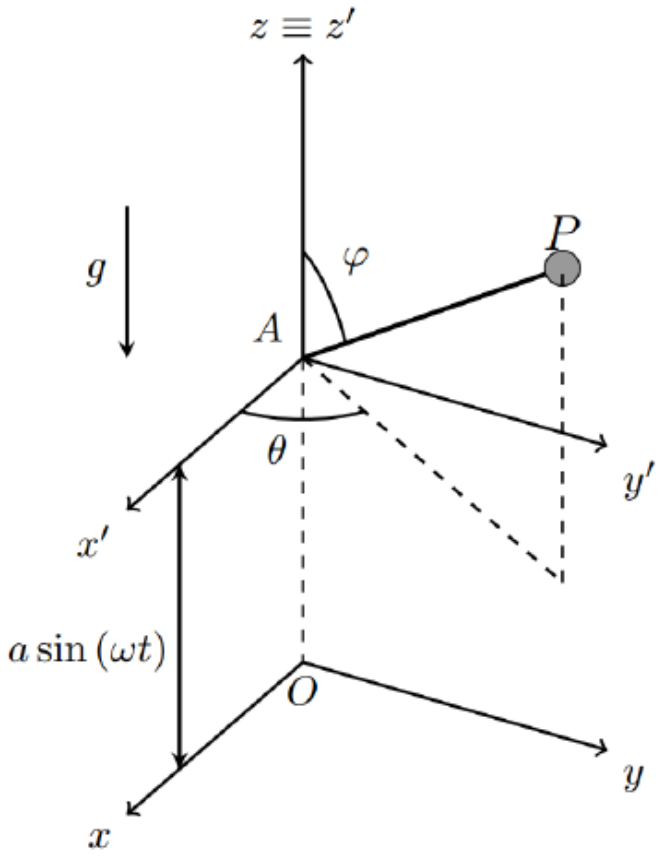


Figure: Point A oscillating vertically as a sin(ωt) above O; translating axes Ax'y'z'; particle P on a cord from A, located by the angle φ from Az' and the azimuth θ (Problems.pdf, p. 16).

At $t = 0$ the relative position and relative velocity of the particle are defined by the values

$$\theta_0 = 0, \qquad \varphi_0 = 30^\circ, \qquad \dot{\theta}_0 = \sqrt{\frac{2g}{\sqrt{3}L}}, \qquad \dot{\varphi}_0 = 0,$$

where g is the gravitational acceleration. We define the following unit vectors that define a spherical coordinate system

$$\mathbf{e}_r = \sin\varphi\cos\theta\,\mathbf{i} + \sin\varphi\sin\theta\,\mathbf{j} + \cos\varphi\,\mathbf{k}, \qquad \mathbf{e}_\varphi = \cos\varphi\cos\theta\,\mathbf{i} + \cos\varphi\sin\theta\,\mathbf{j} - \sin\varphi\,\mathbf{k}, \qquad \mathbf{e}_\theta = -\sin\theta\,\mathbf{i} + \cos\theta\,\mathbf{j}.$$

The partial derivatives of these vectors with respect to φ and θ are

$$\frac{\partial \mathbf{e}_r}{\partial \varphi} = \mathbf{e}_\varphi, \quad \frac{\partial \mathbf{e}_\varphi}{\partial \varphi} = -\mathbf{e}_r, \quad \frac{\partial \mathbf{e}_\theta}{\partial \varphi} = \mathbf{0}, \quad \frac{\partial \mathbf{e}_r}{\partial \theta} = \sin\varphi\,\mathbf{e}_\theta, \quad \frac{\partial \mathbf{e}_\varphi}{\partial \theta} = \cos\varphi\,\mathbf{e}_\theta, \quad \frac{\partial \mathbf{e}_\theta}{\partial \theta} = -\sin\varphi\,\mathbf{e}_r - \cos\varphi\,\mathbf{e}_\varphi.$$

(a) Show that the relative acceleration of P with respect to $Ax'y'z'$ can be expressed as $\mathbf{a}'_P = -L\left(\dot{\varphi}^2 + \dot{\theta}^2\sin^2\varphi\right)\mathbf{e}_r + L\left(\ddot{\varphi} - \dot{\theta}^2\sin\varphi\cos\varphi\right)\mathbf{e}_\varphi + L\left(\ddot{\theta}\sin\varphi + 2\dot{\theta}\dot{\varphi}\cos\varphi\right)\mathbf{e}_\theta$.

(b) Write down a system of three ordinary differential equations that once solved would yield the angles θ and φ, and the tension on the cord as a function of time (you do not need to solve it).

(c) At $t = 0$, determine the relative acceleration of P with respect to $Ax'y'z'$.

(d) At $t = 0$, determine the value of the tension on the cord.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** a particle P (mass m) hangs from a massless cord of length L fixed at A ; A moves on the vertical axis with $z_A = a\sin\omega t$. In the frame $Ax'y'z'$ the particle is located by the spherical angles φ (from the upward axis Az' to AP, figure) and θ (azimuth from Ax').
- **Hypotheses:** $Oxyz$ is inertial with Oz upwards and gravity $-g\,\mathbf{k}$; the cord is massless and taut, so it acts on P with a tension $\mathbf{T} = -T\,\mathbf{e}_r, T \geq 0$, directed from P to A; no friction or drag. Here ω is the frequency of the oscillation of A, not an angular velocity of the frame.
- **Why a moving frame:** in $Ax'y'z'$ the cord has fixed length, so P moves on a sphere of radius L described by two angles. The frame does not rotate, so the only inertia force is the one due to the acceleration of its origin.
- **Degrees of freedom:** 3 coordinates minus the constraint $|\mathbf{AP}| = L$: CDOF $= 2$, generalized coordinates (φ, θ). The motion of A is prescribed.
- **Initial conditions:** $\theta_0 = 0, \varphi_0 = 30^\circ = \pi/6, \dot\theta_0 = \sqrt{2g/(\sqrt{3}L)}, \dot\varphi_0 = 0$ (relative values).

Symbol	Meaning	SI unit
m	mass of P	kg
L	cord length	m
a	amplitude of the motion of A	m
ω	frequency of the motion of A	rad/s
g	gravitational acceleration	m/s^2
φ, θ	polar and azimuthal angles of AP in $Ax'y'z'$	rad
T	cord tension	N

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial frame:** $S_0 = Oxyz = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$.
- **Moving frame:** $S_1 = Ax'y'z' = \{A; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$: its axes stay parallel to those of $Oxyz$ (figure), so the frame only translates and $[{}_0R_1] = I$ ($\det I = 1$, $II^T = I$). In the notation of the theory, $\mathbf{a}'_P \equiv \mathbf{a}_1^P$.
- **Motion of S_1: $\boldsymbol{\omega}_{10} = \mathbf{0}$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$** and $\mathbf{r}_0^A = a \sin \omega t \, \mathbf{k}$. By the chain rule, $\frac{d}{dt} \sin \omega t = \omega \cos \omega t$ and $\frac{d}{dt} \cos \omega t = -\omega \sin \omega t$:

$$\mathbf{v}_0^A = a\omega \cos \omega t \, \mathbf{k}, \qquad \mathbf{a}_0^A = -a\omega^2 \sin \omega t \, \mathbf{k}$$

- **Spherical basis** $\{\mathbf{e}_r, \mathbf{e}_\varphi, \mathbf{e}_\theta\}$ of the statement. The matrix with these columns (components in $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$)

$$[{}_1R_s] = \begin{pmatrix} \sin\varphi\cos\theta & \cos\varphi\cos\theta & -\sin\theta \\ \sin\varphi\sin\theta & \cos\varphi\sin\theta & \cos\theta \\ \cos\varphi & -\sin\varphi & 0 \end{pmatrix}$$

has orthonormal columns ($|\mathbf{e}_r|^2 = \sin^2\varphi(\cos^2\theta + \sin^2\theta) + \cos^2\varphi = 1$, $\mathbf{e}_r \cdot \mathbf{e}_\varphi = \sin\varphi\cos\varphi - \cos\varphi\sin\varphi = 0$, $\mathbf{e}_r \cdot \mathbf{e}_\theta = 0$, $\mathbf{e}_\varphi \cdot \mathbf{e}_\theta = 0$), so $[{}_1R_s]^T[{}_1R_s] = I$; and $\mathbf{e}_r \times \mathbf{e}_\varphi = \mathbf{e}_\theta$, so $\det[{}_1R_s] = 1$ (right-handed). The last row gives the vertical: $\mathbf{k} = \cos\varphi\,\mathbf{e}_r - \sin\varphi\,\mathbf{e}_\varphi$.

- **Relative position:** $\mathbf{r}_1^P = L\,\mathbf{e}_r$. Since $\varphi < \pi/2$ means $\mathbf{e}_r \cdot \mathbf{k} > 0$, at $t = 0$ ($\varphi_0 = 30^\circ$) the particle is **above** A, as drawn in the figure.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Relative acceleration

> **Pedagogical Justification:** $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are constant in S_1; the spherical vectors depend on time only through $\varphi(t)$ and $\theta(t)$, so their derivative in S_1 follows from the multivariable chain rule $\dfrac{d\mathbf{e}}{dt} = \dfrac{\partial \mathbf{e}}{\partial \varphi}\dot{\varphi} + \dfrac{\partial \mathbf{e}}{\partial \theta}\dot{\theta}$ with the partial derivatives of the statement.

Time derivatives of the basis (in S_1):

$$\dot{\mathbf{e}}_r = \dot\varphi\,\mathbf{e}_\varphi + \dot\theta\sin\varphi\,\mathbf{e}_\theta, \qquad \dot{\mathbf{e}}_\varphi = -\dot\varphi\,\mathbf{e}_r + \dot\theta\cos\varphi\,\mathbf{e}_\theta, \qquad \dot{\mathbf{e}}_\theta = -\dot\theta\,(\sin\varphi\,\mathbf{e}_r + \cos\varphi\,\mathbf{e}_\varphi)$$

Relative velocity (L constant):

$$\mathbf{v}_1^P = L\,\dot{\mathbf{e}}_r = L\dot\varphi\,\mathbf{e}_\varphi + L\dot\theta\sin\varphi\,\mathbf{e}_\theta$$

Relative acceleration. Product rule on each term: $\frac{d}{dt}\left(L\dot\varphi\,\mathbf{e}_\varphi\right) = L\ddot\varphi\,\mathbf{e}_\varphi + L\dot\varphi\,\dot{\mathbf{e}}_\varphi$ and $\frac{d}{dt}\left(L\dot\theta\sin\varphi\,\mathbf{e}_\theta\right) = L\ddot\theta\sin\varphi\,\mathbf{e}_\theta + L\dot\theta\cos\varphi\,\dot\varphi\,\mathbf{e}_\theta + L\dot\theta\sin\varphi\,\dot{\mathbf{e}}_\theta$ (chain rule $\frac{d}{dt}\sin\varphi = \cos\varphi\,\dot\varphi$). Substituting the basis derivatives:

$$\mathbf{a}_1^P = L\ddot\varphi\,\mathbf{e}_\varphi + L\dot\varphi\left(-\dot\varphi\,\mathbf{e}_r + \dot\theta\cos\varphi\,\mathbf{e}_\theta\right) + L\ddot\theta\sin\varphi\,\mathbf{e}_\theta + L\dot\theta\dot\varphi\cos\varphi\,\mathbf{e}_\theta - L\dot\theta^2\sin\varphi\,(\sin\varphi\,\mathbf{e}_r + \cos\varphi\,\mathbf{e}_\varphi)$$

Collecting along $\mathbf{e}_r$ ($-L\dot\varphi^2 - L\dot\theta^2\sin^2\varphi$), $\mathbf{e}_\varphi$ ($L\ddot\varphi - L\dot\theta^2\sin\varphi\cos\varphi$) and $\mathbf{e}_\theta$ ($L\dot\varphi\dot\theta\cos\varphi + L\ddot\theta\sin\varphi + L\dot\theta\dot\varphi\cos\varphi$):

$$\mathbf{a}'_P \equiv \mathbf{a}_1^P = -L\left(\dot\varphi^2 + \dot\theta^2\sin^2\varphi\right)\mathbf{e}_r + L\left(\ddot\varphi - \dot\theta^2\sin\varphi\cos\varphi\right)\mathbf{e}_\varphi + L\left(\ddot\theta\sin\varphi + 2\dot\theta\dot\varphi\cos\varphi\right)\mathbf{e}_\theta$$

which is the expression of the statement.

(b) Equations of motion and tension

> **Pedagogical Justification:** in the translating frame, Notes Eq. 4.63 with $\boldsymbol{\omega}_{10} = \boldsymbol{\alpha}_{10} = \mathbf{0}$ reduces to $m\mathbf{a}_1^P = \mathbf{F} - m\mathbf{a}_0^A$ (equivalently Notes Eq. 3.29, $\mathbf{a}_0^P = \mathbf{a}_0^A + \mathbf{a}_1^P$, inserted in $m\mathbf{a}_0^P = \mathbf{F}$). Projecting on the spherical basis separates the tension (only along $\mathbf{e}_r$) from the two equations of motion.

Forces in S_1. Tension $-T\,\mathbf{e}_r$; weight $-mg\,\mathbf{k}$; inertia force of the origin $\mathbf{F}_A = -m\mathbf{a}_0^A = ma\omega^2\sin\omega t\,\mathbf{k}$. Weight and $\mathbf{F}_A$ combine into an effective, time-dependent gravity:

$$-mg\,\mathbf{k} + ma\omega^2\sin\omega t\,\mathbf{k} = -m\,g_e(t)\,\mathbf{k}, \qquad g_e(t) = g - a\omega^2\sin\omega t$$

With $\mathbf{k} = \cos\varphi\,\mathbf{e}_r - \sin\varphi\,\mathbf{e}_\varphi$, this force is $-mg_e\cos\varphi\,\mathbf{e}_r + mg_e\sin\varphi\,\mathbf{e}_\varphi$.

Three components of $m\mathbf{a}_1^P = -T\,\mathbf{e}_r - mg_e\,\mathbf{k}$:

$$\mathbf{e}_r: \quad -mL\left(\dot\varphi^2+\dot\theta^2\sin^2\varphi\right)=-T-mg_e(t)\cos\varphi$$

$$\mathbf{e}_\varphi: \quad mL\left(\ddot\varphi-\dot\theta^2\sin\varphi\cos\varphi\right)=mg_e(t)\sin\varphi$$

$$\mathbf{e}_\theta: \quad mL\left(\ddot\theta\sin\varphi+2\dot\theta\dot\varphi\cos\varphi\right)=0$$

The requested system (solved for the unknowns):

$$\begin{cases}\ddot\varphi=\dot\theta^2\sin\varphi\cos\varphi+\dfrac{g-a\omega^2\sin\omega t}{L}\sin\varphi\\ \ddot\theta\sin\varphi+2\dot\theta\dot\varphi\cos\varphi=0\\ T=mL\left(\dot\varphi^2+\dot\theta^2\sin^2\varphi\right)-m\left(g-a\omega^2\sin\omega t\right)\cos\varphi\end{cases}$$

The first two equations, with the initial conditions, give $\varphi(t)$ and $\theta(t)$; the third gives $T(t)$. Multiplying the second by $\sin\varphi$, product and chain rules show it is $\frac{d}{dt}\left(\dot\theta\sin^2\varphi\right)=\ddot\theta\sin^2\varphi+2\dot\theta\sin\varphi\cos\varphi\,\dot\varphi=0$: the vertical component of the angular momentum about A, $mL^2\dot\theta\sin^2\varphi$, is conserved, because tension, weight and $\mathbf{F}_A$ have no moment about the vertical axis through A (Topic 4).

(c) Relative acceleration at $t=0$

Data: $\sin\omega\cdot0=0$, so $g_e(0)=g$; $\sin\varphi_0=\frac12$, $\cos\varphi_0=\frac{\sqrt3}{2}$, $\dot\theta_0^2=\dfrac{2g}{\sqrt3L}$, $\dot\varphi_0=0$.

Angular accelerations. From the first equation of (b):

$$\ddot\varphi_0=\frac{2g}{\sqrt3L}\cdot\frac12\cdot\frac{\sqrt3}{2}+\frac{g}{L}\cdot\frac12=\frac{g}{2L}+\frac{g}{2L}=\frac{g}{L}$$

From the second, $\ddot\theta_0\sin\varphi_0=-2\dot\theta_0\cdot0\cdot\cos\varphi_0=0$, so $\ddot\theta_0=0$.

Components of $\mathbf{a}_1^P(0)$. $\dot\theta_0^2\sin^2\varphi_0=\dfrac{2g}{\sqrt3L}\cdot\dfrac14=\dfrac{g}{2\sqrt3L}$ and $\dot\theta_0^2\sin\varphi_0\cos\varphi_0=\dfrac{2g}{\sqrt3L}\cdot\dfrac{\sqrt3}{4}=\dfrac{g}{2L}$:

$$\mathbf{a}_1^P(0) = -\frac{g}{2\sqrt{3}}\,\mathbf{e}_r + L\left(\frac{g}{L} - \frac{g}{2L}\right)\mathbf{e}_\varphi + 0\,\mathbf{e}_\theta = -\frac{g}{2\sqrt{3}}\,\mathbf{e}_r + \frac{g}{2}\,\mathbf{e}_\varphi$$

At $\theta = 0$, $\varphi = 30^\circ$: $\mathbf{e}_r = \tfrac{1}{2}\mathbf{i} + \tfrac{\sqrt{3}}{2}\mathbf{k}$ and $\mathbf{e}_\varphi = \tfrac{\sqrt{3}}{2}\mathbf{i} - \tfrac{1}{2}\mathbf{k}$, so the $\mathbf{i}$ component is $-\frac{g}{4\sqrt{3}} + \frac{\sqrt{3}g}{4} = \frac{(3-1)g}{4\sqrt{3}} = \frac{g}{2\sqrt{3}}$ and the $\mathbf{k}$ component is $-\frac{g}{4} - \frac{g}{4} = -\frac{g}{2}$:

$$\mathbf{a}_1^P(0) = \frac{g}{2\sqrt{3}}\,\mathbf{i} - \frac{g}{2}\,\mathbf{k}, \qquad \left|\mathbf{a}_1^P(0)\right| = g\sqrt{\frac{1}{12} + \frac{1}{4}} = \frac{g}{\sqrt{3}}$$

(d) Tension at $t = 0$

From the third equation of (b):

$$T(0) = mL\left(0 + \frac{g}{2\sqrt{3}L}\right) - mg\,\frac{\sqrt{3}}{2} = mg\left(\frac{1}{2\sqrt{3}} - \frac{3}{2\sqrt{3}}\right) = -\frac{mg}{\sqrt{3}}$$

Interpretation. The value is negative: at $t = 0$ the cord would have to push the particle outwards. A cord can only pull, so with the data as stated the cord is not taut at $t = 0$ and the taut-cord model of (a)–(c) does not apply. The actual motion starts as a free motion (the cord goes slack), with $m\mathbf{a}_1^P = -mg_e\,\mathbf{k}$, i.e. $\mathbf{a}_1^P(0) = -g\,\mathbf{k}$ instead of the value of (c). The result $T(0) = -mg/\sqrt{3}$ would be meaningful as a compression if P were held by a massless rigid rod instead of a cord.

Data inconsistency in the statement (no official key)
With φ measured from the upward axis Az' (figure and $\mathbf{e}_r \cdot \mathbf{k} = \cos\varphi$), the given $\dot\theta_0$ cannot keep the cord taut: $T(0) = -mg/\sqrt{3} < 0$. The given value $\dot\theta_0^2 = 2g/(\sqrt{3}L) = g/(L\cos 30^\circ)$ is exactly the rate of a steady **conical pendulum hanging below** A at 30° from the downward vertical. Under that alternative reading (which contradicts the figure and the given $\mathbf{e}_r$, so it is only a remark), the same method gives $\ddot\varphi_0 = 0$, $\mathbf{a}_1^P(0) = -\frac{g}{\sqrt{3}}\,\mathbf{i}$ (purely centripetal, towards the axis) and $T(0) = mL\dot\theta_0^2\sin^2\varphi_0 + mg\cos\varphi_0 = \frac{mg}{2\sqrt{3}} + \frac{\sqrt{3}mg}{2} = \frac{2mg}{\sqrt{3}}$. The answers above follow the statement as written.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Effective gravity.** In the translating frame of A the oscillation of the support only modulates gravity, $g_e(t) = g - a\omega^2 \sin \omega t$ (Slide 28: the force due to the acceleration of the origin is indistinguishable from a gravitational field). When $a\omega^2 > g$, g_e changes sign periodically and the upper position can be dynamically stabilised (inverted pendulum with a vibrating support; extension, not course material).
- **Limits.** $a \to 0$ or $\omega \to 0$: $g_e = g$ and the system is the classical spherical pendulum written with φ from the upward vertical (equilibrium $\varphi = 0$ unstable, $\varphi = \pi$ stable). $\dot\theta = 0$: the θ equation is satisfied identically and the motion is planar.
- **Signs at $t = 0$.** $\ddot\varphi_0 = g/L > 0$: the particle starts to fall away from the upward vertical, since both the weight and the centrifugal effect of $\dot\theta_0$ push φ to grow.
- **Dimensions:** $[a\omega^2] = \mathrm{m \cdot s^{-2}} = [g]$; $[\dot\theta^2] = [g/L] = \mathrm{s^{-2}}$, so each term of the $\ddot\varphi$ equation is in $\mathrm{s^{-2}}$; $[mL\dot\theta^2] = [mg] = \mathrm{N}$; $[L\dot\varphi^2] = [L\ddot\theta] = \mathrm{m \cdot s^{-2}}$.
- **Symbolic check (sympy, `verify_P28.py`):** orthonormality and handedness of the spherical basis and the six partial derivatives of the statement; $\mathbf{v}_1^P$ and $\mathbf{a}_1^P$ by direct differentiation; $\mathbf{a}_0^P = \mathbf{a}_0^A + \mathbf{a}_1^P$; the three equations of (b) solved from the vector equation; $\ddot\varphi_0 = g/L, \ddot\theta_0 = 0, \mathbf{a}_1^P(0), |\mathbf{a}_1^P(0)| = g/\sqrt{3}, T(0) = -mg/\sqrt{3}$; and the alternative reading ($\ddot\varphi_0 = 0, T(0) = 2mg/\sqrt{3}$): all verified.

Source: Problems.pdf

PROBLEM 30

Particle on a Rotating Quarter Circumference with a Spring

Original statement · Problems.pdf, p. 18

Rotating Frame + Path Variables

A point particle P of mass m is constrained to move on a smooth quarter of circumference as shown in the figure below. We define a moving reference frame $OX_1Y_1Z_1$ that rotates with respect to an inertial reference frame $OXYZ$. The rotation of the moving reference frame is performed along a vertical axis defined as $x_1 = -R$ and $y_1 = 0$; the angular velocity is $\boldsymbol{\omega} = \omega\mathbf{k} = \omega\mathbf{k}_1$. The particle is attached to a linear spring with the other end attached to the point B of the reference frame $OX_1Y_1Z_1$ with coordinates $(-2R, 0, 0)$. The constant of the spring is $K = \alpha m\omega^2$. The angle between $\overline{OP}$ and the negative side of the z_1 axis, ϕ, is used to define the position of the particle. Neglect the action of gravity on the particle.

Initially the particle is located at $\phi = 0$ and it is set to motion with a relative velocity $-\beta\omega R\,\mathbf{i}_1$. Both α and β are positive nondimensional parameters.

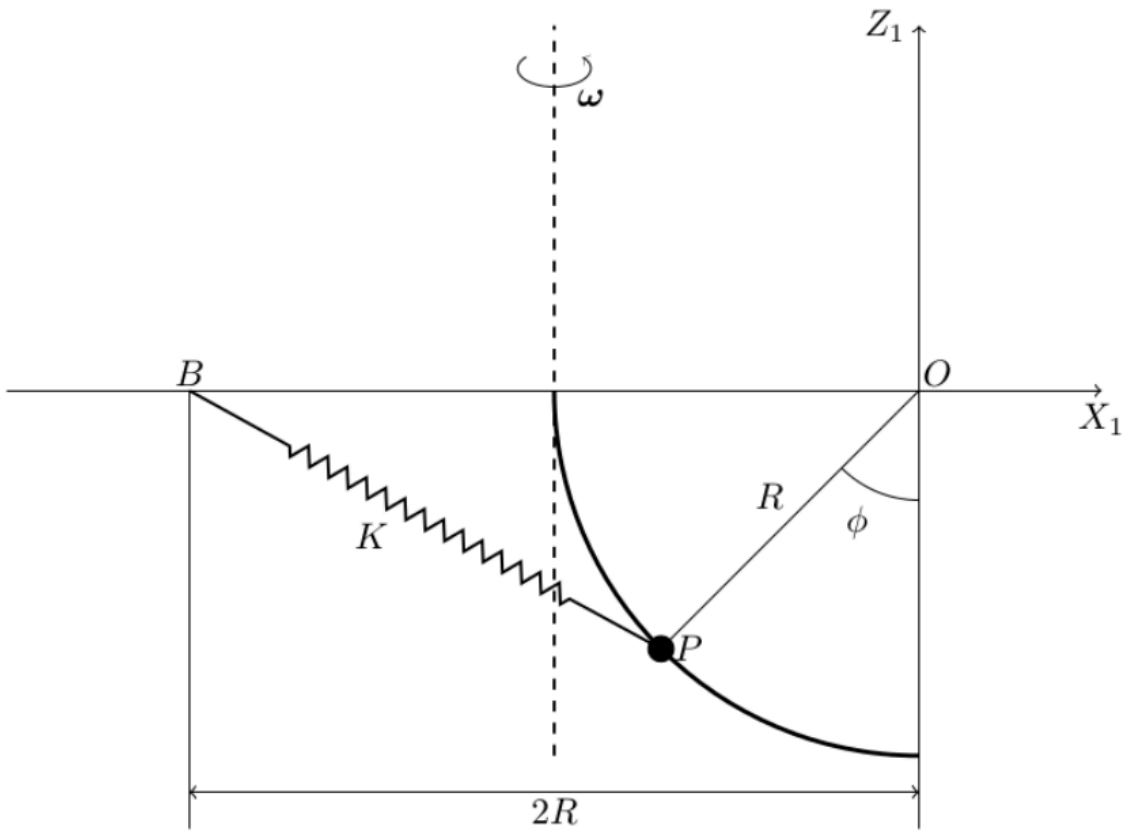


Figure: Quarter circumference of radius R centred at O in the plane X1Z1, rotation axis x1 = −R (dashed), spring of constant K from B = (−2R, 0, 0) to P, angle φ measured from the negative z1 axis (Problems.pdf, p. 18).

(a) Compute the relative velocity and the relative acceleration of the particle in the path variable reference frame $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$ as functions of the coordinate ϕ and their derivatives.

(b) Write down the three components of Newton's second law expressed in the path variables reference frame $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$.

(c) Express the energy conservation equation in the relative reference frame. Discuss if this equation is equivalent to one of the above ones.

Let us now consider a different problem. Depending on the values of the parameters, we may find the equilibrium of the forces at different positions.

(d) Obtain the value of α to have an equilibrium position when the particle is positioned at ϕ equal to $\pi/6$.

For this value of α, the particle is again located at $\phi = 0$ and is set to motion with the relative velocity defined before.

(e) Describe the possible types of motion for the different values of β. In particular, obtain the critical value of β that makes the particle reach the position $\phi = \pi/2$ and the reaction force of the curve to the particle at that instant.

(f) For a value of β higher than the critical one, the particle leaves the arc of circumference and its motion is not anymore constrained. Under these conditions compute the maximum height reached by the particle, z_{max}, as a function of β. Note that the particle is still under the action of the spring.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **System:** a particle P of mass m slides on a smooth quarter circumference of radius R and centre O, drawn in the vertical plane X_1Z_1 of a frame that rotates with constant angular velocity $\omega\mathbf{k}_1$ about the vertical axis $x_1 = -R$, $y_1 = 0$. A spring of constant $K = \alpha m\omega^2$

joins P to the point $B = (-2R, 0, 0)$ of the rotating frame. Gravity is neglected.

- **Why a rotating frame.** The wire and the spring anchor B are fixed in $OX_1Y_1Z_1$, so the constraint is a fixed circle there and the position of P is described by one angle. The price is the inertia force of Notes Eqs. 4.62–4.63, which is easy to compute because $\boldsymbol{\omega}_{10}$ is constant. In the inertial frame the wire itself moves, and the constraint would be harder to write.
- **Hypotheses:**
  1. $OXYZ$ is inertial; the frame rotates uniformly, $\boldsymbol{\omega}_{10} = \omega \mathbf{k}_1$ with ω constant.
  2. The arc is smooth (no friction) and acts as a bilateral constraint (a bead on a wire): the reaction has no tangential component but its normal and binormal components can take either sign.
  3. The spring is linear and, since no natural length is given, of **zero natural length**: $\mathbf{F}_K = -K\,\overrightarrow{BP}$.
  4. Gravity is neglected, as the statement says.
- **Degrees of freedom:** on the arc, P has one degree of freedom, the angle $\phi \in [0, \pi/2]$ between $\overline{OP}$ and the negative z_1 axis. After leaving the arc in part (f), P is free: three degrees of freedom (x_1, y_1, z_1).
- **Initial conditions:** $\phi(0) = 0$, $\mathbf{v}_1^P(0) = -\beta\omega R\,\mathbf{i}_1$.

Symbol	Meaning	SI unit
m	mass of P	kg
R	radius of the arc and distance from O to the axis	m
ω	constant angular velocity of the frame	rad/s
$K = \alpha m \omega^2$	spring constant	N/m = kg/s^2
α, β	positive nondimensional parameters	1
ϕ	angle between $\overline{OP}$ and the $-z_1$ axis	rad
N_n, N_b	components of the reaction of the arc	N

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Fixed frame.** The only points that do not move are those of the rotation axis. We take the inertial frame $S_0 = \{Q; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ with origin at the point Q of the axis at the height of O (the point $x_1 = -R$, $y_1 = z_1 = 0$, the upper end of the arc). It plays the role of the statement's $OXYZ$; moving the origin of an inertial frame to another fixed point does not change any velocity or acceleration. $\mathbf{k}_0 = \mathbf{k}$ is vertical.
- **Moving frame.** $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ is the statement's $OX_1Y_1Z_1$, attached to the arc, with $\mathbf{k}_1 = \mathbf{k}_0$. Its origin O is at distance R from the axis, so it moves on a horizontal circle.
- **Change of basis.** S_1 turns about $\mathbf{k}_0$ through the angle ωt (we choose $\mathbf{i}_1 = \mathbf{i}_0$ at $t = 0$). The columns of $[{}_0R_1]$ are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- **Checks.** $\det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1$. The columns have unit length ($\cos^2\omega t + \sin^2\omega t = 1$) and the first two are orthogonal ($-\cos\omega t\sin\omega t + \sin\omega t\cos\omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.
- **Angular motion of S_1:** $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$ and, ω being constant, $\boldsymbol{\alpha}_{10} = \left.\dfrac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \mathbf{0}$.
- **Motion of the origin O.** $\mathbf{r}_0^O = \overrightarrow{QO} = R\,\mathbf{i}_1$. By the Coriolis theorem (Notes Eq. 3.8), since $R\,\mathbf{i}_1$ is constant in S_1:

$$\mathbf{v}_0^O = \left.\frac{d(R\mathbf{i}_1)}{dt}\right|_1 + \boldsymbol{\omega}_{10}\times R\,\mathbf{i}_1 = \mathbf{0} + \omega R\,(\mathbf{k}_1\times\mathbf{i}_1) = \omega R\,\mathbf{j}_1$$

$$\mathbf{a}_0^O = \left.\frac{d(\omega R\mathbf{j}_1)}{dt}\right|_1 + \boldsymbol{\omega}_{10}\times\omega R\,\mathbf{j}_1 = \omega^2 R\,(\mathbf{k}_1\times\mathbf{j}_1) = -\omega^2 R\,\mathbf{i}_1$$

- **Path (intrinsic) basis.** On the arc, $\mathbf{r}_1^P = -R\sin\phi\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1$. We take the arc length $s = R\phi$ (increasing with ϕ), the tangent $\mathbf{e}_t$ in the direction of increasing ϕ, the principal normal $\mathbf{e}_n$ towards the centre of curvature O, and $\mathbf{e}_b = \mathbf{e}_t\times\mathbf{e}_n$:

$$\mathbf{e}_t = \frac{1}{R}\frac{d\mathbf{r}_1^P}{d\phi} = -\cos\phi\,\mathbf{i}_1 + \sin\phi\,\mathbf{k}_1, \qquad \mathbf{e}_n = -\frac{\mathbf{r}_1^P}{R} = \sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{k}_1$$

$$\mathbf{e}_b = \mathbf{e}_t\times\mathbf{e}_n = \begin{vmatrix} \mathbf{i}_1 & \mathbf{j}_1 & \mathbf{k}_1 \\ -\cos\phi & 0 & \sin\phi \\ \sin\phi & 0 & \cos\phi \end{vmatrix} = (0-0)\,\mathbf{i}_1 - (-\cos^2\phi - \sin^2\phi)\,\mathbf{j}_1 + (0-0)\,\mathbf{k}_1 = \mathbf{j}_1$$

- The matrix whose columns are $\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b$ in $\mathcal{B}_1$ is

$$[{}_1R_t] = \begin{pmatrix} -\cos\phi & \sin\phi & 0 \\ 0 & 0 & 1 \\ \sin\phi & \cos\phi & 0 \end{pmatrix}, \qquad \det[{}_1R_t] = -\cos\phi\,(0-\cos\phi) - \sin\phi\,(0-\sin\phi) = 1$$

and its columns are unit and mutually orthogonal ($\mathbf{e}_t \cdot \mathbf{e}_n = -\cos\phi\sin\phi + \sin\phi\cos\phi = 0$), so $[{}_1R_t][{}_1R_t]^T = I$.

- **Useful projections:** $\mathbf{i}_1 \cdot \mathbf{e}_t = -\cos\phi$, $\mathbf{i}_1 \cdot \mathbf{e}_n = \sin\phi$, $\mathbf{k}_1 \cdot \mathbf{e}_t = \sin\phi$, $\mathbf{k}_1 \cdot \mathbf{e}_n = \cos\phi$, $\mathbf{j}_1 = \mathbf{e}_b$.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Relative velocity and acceleration in the path basis

Pedagogical Justification: The relative velocity and acceleration are the first and second derivatives of $\mathbf{r}_1^P$ taken in S_1, where $\mathbf{i}_1, \mathbf{k}_1$ are constant. Only ϕ depends on time, so each derivative is a chain rule.

Differentiate $\mathbf{r}_1^P = -R\sin\phi\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1$ with $\dfrac{d}{dt}\sin\phi = \cos\phi\,\dot\phi$ and $\dfrac{d}{dt}\cos\phi = -\sin\phi\,\dot\phi$:

$$\mathbf{v}_1^P = \left.\frac{d\mathbf{r}_1^P}{dt}\right|_1 = -R\cos\phi\,\dot\phi\,\mathbf{i}_1 + R\sin\phi\,\dot\phi\,\mathbf{k}_1 = R\dot\phi\,(-\cos\phi\,\mathbf{i}_1 + \sin\phi\,\mathbf{k}_1)$$

$$\boxed{\mathbf{v}_1^P = R\dot\phi\,\mathbf{e}_t}$$

which is $\dot s\,\mathbf{e}_t$ with $s = R\phi$. For the acceleration use the product rule on $R\dot\phi\,\mathbf{e}_t$, and the chain rule on $\mathbf{e}_t(\phi(t))$:

$$\left.\frac{d\mathbf{e}_t}{dt}\right|_1 = \frac{d\mathbf{e}_t}{d\phi}\,\dot\phi = (\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{k}_1)\,\dot\phi = \dot\phi\,\mathbf{e}_n$$

$$\mathbf{a}_1^P = \left.\frac{d\mathbf{v}_1^P}{dt}\right|_1 = R\ddot{\phi}\,\mathbf{e}_t + R\dot{\phi}\left.\frac{d\mathbf{e}_t}{dt}\right|_1 \implies \boxed{\mathbf{a}_1^P = R\ddot{\phi}\,\mathbf{e}_t + R\dot{\phi}^2\,\mathbf{e}_n}$$

The normal part is $\dot{s}^2/\rho$ with radius of curvature $\rho = R$, as expected for a circle. There is no binormal component.

(b) Newton's second law in the path basis

> **Pedagogical Justification:** S_1 is not inertial, so we use Newton's law with inertia forces, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eqs. 4.62–4.63, Slides 27–28). This keeps the simple relative acceleration of part (a) and puts the frame's motion into four forces.

Actual forces.

- Spring (zero natural length): $\overrightarrow{BP} = \mathbf{r}_1^P - \mathbf{r}_1^B = (2R - R\sin\phi)\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1$, so

$$\mathbf{F}_K = -\alpha m\omega^2 R\left[(2-\sin\phi)\,\mathbf{i}_1 - \cos\phi\,\mathbf{k}_1\right]$$

- Reaction of the smooth arc: $\mathbf{N} = N_n\,\mathbf{e}_n + N_b\,\mathbf{e}_b$ (no $\mathbf{e}_t$ component, because the arc is smooth).

Inertia forces (Slide 28), with $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, $\mathbf{a}_0^O = -\omega^2 R\,\mathbf{i}_1$:

- Force due to the acceleration of the origin: $\mathbf{F}_A = -m\mathbf{a}_0^O = m\omega^2 R\,\mathbf{i}_1$.
- Euler force: $\mathbf{F}_E = -m\boldsymbol{\alpha}_{10}\times\mathbf{r}_1^P = \mathbf{0}$.
- Centrifugal force. First $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = \omega\mathbf{k}_1\times(-R\sin\phi\,\mathbf{i}_1 - R\cos\phi\,\mathbf{k}_1) = -\omega R\sin\phi\,\mathbf{j}_1$ (using $\mathbf{k}_1\times\mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$); then $\boldsymbol{\omega}_{10}\times(-\omega R\sin\phi\,\mathbf{j}_1) = -\omega^2 R\sin\phi\,(\mathbf{k}_1\times\mathbf{j}_1) = \omega^2 R\sin\phi\,\mathbf{i}_1$, so $\mathbf{F}_C = -m\omega^2 R\sin\phi\,\mathbf{i}_1$.
- Coriolis force: $\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P = \omega R\dot{\phi}\,\mathbf{k}_1\times(-\cos\phi\,\mathbf{i}_1 + \sin\phi\,\mathbf{k}_1) = -\omega R\dot{\phi}\cos\phi\,\mathbf{j}_1$, so $\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P = 2m\omega R\dot{\phi}\cos\phi\,\mathbf{j}_1$.

The first and third combine into the centrifugal force about the **true** rotation axis, whose distance to P is $x_1 + R = R(1-\sin\phi)$ (Slide 30):

$$\mathbf{F}_A + \mathbf{F}_C = m\omega^2 R\,(1-\sin\phi)\,\mathbf{i}_1$$

Projections. With the projections listed in Phase 2:

$$\mathbf{F}_K \cdot \mathbf{e}_t = -\alpha m\omega^2 R\,[(2-\sin\phi)(-\cos\phi) + (-\cos\phi)(\sin\phi)] = -\alpha m\omega^2 R\,[-2\cos\phi + \sin\phi\cos\phi - \sin\phi\cos\phi] = 2\alpha m\omega^2 R\cos\phi$$

$$\mathbf{F}_K \cdot \mathbf{e}_n = -\alpha m\omega^2 R\left[(2-\sin\phi)\sin\phi - \cos^2\phi\right] = -\alpha m\omega^2 R\,[2\sin\phi - 1] = \alpha m\omega^2 R\,(1-2\sin\phi)$$

$$(\mathbf{F}_A + \mathbf{F}_C)\cdot\mathbf{e}_t = -m\omega^2 R\,(1-\sin\phi)\cos\phi, \qquad (\mathbf{F}_A + \mathbf{F}_C)\cdot\mathbf{e}_n = m\omega^2 R\,(1-\sin\phi)\sin\phi$$

and $\mathbf{F}_K$, $\mathbf{F}_A + \mathbf{F}_C$ have no $\mathbf{e}_b = \mathbf{j}_1$ component, while $\mathbf{F}_{COR}$ lies entirely along $\mathbf{e}_b$.

The three components of $m\mathbf{a}_1^P = \mathbf{F}_K + \mathbf{N} + \mathbf{F}_A + \mathbf{F}_E + \mathbf{F}_C + \mathbf{F}_{COR}$:

$$\mathbf{e}_t: \quad mR\ddot{\phi} = 2\alpha m\omega^2 R\cos\phi - m\omega^2 R\,(1-\sin\phi)\cos\phi$$

$$\mathbf{e}_n: \quad mR\dot{\phi}^2 = N_n + \alpha m\omega^2 R\,(1-2\sin\phi) + m\omega^2 R\,(1-\sin\phi)\sin\phi$$

$$\mathbf{e}_b: \quad 0 = N_b + 2m\omega R\dot{\phi}\cos\phi$$

The tangential component contains no reaction and is the **equation of motion**; the other two give the **reaction**:

$$\boxed{\ddot{\phi} = \omega^2\cos\phi\;(2\alpha - 1 + \sin\phi)}$$

$$\boxed{N_n = mR\dot{\phi}^2 - \alpha m\omega^2 R\,(1-2\sin\phi) - m\omega^2 R\,\sin\phi\,(1-\sin\phi), \qquad N_b = -2m\omega R\dot{\phi}\cos\phi}$$

The binormal reaction balances the Coriolis force: the arc has to push P sideways to keep it in the rotating plane.

(c) Energy conservation in the relative frame

> **Pedagogical Justification:** In S_1 (Slide 29) the mechanical energy E_1^P is conserved if every force that does work in S_1 has a potential. Here: $\mathbf{N}$ does no work in S_1 because the arc is fixed in S_1 and $\mathbf{N} \perp \mathbf{v}_1^P = R\dot{\phi}\,\mathbf{e}_t$; $\mathbf{F}_{COR} \perp \mathbf{v}_1^P$ never does work (Slide 31); $\mathbf{F}_E = \mathbf{0}$; the spring is conservative; and $\mathbf{F}_A + \mathbf{F}_C$ is the centrifugal force about an axis fixed in S_1 with constant ω, which has the potential of Slide 30.

Potentials. Spring: $V_K = \frac{1}{2}K\,|\overrightarrow{BP}|^2$ with

$$|\overrightarrow{BP}|^2 = R^2\left[(2-\sin\phi)^2 + \cos^2\phi\right] = R^2\left[4 - 4\sin\phi + \sin^2\phi + \cos^2\phi\right] = R^2\,(5 - 4\sin\phi)$$

so $V_K = \frac{1}{2}\alpha m\omega^2 R^2\,(5 - 4\sin\phi)$. Centrifugal (Slide 30), with $r_\perp = x_1 + R = R(1-\sin\phi)$:

$$V_C = -\tfrac{1}{2}m\omega^2 r_\perp^2 = -\tfrac{1}{2}m\omega^2 R^2\,(1-\sin\phi)^2$$

(Check: in Cartesian form $V_C = -\frac{1}{2}m\omega^2(x_1+R)^2$ and $-\partial V_C/\partial x_1 = m\omega^2(x_1+R)$, the $\mathbf{i}_1$ component of $\mathbf{F}_A + \mathbf{F}_C$.)

Energy equation:

$$\boxed{E_1^P = \tfrac{1}{2}mR^2\dot{\phi}^2 + \tfrac{1}{2}\alpha m\omega^2 R^2\,(5 - 4\sin\phi) - \tfrac{1}{2}m\omega^2 R^2\,(1-\sin\phi)^2 = \text{const}}$$

Initial value. At $t=0$, $\mathbf{e}_t(0) = -\mathbf{i}_1$, so $\mathbf{v}_1^P(0) = R\dot{\phi}(0)\,(-\mathbf{i}_1) = -\beta\omega R\,\mathbf{i}_1$ gives $\dot{\phi}(0) = \beta\omega$, and $E_1^P(0) = \frac{1}{2}m\omega^2R^2\,(\beta^2 + 5\alpha - 1)$. Equating and dividing by $\frac{1}{2}mR^2$:

$$\dot{\phi}^2 = \omega^2\left[\beta^2 + 5\alpha - 1 - \alpha(5 - 4\sin\phi) + (1-\sin\phi)^2\right] \;\Longrightarrow\; \boxed{\dot{\phi}^2 = \omega^2\left[\beta^2 + (4\alpha - 2)\sin\phi + \sin^2\phi\right]}$$

Equivalence. Differentiate E_1^P in time with the chain rule ($\frac{d}{dt}\sin\phi = \cos\phi\,\dot{\phi}$, $\frac{d}{dt}(1-\sin\phi)^2 = 2(1-\sin\phi)(-\cos\phi\,\dot{\phi})$):

$$\frac{dE_1^P}{dt} = mR^2\dot{\phi}\ddot{\phi} - 2\alpha m\omega^2R^2\cos\phi\,\dot{\phi} + m\omega^2R^2(1-\sin\phi)\cos\phi\,\dot{\phi} = mR^2\dot{\phi}\left[\ddot{\phi} - 2\alpha\omega^2\cos\phi + \omega^2(1-\sin\phi)\cos\phi\right]$$

The bracket is zero exactly when the $\mathbf{e}_t$ equation of part (b) holds. So the energy equation is **not new information**: it is the first integral of the tangential component (the $\mathbf{e}_t$ equation multiplied by $R\dot{\phi}$ and integrated once). The $\mathbf{e}_n$ and $\mathbf{e}_b$ components are independent of it and are needed to compute the reaction.

(d) Value of α for equilibrium at $\phi = \pi/6$

> **Pedagogical Justification:** Relative equilibrium in S_1 means $\dot{\phi} = 0$ and $\ddot{\phi} = 0$ for all time. The $\mathbf{e}_t$ equation then requires that the tangential force vanish.

$$\ddot{\phi} = 0 \iff \cos\phi\,(2\alpha - 1 + \sin\phi) = 0$$

The root $\cos\phi = 0$ ($\phi = \pi/2$, the particle on the axis) is an equilibrium for every α. Any other equilibrium must satisfy $\sin\phi_e = 1 - 2\alpha$. For $\phi_e = \pi/6$, $\sin\phi_e = \tfrac{1}{2}$:

$$2\alpha - 1 + \tfrac{1}{2} = 0 \implies \boxed{\alpha = \tfrac{1}{4}}$$

At that equilibrium, from part (b) with $\dot{\phi} = 0$: $N_b = 0$ and $N_n = -\tfrac{1}{4}m\omega^2 R\,(1-1) - m\omega^2 R \cdot \tfrac{1}{2}\cdot\tfrac{1}{2} = -\tfrac{1}{4}m\omega^2 R$, i.e. the arc pulls P away from O with force $\tfrac{1}{4}m\omega^2 R$.

(e) Types of motion for $\alpha = 1/4$, critical β and reaction at $\phi = \pi/2$

> **Pedagogical Justification:** The energy integral of part (c) is a first-order equation $\dot{\phi}^2 = \omega^2[\beta^2 - W(\phi)]$; motion is possible only where $W(\phi) \le \beta^2$, so the shape of W classifies the motions, as with a one-dimensional potential.

With $\alpha = \tfrac{1}{4}$, $4\alpha - 2 = -1$:

$$\dot{\phi}^2 = \omega^2\left[\beta^2 - W(\phi)\right], \qquad W(\phi) = \sin\phi\,(1 - \sin\phi)$$

On $[0, \pi/2]$: $W(0) = 0$, $W(\pi/2) = 0$, and $\dfrac{dW}{d\phi} = \cos\phi\,(1 - 2\sin\phi)$ vanishes inside the interval only at $\sin\phi = \tfrac{1}{2}$, i.e. $\phi = \pi/6$, where W has its maximum $W(\pi/6) = \tfrac{1}{2}\cdot\tfrac{1}{2} = \tfrac{1}{4}$. W plays the role of a potential barrier, so the equilibrium $\phi = \pi/6$ is **unstable**. Three types of motion follow:

- $\beta > \tfrac{1}{2}$**:** $\dot{\phi}^2 \ge \omega^2(\beta^2 - \tfrac{1}{4}) > 0$ on the whole arc, so ϕ increases monotonically, crosses the barrier at $\pi/6$ and reaches $\phi = \pi/2$ with $\dot{\phi}^2 = \omega^2[\beta^2 - 0]$, i.e. $\dot{\phi} = \beta\omega$: the relative speed there equals the initial one, $\beta\omega R$. The particle then leaves the arc at its upper end (part f).

- $\beta = \frac{1}{2}$**:** $\beta^2 - W = \frac{1}{4} - \sin\phi + \sin^2\phi = (\frac{1}{2} - \sin\phi)^2$, so $\dot\phi = \omega(\frac{1}{2} - \sin\phi)$. The particle approaches the unstable equilibrium $\phi = \pi/6$ asymptotically and never reaches it: near it, $\sin\phi \approx \frac{1}{2} + \frac{\sqrt{3}}{2}(\phi - \frac{\pi}{6})$ and $\dot\phi \approx -\frac{\sqrt{3}}{2}\omega\,(\phi - \frac{\pi}{6})$, an exponential approach with rate $\frac{\sqrt{3}}{2}\omega$.
- $\beta < \frac{1}{2}$**:** $\dot\phi$ vanishes at the turning point ϕ^* with $W(\phi^*) = \beta^2$, i.e. $\sin^2\phi^* - \sin\phi^* + \beta^2 = 0$; the root reached first is

$$\sin\phi^* = \frac{1 - \sqrt{1 - 4\beta^2}}{2} < \frac{1}{2}$$

There $\ddot\phi = \omega^2\cos\phi^*\,(\sin\phi^* - \frac{1}{2}) < 0$, so the particle turns back, returns to $\phi = 0$ with $\dot\phi = -\beta\omega$ (relative velocity $+\beta\omega R\,\mathbf{i}_1$, same speed by energy conservation) and leaves the quarter circumference at its lower end.

Critical value. The particle reaches $\phi = \pi/2$ if and only if β exceeds

$$\boxed{\beta_c = \tfrac{1}{2}}$$

Reaction at $\phi = \pi/2$. Here $\cos\phi = 0$, $\sin\phi = 1$, $\dot\phi = \beta\omega$, and $\mathbf{e}_n(\pi/2) = \mathbf{i}_1$. From part (b) with $\alpha = \frac{1}{4}$:

$$N_b = -2m\omega R\,\beta\omega\cdot 0 = 0, \qquad N_n = mR\beta^2\omega^2 - \tfrac{1}{4}m\omega^2 R\,(1-2) - m\omega^2 R\cdot 1\cdot 0 = m\omega^2 R\left(\beta^2 + \tfrac{1}{4}\right)$$

$$\boxed{\mathbf{N}(\phi = \tfrac{\pi}{2}) = m\omega^2 R\left(\beta^2 + \tfrac{1}{4}\right)\mathbf{i}_1 \xrightarrow{\beta\to\beta_c} \tfrac{1}{2}m\omega^2 R\,\mathbf{i}_1}$$

It points towards O: it supplies the centripetal force $mR\dot\phi^2$ and also balances the spring, which pulls P towards B with $\frac{1}{4}m\omega^2 R$. The centrifugal and Coriolis forces vanish there because P is on the axis and moves parallel to it.

(f) Maximum height after leaving the arc ($\beta > \beta_c$)

> **Pedagogical Justification:** Once free, P has three degrees of freedom, and the horizontal motion is coupled through the Coriolis and centrifugal forces. But every inertia force of a frame rotating about the vertical is horizontal, so the vertical component of $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ involves only the spring and decouples.

Initial state of the free motion (time $\tau = 0$ at separation): $\phi = \pi/2$ gives $\mathbf{r}_1^P = -R\,\mathbf{i}_1$ (on the axis, $z_1 = 0$), and $\mathbf{v}_1^P = R\dot{\phi}\,\mathbf{e}_t(\pi/2) = \beta\omega R\,\mathbf{k}_1$ (vertical, upward).

Vertical equation. For a free position $\mathbf{r}_1^P = x_1\mathbf{i}_1 + y_1\mathbf{j}_1 + z_1\mathbf{k}_1$:

- $\mathbf{k}_1 \cdot \mathbf{F}_K = -K\,\mathbf{k}_1 \cdot [(x_1 + 2R)\mathbf{i}_1 + y_1\mathbf{j}_1 + z_1\mathbf{k}_1] = -\alpha m\omega^2 z_1$;
- $\mathbf{k}_1 \cdot (\mathbf{F}_A + \mathbf{F}_C) = 0$, because $\mathbf{F}_A + \mathbf{F}_C = m\omega^2\,[(x_1 + R)\mathbf{i}_1 + y_1\mathbf{j}_1]$ is horizontal;
- $\mathbf{k}_1 \cdot \mathbf{F}_{COR} = -2m\,\mathbf{k}_1 \cdot (\omega\mathbf{k}_1 \times \mathbf{v}_1^P) = 0$, because $\mathbf{k}_1 \times \mathbf{v}_1^P \perp \mathbf{k}_1$.

Hence, with $\alpha = \frac{1}{4}$,

$$m\ddot{z}_1 = -\alpha m\omega^2 z_1 \implies \ddot{z}_1 + \frac{\omega^2}{4} z_1 = 0$$

a harmonic oscillator of angular frequency $\omega/2$. Its general solution and derivative (chain rule $\frac{d}{d\tau}\sin\frac{\omega\tau}{2} = \frac{\omega}{2}\cos\frac{\omega\tau}{2}$) are

$$z_1 = C_1\cos\frac{\omega\tau}{2} + C_2\sin\frac{\omega\tau}{2}, \qquad \dot{z}_1 = -\frac{\omega}{2}C_1\sin\frac{\omega\tau}{2} + \frac{\omega}{2}C_2\cos\frac{\omega\tau}{2}$$

Initial conditions: $z_1(0) = C_1 = 0$ and $\dot{z}_1(0) = \frac{\omega}{2}C_2 = \beta\omega R$, so $C_2 = 2\beta R$:

$$z_1(\tau) = 2\beta R\,\sin\frac{\omega\tau}{2} \implies \boxed{z_{max} = 2\beta R} \quad \text{reached at } \tau = \pi/\omega$$

Since $\mathbf{k}_1 = \mathbf{k}$, this is also the height in the inertial frame, measured from the horizontal plane through O and B. For a general α the same steps give $z_{max} = \beta R/\sqrt{\alpha}$. (Equivalently, the vertical motion conserves $\frac{1}{2}m\dot{z}_1^2 + \frac{1}{2}Kz_1^2$, and $\frac{1}{2}m(\beta\omega R)^2 = \frac{1}{2}\cdot\frac{1}{4}m\omega^2 z_{max}^2$ gives the same result.) The result assumes that the free particle does not hit the wire again.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Competition of two forces.** Along the arc the spring pulls P towards B (towards larger ϕ) with tangential force $2\alpha m\omega^2 R\cos\phi$, while the centrifugal force about the axis pushes it away from the axis (towards smaller ϕ) with $m\omega^2 R(1 - \sin\phi)\cos\phi$. Both scale with ω^2 because $K = \alpha m\omega^2$, so ω only sets the time scale $1/\omega$ and the dynamics depends on α and β alone.

- **Limiting cases.** For $\alpha > \tfrac{1}{2}$ the factor $2\alpha - 1 + \sin\phi$ is positive on the whole arc and there is no interior equilibrium. For $\beta \to \beta_c^{+}$ the time to cross the barrier grows without bound (logarithmically), consistent with the asymptotic motion at $\beta = \beta_c$. As $\phi \to \pi/2$, both the Coriolis force and N_b vanish, as they must for motion parallel to $\boldsymbol{\omega}_{10}$.
- **Inertial check.** Computing $\mathbf{a}_0^P$ by differentiating $\mathbf{r}_0^P = \mathbf{r}_0^O + {}_0R_1\mathbf{r}_1^P$ twice gives exactly $\mathbf{a}_0^O + \mathbf{a}_1^P + \boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P) + 2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P$ (Notes Eq. 3.27), and $m\mathbf{a}_0^P = \mathbf{F}_K + \mathbf{N}$ holds with the reactions of part (b) (checked symbolically with sympy).
- **Dimensions.** $[K] = [\alpha m\omega^2] = \mathrm{kg}\cdot\mathrm{s}^{-2} = \mathrm{N/m}$; $[\ddot{\phi}] = [\omega^2] = \mathrm{s}^{-2}$; $[N_n] = [m\omega^2 R] = \mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2} = \mathrm{N}$; $[E_1^P] = [m\omega^2R^2] = \mathrm{J}$; $[z_{max}] = [\beta R] = \mathrm{m}$.

Source: Problems.pdf

PROBLEM 33

Two Repelling Particles on a Rotating Parabola

Original statement · Problems.pdf, p. 21

Rotating Wire + Two Particles

Two heavy point particles, P_1 and P_2, of mass m each, slide without friction along a rigid wire parabola given by the equation $z = x^2/(2L)$, where L is a known length. The wire rotates about its axis with a fixed, known angular velocity ω. The two particles repel each other electrically with a force of magnitude $F_E = k/d^2$, where k is a known constant and d is the distance between the particles. We define the rotating reference frame $Oxyz$ attached to the parabola as shown in the figure below.

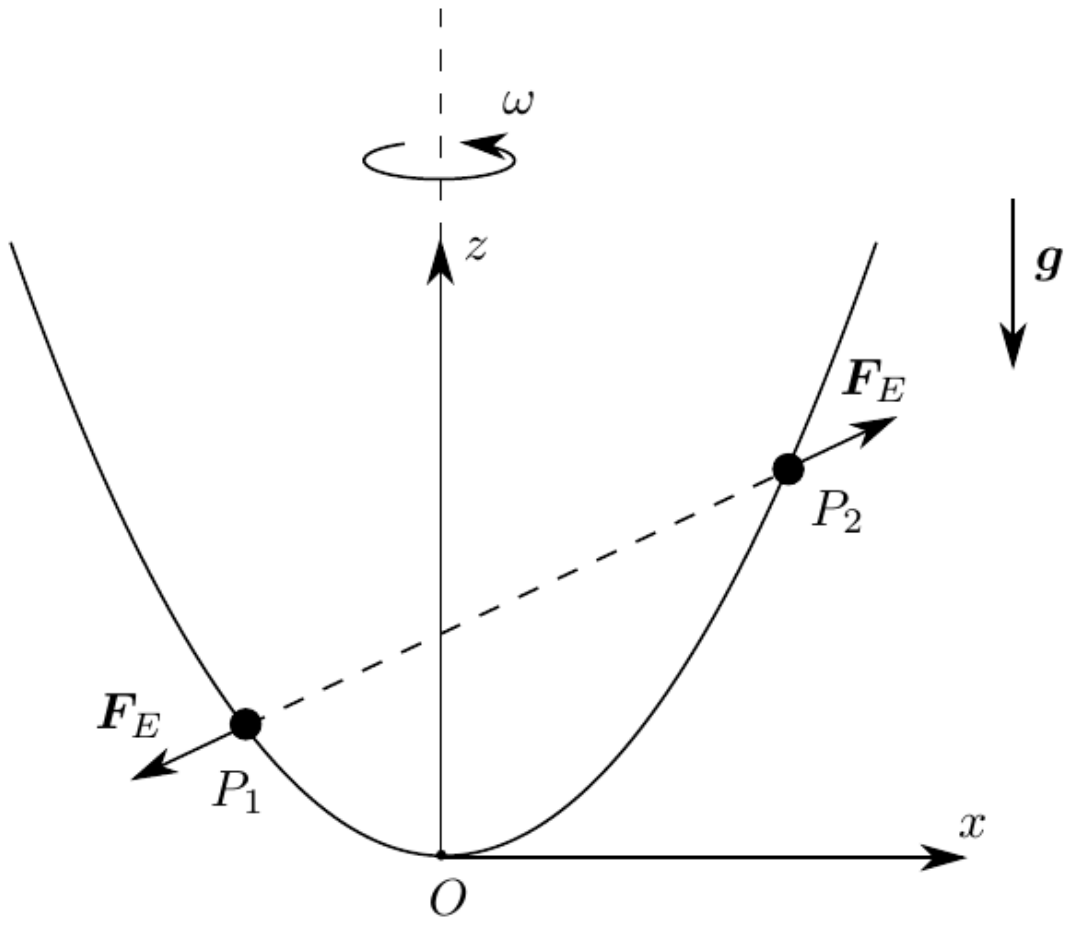


Figure: Parabola z = x²/(2L) in the plane Oxz rotating with angular velocity ω about Oz, particles P1 (left branch) and P2 (right branch) with the repulsive forces F_E along the line P1P2, gravity g downward (Problems.pdf, p. 21).

(a)

Write down the position, velocity, and acceleration vectors of each particle relative to $Oxyz$ in a generic instant of time, as a function of x_1, x_2 (the position of P_1 and P_2 along the Ox axis) and their derivatives.

(b) Write down Newton's second law for each particle in the $Oxyz$ reference frame.

(c) If the rotation of the wire is maintained by a motor, how much torque must it produce in the Oz direction to keep ω constant as a function of the variables of the problem and their derivatives?

(d) Find out the equilibrium configurations of the system in the reference frame $Oxyz$ as a function of the parameters $\gamma = g/(\omega^2 L)$ and $\kappa = k/(m\omega^2 L^3)$.

Consider now that P_1 and P_2 are constrained to move symmetrically on each branch of the parabola, and that $\gamma = \kappa = 2$. Initially, $x_1(0) = L$ and $v_1(0) = L\omega$ (with $\dot{x}_1(0) > 0$).

(e) Write down the equation that gives the minimum and maximum values of x_1 occurring in this case. *Hint: this equation is a third-order polynomial in x_1.*

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **System:** two particles P_1, P_2 of mass m slide on a smooth parabolic wire $z = x^2/(2L)$ that spins about its vertical axis Oz at constant ω. They repel each other with $F_E = k/d^2$ along the line that joins them, and both are heavy.
- **Why the rotating frame $Oxyz$.** In $Oxyz$ the wire is fixed, so each particle is located by a single coordinate x_i and its relative kinematics is that of a point on a fixed plane curve. The frame rotates uniformly about a fixed axis through O, so the inertia forces reduce to the centrifugal and Coriolis forces (Slide 28), and the centrifugal force has a potential (Slide 30).
- **Hypotheses:**
  1. The frame $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, with O the vertex of the parabola and $\mathbf{k}_0$ vertical upward, is inertial.
  2. The wire is smooth and bilateral (each particle is a bead): its reaction on P_i is

perpendicular to the wire, with components in the plane of the wire and along $\mathbf{j}_1$.
3. ω is kept constant by the motor; the axis bearings exert no torque about Oz.
4. Electric repulsion is a central internal force: equal and opposite on the two particles, along $\overrightarrow{P_2P_1}$.

- **Degrees of freedom:** two, the abscissas x_1 and x_2 in $Oxyz$. In part (e) the symmetry constraint $x_2 = -x_1$ leaves one.

Symbol	Meaning	SI unit
m	mass of each particle	kg
L	parameter of the parabola $z = x^2/(2L)$ (the radius of curvature at the vertex)	m
ω	constant angular velocity of the wire	rad/s
g	gravitational acceleration	m/s^2
k	electric repulsion constant, $F_E = k/d^2$	N·m^2
$\gamma = g/(\omega^2 L)$	weight versus centrifugal force	1
$\kappa = k/(m\omega^2 L^3)$	repulsion versus centrifugal force	1

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Frames.** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ inertial; $S_1 = Oxyz = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ attached to the wire, with the wire in the plane Oxz and $\mathbf{k}_1 = \mathbf{k}_0$ along the axis. The two frames share the origin O, which lies on the axis and is fixed.
- **Change of basis** (rotation of angle ωt about $\mathbf{k}_0$, with $\mathbf{i}_1 = \mathbf{i}_0$ at $t = 0$; columns are $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1$$

The columns are unit vectors and mutually orthogonal, so $[{}_0R_1][{}_0R_1]^T = I$.

- **Motion of S_1:** $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$; $\boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$; $\boldsymbol{\alpha}_{10} = \mathbf{0}$ because ω is constant.
- **Local geometry of the wire at P_i.** A tangent vector is $\dfrac{d}{dx}\left(x\,\mathbf{i}_1 + \dfrac{x^2}{2L}\mathbf{k}_1\right) = \mathbf{i}_1 + \dfrac{x}{L}\mathbf{k}_1$. With $s_i = \sqrt{1 + x_i^2/L^2}$, the unit tangent and the in-plane unit normal (towards the concave side) are

$$\mathbf{t}_i = \frac{1}{s_i}\left(\mathbf{i}_1 + \frac{x_i}{L}\mathbf{k}_1\right), \qquad \mathbf{n}_i = \frac{1}{s_i}\left(-\frac{x_i}{L}\mathbf{i}_1 + \mathbf{k}_1\right)$$

They are orthogonal, $\mathbf{t}_i \cdot \mathbf{n}_i = \frac{1}{s_i^2}\left(-\frac{x_i}{L} + \frac{x_i}{L}\right) = 0$, and, using $\mathbf{i}_1 \times \mathbf{k}_1 = -\mathbf{j}_1$ and $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$,

$$\mathbf{t}_i \times \mathbf{n}_i = \frac{1}{s_i^2}\left[\mathbf{i}_1 \times \mathbf{k}_1 - \frac{x_i^2}{L^2}\,\mathbf{k}_1 \times \mathbf{i}_1\right] = \frac{1}{s_i^2}\left[-\mathbf{j}_1 - \frac{x_i^2}{L^2}\mathbf{j}_1\right] = -\mathbf{j}_1$$

so $\{\mathbf{t}_i, \mathbf{n}_i, -\mathbf{j}_1\}$ is a right-handed orthonormal triad attached to the wire at P_i. The reaction of the smooth wire lies in the plane spanned by $\mathbf{n}_i$ and $\mathbf{j}_1$.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Relative position, velocity and acceleration

Pedagogical Justification: Relative kinematics is computed in S_1, where $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ are constant, so only the scalar components are differentiated. Each $z_i = x_i^2/(2L)$ depends on time through $x_i(t)$, so the chain rule is needed.

For $i = 1, 2$, the position on the wire is

$$\mathbf{r}_1^{P_i} = x_i\,\mathbf{i}_1 + \frac{x_i^2}{2L}\,\mathbf{k}_1$$

Chain rule: $\frac{d}{dt}\left(\frac{x_i^2}{2L}\right) = \frac{2x_i}{2L}\,\dot{x}_i = \frac{x_i\dot{x}_i}{L}$. Hence

$$\boxed{\mathbf{v}_1^{P_i} = \dot{x}_i\,\mathbf{i}_1 + \frac{x_i\dot{x}_i}{L}\,\mathbf{k}_1}$$

Product rule on $x_i\dot{x}_i$: $\frac{d}{dt}(x_i\dot{x}_i) = \dot{x}_i\dot{x}_i + x_i\ddot{x}_i$. Hence

$$\boxed{\mathbf{a}_1^{P_i} = \ddot{x}_i\,\mathbf{i}_1 + \frac{\dot{x}_i^2 + x_i\ddot{x}_i}{L}\,\mathbf{k}_1}$$

Note $\mathbf{v}_1^{P_i} = s_i\dot{x}_i\,\mathbf{t}_i$, tangent to the wire, as it must be.

(b) Newton's second law in $Oxyz$

> **Pedagogical Justification:** $Oxyz$ is not inertial; Newton's law holds there if the inertia forces are added, $m\mathbf{a}_1^{P_i} = \mathbf{F} + \mathbf{F}_i$ (Notes Eqs. 4.62–4.63). Projecting on the tangent $\mathbf{t}_i$ eliminates the unknown reaction and gives the equation of motion; the other two projections give the reaction.

Actual forces on P_i:

- Weight: $-mg\,\mathbf{k}_1$.
- Reaction of the smooth wire: $\mathbf{N}_i = N_{in}\,\mathbf{n}_i + N_{iy}\,\mathbf{j}_1$ (so $\mathbf{N}_i \cdot \mathbf{t}_i = 0$).
- Electric repulsion. With $\Delta x = x_1 - x_2$ and $\Delta z = \dfrac{x_1^2 - x_2^2}{2L} = \Delta x\,\dfrac{x_1 + x_2}{2L}$,

$$\overrightarrow{P_2P_1} = \Delta x\left[\mathbf{i}_1 + \frac{x_1 + x_2}{2L}\mathbf{k}_1\right], \qquad d^2 = |\overrightarrow{P_2P_1}|^2 = \Delta x^2\left[1 + \frac{(x_1 + x_2)^2}{4L^2}\right]$$

The force on P_1 has magnitude k/d^2 and the direction of $\overrightarrow{P_2P_1}/d$, and the force on P_2 is opposite:

$$\mathbf{F}_{E1} = \frac{k}{d^3}\,\Delta x\left[\mathbf{i}_1 + \frac{x_1 + x_2}{2L}\mathbf{k}_1\right] = -\mathbf{F}_{E2}$$

Inertia forces ($\mathbf{a}_0^O = \mathbf{0}$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, so $\mathbf{F}_A = \mathbf{F}_E^{\text{Euler}} = \mathbf{0}$):

- Centrifugal: $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^{P_i} = \omega\mathbf{k}_1 \times \left(x_i\mathbf{i}_1 + \frac{x_i^2}{2L}\mathbf{k}_1\right) = \omega x_i\,\mathbf{j}_1$, then $\boldsymbol{\omega}_{10} \times \omega x_i\mathbf{j}_1 = \omega^2 x_i\,(\mathbf{k}_1 \times \mathbf{j}_1) = -\omega^2 x_i\,\mathbf{i}_1$, so $\mathbf{F}_{C,i} = m\omega^2 x_i\,\mathbf{i}_1$ (outward from the axis, Slide 30).
- Coriolis: $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^{P_i} = \omega\mathbf{k}_1 \times \left(\dot{x}_i\mathbf{i}_1 + \frac{x_i\dot{x}_i}{L}\mathbf{k}_1\right) = \omega\dot{x}_i\,\mathbf{j}_1$, so $\mathbf{F}_{COR,i} = -2m\omega\dot{x}_i\,\mathbf{j}_1$.

Newton's second law for each particle (vector form):

$$m\left[\ddot{x}_i\,\mathbf{i}_1 + \frac{\dot{x}_i^2 + x_i\ddot{x}_i}{L}\,\mathbf{k}_1\right] = -mg\,\mathbf{k}_1 + N_{in}\,\mathbf{n}_i + N_{iy}\,\mathbf{j}_1 + \mathbf{F}_{Ei} + m\omega^2 x_i\,\mathbf{i}_1 - 2m\omega\dot{x}_i\,\mathbf{j}_1, \qquad i = 1, 2$$

Projection on $s_i\mathbf{t}_i = \mathbf{i}_1 + \dfrac{x_i}{L}\mathbf{k}_1$ (no reaction). Left side: $m\left[\ddot{x}_i + \dfrac{x_i}{L}\cdot\dfrac{\dot{x}_i^2 + x_i\ddot{x}_i}{L}\right] = m\left[\left(1+\dfrac{x_i^2}{L^2}\right)\ddot{x}_i + \dfrac{x_i\dot{x}_i^2}{L^2}\right]$. Right side: weight $-mg\,x_i/L$; centrifugal $m\omega^2 x_i$; Coriolis 0; electric, for P_1,

$$\mathbf{F}_{E1}\cdot\left(\mathbf{i}_1 + \frac{x_1}{L}\mathbf{k}_1\right) = \frac{k}{d^3}\,(x_1 - x_2)\left[1 + \frac{x_1(x_1+x_2)}{2L^2}\right]$$

and for P_2 the same with the indices exchanged (because $\mathbf{F}_{E2} = -\mathbf{F}_{E1}$ and $-\Delta x = x_2 - x_1$). The **equations of motion** are

$$\boxed{m\left[\left(1+\frac{x_1^2}{L^2}\right)\ddot{x}_1 + \frac{x_1\dot{x}_1^2}{L^2}\right] = m\left(\omega^2 - \frac{g}{L}\right)x_1 + \frac{k}{d^3}\,(x_1 - x_2)\left[1 + \frac{x_1(x_1+x_2)}{2L^2}\right]}$$

$$\boxed{m\left[\left(1+\frac{x_2^2}{L^2}\right)\ddot{x}_2 + \frac{x_2\dot{x}_2^2}{L^2}\right] = m\left(\omega^2 - \frac{g}{L}\right)x_2 + \frac{k}{d^3}\,(x_2 - x_1)\left[1 + \frac{x_2(x_1+x_2)}{2L^2}\right]}$$

with $d^3 = |x_1 - x_2|^3\left[1 + \dfrac{(x_1+x_2)^2}{4L^2}\right]^{3/2}$.

Projection on $\mathbf{j}_1$: $0 = N_{iy} - 2m\omega\dot{x}_i$, so $\boxed{N_{iy} = 2m\omega\dot{x}_i}$: the wire pushes each bead sideways to cancel the Coriolis force.

Projection on $s_i\mathbf{n}_i = -\dfrac{x_i}{L}\mathbf{i}_1 + \mathbf{k}_1$. Left side: $m\left[-\dfrac{x_i\ddot{x}_i}{L} + \dfrac{\dot{x}_i^2 + x_i\ddot{x}_i}{L}\right] = \dfrac{m\dot{x}_i^2}{L}$. Right side: $N_{in}s_i - mg - \dfrac{m\omega^2 x_i^2}{L} + \mathbf{F}_{Ei}\cdot(s_i\mathbf{n}_i)$, where for P_1

$$\mathbf{F}_{E1}\cdot\left(-\frac{x_1}{L}\mathbf{i}_1 + \mathbf{k}_1\right) = \frac{k\,\Delta x}{d^3}\left[-\frac{x_1}{L} + \frac{x_1+x_2}{2L}\right] = \frac{k\,\Delta x}{d^3}\cdot\frac{x_2 - x_1}{2L} = -\frac{k\,\Delta x^2}{2Ld^3}$$

and the same value is found for P_2. Therefore

$$\boxed{N_{in} = \frac{m}{s_i}\left[\frac{\dot{x}_i^2}{L} + g + \frac{\omega^2 x_i^2}{L}\right] + \frac{k\,(x_1 - x_2)^2}{2L\,s_i\,d^3}}, \qquad s_i = \sqrt{1 + \frac{x_i^2}{L^2}}$$

(c) Motor torque that keeps ω constant

> **Pedagogical Justification:** The motor torque acts on the wire; the wire acts on the particles through the reactions $\mathbf{N}_i$, and the particles act back with $-\mathbf{N}_i$. The wire rotates at constant ω about a fixed axis and is symmetric about Oz (its products of inertia I_{xz}, I_{yz} vanish), so its own angular momentum $I_{zz}\omega\,\mathbf{k}_1$ is constant, and the moment about Oz of all actions on it must vanish.

Balance of moments about Oz on the wire: $M + \sum_i \left[\mathbf{r}^{P_i} \times (-\mathbf{N}_i)\right] \cdot \mathbf{k}_1 = 0$. Expand the cross product with $\mathbf{r}^{P_i} = x_i\mathbf{i}_1 + z_i\mathbf{k}_1$ and $\mathbf{N}_i = N_{ix}\mathbf{i}_1 + N_{iy}\mathbf{j}_1 + N_{iz}\mathbf{k}_1$; its $\mathbf{k}_1$ component is $x_i N_{iy} - 0 \cdot N_{ix} = x_i N_{iy}$. With $N_{iy} = 2m\omega\dot{x}_i$ from part (b):

$$\boxed{M = x_1 N_{1y} + x_2 N_{2y} = 2m\omega\left(x_1\dot{x}_1 + x_2\dot{x}_2\right)}$$

Check with the angular momentum of the particles in S_0. By Notes Eq. 3.26, $\mathbf{v}_0^{P_i} = \mathbf{v}_1^{P_i} + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^{P_i} = \dot{x}_i\mathbf{i}_1 + \omega x_i\mathbf{j}_1 + \frac{x_i\dot{x}_i}{L}\mathbf{k}_1$, whose $\mathbf{k}_1$-moment about O is $m\left[x_i(\omega x_i) - 0\cdot\dot{x}_i\right] = m\omega x_i^2$. So $H_{Oz} = m\omega(x_1^2 + x_2^2)$ and, by the product and chain rules, $\dot{H}_{Oz} = 2m\omega(x_1\dot{x}_1 + x_2\dot{x}_2)$. The weights are parallel to Oz and the electric pair is central ($(\mathbf{r}^{P_1} - \mathbf{r}^{P_2}) \times \mathbf{F}_{E1} = \mathbf{0}$), so only the reactions have a moment about Oz, and that moment equals M. Both routes agree with the official key (Problems.pdf, p. 75).

(d) Equilibrium configurations in $Oxyz$

> **Pedagogical Justification:** A relative equilibrium has $\dot{x}_i = \ddot{x}_i = 0$; then the two equations of motion become algebraic. Writing them with $u_i = x_i/L$ and dividing by $m\omega^2 L$ shows that only γ and κ matter.

With $\dot{x}_i = \ddot{x}_i = 0$, $x_i = Lu_i$ and $d = LD$, $D = |u_1 - u_2|\sqrt{1 + (u_1+u_2)^2/4}$, dividing by $m\omega^2 L$:

$$f_1 \equiv (1-\gamma)\,u_1 + \frac{\kappa}{D^3}(u_1 - u_2)\left[1 + \frac{u_1(u_1+u_2)}{2}\right] = 0, \qquad f_2 \equiv (1-\gamma)\,u_2 + \frac{\kappa}{D^3}(u_2 - u_1)\left[1 + \frac{u_2(u_1+u_2)}{2}\right] = 0$$

Sum. With $S = u_1 + u_2$, the electric terms add to $\dfrac{\kappa}{D^3}(u_1 - u_2)\left[\dfrac{u_1 S}{2} - \dfrac{u_2 S}{2}\right] = \dfrac{\kappa\,(u_1-u_2)^2 S}{2D^3}$, so

$$f_1 + f_2 = S\left[(1-\gamma) + \frac{\kappa\,(u_1-u_2)^2}{2D^3}\right] = 0$$

Either $S = 0$ (symmetric configuration) or $\gamma - 1 = \dfrac{\kappa (u_1 - u_2)^2}{2D^3} > 0$.

Case $\gamma < 1$. If $S = 0, u_2 = -u_1, D = 2|u_1|$ and $f_1 = u_1 \left[(1-\gamma) + \dfrac{2\kappa}{8|u_1|^3} \right]$: both terms of the bracket are positive and $u_1 \neq 0$ (the particles cannot coincide), so $f_1 \neq 0$. If $S \neq 0$ the bracket of the sum is positive. **No equilibrium exists.** (Centrifugal force beats gravity everywhere and the repulsion only pushes the particles further out.)

Case $\gamma = 1$. Both alternatives require $\kappa = 0$. For $\kappa > 0$ there is no equilibrium; only without repulsion ($\kappa = 0$) is every configuration an equilibrium (gravity and centrifugal force balance along the wire at every point: $(\omega^2 - g/L) x_i = 0$).

Case $\gamma > 1$, symmetric ($x_2 = -x_1$). $f_1 = u_1 \left[(1-\gamma) + \dfrac{\kappa}{4|u_1|^3} \right] = 0$ gives $|u_1|^3 = \dfrac{\kappa}{4(\gamma-1)}$:

$$\boxed{x_1 = -x_2, \qquad x_1^2 = \left[\frac{\kappa L^3}{4(\gamma-1)} \right]^{2/3}}$$

Case $\gamma > 1$, asymmetric ($S \neq 0$). Insert $\dfrac{\kappa}{D^3} = \dfrac{2(\gamma-1)}{(u_1-u_2)^2}$ from the sum into $f_1 = 0$ and divide by $(\gamma - 1)$:

$$-u_1 + \frac{2}{u_1 - u_2} \left[1 + \frac{u_1(u_1+u_2)}{2} \right] = 0 \implies u_1(u_1-u_2) = 2 + u_1(u_1+u_2) \implies \boxed{x_1 x_2 = -L^2}$$

The particles are on opposite branches and the chord $P_1 P_2$ passes through the focus $(0, L/2)$ of the parabola. Then $u_2 = -1/u_1$, $S = u_1 - 1/u_1$, $u_1 - u_2 = u_1 + 1/u_1$, and $1 + S^2/4 = (u_1 + 1/u_1)^2/4$, so $D^3 = |u_1 + 1/u_1|^6/8$ and the sum condition becomes

$$\gamma - 1 = \frac{4\kappa}{(u_1 + 1/u_1)^4} \implies \left| u_1 + \frac{1}{u_1} \right| = \left(\frac{4\kappa}{\gamma - 1} \right)^{1/4}$$

Since $|u_1 + 1/u_1| \geq 2$, with equality only for $|u_1| = 1$ (which is the symmetric equilibrium), **asymmetric equilibria exist if and only if** $\kappa > 4(\gamma - 1)$, i.e. exactly when the symmetric equilibrium lies at $|x_1| > L$. They come in mirror pairs, $u_1 = \frac{1}{2} \left[q \pm \sqrt{q^2 - 4} \right]$ with $q = (4\kappa/(\gamma-1))^{1/4}$ and $u_2 = -1/u_1$ (plus the configurations with the labels exchanged). No equilibrium has both particles on the same branch.

(e) Symmetric motion with $\gamma = \kappa = 2$: turning points

> **Pedagogical Justification:** In $Oxyz$ the wire (and the symmetry constraint) is fixed, $\boldsymbol{\omega}_{10}$ is constant and the origin is fixed. The reactions and the Coriolis forces do no work in S_1 (Slide 31), and the weights, the electric pair and the centrifugal forces have potentials (Slides 29–30). The mechanical energy of the pair in S_1 is therefore conserved, and its turning points ($\dot{x}_1 = 0$) give the extreme values of x_1.

Put $x_1 = x > 0$, $x_2 = -x$ (the repulsion becomes infinite at $x = 0$, so x_1 stays positive). Then $d = 2x$ and, for each particle, $|\mathbf{v}_1^{P_i}|^2 = \dot{x}^2\left(1 + x^2/L^2\right)$.

- Kinetic energy: $T_1 = 2 \cdot \tfrac{1}{2} m \dot{x}^2 \left(1 + \dfrac{x^2}{L^2}\right) = m\dot{x}^2 \left(1 + \dfrac{x^2}{L^2}\right)$.
- Gravity: $2\,mg\,\dfrac{x^2}{2L} = \dfrac{mgx^2}{L}$. Centrifugal: $2\left(-\tfrac{1}{2}m\omega^2 x^2\right) = -m\omega^2 x^2$. Electric: $\dfrac{k}{d} = \dfrac{k}{2x}$ (its gradient gives the repulsion k/d^2).

$$E_1 = m\dot{x}^2\left(1 + \frac{x^2}{L^2}\right) + m\left(\frac{g}{L} - \omega^2\right)x^2 + \frac{k}{2x} = \text{const}$$

Initial value. $|\mathbf{v}_1^{P_1}(0)| = L\omega$ at $x = L$ gives $2\dot{x}(0)^2 = L^2\omega^2$, so $T_1(0) = mL^2\omega^2$ and $E_1 = mL^2\omega^2 + m(g/L - \omega^2)L^2 + k/(2L)$. Dividing by $m\omega^2 L^2$, with $u = x/L$:

$$\frac{E_1}{m\omega^2 L^2} = \gamma + \frac{\kappa}{2} = 3 \qquad (\gamma = \kappa = 2)$$

Turning points ($\dot{x} = 0$): $(\gamma - 1)u^2 + \dfrac{\kappa}{2u} = \gamma + \dfrac{\kappa}{2}$, i.e. $u^2 + \dfrac{1}{u} = 3$. Multiply by $u > 0$:

$$\boxed{\frac{x_1^3}{L^3} - 3\,\frac{x_1}{L} + 1 = 0}$$

Roots. With $u = 2\cos\theta$ and $4\cos^3\theta - 3\cos\theta = \cos 3\theta$: $8\cos^3\theta - 6\cos\theta + 1 = 2\cos 3\theta + 1 = 0$, so $\cos 3\theta = -\tfrac{1}{2}$ and $\theta = 2\pi/9,\ 4\pi/9,\ 8\pi/9$. The roots are $u = 2\cos(2\pi/9) = 1.532$, $2\cos(4\pi/9) = 0.347$ and $2\cos(8\pi/9) = -1.879$ (rejected, $x_1 > 0$). Since the starting point $u = 1$ lies between the two positive roots (there $u^2 + 1/u = 2 < 3$),

$$x_{1,\min} = 2L\cos\frac{4\pi}{9} \approx 0.347\,L, \qquad x_{1,\max} = 2L\cos\frac{2\pi}{9} \approx 1.532\,L$$

and x_1 oscillates between them. The key also lists $x_1^3/L^3 - 3x_1/L - 1 = 0$ for $x_1 < 0$: that is the mirror image (a start on the negative branch) and does not apply to $x_1(0) = L > 0$.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Effective spring along the wire.** Gravity and the centrifugal force combine into $m(\omega^2 - g/L)x_i$ in the tangential equation: a restoring force if $\gamma > 1$ (gravity wins and pulls the beads to the vertex) and a repelling one if $\gamma < 1$. Equilibrium requires the repulsion to be balanced by this restoring force, hence the condition $\gamma > 1$.
- **Limits.** $\kappa \to 0$: the symmetric equilibrium collapses to $x_1 = x_2 = 0$ (no repulsion to separate the beads). $\gamma \to 1^+$: $x_1^2 \to \infty$, the restoring force vanishes and the beads are pushed to infinity.
- **Torque sign.** $M = \frac{d}{dt}\left[m\omega(x_1^2 + x_2^2)\right]$: when the beads move outward the moment of inertia of the system grows and the motor has to supply angular momentum ($M > 0$); when they move inward it must brake ($M < 0$).
- **Inertial check.** Differentiating $[{}_0R_1]\mathbf{r}_1^{P_i}$ twice reproduces $\mathbf{a}_1^{P_i} + \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^{P_i}) + 2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^{P_i}$ (Notes Eq. 3.27), and $dE_1/dt = 0$ along the equations of motion of part (b) (both checked symbolically with sympy, together with the equilibria of part (d)).
- **Dimensions.** $[k/d^2] = \mathrm{N \cdot m^2/m^2} = \mathrm{N}$; $[m\omega^2 x] = \mathrm{kg \cdot s^{-2} \cdot m} = \mathrm{N}$; $[M] = [m\omega x\dot{x}] = \mathrm{kg \cdot s^{-1} \cdot m \cdot m/s} = \mathrm{N \cdot m}$; $[\gamma] = \dfrac{\mathrm{m/s^2}}{\mathrm{s^{-2}\,m}} = 1$; $[\kappa] = \dfrac{\mathrm{N \cdot m^2}}{\mathrm{kg \cdot s^{-2} \cdot m^3}} = 1$.

Source: Problems.pdf (PDF page 21); key on PDF page 75

PROBLEM 34

Particle on a Circular Wire in an Oscillating Vertical Plane

Original statement · Problems.pdf, p. 22

Oscillating Plane + Euler Force

A heavy point particle P of mass m slides without friction on a circular wire of center C and radius R. The wire is located on a vertical plane Oxy that rotates about the vertical axis Oy with a prescribed angular frequency $\boldsymbol{\Omega}(t) = A\omega\sin(\omega t)\,\mathbf{j}$, where A and ω are known positive constants, as shown in the figure. The angle that the vector $\overrightarrow{CP}$ makes with the Ox axis is denoted φ.

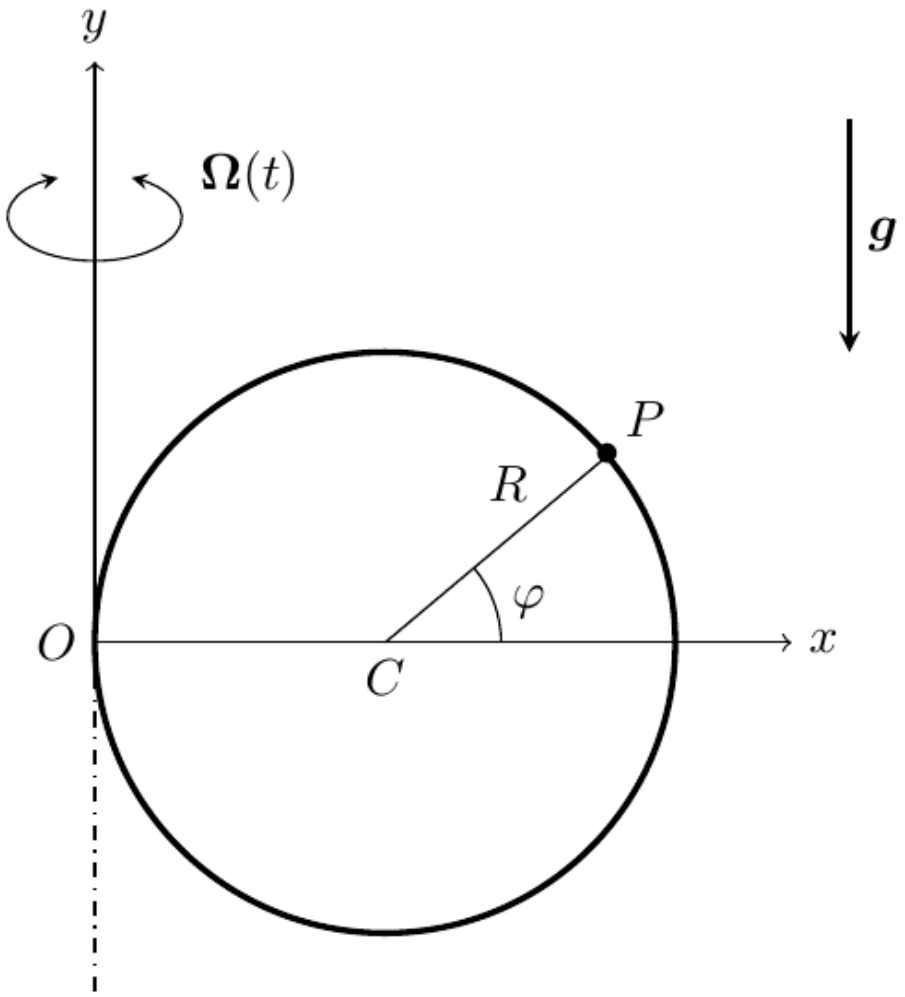


Figure: Circular wire of centre C and radius R in the vertical plane Oxy, passing through O, with C on the Ox axis; P on the wire at angle φ from Ox; the plane rotates about the vertical axis Oy with Ω(t); gravity g downward (Problems.pdf, p. 22).

Answer the following questions in a reasoned way:

(a) Write down the position, velocity and acceleration vectors of P relative to the reference frame $Oxyz$, as a function of φ and its derivatives.

(b) Identify the forces that act on P, and write down their mathematical expression for a generic instant of time t.

(c) Is the mechanical energy of P conserved as seen from the Oxy reference frame? Why?

(d) Write down a single second order differential equation to calculate φ as a function of time. Note that you do not need to solve this equation.

(e) Compute the reaction of the wire on the particle P as a function of φ, $\dot{\varphi}$ and t.

(f) Assume now that $A^2 \ll g/(R\omega^2)$ and that the motion of P is such that $\varphi = \delta - \pi/2$ with $\delta \ll 1$. Around what value of δ does the particle oscillate? Are there any values of ω for which the approximation $\delta \ll 1$ will not hold?

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **System:** a heavy bead P (mass m) slides on a smooth circular wire of radius R whose centre C lies on the Ox axis at distance R from O (the circle passes through O). The vertical plane Oxy of the wire swings about the vertical axis Oy with the prescribed angular velocity $\mathbf{\Omega}(t) = A\omega \sin(\omega t)\,\mathbf{j}$.
- **Why the frame $Oxyz$.** The wire is fixed in $Oxyz$, so P is described by the single angle φ and its relative kinematics is that of circular motion. Because Ω depends on time, the inertia forces now include the **Euler force** $-m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P$ (Slide 28), which has no potential (Slide 31), and the centrifugal potential depends explicitly on time; the energy discussion of part (c) depends on both facts.
- **Hypotheses:**
  1. A frame S_0 with origin O on the fixed vertical axis is inertial; Oy is vertical upward and gravity is $-g\,\mathbf{j}$.
  2. The wire is smooth and bilateral: the reaction is perpendicular to the tangent of the

wire, with a radial component in the plane of the wire and a component along $\mathbf{k}_1$, normal to the plane.
3. The rotation of the plane is prescribed (imposed by an external drive whatever the motion of P).

- **Degrees of freedom:** one, the angle φ of $\overrightarrow{CP}$ measured from Ox.

Symbol	Meaning	SI unit
m	mass of P	kg
R	radius of the wire ($OC = R$)	m
A	amplitude parameter of $\Omega(t)$	1 (rad)
ω	frequency of the oscillation of the plane	rad/s
g	gravitational acceleration	m/s^2
φ	angle of $\overrightarrow{CP}$ from Ox	rad

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Frames.** $S_1 = Oxyz = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ attached to the plane of the wire ($\mathbf{i}_1$ along Ox, $\mathbf{j}_1$ along Oy upward, $\mathbf{k}_1 = \mathbf{i}_1 \times \mathbf{j}_1$ normal to the plane). $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ inertial, with $\mathbf{j}_0 = \mathbf{j}_1$ (the rotation axis) and S_1 coinciding with S_0 at $t = 0$.
- **Angle turned by the plane.** With $\psi(t)$ the angle of rotation about $\mathbf{j}_0$, $\dot{\psi} = \Omega(t)$ and $\psi(0) = 0$. Integrating with the substitution $u = \omega\tau$, $du = \omega\, d\tau$, and Barrow's rule:

$$\psi(t) = \int_0^t A\omega \sin(\omega\tau)\, d\tau = A \int_0^{\omega t} \sin u\, du = A\,[-\cos u]_0^{\omega t} = A\,(1 - \cos\omega t)$$

The plane swings between $\psi = 0$ and $\psi = 2A$: it oscillates, it does not spin.

- **Change of basis** (simple rotation of angle ψ about $\mathbf{j}_0$, Notes Eqs. 1.18–1.20; columns are $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\psi & 0 & \sin\psi \\ 0 & 1 & 0 \\ -\sin\psi & 0 & \cos\psi \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\psi\cdot\cos\psi - 0 + \sin\psi\cdot\sin\psi = 1$$

The columns are unit vectors and pairwise orthogonal ($\cos\psi\sin\psi - \sin\psi\cos\psi = 0$), so $[{}_0R_1][{}_0R_1]^T = I$. Poisson check: $\left.\dfrac{d\mathbf{i}_1}{dt}\right|_0 = \dot{\psi}\,(-\sin\psi\,\mathbf{i}_0 - \cos\psi\,\mathbf{k}_0) = -\dot{\psi}\,\mathbf{k}_1$, and indeed $\Omega\,\mathbf{j}_1 \times \mathbf{i}_1 = -\Omega\,\mathbf{k}_1$.

- **Motion of S_1:** origin fixed on the axis, $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$;

$$\boldsymbol{\omega}_{10} = \Omega(t)\,\mathbf{j}_1 = A\omega\sin(\omega t)\,\mathbf{j}_1, \qquad \boldsymbol{\alpha}_{10} = \left.\frac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \dot{\Omega}\,\mathbf{j}_1 = A\omega^2\cos(\omega t)\,\mathbf{j}_1$$

($\mathbf{j}_1 = \mathbf{j}_0$ is fixed, and the chain rule gives $\frac{d}{dt}\sin\omega t = \omega\cos\omega t$.)

- **Polar basis of the wire** (fixed in shape in S_1): $\mathbf{e}_r = \cos\varphi\,\mathbf{i}_1 + \sin\varphi\,\mathbf{j}_1$ (along $\overrightarrow{CP}$), $\mathbf{e}_\varphi = -\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1$ (tangent), and $\mathbf{k}_1 = \mathbf{e}_r \times \mathbf{e}_\varphi$ ($\cos^2\varphi + \sin^2\varphi = 1$). Its matrix in $\mathcal{B}_1$ is a rotation about $\mathbf{k}_1$, with determinant $\cos^2\varphi + \sin^2\varphi = 1$.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Relative position, velocity and acceleration

Pedagogical Justification: Relative quantities are derivatives in S_1, where $\mathbf{i}_1, \mathbf{j}_1$ are constant; only φ depends on time, so each component is differentiated with the chain rule.

$\mathbf{r}_1^P = \overrightarrow{OC} + \overrightarrow{CP}$ with $\overrightarrow{OC} = R\,\mathbf{i}_1$ and $\overrightarrow{CP} = R\,\mathbf{e}_r$:

$$\boxed{\mathbf{r}_1^P = R\,(1+\cos\varphi)\,\mathbf{i}_1 + R\sin\varphi\,\mathbf{j}_1}$$

With $\frac{d}{dt}\cos\varphi = -\sin\varphi\,\dot{\varphi}$ and $\frac{d}{dt}\sin\varphi = \cos\varphi\,\dot{\varphi}$:

$$\boxed{\mathbf{v}_1^P = -R\dot{\varphi}\sin\varphi\,\mathbf{i}_1 + R\dot{\varphi}\cos\varphi\,\mathbf{j}_1 = R\dot{\varphi}\,\mathbf{e}_\varphi}$$

Product rule on each component, e.g. $\frac{d}{dt}\left(-R\dot{\varphi}\sin\varphi\right) = -R\ddot{\varphi}\sin\varphi - R\dot{\varphi}^2\cos\varphi$:

$$\boxed{\mathbf{a}_1^P = -R\left(\ddot{\varphi}\sin\varphi + \dot{\varphi}^2\cos\varphi\right)\mathbf{i}_1 + R\left(\ddot{\varphi}\cos\varphi - \dot{\varphi}^2\sin\varphi\right)\mathbf{j}_1 = R\ddot{\varphi}\,\mathbf{e}_\varphi - R\dot{\varphi}^2\,\mathbf{e}_r}$$

(b) Forces acting on P

> **Pedagogical Justification:** In the non-inertial frame $Oxyz$ we use $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eqs. 4.62–4.63). The actual forces are the weight and the reaction of the wire; the inertia forces follow from Slide 28 with $\mathbf{a}_0^O = \mathbf{0}, \boldsymbol{\omega}_{10} = \Omega\mathbf{j}_1, \boldsymbol{\alpha}_{10} = \dot{\Omega}\mathbf{j}_1$.

Actual forces:

- Weight: $\mathbf{W} = -mg\,\mathbf{j}_1$.
- Reaction of the smooth wire (no component along the tangent $\mathbf{e}_\varphi$): $\mathbf{N} = N_1\,\mathbf{e}_r + N_2\,\mathbf{k}_1 = N_1\cos\varphi\,\mathbf{i}_1 + N_1\sin\varphi\,\mathbf{j}_1 + N_2\,\mathbf{k}_1$.

Inertia forces (cross products with $\mathbf{j}_1 \times \mathbf{i}_1 = -\mathbf{k}_1, \mathbf{j}_1 \times \mathbf{j}_1 = \mathbf{0}, \mathbf{j}_1 \times \mathbf{k}_1 = \mathbf{i}_1$):

- Origin: $\mathbf{F}_A = -m\mathbf{a}_0^O = \mathbf{0}$.
- Euler: $\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P = \dot{\Omega}\,\mathbf{j}_1 \times [R(1+\cos\varphi)\mathbf{i}_1 + R\sin\varphi\,\mathbf{j}_1] = -\dot{\Omega}R(1+\cos\varphi)\,\mathbf{k}_1$, so

$$\mathbf{F}_E = -m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P = mA\omega^2 R\cos(\omega t)\,(1+\cos\varphi)\,\mathbf{k}_1$$

- Centrifugal: $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = -\Omega R(1+\cos\varphi)\,\mathbf{k}_1$ and $\boldsymbol{\omega}_{10} \times (-\Omega R(1+\cos\varphi)\mathbf{k}_1) = -\Omega^2 R(1+\cos\varphi)\,\mathbf{i}_1$, so (Slide 30, distance to the axis $x = R(1+\cos\varphi)$)

$$\mathbf{F}_C = m\Omega^2 R\,(1+\cos\varphi)\,\mathbf{i}_1 = mA^2\omega^2 R\sin^2(\omega t)\,(1+\cos\varphi)\,\mathbf{i}_1$$

- Coriolis: $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = \Omega R\dot{\varphi}\,\mathbf{j}_1 \times (-\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1) = \Omega R\dot{\varphi}\sin\varphi\,\mathbf{k}_1$, so

$$\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = -2mA\omega R\dot{\varphi}\sin(\omega t)\sin\varphi\,\mathbf{k}_1$$

(c) Is the mechanical energy conserved in Oxy?

> **Pedagogical Justification:** By Slide 29, E_1^P is conserved only if every force that does work in S_1 derives from a potential that does not depend explicitly on time. Compute the power of each force in S_1.

- $\mathbf{N} \cdot \mathbf{v}_1^P = R\dot{\varphi}\,(N_1\mathbf{e}_r + N_2\mathbf{k}_1)\cdot\mathbf{e}_\varphi = 0$ (the wire is fixed in S_1).
- $\mathbf{F}_E$ and $\mathbf{F}_{COR}$ are along $\mathbf{k}_1$, normal to the plane in which $\mathbf{v}_1^P$ lies: they do no work. (In general the Euler force has no potential, Slide 31; here it happens to be powerless.)
- The weight has the potential $mgy = mgR\sin\varphi$.
- The centrifugal force has the potential $V_C = -\tfrac{1}{2}m\Omega(t)^2x^2$, but **it depends explicitly on time** because Ω does.

Define $E_1^P = \tfrac{1}{2}mR^2\dot{\varphi}^2 + mgR\sin\varphi - \tfrac{1}{2}m\Omega^2R^2(1+\cos\varphi)^2$. Differentiating with the chain and product rules,

$$\frac{dE_1^P}{dt} = mR^2\dot{\varphi}\ddot{\varphi} + mgR\cos\varphi\,\dot{\varphi} + m\Omega^2R^2(1+\cos\varphi)\sin\varphi\,\dot{\varphi} - m\Omega\dot{\Omega}R^2(1+\cos\varphi)^2$$

The first three terms are $mR^2\dot{\varphi}$ times the equation of motion of part (d), so they cancel, leaving

$$\frac{dE_1^P}{dt} = -m\Omega\dot{\Omega}\,R^2(1+\cos\varphi)^2 = -mA^2\omega^3R^2\sin(\omega t)\cos(\omega t)\,(1+\cos\varphi)^2 \neq 0$$

No: the mechanical energy of P in Oxy is not conserved, because the centrifugal potential changes with time (the frame does not rotate at a constant rate). It would be conserved if Ω were constant. Nor is the energy conserved in S_0: there the wire moves and its reaction $N_2\mathbf{k}_1$ does work.

(d) Equation of motion

> **Pedagogical Justification:** The tangential direction $\mathbf{e}_\varphi$ is the only one free of the unknown reaction ($\mathbf{N}\perp\mathbf{e}_\varphi$), and the Euler and Coriolis forces (along $\mathbf{k}_1$) also drop out.

Projections on $\mathbf{e}_\varphi = -\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1$: $m\mathbf{a}_1^P\cdot\mathbf{e}_\varphi = mR\ddot{\varphi}$; $\mathbf{W}\cdot\mathbf{e}_\varphi = -mg\cos\varphi$; $\mathbf{F}_C\cdot\mathbf{e}_\varphi = -m\Omega^2R(1+\cos\varphi)\sin\varphi$. Hence $mR\ddot{\varphi} = -mg\cos\varphi - m\Omega^2R(1+\cos\varphi)\sin\varphi$, i.e.

$$\boxed{\ddot{\varphi} + \frac{g}{R}\cos\varphi + A^2\omega^2\sin^2(\omega t)\,(1+\cos\varphi)\sin\varphi = 0}$$

a nonlinear equation with a time-periodic coefficient (parametric forcing). It agrees with the official key.

(e) Reaction of the wire

> **Pedagogical Justification:** The remaining two components of $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$, along $\mathbf{e}_r$ and $\mathbf{k}_1$, each contain one reaction component.

Along $\mathbf{e}_r$: $m\mathbf{a}_1^P \cdot \mathbf{e}_r = -mR\dot{\varphi}^2$; $\mathbf{W} \cdot \mathbf{e}_r = -mg\sin\varphi$; $\mathbf{F}_C \cdot \mathbf{e}_r = m\Omega^2 R(1+\cos\varphi)\cos\varphi$. So $-mR\dot{\varphi}^2 = N_1 - mg\sin\varphi + m\Omega^2 R(1+\cos\varphi)\cos\varphi$:

$$\boxed{N_1 = mg\sin\varphi - mA^2R\omega^2\,(1+\cos\varphi)\sin^2(\omega t)\cos\varphi - mR\dot{\varphi}^2}$$

Along $\mathbf{k}_1$: $m\mathbf{a}_1^P \cdot \mathbf{k}_1 = 0 = N_2 + mA\omega^2 R\cos(\omega t)(1+\cos\varphi) - 2mA\omega R\dot{\varphi}\sin(\omega t)\sin\varphi$:

$$\boxed{N_2 = 2mAR\omega\,\dot{\varphi}\sin\varphi\sin(\omega t) - mAR\omega^2\,(1+\cos\varphi)\cos(\omega t)}$$

and $\mathbf{N} = N_1\cos\varphi\,\mathbf{i}_1 + N_1\sin\varphi\,\mathbf{j}_1 + N_2\,\mathbf{k}_1$. Both components agree with the official key. The normal component N_2 is what the wire must supply to drag P along with the swinging plane (it balances the Euler and Coriolis forces).

(f) Small oscillations about the lowest point

> **Pedagogical Justification:** For $\delta \ll 1$ the trigonometric functions are replaced by their first-order Taylor expansions; the condition $A^2 \ll g/(R\omega^2)$, i.e. $A^2\omega^2 \ll g/R$, lets us drop the parametric term $A^2\omega^2\sin^2(\omega t)\,\delta$ against $(g/R)\,\delta$. The result is a linear forced oscillator.

With $\varphi = \delta - \pi/2$: $\cos\varphi = \cos(\delta - \pi/2) = \sin\delta \approx \delta$, $\sin\varphi = \sin(\delta - \pi/2) = -\cos\delta \approx -1$, $\ddot{\varphi} = \ddot{\delta}$. Substituting into the equation of part (d) and keeping first-order terms:

$$\ddot{\delta} + \frac{g}{R}\,\delta - A^2\omega^2\sin^2(\omega t)\,(1+\delta) = 0$$

Dropping $A^2\omega^2\sin^2(\omega t)\,\delta$ (small compared with $(g/R)\,\delta$) and using $\sin^2(\omega t) = \frac{1}{2}\,[1-\cos(2\omega t)]$:

$$\boxed{\ddot{\delta} + \frac{g}{R}\,\delta = \frac{A^2\omega^2}{2}\left[1 - \cos(2\omega t)\right]}$$

Solution. With $\omega_n = \sqrt{g/R}$, a particular solution is a constant plus a term at 2ω. The constant δ_0 satisfies $\omega_n^2 \delta_0 = A^2\omega^2/2$. For $\delta_p = C\cos(2\omega t)$, $\ddot{\delta}_p = -4\omega^2 C\cos(2\omega t)$ and $(\omega_n^2 - 4\omega^2)\,C = -A^2\omega^2/2$. Hence

$$\delta(t) = \frac{A^2 R\omega^2}{2g} - \frac{A^2\omega^2}{2\left(\omega_n^2 - 4\omega^2\right)}\cos(2\omega t) + a\cos\omega_n t + b\sin\omega_n t$$

(a, b set by the initial conditions). The particle oscillates, with natural frequency $\omega_n = \sqrt{g/R}$, about

$$\boxed{\delta_0 = \frac{A^2 R\omega^2}{2g}}$$

which is small, consistent with $A^2 \ll g/(R\omega^2)$. It is positive: on average the bead is displaced slightly outward, away from the axis, by the mean centrifugal force.

Breakdown. The forced amplitude $|C| = \dfrac{A^2\omega^2}{2\,|\omega_n^2 - 4\omega^2|}$ grows without bound when $2\omega \to \omega_n$, i.e. for

$$\boxed{\omega \approx \frac{1}{2}\sqrt{\frac{g}{R}}}$$

At exact resonance the particular solution is $\delta_p = -\dfrac{A^2\omega^2}{4\omega_n}\,t\sin(\omega_n t)$ (check: $\ddot{\delta}_p + \omega_n^2\delta_p = -\frac{A^2\omega^2}{2}\cos\omega_n t$), whose amplitude grows linearly in time. The assumption $\delta \ll 1$ then fails, as stated in the key.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Role of each inertia force.** The centrifugal force (along $\mathbf{i}_1$, in the plane) is the only inertia force that enters the equation of motion. The Euler and Coriolis forces are normal to the plane: they do not change φ, but they load the wire through N_2.
- **Limits.** $A \to 0$: $\ddot{\varphi} + (g/R)\cos\varphi = 0$, the simple pendulum about $\varphi = -\pi/2$ in a fixed plane, and $N_2 = 0$. $\omega \to 0$ with A fixed: $\Omega \to 0$ and the same pendulum is recovered.
- **Energy.** dE_1^P/dt averages to zero over a period of the forcing ($\sin\omega t \cos\omega t$ has zero mean) but is not zero instant by instant; near resonance the drive pumps energy into the oscillation.
- **Inertial check.** Differentiating $[{}_0R_1]\mathbf{r}_1^P$ twice reproduces $\mathbf{a}_1^P + \boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P + \boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P) + 2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$ (Notes Eq. 3.27), and $m\mathbf{a}_0^P = \mathbf{W} + \mathbf{N}$ holds with the reaction of part (e) (checked symbolically with sympy).
- **Dimensions.** $[g/R] = [A^2\omega^2] = \mathrm{s}^{-2}$ (same units as $\ddot{\varphi}$); $[mA^2R\omega^2] = [mR\dot{\varphi}^2] = [mg] = \mathrm{N}$; $[mAR\omega\dot{\varphi}] = \mathrm{kg}\cdot\mathrm{m}\cdot\mathrm{s}^{-2} = \mathrm{N}$; $[A^2R\omega^2/g] = \dfrac{\mathrm{m}\cdot\mathrm{s}^{-2}}{\mathrm{m}\cdot\mathrm{s}^{-2}} = 1$, as an angle must be.

Source: Problems.pdf (PDF page 22); key on PDF page 75

PROBLEM 36

Pendulum String Dragged by a Rotating Bar

Original statement · Problems.pdf, p. 24

Rotating Frame + String Constraint

A heavy point particle P with mass m hangs from a fixed point A on the ceiling with a massless string of constant length L. The particle is initially at rest. A bar b rotates about its center O in a horizontal plane at a distance h from the ceiling, with a known, constant angular velocity ω. The horizontal distance between O and A is R. At time $t = 0$, the bar collides with the string and drags it along its motion. The string slides over the bar without friction. We call C the instantaneous point of contact of the string with the bar. We define a reference frame S_1 that rotates with the bar as shown in the figure. We call ξ the distance between O and C; ℓ the length of the string segment between C and P; θ the angle between the vertical plane Ox_1z_1 and the plane defined by the bar and P; and φ the angle formed between the string segment CP and the bar. You are asked to study only the part of the motion in which the string remains tense and in contact with the bar.

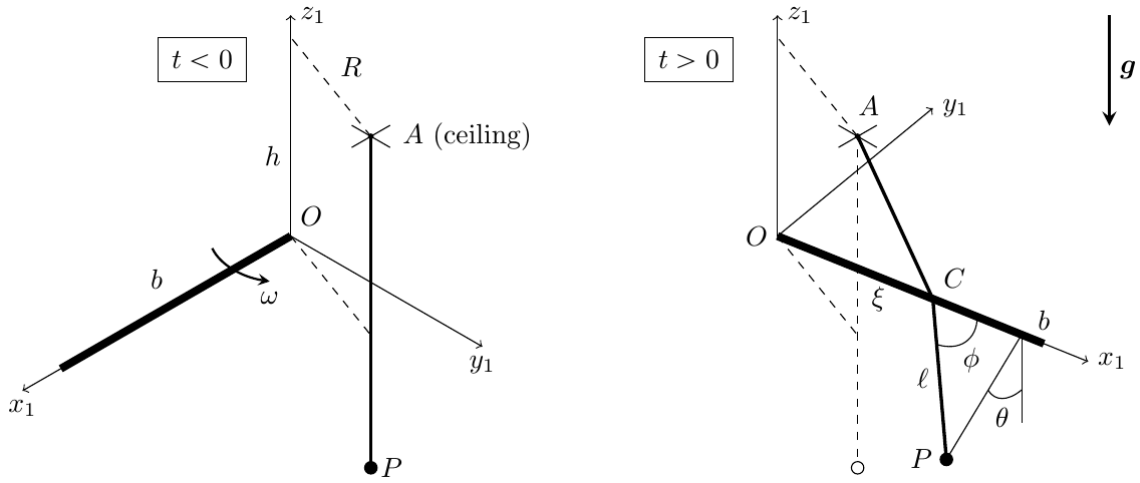


Figure: Left, before the impact (t negative): bar b rotating with ω about O in the horizontal plane x1y1, ceiling point A at height h above that plane and horizontal distance R from O, P hanging below A. Right, after the impact (t positive): frame S1 = Ox1y1z1 attached to the bar, contact point C at distance ξ from O, string A–C–P with segment CP of length ℓ at angle φ from the bar, θ the angle of the plane bar–P from the vertical; gravity g downward (Problems.pdf, p. 24).

(a) Write down the position and velocity of point A in S_1 as a function of time.

(b) Obtain a general expression for the position, velocity, and acceleration of P in S_1 as a function of $\xi, \ell, \theta, \varphi$, their derivatives, and time.

(c) Express AC, ℓ, and φ as a function of ξ and time.

(d) Identify and express all the forces acting on P.

(e) Write down a set of second order differential equations for the evolution of the unknowns of the problem.

(f) Compute the initial conditions for the previous equations at $t = 0$.

(g) Is the mechanical energy of the particle conserved in time? Justify your answer.

(h) If/when the string separates from the bar, express the geometric condition as a relation between θ and time.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **System:** a heavy particle P (mass m) hangs from the fixed ceiling point A by a string of length L. A bar turning at constant rate ω about the vertical axis through its centre O, in a horizontal plane a distance h below the ceiling, hits the string at $t = 0$ and drags it. The string then runs $A \to C \to P$, bent over the bar at the contact point C.
- **Why the frame S_1 attached to the bar.** In S_1 the bar is the fixed axis Ox_1, so the contact point is simply $C = (\xi, 0, 0)$ and the string geometry is described by ξ, ℓ, φ, θ. The price is that A moves in S_1 and that the centrifugal and Coriolis forces (Slides 27–28) have to be added. Since $\boldsymbol{\omega}_{10}$ is constant and O is fixed, the Euler force and the force of the origin's acceleration vanish, and the centrifugal force has a potential (Slide 30).
- **Hypotheses:**
  1. A frame S_0 with origin O and vertical axis $\mathbf{k}_0$ (upward) is inertial; gravity is $-g\,\mathbf{k}_0$.
  2. The string is massless and inextensible, and stays tense ($T > 0$) and in contact with the bar (the only phase studied, as the statement asks).
  3. The bar is thin (a straight line) and long enough for C to stay on it; the contact is

frictionless.

4. The rotation of the bar is prescribed: ω is kept constant whatever the string does.

5. The bar reaches the string at $t = 0$ while P is at rest below A, so $L > h$.

- **Degrees of freedom:** P moves in space with the single constraint "string tense", so it has **two** degrees of freedom. We take ξ and θ; ℓ and φ are fixed by ξ and t through the constraints of part (c). The tension T is the additional unknown.

Symbol	Meaning	SI unit
m	mass of P	kg
L	total length of the string	m
h	height of the ceiling above the plane of the bar	m
R	horizontal distance between O and A	m
ω	angular velocity of the bar (constant)	rad/s
g	gravitational acceleration	m/s^2
ξ	distance OC along the bar	m
ℓ	length of the segment CP	m
φ	angle between $\overrightarrow{CP}$ and the bar ($\mathbf{i}_1$)	rad
θ	angle between the plane Ox_1z_1 and the plane bar–P	rad
T	tension of the string	N

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Frames.** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ inertial, $\mathbf{k}_0$ vertical upward and $\mathbf{i}_0$ pointing horizontally from O towards the vertical of A, so that $\mathbf{r}_0^A = R\,\mathbf{i}_0 + h\,\mathbf{k}_0$. $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ attached to the bar, $\mathbf{i}_1$ along the bar, $\mathbf{k}_1 = \mathbf{k}_0$. At $t = 0$ the bar is below A, so S_1 coincides with S_0 at that instant.
- **Sense of rotation.** In the left drawing ($t < 0$) the vertical projection of A lies at positive x_1 and positive y_1, so the bar reaches it turning from $\mathbf{i}_1$ towards $\mathbf{j}_1$: $\boldsymbol{\omega}_{10} = +\omega\,\mathbf{k}_1$. (The same sign is implied by the Coriolis force of the official key.) The angle turned by the bar is ωt.
- **Change of basis** (rotation of angle ωt about $\mathbf{k}_0$; columns are $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1$$

The columns are unit vectors and pairwise orthogonal ($-\cos\omega t\sin\omega t + \sin\omega t\cos\omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.

- **Motion of S_1:** $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$ (the centre of the bar is fixed), $\boldsymbol{\omega}_{10} = \omega\,\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$ (ω constant and $\mathbf{k}_1 = \mathbf{k}_0$ fixed).
- **String basis $\mathcal{B}_2 = \{\mathbf{e}_{CP}, \mathbf{e}_\theta, \mathbf{e}_\varphi\}$.** The segment CP makes the angle φ with $\mathbf{i}_1$, and its component perpendicular to the bar lies in the plane bar–P, rotated by θ from the downward vertical; $\theta > 0$ puts P on the side $y_1 < 0$, the side of A for $0 < \omega t < \pi$ (this is the sign convention of the official key):

$$\mathbf{e}_{CP} = \cos\varphi\,\mathbf{i}_1 - \sin\varphi\sin\theta\,\mathbf{j}_1 - \sin\varphi\cos\theta\,\mathbf{k}_1$$

$$\mathbf{e}_\varphi = \frac{\partial \mathbf{e}_{CP}}{\partial \varphi} = -\sin\varphi\,\mathbf{i}_1 - \cos\varphi\sin\theta\,\mathbf{j}_1 - \cos\varphi\cos\theta\,\mathbf{k}_1, \qquad \mathbf{e}_\theta = \frac{1}{\sin\varphi}\frac{\partial \mathbf{e}_{CP}}{\partial \theta} = -\cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1$$

The matrix with columns $\mathbf{e}_{CP}, \mathbf{e}_\theta, \mathbf{e}_\varphi$ in $\mathcal{B}_1$ is

$$[{}_1R_2] = \begin{pmatrix} \cos\varphi & 0 & -\sin\varphi \\ -\sin\varphi\sin\theta & -\cos\theta & -\cos\varphi\sin\theta \\ -\sin\varphi\cos\theta & \sin\theta & -\cos\varphi\cos\theta \end{pmatrix}$$

Expanding along the first row: $\det[{}_1R_2] = \cos\varphi\left(\cos\varphi\cos^2\theta + \cos\varphi\sin^2\theta\right) - \sin\varphi\left(-\sin\varphi\sin^2\theta - \sin\varphi\cos^2\theta\right) = \cos^2\varphi + \sin^2\varphi = 1$. Each column has unit norm, and $\mathbf{e}_{CP}\cdot\mathbf{e}_\theta = \sin\varphi\sin\theta\cos\theta - \sin\varphi\cos\theta\sin\theta = 0$, $\mathbf{e}_{CP}\cdot\mathbf{e}_\varphi = -\cos\varphi\sin\varphi + \sin\varphi\cos\varphi\left(\sin^2\theta + \cos^2\theta\right) = 0$, $\mathbf{e}_\theta\cdot\mathbf{e}_\varphi = \cos\theta\cos\varphi\sin\theta - \sin\theta\cos\varphi\cos\theta = 0$, so $[{}_1R_2][{}_1R_2]^T = I$ and $\mathbf{e}_{CP}\times\mathbf{e}_\theta = \mathbf{e}_\varphi$.

- **Derivatives of $\mathcal{B}_2$ in S_1** (chain rule, $\frac{d}{dt}\mathbf{e}(\varphi(t),\theta(t)) = \frac{\partial \mathbf{e}}{\partial\varphi}\dot\varphi + \frac{\partial \mathbf{e}}{\partial\theta}\dot\theta$, with $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ constant in S_1). Using $\partial\mathbf{e}_\varphi/\partial\varphi = \cos\varphi\,(-\mathbf{i}_1) + \sin\varphi\sin\theta\,\mathbf{j}_1 + \sin\varphi\cos\theta\,\mathbf{k}_1 = -\mathbf{e}_{CP}$, $\partial\mathbf{e}_\varphi/\partial\theta = -\cos\varphi\cos\theta\,\mathbf{j}_1 + \cos\varphi\sin\theta\,\mathbf{k}_1 = \cos\varphi\,\mathbf{e}_\theta$ and $\partial\mathbf{e}_\theta/\partial\theta = \sin\theta\,\mathbf{j}_1 + \cos\theta\,\mathbf{k}_1 = -\left(\sin\varphi\,\mathbf{e}_{CP} + \cos\varphi\,\mathbf{e}_\varphi\right)$:

$$\left.\frac{d\mathbf{e}_{CP}}{dt}\right|_1 = \dot\varphi\,\mathbf{e}_\varphi + \dot\theta\sin\varphi\,\mathbf{e}_\theta, \qquad \left.\frac{d\mathbf{e}_\varphi}{dt}\right|_1 = -\dot\varphi\,\mathbf{e}_{CP} + \dot\theta\cos\varphi\,\mathbf{e}_\theta, \qquad \left.\frac{d\mathbf{e}_\theta}{dt}\right|_1 = -\dot\theta\left(\sin\varphi\,\mathbf{e}_{CP} + \cos\varphi\,\mathbf{e}_\varphi\right)$$

PHASE 3

Step-by-Step Mathematical Deduction

(a) Position and velocity of A in S_1

> **Pedagogical Justification:** A is fixed in S_0; its coordinates in S_1 follow from $\mathbf{r}_1^A = [{}_0R_1]^T \mathbf{r}_0^A$ (the origins coincide), and its relative velocity is the derivative of those coordinates in S_1.

$$[{}_0R_1]^T \begin{pmatrix} R \\ 0 \\ h \end{pmatrix} = \begin{pmatrix} R\cos\omega t \\ -R\sin\omega t \\ h \end{pmatrix} \quad \Longrightarrow \quad \boxed{\mathbf{r}_1^A = R\cos\omega t\,\mathbf{i}_1 - R\sin\omega t\,\mathbf{j}_1 + h\,\mathbf{k}_1}$$

With the chain rule $\frac{d}{dt}\cos\omega t = -\omega\sin\omega t$ and $\frac{d}{dt}\sin\omega t = \omega\cos\omega t$:

$$\boxed{\mathbf{v}_1^A = \left.\frac{d\mathbf{r}_1^A}{dt}\right|_1 = -R\omega\sin\omega t\,\mathbf{i}_1 - R\omega\cos\omega t\,\mathbf{j}_1}$$

Check with the velocity composition (Notes Eq. 3.26): $\mathbf{0} = \mathbf{v}_0^A = \mathbf{v}_1^A + \boldsymbol{\omega}_{10} \times \mathbf{r}_1^A$, and $\omega\mathbf{k}_1 \times \mathbf{r}_1^A = R\omega\cos\omega t\,\mathbf{j}_1 + R\omega\sin\omega t\,\mathbf{i}_1 = -\mathbf{v}_1^A$. In S_1, A describes a circle of radius R at height h with angular velocity $-\omega$.

(b) Position, velocity and acceleration of P in S_1

> **Pedagogical Justification:** $\overrightarrow{OP} = \overrightarrow{OC} + \overrightarrow{CP}$ with $\overrightarrow{OC} = \xi\,\mathbf{i}_1$ along the bar and $\overrightarrow{CP} = \ell\,\mathbf{e}_{CP}$. Derivatives are taken in S_1 with the product rule and the basis derivatives of Phase 2.

Position:

$$\boxed{\mathbf{r}_1^P = \xi\,\mathbf{i}_1 + \ell\,\mathbf{e}_{CP} = (\xi + \ell\cos\varphi)\,\mathbf{i}_1 - \ell\sin\varphi\sin\theta\,\mathbf{j}_1 - \ell\sin\varphi\cos\theta\,\mathbf{k}_1}$$

Velocity. Product rule, $\frac{d}{dt}(\ell\,\mathbf{e}_{CP}) = \dot\ell\,\mathbf{e}_{CP} + \ell\left(\dot\varphi\,\mathbf{e}_\varphi + \dot\theta\sin\varphi\,\mathbf{e}_\theta\right)$:

$$\boxed{\mathbf{v}_1^P = \dot\xi\,\mathbf{i}_1 + \dot\ell\,\mathbf{e}_{CP} + \ell\dot\varphi\,\mathbf{e}_\varphi + \ell\dot\theta\sin\varphi\,\mathbf{e}_\theta}$$

In $\mathcal{B}_1$ (differentiating each component; e.g. $\frac{d}{dt}\left(-\ell\sin\varphi\sin\theta\right) = -\dot{\ell}\sin\varphi\sin\theta - \ell\dot{\varphi}\cos\varphi\sin\theta - \ell\dot{\theta}\sin\varphi\cos\theta$):

$$\mathbf{v}_1^P = \left(\dot{\xi} + \dot{\ell}\cos\varphi - \ell\dot{\varphi}\sin\varphi\right)\mathbf{i}_1 - \left(\dot{\ell}\sin\varphi\sin\theta + \ell\dot{\varphi}\cos\varphi\sin\theta + \ell\dot{\theta}\sin\varphi\cos\theta\right)\mathbf{j}_1 - \left(\dot{\ell}\sin\varphi\cos\theta + \ell\dot{\varphi}\cos\varphi\cos\theta - \ell\dot{\theta}\sin\varphi\sin\theta\right)\mathbf{k}_1$$

Acceleration. Differentiate term by term:

- $\frac{d}{dt}(\dot{\xi}\,\mathbf{i}_1) = \ddot{\xi}\,\mathbf{i}_1$;
- $\frac{d}{dt}(\dot{\ell}\,\mathbf{e}_{CP}) = \ddot{\ell}\,\mathbf{e}_{CP} + \dot{\ell}\dot{\varphi}\,\mathbf{e}_\varphi + \dot{\ell}\dot{\theta}\sin\varphi\,\mathbf{e}_\theta$;
- $\frac{d}{dt}(\ell\dot{\varphi}\,\mathbf{e}_\varphi) = (\dot{\ell}\dot{\varphi} + \ell\ddot{\varphi})\,\mathbf{e}_\varphi + \ell\dot{\varphi}\left(-\dot{\varphi}\,\mathbf{e}_{CP} + \dot{\theta}\cos\varphi\,\mathbf{e}_\theta\right)$;
- $\frac{d}{dt}(\ell\dot{\theta}\sin\varphi\,\mathbf{e}_\theta) = \left(\dot{\ell}\dot{\theta}\sin\varphi + \ell\ddot{\theta}\sin\varphi + \ell\dot{\theta}\dot{\varphi}\cos\varphi\right)\mathbf{e}_\theta - \ell\dot{\theta}^2\sin\varphi\,(\sin\varphi\,\mathbf{e}_{CP} + \cos\varphi\,\mathbf{e}_\varphi)$.

Collecting along each unit vector:

$$\boxed{\mathbf{a}_1^P = \ddot{\xi}\,\mathbf{i}_1 + \left(\ddot{\ell} - \ell\dot{\varphi}^2 - \ell\dot{\theta}^2\sin^2\varphi\right)\mathbf{e}_{CP} + \left(\ell\ddot{\varphi} + 2\dot{\ell}\dot{\varphi} - \ell\dot{\theta}^2\sin\varphi\cos\varphi\right)\mathbf{e}_\varphi + \left(\ell\ddot{\theta}\sin\varphi + 2\dot{\ell}\dot{\theta}\sin\varphi + 2\ell\dot{\theta}\dot{\varphi}\cos\varphi\right)\mathbf{e}_\theta}$$

with $\mathbf{e}_{CP}, \mathbf{e}_\varphi, \mathbf{e}_\theta$ given in $\mathcal{B}_1$ in Phase 2. The bracketed terms are the spherical-coordinate acceleration of P about the moving centre C, plus the acceleration $\ddot{\xi}\,\mathbf{i}_1$ of the centre along the bar. Time enters through $\xi, \ell, \varphi, \theta$ only; the explicit time dependence appears when ℓ and φ are replaced by the constraints of part (c).

(c) AC, ℓ and φ as functions of ξ and t

> **Pedagogical Justification:** The string has constant length, which gives ℓ. The angle φ follows from the frictionless contact: the massless piece of string at C must be in equilibrium under the two tensions and the bar's reaction, which has no component along the bar.

Vector $\overrightarrow{AC}$: with $\mathbf{r}_1^C = \xi\,\mathbf{i}_1$ and part (a),

$$\boxed{\overrightarrow{AC} = \mathbf{r}_1^C - \mathbf{r}_1^A = (\xi - R\cos\omega t)\,\mathbf{i}_1 + R\sin\omega t\,\mathbf{j}_1 - h\,\mathbf{k}_1}$$

$$AC^2 = (\xi - R\cos\omega t)^2 + R^2\sin^2\omega t + h^2 = \xi^2 - 2R\xi\cos\omega t + R^2\left(\cos^2\omega t + \sin^2\omega t\right) + h^2 \quad\Longrightarrow\quad \boxed{AC = \sqrt{\xi^2 - 2R\xi\cos\omega t + R^2 + h^2}}$$

Length ℓ. $AC + CP = L$:

$$\boxed{\ell = L - \sqrt{\xi^2 - 2R\xi\cos\omega t + R^2 + h^2}}$$

Angle φ. With no friction the tension is the same, T, on both sides of C. Equilibrium of the massless string element at C: $T\,\mathbf{e}_{CA} + T\,\mathbf{e}_{CP} + \mathbf{N}_b = \mathbf{0}$ with $\mathbf{N}_b \cdot \mathbf{i}_1 = 0$. Projecting on $\mathbf{i}_1$: $\mathbf{e}_{CP}\cdot\mathbf{i}_1 = -\mathbf{e}_{CA}\cdot\mathbf{i}_1$, where $\mathbf{e}_{CA} = -\overrightarrow{AC}/AC$. Hence

$$\cos\varphi = \frac{\xi - R\cos\omega t}{AC} \quad\Longrightarrow\quad \boxed{\varphi = \arccos\frac{\xi - R\cos\omega t}{\sqrt{\xi^2 - 2R\xi\cos\omega t + R^2 + h^2}}}, \qquad \sin\varphi = \frac{\sqrt{R^2\sin^2\omega t + h^2}}{AC}$$

($\varphi \in (0,\pi)$, so $\sin\varphi > 0$). The two segments make equal angles with the bar, on opposite sides of the plane perpendicular to it. (Extension: this is also the condition for the path $A \to C \to P$ to be the shortest one for fixed P, $\frac{\partial}{\partial\xi}(AC + CP) = \cos\angle(\overrightarrow{AC}, \mathbf{i}_1) - \cos\varphi = 0$.)

Compact notation for part (e). Let $q = \xi - R\cos\omega t$, $\rho = \sqrt{R^2\sin^2\omega t + h^2}$ and $D = AC = \sqrt{q^2 + \rho^2}$. Then $q = D\cos\varphi$, $\rho = D\sin\varphi$: (D,φ) are the polar coordinates of the point (q,ρ), and $\ell = L - D$. Their derivatives:

- $\dot q = \dot\xi + R\omega\sin\omega t$ and $\ddot q = \ddot\xi + R\omega^2\cos\omega t$ (chain rule on $\cos\omega t$).
- From $\rho^2 = R^2\sin^2\omega t + h^2$: $2\rho\dot\rho = 2R^2\omega\sin\omega t\cos\omega t$, so $\dot\rho = \dfrac{R^2\omega\sin 2\omega t}{2\rho}$.
  Differentiating $\rho\dot\rho = \frac12 R^2\omega\sin 2\omega t$ (product rule): $\dot\rho^2 + \rho\ddot\rho = R^2\omega^2\cos 2\omega t$, so $\ddot\rho = \dfrac{R^2\omega^2\cos 2\omega t - \dot\rho^2}{\rho}$.
- Differentiating $q = D\cos\varphi$, $\rho = D\sin\varphi$: $\dot q = \dot D\cos\varphi - D\dot\varphi\sin\varphi$, $\dot\rho = \dot D\sin\varphi + D\dot\varphi\cos\varphi$. Solving this rotation-type system:

$$\dot D = \dot q\cos\varphi + \dot\rho\sin\varphi, \qquad D\dot\varphi = \dot\rho\cos\varphi - \dot q\sin\varphi$$

- Differentiating once more, $\ddot q = (\ddot D - D\dot\varphi^2)\cos\varphi - (D\ddot\varphi + 2\dot D\dot\varphi)\sin\varphi$ and $\ddot\rho = (\ddot D - D\dot\varphi^2)\sin\varphi + (D\ddot\varphi + 2\dot D\dot\varphi)\cos\varphi$, so

$$\ddot D - D\dot\varphi^2 = \ddot q\cos\varphi + \ddot\rho\sin\varphi, \qquad D\ddot\varphi + 2\dot D\dot\varphi = \ddot\rho\cos\varphi - \ddot q\sin\varphi$$

- Finally $\dot\ell = -\dot D$, $\ddot\ell = -\ddot D$. Since $\dot\rho\sin\varphi = \dfrac{R^2\omega\sin\omega t\cos\omega t}{\rho}\,\dfrac{\rho}{D}$ and $\cos\varphi = q/D$,

$$\dot{\ell} = -(\dot{\xi} + R\omega\sin\omega t)\frac{\xi - R\cos\omega t}{D} - \frac{R^2\omega\sin\omega t\cos\omega t}{D} \quad\Longrightarrow\quad \boxed{\dot{\ell} + \dot{\xi}\cos\varphi = -\frac{R\omega\,\xi\sin\omega t}{D}}$$

(d) Forces acting on P

> **Pedagogical Justification:** In the non-inertial frame S_1, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eqs. 4.62–4.63). The actual forces on P are only the weight and the pull of the string (the bar touches the string, not P). The inertia forces are those of Slide 28 with $\mathbf{a}_0^O = \mathbf{0}, \boldsymbol{\alpha}_{10} = \mathbf{0}, \boldsymbol{\omega}_{10} = \omega\mathbf{k}_1$.

Actual forces:

- Weight: $\mathbf{W} = -mg\,\mathbf{k}_1$.
- Tension, along the string towards C: $\mathbf{T} = -T\,\mathbf{e}_{CP} = -T\cos\varphi\,\mathbf{i}_1 + T\sin\varphi\sin\theta\,\mathbf{j}_1 + T\sin\varphi\cos\theta\,\mathbf{k}_1$, with $T > 0$.

Inertia forces: write $\mathbf{r}_1^P = x\,\mathbf{i}_1 + y\,\mathbf{j}_1 + z\,\mathbf{k}_1$ with $x = \xi + \ell\cos\varphi$, $y = -\ell\sin\varphi\sin\theta$, and $\mathbf{v}_1^P = v_x\mathbf{i}_1 + v_y\mathbf{j}_1 + v_z\mathbf{k}_1$ from part (b). With $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1, \mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1, \mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$:

- Origin and Euler: $\mathbf{F}_A = -m\mathbf{a}_0^O = \mathbf{0}, \mathbf{F}_E = -m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P = \mathbf{0}$.
- Centrifugal: $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \omega\,(x\,\mathbf{j}_1 - y\,\mathbf{i}_1), \boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) = \omega^2\,(-x\,\mathbf{i}_1 - y\,\mathbf{j}_1)$, so (Slide 30, $\mathbf{r}_\perp = x\,\mathbf{i}_1 + y\,\mathbf{j}_1$)

$$\boxed{\mathbf{F}_C = m\omega^2(\xi + \ell\cos\varphi)\,\mathbf{i}_1 - m\omega^2\ell\sin\varphi\sin\theta\,\mathbf{j}_1}$$

- Coriolis: $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = \omega\,(v_x\,\mathbf{j}_1 - v_y\,\mathbf{i}_1)$, so $\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = 2m\omega v_y\,\mathbf{i}_1 - 2m\omega v_x\,\mathbf{j}_1$:

$$\boxed{\mathbf{F}_{COR} = -2m\omega\left(\dot{\ell}\sin\varphi\sin\theta + \ell\dot{\varphi}\cos\varphi\sin\theta + \ell\dot{\theta}\sin\varphi\cos\theta\right)\mathbf{i}_1 - 2m\omega\left(\dot{\xi} + \dot{\ell}\cos\varphi - \ell\dot{\varphi}\sin\varphi\right)\mathbf{j}_1}$$

Both inertia forces agree with the official key.

(e) Equations of motion

> **Pedagogical Justification:** Project $m\mathbf{a}_1^P = \mathbf{W} + \mathbf{T} + \mathbf{F}_C + \mathbf{F}_{COR}$ on $\mathcal{B}_2$. The tension is along $\mathbf{e}_{CP}$, so the projections on $\mathbf{e}_\theta$ and $\mathbf{e}_\varphi$ are free of the unknown T: they are the two equations of motion for the two degrees of freedom ξ, θ. The projection on $\mathbf{e}_{CP}$ then gives T .

Projections of each term (using $\mathbf{i}_1 \cdot \mathbf{e}_{CP} = \cos\varphi, \mathbf{i}_1 \cdot \mathbf{e}_\varphi = -\sin\varphi, \mathbf{i}_1 \cdot \mathbf{e}_\theta = 0$, and the components of Phase 2):

Term	$\cdot\,\mathbf{e}_\theta$	$\cdot\,\mathbf{e}_\varphi$	$\cdot\,\mathbf{e}_{CP}$	
$\mathbf{a}_1^P$	$\ell\ddot\theta\sin\varphi + 2\dot\ell\dot\theta\sin\varphi + 2\ell\dot\theta\dot\varphi\cos\varphi$	$\ell\ddot\varphi + 2\dot\ell\dot\varphi - \ddot\xi\sin\varphi - \ell\dot\theta^2\sin\varphi\cos\varphi$	$\ddot\xi\cos\varphi + \ddot\ell - \ell\dot\varphi^2 - \ell\dot\theta^2\sin^2\varphi$	
$\mathbf{W}$	$-mg\sin\theta$	$mg\cos\varphi\cos\theta$	$mg\sin\varphi\cos\theta$	
$\mathbf{T}$	0	0	$-T$	
$\mathbf{F}_C$	$m\omega^2\ell\sin\varphi\sin\theta\cos\theta$	$-m\omega^2\sin\varphi\left(\xi+\ell\cos\varphi\cos^2\theta\right)$	$m\omega^2\left[(\xi+\ell\cos\varphi)\cos\varphi+\ell\sin^2\varphi\sin^2\theta\right]$	
$\mathbf{F}_{COR}$	$2m\omega\cos\theta\,v_x$	$2m\omega\left[\sin\theta(\dot\ell+\dot\xi\cos\varphi)+\ell\dot\theta\sin^2\varphi\cos\theta\right]$	$2m\omega\left[\sin\theta(\dot\xi\sin\varphi-\ell\dot\varphi)-\ell\dot\theta\sin\varphi\cos\varphi\cos\theta\right]$	

Details of the less obvious entries:

- $\mathbf{F}_C\cdot\mathbf{e}_\varphi = m\omega^2\left[-x\sin\varphi - y\cos\varphi\sin\theta\right] = m\omega^2\left[-(\xi+\ell\cos\varphi)\sin\varphi+\ell\sin\varphi\cos\varphi\sin^2\theta\right] = -m\omega^2\sin\varphi\left[\xi+\ell\cos\varphi(1-\sin^2\theta)\right]$.
- $\mathbf{F}_{COR}\cdot\mathbf{e}_\varphi = 2m\omega\left[-v_y\sin\varphi + v_x\cos\varphi\sin\theta\right]$, where $-v_y\sin\varphi = \dot\ell\sin^2\varphi\sin\theta + \ell\dot\varphi\sin\varphi\cos\varphi\sin\theta + \ell\dot\theta\sin^2\varphi\cos\theta$ and $v_x\cos\varphi\sin\theta = \dot\xi\cos\varphi\sin\theta + \dot\ell\cos^2\varphi\sin\theta - \ell\dot\varphi\sin\varphi\cos\varphi\sin\theta$; the $\dot\varphi$ terms cancel and $\sin^2\varphi+\cos^2\varphi=1$.
- $\mathbf{F}_{COR}\cdot\mathbf{e}_{CP} = 2m\omega\left[v_y\cos\varphi + v_x\sin\varphi\sin\theta\right]$: the $\dot\ell$ terms cancel and the $\dot\varphi$ terms add up to $-\ell\dot\varphi\sin\theta\left(\cos^2\varphi+\sin^2\varphi\right)$.
- $\mathbf{F}_{COR}\cdot\mathbf{e}_\theta = 2m\omega\left[-v_x\left(-\cos\theta\right)\right] = 2m\omega\cos\theta\,v_x$, with $v_x = \dot\xi + \dot\ell\cos\varphi - \ell\dot\varphi\sin\varphi$.

Equation along $\mathbf{e}_\theta$ (divided by m):

$$\boxed{\ell\sin\varphi\,\ddot\theta + 2\dot\ell\dot\theta\sin\varphi + 2\ell\dot\theta\dot\varphi\cos\varphi = -g\sin\theta + \omega^2\ell\sin\varphi\sin\theta\cos\theta + 2\omega\cos\theta\left(\dot\xi+\dot\ell\cos\varphi-\ell\dot\varphi\sin\varphi\right)}$$

Equation along $\mathbf{e}_\varphi$ (divided by m):

$$\boxed{\ell\ddot\varphi + 2\dot\ell\dot\varphi - \ddot\xi\sin\varphi - \ell\dot\theta^2\sin\varphi\cos\varphi = g\cos\varphi\cos\theta - \omega^2\sin\varphi\left(\xi+\ell\cos\varphi\cos^2\theta\right) + 2\omega\left[\sin\theta\left(\dot\ell+\dot\xi\cos\varphi\right)+\ell\dot\theta\sin^2\varphi\cos\theta\right]}$$

Closing the system. ℓ, φ and their derivatives are the functions of $(\xi,\dot\xi,\ddot\xi,t)$ of part (c). In particular $\ell\ddot\varphi = \frac{\ell}{D}\left(\ddot\rho\cos\varphi - \ddot q\sin\varphi - 2\dot D\dot\varphi\right)$ with $\ddot q = \ddot\xi + R\omega^2\cos\omega t$, and $2\dot\ell\dot\varphi = -2\dot D\dot\varphi$, so with $\ell + D = L$ the left-hand side of the $\mathbf{e}_\varphi$ equation becomes

$$\ell\ddot\varphi + 2\dot\ell\dot\varphi - \ddot\xi\sin\varphi = -\frac{L}{D}\sin\varphi\,\ddot\xi + \frac{\ell}{D}\left(\ddot\rho\cos\varphi - R\omega^2\cos\omega t\sin\varphi\right) - \frac{2L}{D}\dot D\dot\varphi$$

The coefficient of $\ddot{\xi}$, $-(L/D)\sin\varphi$, is non-zero while the string is bent ($\sin\varphi > 0$), so the $\mathbf{e}_\varphi$ equation gives $\ddot{\xi}$ and the $\mathbf{e}_\theta$ equation gives $\ddot{\theta}$: **two second-order ODEs for $\xi(t)$ and $\theta(t)$**, non-autonomous through ωt (the motion of A in S_1). In the $\mathbf{e}_\varphi$ equation the Coriolis term can also be written with $\dot{\ell} + \dot{\xi}\cos\varphi = -R\omega\xi\sin\omega t/D$.

Tension (projection on $\mathbf{e}_{CP}$, solved for T):

$$\boxed{T = -m\left(\ddot{\xi}\cos\varphi + \ddot{\ell} - \ell\dot{\varphi}^2 - \ell\dot{\theta}^2\sin^2\varphi\right) + mg\sin\varphi\cos\theta + m\omega^2\left[(\xi + \ell\cos\varphi)\cos\varphi + \ell\sin^2\varphi\sin^2\theta\right] + 2m\omega\left[\sin\theta\left(\dot{\xi}\sin\varphi - \ell\dot{\varphi}\right) - \ell\dot{\theta}\sin\varphi\cos\varphi\cos\theta\right]}$$

The solution is valid while $T > 0$ (string tense).

(f) Initial conditions at $t = 0$

> **Pedagogical Justification:** At $t = 0^-$ the particle hangs at rest below A, and the string is straight and vertical. The bar touches it at the point of the string at the height of the bar. No impulsive force acts on P: the string is perpendicular to the bar, so displacing C sideways does not change the path length to first order ($\dot{D}(0) = 0$ below). Position and velocity of P are therefore continuous at $t = 0$.

Positions. At $t = 0$ the bar lies along $\mathbf{i}_0$, and the string meets it directly below A: $\xi(0) = R$. Then $q(0) = R - R = 0$, $\rho(0) = h$, $D(0) = h$, so

$$\xi(0) = R, \qquad \ell(0) = L - h, \qquad \cos\varphi(0) = 0 \;\Rightarrow\; \varphi(0) = \frac{\pi}{2}, \qquad \theta(0) = 0$$

(P is directly below the bar in the plane Ox_1z_1). Check: $\mathbf{r}_1^P(0) = R\,\mathbf{i}_1 - (L-h)\,\mathbf{k}_1$, i.e. P at depth $L - h$ below the bar, L below A.

Velocities. $\mathbf{v}_0^P(0) = \mathbf{0}$, and by Notes Eq. 3.26

$$\mathbf{v}_1^P(0) = \mathbf{v}_0^P - \boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = -\omega\,\mathbf{k}_1\times[R\,\mathbf{i}_1 - (L-h)\,\mathbf{k}_1] = -\omega R\,\mathbf{j}_1$$

At $\varphi = \pi/2, \theta = 0$: $\mathbf{e}_{CP} = -\mathbf{k}_1$, $\mathbf{e}_\varphi = -\mathbf{i}_1$, $\mathbf{e}_\theta = -\mathbf{j}_1$, so the velocity of part (b) is $\mathbf{v}_1^P(0) = \left(\dot{\xi} - \ell\dot{\varphi}\right)\mathbf{i}_1 - \ell\dot{\theta}\,\mathbf{j}_1 - \dot{\ell}\,\mathbf{k}_1$. Matching components:

- $\mathbf{k}_1$: $\dot{\ell}(0) = 0$, consistent with the constraint: $\dot{D}(0) = \dot{q}\cos\frac{\pi}{2} + \dot{\rho}(0)\sin\frac{\pi}{2} = \dot{\rho}(0) = \dfrac{R^2\omega\sin 0}{2h} = 0$.
- $\mathbf{j}_1$: $(L-h)\,\dot{\theta}(0) = \omega R$, so $\dot{\theta}(0) = \dfrac{\omega R}{L-h}$.

- $\mathbf{i}_1$: $\dot{\xi}(0) = (L-h)\,\dot{\varphi}(0)$. From part (c), $D\dot{\varphi} = \dot{\rho}\cos\varphi - \dot{q}\sin\varphi$ gives $h\,\dot{\varphi}(0) = -\dot{q}(0) = -\left(\dot{\xi}(0) + R\omega\sin 0\right) = -\dot{\xi}(0)$. Substituting, $\dot{\xi}(0)\left(1 + \dfrac{L-h}{h}\right) = \dot{\xi}(0)\dfrac{L}{h} = 0$, so $\dot{\xi}(0) = 0$ and $\dot{\varphi}(0) = 0$.

$$\boxed{\xi(0) = R, \quad \dot{\xi}(0) = 0, \quad \theta(0) = 0, \quad \dot{\theta}(0) = \frac{\omega R}{L-h}} \qquad \left(\ell(0) = L-h,\ \dot{\ell}(0) = 0,\ \varphi(0) = \frac{\pi}{2},\ \dot{\varphi}(0) = 0\right)$$

In S_1 the particle starts moving backwards ($-\mathbf{j}_1$) with the speed ωR of the point of the bar that hits the string: P lags behind the bar, θ grows from zero.

(g) Is the mechanical energy conserved?

> **Pedagogical Justification:** By Slide 29, E_1^P is conserved only if every force that does work in S_1 derives from a potential that does not depend explicitly on time. Here $\mathbf{W}$ and $\mathbf{F}_C$ have potentials (mgz and $-\frac{1}{2}m\omega^2 r_\perp^2$, ω constant, Slide 30) and $\mathbf{F}_{COR}$ does no work (Slide 31). The question is the power of the tension, which is a constraint force of a constraint that moves.

In S_1. Define $E_1^P = \frac{1}{2}m\left|\mathbf{v}_1^P\right|^2 + mgz - \frac{1}{2}m\omega^2\left(x^2+y^2\right)$. By the work–energy theorem in S_1, $\dfrac{dE_1^P}{dt} = \mathbf{T}\cdot\mathbf{v}_1^P = -T\,\mathbf{e}_{CP}\cdot\mathbf{v}_1^P$. From part (b), $\mathbf{e}_{CP}\cdot\mathbf{v}_1^P = \dot{\xi}\cos\varphi + \dot{\ell}$, and part (c) gives $\dot{\ell} + \dot{\xi}\cos\varphi = -R\omega\xi\sin\omega t/D$. Therefore

$$\boxed{\frac{dE_1^P}{dt} = \frac{T\,R\,\omega\,\xi\sin\omega t}{\sqrt{\xi^2 - 2R\xi\cos\omega t + R^2 + h^2}}}$$

which is positive for $0 < \omega t < \pi$ (with $T > 0, \xi > 0$): E_1^P **is not conserved.** The reason is that the end A of the string moves in S_1 (part (a)): the constraint "string tense" depends explicitly on time in S_1, and the tension does work on P.

In S_0. $E_0^P = \frac{1}{2}m\left|\mathbf{v}_0^P\right|^2 + mgz$ changes at the rate $\mathbf{T}\cdot\mathbf{v}_0^P$ with $\mathbf{v}_0^P = \mathbf{v}_1^P + \boldsymbol{\omega}_{10}\times\mathbf{r}_1^P$ (Notes Eq. 3.26). The extra term is $-T\,\mathbf{e}_{CP}\cdot\omega\,(x\,\mathbf{j}_1 - y\,\mathbf{i}_1) = -T\omega\,(-x\sin\varphi\sin\theta - y\cos\varphi) = T\omega\sin\varphi\sin\theta\,(\xi + \ell\cos\varphi - \ell\cos\varphi) = T\omega\xi\sin\varphi\sin\theta$, so

$$\frac{dE_0^P}{dt} = T\,\omega\,\xi\left(\frac{R\sin\omega t}{D} + \sin\varphi\sin\theta\right) \neq 0$$

No: the mechanical energy of the particle is not conserved in either frame. The motor that keeps the bar at constant ω feeds energy into the particle through the string.

(h) Separation condition

> **Pedagogical Justification:** The string can only be pushed by the bar. The bar's force on the string, $\mathbf{N}_b = -T\,(\mathbf{e}_{CA} + \mathbf{e}_{CP})$, vanishes when the string is straight; beyond that instant the bend would have to reverse, which a bar pushing from one side cannot impose. Separation is therefore the instant at which A, C and P become collinear: $\mathbf{e}_{CP} = -\mathbf{e}_{CA}$.

From part (c), $-\mathbf{e}_{CA} = \overrightarrow{AC}/D = \dfrac{1}{D}\left[(\xi - R\cos\omega t)\,\mathbf{i}_1 + R\sin\omega t\,\mathbf{j}_1 - h\,\mathbf{k}_1\right]$. The $\mathbf{i}_1$ components of $\mathbf{e}_{CP}$ and $-\mathbf{e}_{CA}$ are already equal (frictionless condition, $\cos\varphi = q/D$). Equating the other two, with $\sin\varphi = \rho/D$:

$$-\frac{\rho}{D}\sin\theta = \frac{R\sin\omega t}{D}, \qquad -\frac{\rho}{D}\cos\theta = -\frac{h}{D} \quad\Longrightarrow\quad \sin\theta = -\frac{R\sin\omega t}{\rho}, \quad \cos\theta = \frac{h}{\rho}$$

Dividing the two relations ($\cos\theta = h/\rho > 0$):

$$\boxed{\tan\theta = -\frac{R\sin\omega t}{h}} \qquad \text{i.e.} \quad \theta_{sep}(t) = -\arctan\left(\frac{R\sin\omega t}{h}\right)$$

The string leaves the bar at the first instant at which $\theta(t)$, obtained from the equations of part (e), reaches this value. For $0 < \omega t < \pi$ it is negative: P must have swung forward, past the vertical plane of the bar, to the side opposite to A, so that the straight line from A through C passes through P.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Mechanism.** In S_1 the anchor A circles backwards at radius R and speed ωR, while the string is held on the rotating bar. The string's end at A and the bar together act as a moving constraint that pumps energy into the pendulum ($dE_1^P/dt > 0$ for $0 < \omega t < \pi$). The centrifugal force pushes P outward along the bar ($+\mathbf{i}_1$); the Coriolis force deflects it sideways and couples ξ and θ.
- **Initial kick.** $\dot\theta(0) = \omega R/(L-h)$: in S_1 the particle starts swinging backwards about the contact point with the speed of the point of the bar that hits the string, a pendulum of length $L - h$.
- **Limits.** $R \to 0$: A lies on the rotation axis and is fixed in S_1, the constraint becomes time-independent, and $dE_1^P/dt = TR\omega\xi\sin\omega t/D \to 0$: E_1^P becomes a first integral (Slide 29). $\omega \to 0$: the inertia forces vanish and the bar would never reach the string; the problem

reduces to a pendulum at rest. Separation: $h \to \infty$ gives $\theta_{sep} \to 0$; $\omega t = \pi$ (anchor on the opposite side of the bar) gives $\theta_{sep} = 0$.

- **Agreement with the key.** The centrifugal and Coriolis forces of part (d) coincide term by term with the official key (the key gives only these two forces).
- **Inertial check.** Differentiating $[{}_0R_1]\mathbf{r}_1^P$ twice reproduces Notes Eq. 3.27; at random states, with $\ddot{\xi}, \ddot{\theta}$ taken from the two equations of part (e), $m\mathbf{a}_0^P - \mathbf{W}$ is parallel to $\mathbf{e}_{CP}$ (only the tension is left), and $dE_1^P/dt = -T\,\mathbf{e}_{CP} \cdot \mathbf{v}_1^P$; the constraint derivatives of part (c), the initial conditions and the separation condition were also checked symbolically with sympy.
- **Dimensions.** $[AC] = \sqrt{\mathrm{m}^2} = \mathrm{m}$; $[\omega^2 \ell] = [\omega\dot{\xi}] = [g] = \mathrm{m \cdot s^{-2}}$, the same as $[\ell\ddot{\theta}] = [\ddot{\xi}]$ in the equations of part (e); $[m\omega^2\xi] = [m\omega\dot{\ell}] = \mathrm{kg \cdot m \cdot s^{-2}} = \mathrm{N}$; $[\dot{\theta}(0)] = \dfrac{\mathrm{s^{-1} \cdot m}}{\mathrm{m}} = \mathrm{s^{-1}}$; $[TR\omega\xi/D] = \dfrac{\mathrm{N \cdot m \cdot s^{-1} \cdot m}}{\mathrm{m}} = \mathrm{W}$, a power; $[R\sin\omega t/h] = 1$, as $\tan\theta$ must be.

Source: Problems.pdf (PDF page 24); key on PDF page 75

PROBLEM 37

Particle on a Spinning Elliptic Cone

Original statement · Problems.pdf, p. 25

Rotating Constraint Surface

A heavy point particle P with mass m moves without friction on a cone of elliptic cross-section as shown in the figure. We define an auxiliary reference frame $S_1 : \{O; \mathcal{B}_1\}$ attached to the cone. The cone is defined in terms of two parameters ζ, β as

$$\text{Elliptic cone:} \quad \begin{cases} x_1 = \zeta \cos\beta \\ y_1 = 2\zeta \sin\beta \\ z_1 = \zeta \end{cases} \qquad (5)$$

The cone spins about its Oz_1 axis at a constant angular rate Ω with respect to the inertial reference frame $S_0 : \{O; \mathcal{B}_0\}$. At time $t = 0$, the reference frame S_1 coincides with S_0, the particle P is located at $\mathbf{r}^P_{1,0} = 2\mathbf{j}_1 + \mathbf{k}_1$, and has a velocity $\mathbf{v}^P_{1,0} = \mathbf{i}_1$.

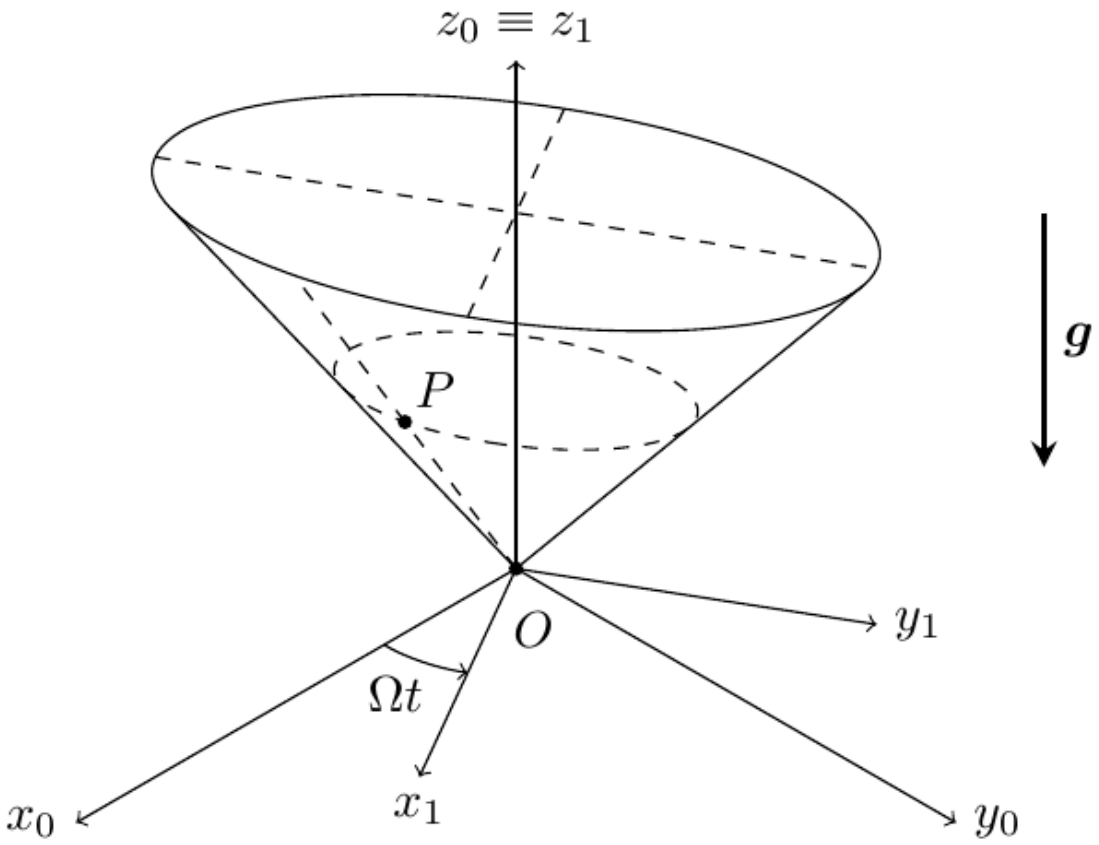


Figure: Elliptic cone with vertex O and vertical axis z0 = z1, frame S1 rotated by the angle Omega t about z0, particle P on the cone, gravity g downward (Problems.pdf, p. 25).

(a) How many degrees of freedom does the particle motion have?

(b) Obtain the general expression for the position, velocity, and acceleration of P **in reference frame** S_1 as a function of ζ, β and their derivatives. Remember to define any unit vectors that you use in terms of the vectors of $\mathcal{B}_0$.

(c) Identify and express mathematically all the forces acting on P as seen from S_1.

(d) Obtain two second-order differential equations for the evolution of ζ, β in time, which do not depend on any other unknowns in the problem.

(e) Find, justify, and write down one conservation law in this problem, including the value of any constants there might be.

(f) Find the equilibrium positions for P in S_1 for $\zeta > 0$ as a function of the constants of the problem.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** a heavy point particle P of mass m slides on the inner surface of an elliptic cone with vertex O and vertical axis $Oz_1 \equiv Oz_0$. The cone spins about that axis with constant angular rate Ω relative to the inertial frame S_0.
- **Hypotheses:**
  1. $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial; gravity is $\mathbf{g} = -g\,\mathbf{k}_0$.
  2. The contact is smooth (no friction) and bilateral: the reaction of the cone is normal to its surface.
  3. Ω is constant and the vertex O is fixed, so O is a common origin of S_0 and S_1.
  4. The data $\mathbf{r}^P_{1,0} = 2\mathbf{j}_1 + \mathbf{k}_1$ and $\mathbf{v}^P_{1,0} = \mathbf{i}_1$ are read in SI units (m and m/s).
- **Why a moving frame:** in S_0 the surface of the cone moves, so the constraint depends explicitly on time. In the frame S_1 attached to the cone the surface is fixed and is described by the two parameters (ζ, β) of Eq. (5). The price is that S_1 is not inertial, so Newton's law in S_1 needs the inertia forces of Notes Eqs. 4.62–4.63 (Slides 27–28); they are known functions of the relative position and velocity.

Symbol	Meaning	SI unit
m	mass of P	kg
g	gravitational acceleration	m/s^2
Ω	constant spin rate of the cone	rad/s
ζ	height of P above the vertex ($z_1 = \zeta$)	m
β	angular parameter of the elliptic section	rad
N	magnitude of the normal reaction	N

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial frame:** $S_0 = \{O; \mathcal{B}_0\}$, $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, with $\mathbf{k}_0$ vertical and upward.
- **Cone frame:** $S_1 = \{O; \mathcal{B}_1\}$, $\mathcal{B}_1 = \{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, attached to the cone. It is obtained from S_0 by a simple rotation of angle Ωt about $\mathbf{k}_0$ (the figure measures Ωt from x_0 to x_1), and coincides with S_0 at $t = 0$:

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i}_0 + \sin\Omega t\,\mathbf{j}_0, \qquad \mathbf{j}_1 = -\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0, \qquad \mathbf{k}_1 = \mathbf{k}_0$$

- **Change-of-basis matrix** (columns are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$):

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & -\sin\Omega t & 0 \\ \sin\Omega t & \cos\Omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- **Checks:** $\det[{}_0R_1] = \cos^2\Omega t + \sin^2\Omega t = 1$. The rows have unit norm ($\cos^2\Omega t + \sin^2\Omega t = 1$, and 1 for the third) and are mutually orthogonal ($-\cos\Omega t\sin\Omega t + \sin\Omega t\cos\Omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.
- **Motion of S_1:** the origin is fixed, $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$. For a simple rotation about the fixed vector $\mathbf{k}_0$ (Slide 16), $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_1$. Check with the chain rule: $\left.\dfrac{d\mathbf{i}_1}{dt}\right|_0 = \dfrac{d\cos\Omega t}{d(\Omega t)}\,\Omega\,\mathbf{i}_0 + \dfrac{d\sin\Omega t}{d(\Omega t)}\,\Omega\,\mathbf{j}_0 = \Omega\,(-\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0) = \Omega\,\mathbf{j}_1 = \Omega\,\mathbf{k}_1 \times \mathbf{i}_1$. Since Ω and $\mathbf{k}_1 = \mathbf{k}_0$ are constant, $\boldsymbol{\alpha}_{10} = \left.\dfrac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \mathbf{0}$.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Degrees of freedom

> **Pedagogical Justification:** a free particle has 3 coordinates; each independent holonomic constraint removes one.

Eliminating the parameters of Eq. (5): $\cos\beta = x_1/\zeta$, $\sin\beta = y_1/(2\zeta)$ and $\zeta = z_1$, so $\cos^2\beta + \sin^2\beta = 1$ gives the single constraint equation of the surface, fixed in S_1:

$$\frac{x_1^2}{z_1^2}+\frac{y_1^2}{4z_1^2}=1 \iff x_1^2+\frac{y_1^2}{4}-z_1^2=0$$

Hence $\mathrm{DOF}=3-1=2$, and (ζ,β) are suitable generalized coordinates. Each horizontal section $z_1=\zeta$ is an ellipse with semi-axes ζ (along $\mathbf{i}_1$) and 2ζ (along $\mathbf{j}_1$).

(b) Position, velocity and acceleration in S_1

> **Pedagogical Justification:** O is the origin of S_1, so $\mathbf{r}_1^P=\overrightarrow{OP}$ (Notes Eq. 3.16 with $\mathbf{r}_0^O=\mathbf{0}$). The relative derivatives $\frac{d}{dt}\big|_1$ treat $\mathbf{i}_1,\mathbf{j}_1,\mathbf{k}_1$ as constants, so only the components are differentiated (product and chain rules).

$$\mathbf{r}_1^P=\zeta\cos\beta\,\mathbf{i}_1+2\zeta\sin\beta\,\mathbf{j}_1+\zeta\,\mathbf{k}_1$$

Component by component, with $\frac{d}{dt}\cos\beta=\frac{d\cos\beta}{d\beta}\dot\beta=-\dot\beta\sin\beta$ and $\frac{d}{dt}\sin\beta=\dot\beta\cos\beta$:

$$\frac{d}{dt}(\zeta\cos\beta)=\dot\zeta\cos\beta-\zeta\dot\beta\sin\beta,\qquad \frac{d}{dt}(2\zeta\sin\beta)=2\dot\zeta\sin\beta+2\zeta\dot\beta\cos\beta,\qquad \frac{d}{dt}\zeta=\dot\zeta$$

$$\mathbf{v}_1^P=\Big(\dot\zeta\cos\beta-\zeta\dot\beta\sin\beta\Big)\,\mathbf{i}_1+2\Big(\dot\zeta\sin\beta+\zeta\dot\beta\cos\beta\Big)\,\mathbf{j}_1+\dot\zeta\,\mathbf{k}_1$$

Differentiating again. For the $\mathbf{i}_1$ component, $\frac{d}{dt}(\dot\zeta\cos\beta)=\ddot\zeta\cos\beta-\dot\zeta\dot\beta\sin\beta$ and $\frac{d}{dt}(\zeta\dot\beta\sin\beta)=\dot\zeta\dot\beta\sin\beta+\zeta\ddot\beta\sin\beta+\zeta\dot\beta^2\cos\beta$; for the $\mathbf{j}_1$ component, $\frac{d}{dt}(\dot\zeta\sin\beta)=\ddot\zeta\sin\beta+\dot\zeta\dot\beta\cos\beta$ and $\frac{d}{dt}(\zeta\dot\beta\cos\beta)=\dot\zeta\dot\beta\cos\beta+\zeta\ddot\beta\cos\beta-\zeta\dot\beta^2\sin\beta$. Collecting terms:

$$\mathbf{a}_1^P=\Big[\Big(\ddot\zeta-\zeta\dot\beta^2\Big)\cos\beta-\Big(2\dot\zeta\dot\beta+\zeta\ddot\beta\Big)\sin\beta\Big]\,\mathbf{i}_1+2\Big[\Big(\ddot\zeta-\zeta\dot\beta^2\Big)\sin\beta+\Big(2\dot\zeta\dot\beta+\zeta\ddot\beta\Big)\cos\beta\Big]\,\mathbf{j}_1+\ddot\zeta\,\mathbf{k}_1$$

The unit vectors $\mathbf{i}_1,\mathbf{j}_1,\mathbf{k}_1$ are defined in terms of $\mathcal{B}_0$ in Phase 2. For later use we write $\mathbf{r}_1^P=x_1\mathbf{i}_1+y_1\mathbf{j}_1+z_1\mathbf{k}_1$ and $\mathbf{v}_1^P=\dot x_1\mathbf{i}_1+\dot y_1\mathbf{j}_1+\dot z_1\mathbf{k}_1$ with the components above.

(c) Forces acting on P as seen from S_1

> **Pedagogical Justification:** in the non-inertial frame S_1 Newton's law reads $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63), where $\mathbf{F}$ are the actual forces and $\mathbf{F}_i$ the inertia forces of Notes Eq. 4.62 (Slide 28).

1. Weight: $\mathbf{W} = -mg\,\mathbf{k}_0 = -mg\,\mathbf{k}_1$.

2. Normal reaction. Two tangent vectors to the cone at P are the partial derivatives of $\mathbf{r}_1^P$ (they are not unit vectors):

$$\mathbf{e}_\zeta \equiv \frac{\partial \mathbf{r}_1^P}{\partial \zeta} = \cos\beta\,\mathbf{i}_1 + 2\sin\beta\,\mathbf{j}_1 + \mathbf{k}_1, \qquad \frac{\partial \mathbf{r}_1^P}{\partial \beta} = \zeta\,\mathbf{e}_\beta, \quad \mathbf{e}_\beta \equiv -\sin\beta\,\mathbf{i}_1 + 2\cos\beta\,\mathbf{j}_1$$

A normal vector is their cross product, expanded component by component:

$$\mathbf{e}_\zeta \times \mathbf{e}_\beta = \begin{vmatrix} \mathbf{i}_1 & \mathbf{j}_1 & \mathbf{k}_1 \\ \cos\beta & 2\sin\beta & 1 \\ -\sin\beta & 2\cos\beta & 0 \end{vmatrix} = (0 - 2\cos\beta)\,\mathbf{i}_1 - (0 + \sin\beta)\,\mathbf{j}_1 + \left(2\cos^2\beta + 2\sin^2\beta\right)\mathbf{k}_1 = -2\cos\beta\,\mathbf{i}_1 - \sin\beta\,\mathbf{j}_1 + 2\,\mathbf{k}_1$$

Its squared norm is $4\cos^2\beta + \sin^2\beta + 4 = 5 + 3\cos^2\beta$, so the smooth reaction is

$$\mathbf{N} = N\,\mathbf{n}, \qquad \mathbf{n} = \frac{-2\cos\beta\,\mathbf{i}_1 - \sin\beta\,\mathbf{j}_1 + 2\,\mathbf{k}_1}{\sqrt{5 + 3\cos^2\beta}}$$

with N an unknown scalar. (Check: the gradient of $x_1^2 + y_1^2/4 - z_1^2$ is $(2x_1, y_1/2, -2z_1) = \zeta(2\cos\beta, \sin\beta, -2)$, parallel to $\mathbf{n}$.)

3. Inertia forces (Notes Eq. 4.62). Since $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$, the force due to the acceleration of the origin and the Euler force vanish: $\mathbf{F}_A = -m\mathbf{a}_0^O = \mathbf{0}, \mathbf{F}_E = -m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P = \mathbf{0}$.

Centrifugal force. With $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1, \mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1, \mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$:

$$\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \Omega\,\mathbf{k}_1 \times (x_1\mathbf{i}_1 + y_1\mathbf{j}_1 + z_1\mathbf{k}_1) = \Omega\,(x_1\mathbf{j}_1 - y_1\mathbf{i}_1)$$

$$\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) = \Omega\,\mathbf{k}_1 \times \Omega\,(x_1\mathbf{j}_1 - y_1\mathbf{i}_1) = -\Omega^2\,(x_1\mathbf{i}_1 + y_1\mathbf{j}_1)$$

$$\mathbf{F}_C = -m\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) = m\Omega^2 \left(x_1 \mathbf{i}_1 + y_1 \mathbf{j}_1\right) = m\Omega^2 \zeta \left(\cos\beta\, \mathbf{i}_1 + 2\sin\beta\, \mathbf{j}_1\right)$$

which is $m\Omega^2 \mathbf{r}_\perp$, radially outward from the axis (Slide 30).

Coriolis force.

$$\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = -2m\Omega\, \mathbf{k}_1 \times \left(\dot{x}_1 \mathbf{i}_1 + \dot{y}_1 \mathbf{j}_1 + \dot{z}_1 \mathbf{k}_1\right) = 2m\Omega \left(\dot{y}_1\, \mathbf{i}_1 - \dot{x}_1\, \mathbf{j}_1\right)$$

$$\mathbf{F}_{COR} = 4m\Omega \left(\dot{\zeta}\sin\beta + \zeta\dot{\beta}\cos\beta\right) \mathbf{i}_1 - 2m\Omega \left(\dot{\zeta}\cos\beta - \zeta\dot{\beta}\sin\beta\right) \mathbf{j}_1$$

Equation of motion in S_1 (Notes Eq. 4.63):

$$m\mathbf{a}_1^P = -mg\, \mathbf{k}_1 + N\, \mathbf{n} + \mathbf{F}_C + \mathbf{F}_{COR}$$

(d) Two equations of motion free of the reaction

> **Pedagogical Justification:** the only unknown besides ζ and β is N, and $\mathbf{N}$ is perpendicular to the surface. Projecting the vector equation on the two tangent vectors $\mathbf{e}_\zeta$ and $\mathbf{e}_\beta$ eliminates it: $(-2\cos\beta)\cos\beta + (-\sin\beta)(2\sin\beta) + 2 = 0$ and $(-2\cos\beta)(-\sin\beta) + (-\sin\beta)(2\cos\beta) = 0$.

Projection on $\mathbf{e}_\zeta = (\cos\beta, 2\sin\beta, 1)$. Left-hand side:

$$\mathbf{a}_1^P \cdot \mathbf{e}_\zeta = \left(\ddot{\zeta} - \zeta\dot{\beta}^2\right)\left(\cos^2\beta + 4\sin^2\beta\right) + \left(2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta}\right)\left(-\sin\beta\cos\beta + 4\sin\beta\cos\beta\right) + \ddot{\zeta}$$

$$= \ddot{\zeta} + \left(\ddot{\zeta} - \zeta\dot{\beta}^2\right)\left(1 + 3\sin^2\beta\right) + 3\left(2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta}\right)\sin\beta\cos\beta$$

Right-hand side, term by term: $-mg\, \mathbf{k}_1 \cdot \mathbf{e}_\zeta = -mg$; $\mathbf{F}_C \cdot \mathbf{e}_\zeta = m\Omega^2\zeta\left(\cos^2\beta + 4\sin^2\beta\right) = m\Omega^2\zeta\left(1 + 3\sin^2\beta\right)$;

$$\mathbf{F}_{COR}\cdot\mathbf{e}_\zeta = 4m\Omega\left(\dot\zeta\sin\beta+\zeta\dot\beta\cos\beta\right)\cos\beta - 4m\Omega\left(\dot\zeta\cos\beta-\zeta\dot\beta\sin\beta\right)\sin\beta = 4m\Omega\zeta\dot\beta\left(\cos^2\beta+\sin^2\beta\right) = 4m\Omega\zeta\dot\beta$$

Dividing by m:

$$\boxed{\ddot\zeta + \left(\ddot\zeta-\zeta\dot\beta^2\right)\left(1+3\sin^2\beta\right) + 3\left(2\dot\zeta\dot\beta+\zeta\ddot\beta\right)\cos\beta\sin\beta = -g+\Omega^2\zeta\left(1+3\sin^2\beta\right)+4\Omega\zeta\dot\beta}$$

Projection on $\mathbf{e}_\beta = (-\sin\beta, 2\cos\beta, 0)$. Left-hand side:

$$\mathbf{a}_1^P\cdot\mathbf{e}_\beta = \left(\ddot\zeta-\zeta\dot\beta^2\right)\left(-\sin\beta\cos\beta+4\sin\beta\cos\beta\right)+\left(2\dot\zeta\dot\beta+\zeta\ddot\beta\right)\left(\sin^2\beta+4\cos^2\beta\right) = 3\left(\ddot\zeta-\zeta\dot\beta^2\right)\sin\beta\cos\beta+\left(2\dot\zeta\dot\beta+\zeta\ddot\beta\right)\left(1+3\cos^2\beta\right)$$

Right-hand side: $-mg\,\mathbf{k}_1\cdot\mathbf{e}_\beta = 0$; $\mathbf{F}_C\cdot\mathbf{e}_\beta = m\Omega^2\zeta\left(-\cos\beta\sin\beta+4\sin\beta\cos\beta\right) = 3m\Omega^2\zeta\sin\beta\cos\beta$;

$$\mathbf{F}_{COR}\cdot\mathbf{e}_\beta = -4m\Omega\left(\dot\zeta\sin\beta+\zeta\dot\beta\cos\beta\right)\sin\beta - 4m\Omega\left(\dot\zeta\cos\beta-\zeta\dot\beta\sin\beta\right)\cos\beta = -4m\Omega\dot\zeta$$

Dividing by m:

$$\boxed{3\left(\ddot\zeta-\zeta\dot\beta^2\right)\cos\beta\sin\beta+\left(2\dot\zeta\dot\beta+\zeta\ddot\beta\right)\left(1+3\cos^2\beta\right) = 3\Omega^2\zeta\cos\beta\sin\beta-4\Omega\dot\zeta}$$

These two equations contain only ζ, β and their derivatives. The initial conditions follow from the data: $\mathbf{r}_{1,0}^P = 2\mathbf{j}_1+\mathbf{k}_1$ gives $\zeta_0 = 1$, $\zeta_0\cos\beta_0 = 0$ and $2\zeta_0\sin\beta_0 = 2$, so $\beta_0 = \pi/2$; $\mathbf{v}_{1,0}^P = \mathbf{i}_1$ gives $\dot\zeta_0 = 0$ ($\mathbf{k}_1$ component) and $\dot\zeta_0\cos\beta_0-\zeta_0\dot\beta_0\sin\beta_0 = -\dot\beta_0 = 1$, so $\dot\beta_0 = -1$ rad/s; the $\mathbf{j}_1$ component $2(\dot\zeta_0\sin\beta_0+\zeta_0\dot\beta_0\cos\beta_0) = 0$ is consistent (the initial velocity is tangent to the cone).

(e) Conservation law: mechanical energy in S_1

> **Pedagogical Justification:** E_1^P is conserved when every force that does work in S_1, inertia forces included, derives from a potential (Slide 29). Here: $\mathbf{N}\cdot\mathbf{v}_1^P = 0$ because the cone is fixed in S_1 and $\mathbf{v}_1^P$ is tangent to it; $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$ always (Slide 31); $\mathbf{F}_A = \mathbf{F}_E = \mathbf{0}$; the weight is a constant force in S_1 ($\mathbf{k}_1 = \mathbf{k}_0$) with potential mgz_1; and, since $\boldsymbol{\omega}_{10}$ is constant, the centrifugal force has potential $-\frac{1}{2}m\Omega^2 r_\perp^2$ (Slide 30).

Kinetic energy in S_1, expanding the squares of the components of $\mathbf{v}_1^P$:

$$|\mathbf{v}_1^P|^2 = \left(\dot\zeta^2\cos^2\beta - 2\zeta\dot\zeta\dot\beta\sin\beta\cos\beta + \zeta^2\dot\beta^2\sin^2\beta\right) + 4\left(\dot\zeta^2\sin^2\beta + 2\zeta\dot\zeta\dot\beta\sin\beta\cos\beta + \zeta^2\dot\beta^2\cos^2\beta\right) + \dot\zeta^2$$

$$T_1^P = \frac{m}{2}\left[\left(2+3\sin^2\beta\right)\dot\zeta^2 + \left(1+3\cos^2\beta\right)\zeta^2\dot\beta^2 + 6\zeta\dot\zeta\dot\beta\sin\beta\cos\beta\right]$$

Potential in S_1, with $z_1 = \zeta$ and $r_\perp^2 = x_1^2 + y_1^2 = \zeta^2\left(\cos^2\beta + 4\sin^2\beta\right) = \zeta^2\left(1+3\sin^2\beta\right)$:

$$V_1 = mg\zeta - \frac{1}{2}m\Omega^2\zeta^2\left(1+3\sin^2\beta\right)$$

Direct check that $E_1^P = T_1^P + V_1$ is constant. Since $\mathbf{v}_1^P = \dot\zeta\,\mathbf{e}_\zeta + \zeta\dot\beta\,\mathbf{e}_\beta$ and $\dfrac{dV_1}{dt} = -\left(\mathbf{W}+\mathbf{F}_C\right)\cdot\mathbf{v}_1^P$ (chain rule on $V_1(\mathbf{r}_1^P(t))$):

$$\frac{dE_1^P}{dt} = \left(m\mathbf{a}_1^P - \mathbf{W} - \mathbf{F}_C\right)\cdot\mathbf{v}_1^P = \dot\zeta\left(m\mathbf{a}_1^P - \mathbf{W} - \mathbf{F}_C\right)\cdot\mathbf{e}_\zeta + \zeta\dot\beta\left(m\mathbf{a}_1^P - \mathbf{W} - \mathbf{F}_C\right)\cdot\mathbf{e}_\beta$$

and by the two equations of part (d) the brackets equal $4m\Omega\zeta\dot\beta$ and $-4m\Omega\dot\zeta$, so

$$\frac{dE_1^P}{dt} = \dot\zeta\left(4m\Omega\zeta\dot\beta\right) + \zeta\dot\beta\left(-4m\Omega\dot\zeta\right) = 0$$

Conservation law:

$$\boxed{\frac{m}{2}\left[\left(2+3\sin^2\beta\right)\dot\zeta^2 + \left(1+3\cos^2\beta\right)\zeta^2\dot\beta^2 + 6\zeta\dot\zeta\dot\beta\sin\beta\cos\beta\right] + mg\zeta - \frac{m}{2}\Omega^2\zeta^2\left(1+3\sin^2\beta\right) = E_1^P = m\left(\frac{1}{2} + g - 2\Omega^2\right)}$$

The constant comes from $t=0$: $\zeta_0 = 1$ m, $\beta_0 = \pi/2$, $\dot\zeta_0 = 0$, $\dot\beta_0 = -1$ rad/s give $T_1^P(0) = \frac{m}{2}(1+0)(1)^2(-1)^2 = \frac{m}{2}$ (that is, $\frac{1}{2}m|\mathbf{v}_{1,0}^P|^2$ with $|\mathbf{v}_{1,0}^P| = 1$ m/s) and $V_1(0) = mg(1) - \frac{1}{2}m\Omega^2(1)^2(1+3) = m\left(g - 2\Omega^2\right)$, in joules when m, g and Ω are in SI units.

The angular momentum about Oz_0 is **not** conserved: the moment of $\mathbf{N}$ about the axis is $\left(\mathbf{r}_1^P \times \mathbf{n}\right) \cdot \mathbf{k}_1 \, N = \left(x_1 n_y - y_1 n_x\right) N = \dfrac{\zeta\left(-\cos\beta\sin\beta + 4\sin\beta\cos\beta\right)}{\sqrt{5+3\cos^2\beta}} N = \dfrac{3\zeta\sin\beta\cos\beta}{\sqrt{5+3\cos^2\beta}} N$, which does not vanish in general because the section is not circular.

(f) Equilibrium positions in S_1

> **Pedagogical Justification:** an equilibrium in S_1 is a solution with ζ and β constant, so $\dot\zeta = \dot\beta = \ddot\zeta = \ddot\beta = 0$ in the two equations of part (d); the Coriolis terms vanish automatically.

The $\mathbf{e}_\beta$ equation reduces to $0 = 3\Omega^2\zeta\cos\beta\sin\beta = \frac{3}{2}\Omega^2\zeta\sin 2\beta$. With $\zeta > 0$ and $\Omega \neq 0$, $\sin 2\beta = 0$:

$$\beta \in \left\{0,\ \frac{\pi}{2},\ \pi,\ \frac{3\pi}{2}\right\}$$

The $\mathbf{e}_\zeta$ equation reduces to $0 = -g + \Omega^2\zeta\left(1 + 3\sin^2\beta\right)$, so

$$\zeta_{eq} = \frac{g}{\Omega^2\left(1+3\sin^2\beta\right)}$$

β	ζ_{eq}	**Position $\mathbf{r}_1^P$**
0	g/Ω^2	$\frac{g}{\Omega^2}\left(\mathbf{i}_1 + \mathbf{k}_1\right)$
π	g/Ω^2	$\frac{g}{\Omega^2}\left(-\mathbf{i}_1 + \mathbf{k}_1\right)$
$\pi/2$	$g/(4\Omega^2)$	$\frac{g}{2\Omega^2}\mathbf{j}_1 + \frac{g}{4\Omega^2}\mathbf{k}_1$
$3\pi/2$	$g/(4\Omega^2)$	$-\frac{g}{2\Omega^2}\mathbf{j}_1 + \frac{g}{4\Omega^2}\mathbf{k}_1$

Force-balance check at $\beta = 0$: $\mathbf{n} = (-2,0,2)/\sqrt{8}$, so $\mathbf{0} = -mg\,\mathbf{k}_1 + m\Omega^2\zeta\,\mathbf{i}_1 + \dfrac{N}{\sqrt{2}}\left(-\mathbf{i}_1 + \mathbf{k}_1\right)$ gives $N = \sqrt{2}\,mg$ and $\Omega^2\zeta = g$. **At $\beta = \pi/2$:** $\mathbf{n} = (0,-1,2)/\sqrt{5}$, so $\mathbf{0} = -mg\,\mathbf{k}_1 + 2m\Omega^2\zeta\,\mathbf{j}_1 + \dfrac{N}{\sqrt{5}}\left(-\mathbf{j}_1 + 2\mathbf{k}_1\right)$ gives $N = \frac{\sqrt{5}}{2}mg$ and $2\Omega^2\zeta = \frac{g}{2}$, i.e. $\zeta = g/(4\Omega^2)$. Both agree with the table.

Extension (stability, not asked): the Hessian of $V_1(\zeta,\beta)$ is diag $\left(-m\Omega^2,\ -3mg^2/\Omega^2\right)$ at $\beta = 0, \pi$ and diag $\left(-4m\Omega^2,\ 3mg^2/(16\Omega^2)\right)$ at $\beta = \pi/2, 3\pi/2$. V_1 has no minimum (along each generator it is a downward parabola in ζ), so the energy criterion cannot certify the stability of any of these equilibria.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Meaning of the equilibria:** they lie on the four generators contained in the principal planes $x_1 z_1$ and $y_1 z_1$ of the cone. Only there does the normal $\mathbf{n}$ lie in the vertical plane through the axis, so that it can balance the weight and the (radial) centrifugal force without a tangential remainder. On the steeper generators ($\beta = 0, \pi$, slope 45°) the particle must be higher, $\zeta = g/\Omega^2$, than on the flatter ones ($\beta = \pm\pi/2$, $\zeta = g/(4\Omega^2)$).
- **Gyroscopic coupling:** the Coriolis terms $4\Omega\zeta\dot\beta$ and $-4\Omega\dot\zeta$ couple the two equations antisymmetrically, which is why they cancel in dE_1^P/dt (Coriolis never does work).
- **Limit $\Omega \to 0$:** the centrifugal and Coriolis terms vanish, E_1^P becomes the ordinary energy $\frac{1}{2}mv^2 + mgz$, and $\zeta_{eq} \to \infty$: a fixed, vertex-down smooth cone has no equilibrium position with $\zeta > 0$.
- **Dimensions:** in the $\mathbf{e}_\zeta$ equation, $[\ddot\zeta] = [\zeta\dot\beta^2] = [\dot\zeta\dot\beta] = [\zeta\ddot\beta] = \mathrm{m/s^2}$, $[g] = \mathrm{m/s^2}$, $[\Omega^2\zeta] = \mathrm{s^{-2}\,m}$, $[\Omega\zeta\dot\beta] = \mathrm{s^{-1}\,m\,s^{-1}}$; the $\mathbf{e}_\beta$ equation has the same terms. $[E_1^P] = \mathrm{kg\,m^2/s^2} = \mathrm{J}$; $[g/\Omega^2] = \mathrm{(m/s^2)/s^{-2}} = \mathrm{m}$.
- **Official key:** both equations of part (d) coincide with the official key (Problems.pdf, p. 75) term by term.
- **Symbolic check (sympy):** $\mathbf{a}_0^P$ obtained by differentiating $[{}_0R_1]\,\mathbf{r}_1^P$ twice equals $\mathbf{a}_1^P + \boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P) + 2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P$ (Notes Eq. 3.27); both projections match the key; $dE_1^P/dt = 0$ along the equations of motion; the four equilibria satisfy both equations.

Source: Problems.pdf (PDF page 25); key on PDF page 75

PROBLEM 40

Spring-Coupled Particles on a Vertical Axis and a Revolving Ring

Original statement · Problems.pdf, p. 28

Two Particles in a Rotating Frame

Two heavy particles P and Q, each of mass m, are connected with a spring of stiffness k as shown in the figure. An inertial reference frame S_0 is defined with origin O and vector basis $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$. Particle P slides without friction along the vertical axis, while particle Q slides without friction on a ring of radius R contained in the horizontal plane defined by O, $\mathbf{i}_0$ and $\mathbf{j}_0$. The center D of this ring is located a distance ℓ from O and rotates around the vertical axis of S_0 with a known constant angular rate Ω. A rotating reference frame S_1 is defined so that the $\mathbf{j}_1$ vector follows the position of D. We define the generalized coordinates ζ (vertical position of P) and ϕ (angle of Q on the circle from $\mathbf{j}_1$) as shown.

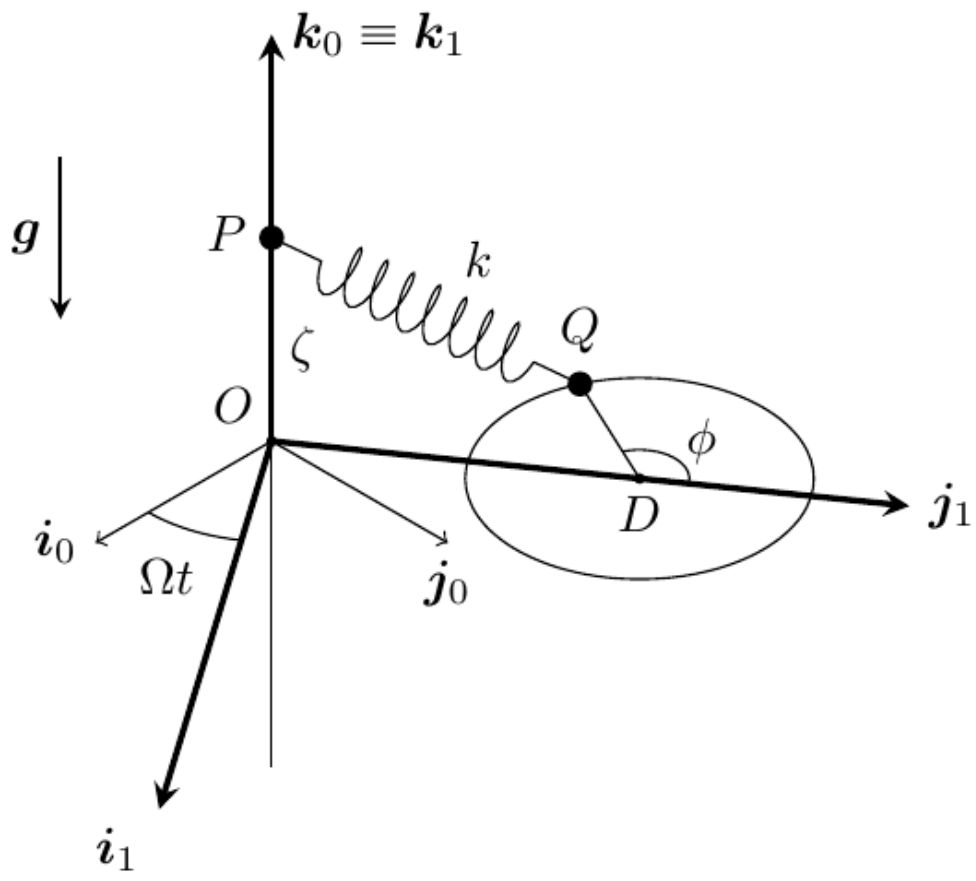


Figure: Particle P on the vertical axis k0 = k1, particle Q on a horizontal ring of centre D on the j1 axis, spring k between P and Q, frame S1 rotated by Omega t about k0 (Problems.pdf, p. 28).

(a) How many degrees of freedom does the system have? Define the vectors $\mathbf{i}_1, \mathbf{j}_1$ and $\mathbf{k}_1$ in terms of the vectors of $\mathcal{B}_0$.

(b) Write down the general expression of the position, velocity, and acceleration of P and Q with respect to S_1 in terms of ζ, ϕ and their derivatives.

(c) Identify and write down the general expression of all the forces acting on P and Q.

(d) Using S_1 for your analysis, write down the scalar equations of motion of P and Q.

(e) Starting from these equations, write down a system of two differential equations for ζ and ϕ, that do not depend on any other unknowns of the problem.

(f) Find one conservation law of the problem (i.e., a first integral of motion) in reference frame S_1.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** two heavy particles of mass m. P slides on the fixed vertical axis Oz_0; Q slides on a horizontal ring of radius R in the plane $z_0 = 0$ whose centre D, at distance ℓ from O, revolves about Oz_0 at constant rate Ω. A spring of stiffness k joins P and Q.
- **Hypotheses:**
  1. $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial, $\mathbf{k}_0$ vertical upward, gravity $-g\,\mathbf{k}_0$.
  2. Both contacts are smooth and bilateral: the reactions have no component along the direction in which each particle can move.
  3. The spring is linear and massless, with zero natural length (no natural length is given), so the force it exerts on P is $k\left(\mathbf{r}^Q - \mathbf{r}^P\right)$ and on Q the opposite.
  4. The ring moves rigidly with S_1 (only its centre's motion is prescribed; it does not spin relative to S_1, which is irrelevant anyway for a smooth circular wire).
  5. ϕ is measured from $\mathbf{j}_1$ in the positive sense about $\mathbf{k}_1$ (from $\mathbf{j}_1$ towards $-\mathbf{i}_1$), as in the figure; this is also the convention of the official key, whose ring reaction contains the radial vector $-\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1$.

- **Why S_1:** in S_1 the ring is fixed (centre at $\ell\,\mathbf{j}_1$), so Q moves on a fixed circle described by ϕ alone and P stays on the axis. The inertia forces (Notes Eq. 4.62) are the price; for P, on the rotation axis, they vanish.

Symbol	Meaning	SI unit
m	mass of each particle	kg
k	spring stiffness	N/m
R	radius of the ring	m
ℓ	distance OD	m
Ω	constant angular rate of D	rad/s
g	gravitational acceleration	m/s^2
ζ, ϕ	height of P, angle of Q on the ring	m, rad
$N_1, \dots, N_4$	reaction components	N

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Frames:** $S_0 = \{O; \mathcal{B}_0\}$ inertial; $S_1 = \{O; \mathcal{B}_1\}$ with the same origin, $\mathbf{k}_1 = \mathbf{k}_0$ and $\mathbf{j}_1$ pointing from O to D. S_1 is rotated by Ωt about $\mathbf{k}_0$ (the figure measures Ωt from $\mathbf{i}_0$ to $\mathbf{i}_1$), and we take $S_1 \equiv S_0$ at $t = 0$:

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i}_0 + \sin\Omega t\,\mathbf{j}_0, \qquad \mathbf{j}_1 = -\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0, \qquad \mathbf{k}_1 = \mathbf{k}_0$$

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & -\sin\Omega t & 0 \\ \sin\Omega t & \cos\Omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\Omega t + \sin^2\Omega t = 1$$

- **Orthogonality:** each row has norm $\cos^2\Omega t + \sin^2\Omega t = 1$ (or 1), and the first two rows have dot product $\cos\Omega t\sin\Omega t - \sin\Omega t\cos\Omega t = 0$, so $[{}_0R_1][{}_0R_1]^T = I$.
- **Motion of S_1:** $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$; simple rotation about $\mathbf{k}_0$ (Slide 16): $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_1$, and $\boldsymbol{\alpha}_{10} = \mathbf{0}$ because Ω and $\mathbf{k}_0$ are constant.
- **Ring basis** (attached to Q, defined in $\mathcal{B}_1$):

$$\mathbf{e}_r = -\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1, \qquad \mathbf{e}_\phi = \frac{d\mathbf{e}_r}{d\phi} = -\cos\phi\,\mathbf{i}_1 - \sin\phi\,\mathbf{j}_1, \qquad \mathbf{k}_1$$

$\mathbf{e}_r$ points from D to Q and $\mathbf{e}_\phi$ is tangent to the ring. Check: $\mathbf{e}_r \times \mathbf{e}_\phi = [(-\sin\phi)(-\sin\phi) - (\cos\phi)(-\cos\phi)]\,\mathbf{k}_1 = \mathbf{k}_1$, so $\{\mathbf{e}_r, \mathbf{e}_\phi, \mathbf{k}_1\}$ is right-handed and orthonormal. The inverse relations used below are $\mathbf{j}_1 = \cos\phi\,\mathbf{e}_r - \sin\phi\,\mathbf{e}_\phi$ and $\mathbf{i}_1 = -\sin\phi\,\mathbf{e}_r - \cos\phi\,\mathbf{e}_\phi$.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Degrees of freedom and the basis $\mathcal{B}_1$

Pedagogical Justification: count coordinates minus independent constraints for each particle.

P is free in space (3 coordinates) but constrained to the vertical axis ($x = y = 0$): 1 DOF, the height ζ. Q (3 coordinates) is constrained to the plane $z_1 = 0$ and to the circle $\left|\mathbf{r}^Q - \overrightarrow{OD}\right| = R$: 1 DOF, the angle ϕ. The motion of D is prescribed, so it adds no unknown. Hence

$$\text{DOF} = 6 - 4 = 2, \qquad \text{generalized coordinates } (\zeta, \phi)$$

The vectors of $\mathcal{B}_1$ in terms of $\mathcal{B}_0$ are those given in Phase 2.

(b) Relative kinematics of P and Q in S_1

Pedagogical Justification: O is the origin of S_1, so $\mathbf{r}_1 = \overrightarrow{OP}$ or $\overrightarrow{OQ}$ (Notes Eq. 3.16). The vectors $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ are constant in S_1; $\mathbf{e}_r$ and $\mathbf{e}_\phi$ change only through $\phi(t)$, so the chain rule gives $\left.\dfrac{d\mathbf{e}_r}{dt}\right|_1 = \dfrac{d\mathbf{e}_r}{d\phi}\dot\phi = \dot\phi\,\mathbf{e}_\phi$ and $\left.\dfrac{d\mathbf{e}_\phi}{dt}\right|_1 = \dfrac{d\mathbf{e}_\phi}{d\phi}\dot\phi = \dot\phi\,(\sin\phi\,\mathbf{i}_1 - \cos\phi\,\mathbf{j}_1) = -\dot\phi\,\mathbf{e}_r$.

Particle P:

$$\mathbf{r}_1^P = \zeta\,\mathbf{k}_1, \qquad \mathbf{v}_1^P = \dot\zeta\,\mathbf{k}_1, \qquad \mathbf{a}_1^P = \ddot\zeta\,\mathbf{k}_1$$

Particle Q: $\overrightarrow{OQ} = \overrightarrow{OD} + \overrightarrow{DQ}$ with $\overrightarrow{OD} = \ell\,\mathbf{j}_1$ (constant in S_1):

$$\mathbf{r}_1^Q = \ell\,\mathbf{j}_1 + R\,\mathbf{e}_r = -R\sin\phi\,\mathbf{i}_1 + (\ell + R\cos\phi)\,\mathbf{j}_1$$

$$\mathbf{v}_1^Q = R\left.\frac{d\mathbf{e}_r}{dt}\right|_1 = R\dot\phi\,\mathbf{e}_\phi = -R\dot\phi\cos\phi\,\mathbf{i}_1 - R\dot\phi\sin\phi\,\mathbf{j}_1$$

By the product rule, $\mathbf{a}_1^Q = R\ddot\phi\,\mathbf{e}_\phi + R\dot\phi\left.\frac{d\mathbf{e}_\phi}{dt}\right|_1$:

$$\mathbf{a}_1^Q = R\ddot\phi\,\mathbf{e}_\phi - R\dot\phi^2\,\mathbf{e}_r$$

(c) Forces on P and Q (actual and inertia forces in S_1)

> **Pedagogical Justification:** in S_1 we apply $m\mathbf{a}_1 = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63) with $\mathbf{F}_i$ from Notes Eq. 4.62. Here $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$, so only the centrifugal and Coriolis forces survive (Slide 28).

Spring. The relative position is

$$\mathbf{r}^Q - \mathbf{r}^P = \ell\,\mathbf{j}_1 + R\,\mathbf{e}_r - \zeta\,\mathbf{k}_1 = -R\sin\phi\,\mathbf{i}_1 + (\ell + R\cos\phi)\,\mathbf{j}_1 - \zeta\,\mathbf{k}_1$$

so the spring force on P is $\mathbf{F}_s^P = k\left(\mathbf{r}^Q - \mathbf{r}^P\right)$ and on Q it is $\mathbf{F}_s^Q = -k\left(\mathbf{r}^Q - \mathbf{r}^P\right)$. In the ring basis, using $\mathbf{j}_1 = \cos\phi\,\mathbf{e}_r - \sin\phi\,\mathbf{e}_\phi$:

$$\mathbf{F}_s^Q = -k\left(\ell\cos\phi + R\right)\mathbf{e}_r + k\ell\sin\phi\,\mathbf{e}_\phi + k\zeta\,\mathbf{k}_1$$

Forces on P:

- Weight $-mg\,\mathbf{k}_1$.
- Reaction of the smooth axis, perpendicular to $\mathbf{k}_1$: $\mathbf{N}_P = N_1\,\mathbf{i}_1 + N_2\,\mathbf{j}_1$.
- Spring $\mathbf{F}_s^P = k\left[-R\sin\phi\,\mathbf{i}_1 + (\ell + R\cos\phi)\,\mathbf{j}_1 - \zeta\,\mathbf{k}_1\right]$.
- Inertia forces: P is on the rotation axis. $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = \Omega\zeta\,\mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$, so $\mathbf{F}_C^P = \mathbf{0}$; $\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P = \Omega\dot\zeta\,\mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$, so $\mathbf{F}_{COR}^P = \mathbf{0}$.

Forces on Q:

- Weight $-mg\,\mathbf{k}_1$.
- Reaction of the smooth ring, perpendicular to the tangent $\mathbf{e}_\phi$: $\mathbf{N}_Q = N_3\,\mathbf{e}_r + N_4\,\mathbf{k}_1 = N_3\left(-\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1\right) + N_4\,\mathbf{k}_1$.
- Spring $\mathbf{F}_s^Q$ above.

- Centrifugal force. Since $\mathbf{r}_1^Q$ has no $\mathbf{k}_1$ component, $\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^Q\right) = -\Omega^2 \mathbf{r}_1^Q$ (from $\mathbf{k}_1 \times (\mathbf{k}_1 \times \mathbf{i}_1) = -\mathbf{i}_1$ and $\mathbf{k}_1 \times (\mathbf{k}_1 \times \mathbf{j}_1) = -\mathbf{j}_1$), hence

$$\mathbf{F}_C^Q = m\Omega^2 \mathbf{r}_1^Q = m\Omega^2 \left(\ell\, \mathbf{j}_1 + R\, \mathbf{e}_r\right) = m\Omega^2 \left(\ell \cos\phi + R\right) \mathbf{e}_r - m\Omega^2 \ell \sin\phi\, \mathbf{e}_\phi$$

- Coriolis force. From $\mathbf{e}_r \times \mathbf{e}_\phi = \mathbf{k}_1$ (cyclic: $\mathbf{e}_\phi \times \mathbf{k}_1 = \mathbf{e}_r$), $\mathbf{k}_1 \times \mathbf{e}_\phi = -\mathbf{e}_r$:

$$\mathbf{F}_{COR}^Q = -2m\Omega\, \mathbf{k}_1 \times R\dot{\phi}\, \mathbf{e}_\phi = 2m\Omega R\dot{\phi}\, \mathbf{e}_r$$

- Euler force and force due to the origin's acceleration: zero for both particles.

(d) Scalar equations of motion in S_1

> **Pedagogical Justification:** project $m\mathbf{a}_1 = \mathbf{F} + \mathbf{F}_i$ for each particle on an orthonormal basis adapted to its constraint: $\{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ for P and $\{\mathbf{e}_r, \mathbf{e}_\phi, \mathbf{k}_1\}$ for Q. This gives $3 + 3 = 6$ scalar equations for the 6 unknowns $\zeta, \phi, N_1, N_2, N_3, N_4$.

Particle P ($m\ddot{\zeta}\, \mathbf{k}_1 = -mg\, \mathbf{k}_1 + N_1 \mathbf{i}_1 + N_2 \mathbf{j}_1 + \mathbf{F}_s^P$):

$$\mathbf{i}_1 : \quad 0 = N_1 - kR\sin\phi$$

$$\mathbf{j}_1 : \quad 0 = N_2 + k\left(\ell + R\cos\phi\right)$$

$$\mathbf{k}_1 : \quad m\ddot{\zeta} = -mg - k\zeta$$

Particle Q ($m\left(R\ddot{\phi}\, \mathbf{e}_\phi - R\dot{\phi}^2 \mathbf{e}_r\right) = -mg\, \mathbf{k}_1 + N_3 \mathbf{e}_r + N_4 \mathbf{k}_1 + \mathbf{F}_s^Q + \mathbf{F}_C^Q + \mathbf{F}_{COR}^Q$):

$$\mathbf{e}_r : \quad -mR\dot{\phi}^2 = N_3 - k\left(\ell\cos\phi + R\right) + m\Omega^2\left(\ell\cos\phi + R\right) + 2m\Omega R\dot{\phi}$$

$$\mathbf{e}_\phi: \quad mR\ddot{\phi} = k\ell\sin\phi - m\Omega^2\ell\sin\phi$$

$$\mathbf{k}_1: \quad 0 = -mg + N_4 + k\zeta$$

(e) Two equations free of the reactions

> **Pedagogical Justification:** the reactions appear only in the projections perpendicular to the allowed motions; the projections along the allowed motions ($\mathbf{k}_1$ for P, $\mathbf{e}_\phi$ for Q) are free of them.

Dividing the $\mathbf{k}_1$ equation of P and the $\mathbf{e}_\phi$ equation of Q by m and mR:

$$\boxed{\ddot{\zeta} + \frac{k}{m}\zeta = -g, \qquad \ddot{\phi} = \frac{\ell}{R}\left(\frac{k}{m} - \Omega^2\right)\sin\phi}$$

The other four equations then give the reactions once $\zeta(t)$, $\phi(t)$ are known: $N_1 = kR\sin\phi$, $N_2 = -k\left(\ell + R\cos\phi\right)$, $N_3 = -mR\dot{\phi}^2 + \left(k - m\Omega^2\right)\left(R + \ell\cos\phi\right) - 2m\Omega R\dot{\phi}$, $N_4 = mg - k\zeta$. In particular the ring reaction has the form $\mathbf{N}_Q = N_3\left(-\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1\right) + N_4\,\mathbf{k}_1$ of the official key.

The two equations are **decoupled**: the vertical spring component acts only on P's coordinate, and the horizontal one only on Q's, because the spring force is linear in $\mathbf{r}^Q - \mathbf{r}^P$ and P moves along $\mathbf{k}_1$ while Q moves in the plane $\perp \mathbf{k}_1$.

(f) A conservation law in S_1

> **Pedagogical Justification:** in S_1 the constraints (axis and ring) are fixed and smooth, so the reactions do no work; the Coriolis force on Q is perpendicular to $\mathbf{v}_1^Q$ (Slide 31); the weight and spring are conservative; and with $\boldsymbol{\omega}_{10}$ constant the centrifugal force has potential $-\tfrac{1}{2}m\Omega^2 r_\perp^2$ (Slide 30). Hence the mechanical energy of the system in S_1 is conserved (Slide 29).

Potential energies in S_1: weight of P, $mg\zeta$ (weight of Q: $mg\cdot 0$); spring $\tfrac{1}{2}k\left|\mathbf{r}^Q - \mathbf{r}^P\right|^2$; centrifugal on Q, $-\tfrac{1}{2}m\Omega^2\left|\mathbf{r}_1^Q\right|^2$ (here $r_\perp^Q = |\mathbf{r}_1^Q|$). With

$$|\mathbf{r}^Q-\mathbf{r}^P|^2=R^2\sin^2\phi+(\ell+R\cos\phi)^2+\zeta^2=\ell^2+R^2+2\ell R\cos\phi+\zeta^2,\qquad \left|\mathbf{r}_1^Q\right|^2=\ell^2+R^2+2\ell R\cos\phi$$

and dropping the constant $\frac{1}{2}\left(k-m\Omega^2\right)\left(\ell^2+R^2\right)$:

$$\boxed{E_1=\frac{1}{2}m\left(\dot\zeta^2+R^2\dot\phi^2\right)+mg\zeta+\frac{1}{2}k\zeta^2+\left(k-m\Omega^2\right)\ell R\cos\phi=\text{const}}$$

Check with the equations of part (e) (chain rule $\frac{d}{dt}\cos\phi=-\dot\phi\sin\phi$, $\frac{d}{dt}\zeta^2=2\zeta\dot\zeta$):

$$\frac{dE_1}{dt}=m\dot\zeta\ddot\zeta+mR^2\dot\phi\ddot\phi+mg\dot\zeta+k\zeta\dot\zeta-\left(k-m\Omega^2\right)\ell R\dot\phi\sin\phi=m\dot\zeta\left(\ddot\zeta+g+\frac{k}{m}\zeta\right)+mR\dot\phi\left(R\ddot\phi-\left(\frac{k}{m}-\Omega^2\right)\ell\sin\phi\right)=0$$

Because the equations are decoupled, each bracket vanishes separately, so in fact two first integrals exist: $\frac{1}{2}m\dot\zeta^2+mg\zeta+\frac{1}{2}k\zeta^2=\text{const}$ and $\frac{1}{2}mR^2\dot\phi^2+\left(k-m\Omega^2\right)\ell R\cos\phi=\text{const}$. The constants are fixed by initial conditions, which the statement does not give.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Particle P:** $\ddot\zeta+(k/m)\zeta=-g$ is a harmonic oscillator about $\zeta_{eq}=-mg/k$ with angular frequency $\sqrt{k/m}$, unaffected by the rotation (it lies on the axis, where centrifugal and Coriolis forces vanish).
- **Particle Q:** $\ddot\phi=\frac{\ell}{R}\left(\frac{k}{m}-\Omega^2\right)\sin\phi$ is a pendulum equation. The spring pulls Q towards the axis (towards $\phi=\pi$, the point of the ring nearest O) and the centrifugal force pushes it outward (towards $\phi=0$). If $\Omega^2<k/m$ the stable position is $\phi=\pi$; if $\Omega^2>k/m$ it is $\phi=0$; at $\Omega^2=k/m$ the two effects cancel exactly and $\dot\phi$ is constant. The Coriolis force only loads the ring radially (N_3).
- **Limits:** $\Omega\to0$ gives a fixed ring and $\ddot\phi=\frac{k\ell}{mR}\sin\phi$; $\ell\to0$ (ring centred on the axis) gives $\ddot\phi=0$, uniform motion, since every point of the ring is then equidistant from the axis.
- **Dimensions:** $[k/m]=\mathrm{N/(m\,kg)}=\mathrm{s^{-2}}$, so $[(k/m)\zeta]=\mathrm{m/s^2}=[g]=[\ddot\zeta]$; $[(\ell/R)(k/m-\Omega^2)]=\mathrm{s^{-2}}=[\ddot\phi]$; $[N_3]$: $[mR\dot\phi^2]=[kR]=[m\Omega R\dot\phi]=\mathrm{N}$; $[E_1]=\mathrm{kg\,m^2/s^2}=[k\zeta^2]=[k\ell R]=\mathrm{J}$.
- **Official key** (Problems.pdf, p. 76): "One of the forces acting on Q is the reaction of the ring, $\mathbf{N}_Q=N_3(-\sin\phi\,\mathbf{i}_1+\cos\phi\,\mathbf{j}_1)+N_4\,\mathbf{k}_1$", which matches part (c).

- **Symbolic check (sympy):** $\mathbf{a}_0$ of both particles obtained by differentiating $[{}_0R_1]\,\mathbf{r}_1$ twice matches Notes Eq. 3.27; the six scalar equations, the four reactions, and $dE_1/dt = 0$ along the motion are confirmed.

Source: Problems.pdf (PDF page 28); key on PDF page 76

PROBLEM 41

Satellite Solar Panel: Composition of Rotations

Original statement · Problems.pdf, p. 29

Relative Kinematics

The angle θ describing the rotation of the solar panels of a reconnaissance satellite about the body-fixed axis x is an arbitrary function of time. The satellite spins about the z axis at the constant angular rate ω. Derive expressions for the absolute velocity and acceleration of point B in the figure relative to an inertial reference frame centered in A.

(Taken from Ginsberg, "Engineering Dynamics", Cambridge University Press, 2008.)

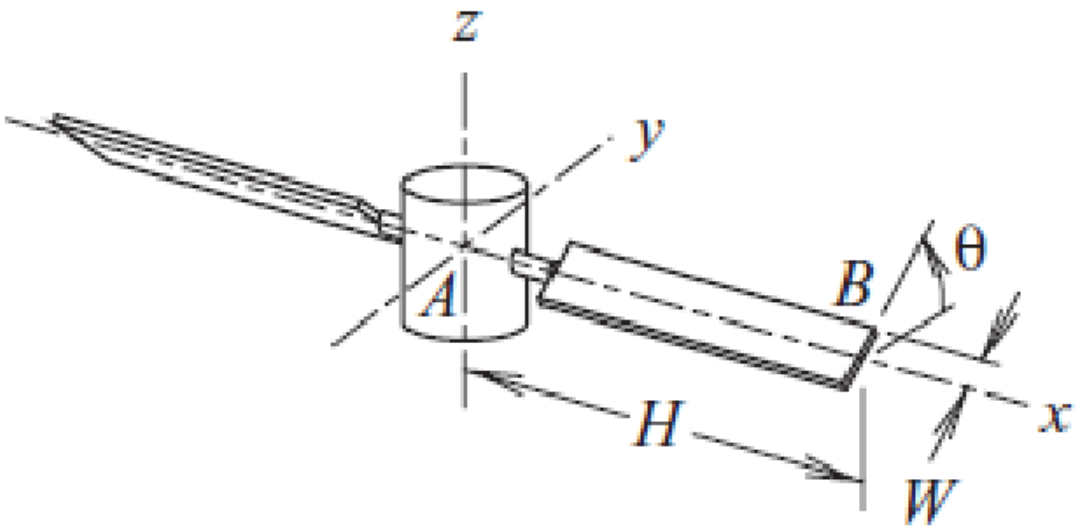


Figure: Satellite body spinning about z with a solar panel rotated by the angle θ about the body-fixed x axis, points A and B, dimensions H and W (Problems.pdf, p. 29).

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** a satellite body spins about its z axis at constant rate ω; a solar panel turns about the body-fixed x axis through the prescribed angle $\theta(t)$. Point B is a corner of the panel.
- **Why relative motion.** The motion of B is the superposition of two simple motions: in the body frame S_1, B just describes a circle of radius W about the x axis; and S_1 itself just spins about z. Notes Eqs. 3.26–3.27 combine the two without writing the inertial position explicitly. A second route uses the panel frame S_2, in which B is fixed, and the composition of angular velocities (Notes Eqs. 3.13–3.15).
- **Hypotheses:**
  1. The frame centred at A with non-rotating axes is inertial, so the centre A of the satellite is fixed: $\mathbf{v}_0^A = \mathbf{a}_0^A = \mathbf{0}$.
  2. Body and panel are rigid; ω is constant; $\theta(t)$ is an arbitrary (known) function.
  3. **Geometry (read from the figure):** B is at distance H from A measured along x, and at distance W from the x axis; θ is measured about $+x$ from the body axis y, so for $\theta = 0$ the segment from the hinge axis to B points along $+y$.
- **Degrees of freedom:** none to determine; the motion is prescribed by ωt and $\theta(t)$ (kinematics only).

Symbol	Meaning	SI unit
H	axial distance from A to B (along x)	m
W	distance from B to the hinge axis x	m
ω	constant spin rate of the satellite about z	rad/s
$\theta(t)$	panel angle about x	rad

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial frame:** $S_0 = \{A; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$.
- **Body frame:** $S_1 = \{A; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ (x, y, z of the figure), a simple rotation of angle ωt about $\mathbf{k}_0 = \mathbf{k}_1$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \boldsymbol{\omega}_{10} = \omega\,\mathbf{k}_1, \qquad \boldsymbol{\alpha}_{10} = \mathbf{0}$$

- **Panel frame:** $S_2 = \{C; \mathbf{i}_2, \mathbf{j}_2, \mathbf{k}_2\}$, where $C = A + H\mathbf{i}_1$ is the point of the hinge axis at the abscissa of B; simple rotation of angle θ about $\mathbf{i}_1 = \mathbf{i}_2$:

$$[{}_1R_2] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}, \qquad \boldsymbol{\omega}_{21} = \dot\theta\,\mathbf{i}_1$$

- **Checks:** $\det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1$ and $\det[{}_1R_2] = \cos^2\theta + \sin^2\theta = 1$; the columns of each matrix are orthonormal, so $RR^T = I$; hence $[{}_0R_2] = [{}_0R_1][{}_1R_2]$ is also a proper rotation (Notes Eq. 1.16).
- **Position of B.** $\mathbf{r}_{CB} = W\,\mathbf{j}_2$ is fixed in S_2; with the second column of $[{}_1R_2]$, $\mathbf{j}_2 = \cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1$, so

$$\mathbf{r}_1^B = \mathbf{AC} + \mathbf{CB} = H\,\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1$$

PHASE 3

Step-by-Step Mathematical Deduction

1. Relative motion of B in the body frame S_1

> **Pedagogical Justification:** the body frame rotates at constant rate with fixed origin, so Notes Eq. 3.27 keeps only three terms; the relative quantities are derivatives in S_1, where $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ are constant.

With the chain rule $\frac{d}{dt}\cos\theta = -\dot\theta\sin\theta$, $\frac{d}{dt}\sin\theta = \dot\theta\cos\theta$, and the product rule $\frac{d}{dt}(\dot\theta\sin\theta) = \ddot\theta\sin\theta + \dot\theta^2\cos\theta$, $\frac{d}{dt}(\dot\theta\cos\theta) = \ddot\theta\cos\theta - \dot\theta^2\sin\theta$:

$$\mathbf{v}_1^B = \left.\frac{d\mathbf{r}_1^B}{dt}\right|_1 = -W\dot\theta\sin\theta\,\mathbf{j}_1 + W\dot\theta\cos\theta\,\mathbf{k}_1$$

$$\mathbf{a}_1^B = \left.\frac{d\mathbf{v}_1^B}{dt}\right|_1 = -W\left(\ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta\right)\mathbf{j}_1 + W\left(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta\right)\mathbf{k}_1$$

(circular motion of radius W about the x axis: tangential part $W\ddot{\theta}\,\mathbf{k}_2$ and normal part $-W\dot{\theta}^2\,\mathbf{j}_2$).

2. Absolute velocity of B

> **Pedagogical Justification:** Notes Eq. 3.26 (Slide 24), $\mathbf{v}_0^B = \mathbf{v}_0^A + \mathbf{v}_1^B + \boldsymbol{\omega}_{10}\times\mathbf{r}_1^B$, with $\mathbf{v}_0^A = \mathbf{0}$.

Cross products: $\mathbf{k}_1\times\mathbf{i}_1 = \mathbf{j}_1, \mathbf{k}_1\times\mathbf{j}_1 = -\mathbf{i}_1, \mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$:

$$\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B = \omega\mathbf{k}_1\times(H\,\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1) = H\omega\,\mathbf{j}_1 - W\omega\cos\theta\,\mathbf{i}_1$$

Adding $\mathbf{v}_1^B$ component by component:

$$\boxed{\mathbf{v}_0^B = -W\omega\cos\theta\,\mathbf{i}_1 + \left(H\omega - W\dot{\theta}\sin\theta\right)\mathbf{j}_1 + W\dot{\theta}\cos\theta\,\mathbf{k}_1}$$

which agrees with the official key (Problems.pdf, p. 76). The components are given in the body basis $\mathcal{B}_1$; in $\mathcal{B}_0$ they follow from $\{\mathbf{v}_0^B\}_0 = [{}_0R_1]\{\mathbf{v}_0^B\}_1$.

3. Absolute acceleration of B

> **Pedagogical Justification:** Notes Eq. 3.27 (Slide 25) with $\mathbf{a}_0^A = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$: $\mathbf{a}_0^B = \mathbf{a}_1^B + \boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B\right) + 2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^B$.

Centripetal term (reusing $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B$ from step 2):

$$\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B\right) = \omega\mathbf{k}_1\times(-W\omega\cos\theta\,\mathbf{i}_1 + H\omega\,\mathbf{j}_1) = -W\omega^2\cos\theta\,\mathbf{j}_1 - H\omega^2\,\mathbf{i}_1$$

i.e. $-\omega^2\mathbf{r}_\perp$ with $\mathbf{r}_\perp = H\,\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1$.

Coriolis term:

$$2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^B = 2\omega\mathbf{k}_1\times\left(-W\dot\theta\sin\theta\,\mathbf{j}_1 + W\dot\theta\cos\theta\,\mathbf{k}_1\right) = 2W\omega\dot\theta\sin\theta\,\mathbf{i}_1$$

Sum, collecting the $\mathbf{i}_1$, $\mathbf{j}_1$ and $\mathbf{k}_1$ components of $\mathbf{a}_1^B$, the centripetal term and the Coriolis term:

$$\boxed{\mathbf{a}_0^B = \left(2W\omega\dot\theta\sin\theta - H\omega^2\right)\mathbf{i}_1 - \left(W\ddot\theta\sin\theta + W\dot\theta^2\cos\theta + W\omega^2\cos\theta\right)\mathbf{j}_1 + \left(W\ddot\theta\cos\theta - W\dot\theta^2\sin\theta\right)\mathbf{k}_1}$$

which agrees with the official key (Problems.pdf, p. 76).

4. Second route: panel frame and composition of angular velocities

> **Pedagogical Justification:** B is fixed in S_2, so $\mathbf{v}_2^B = \mathbf{a}_2^B = \mathbf{0}$ and Notes Eqs. 3.26–3.27 applied to S_2 (origin C) need only $\boldsymbol{\omega}_{20}$, $\boldsymbol{\alpha}_{20}$ and the motion of C. Angular velocities add (Notes Eqs. 3.13–3.15, Slide 19); angular accelerations do not simply add.

$$\boldsymbol{\omega}_{20} = \boldsymbol{\omega}_{21} + \boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{i}_1 + \omega\,\mathbf{k}_1$$

Its derivative in S_0 by the Coriolis theorem (Notes Eq. 3.8), with $\frac{d}{dt}\big|_1\left(\dot\theta\,\mathbf{i}_1 + \omega\,\mathbf{k}_1\right) = \ddot\theta\,\mathbf{i}_1$ (ω constant):

$$\boldsymbol{\alpha}_{20} = \left.\frac{d\boldsymbol{\omega}_{20}}{dt}\right|_1 + \boldsymbol{\omega}_{10}\times\boldsymbol{\omega}_{20} = \ddot\theta\,\mathbf{i}_1 + \omega\mathbf{k}_1\times\dot\theta\,\mathbf{i}_1 = \ddot\theta\,\mathbf{i}_1 + \omega\dot\theta\,\mathbf{j}_1$$

Motion of C (fixed in S_1 at $H\mathbf{i}_1$): $\mathbf{v}_0^C = \omega\mathbf{k}_1\times H\mathbf{i}_1 = H\omega\,\mathbf{j}_1$, $\mathbf{a}_0^C = \omega\mathbf{k}_1\times H\omega\,\mathbf{j}_1 = -H\omega^2\,\mathbf{i}_1$. With $\mathbf{r}_2^B = \mathbf{CB} = W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1$:

$$\boldsymbol{\omega}_{20}\times\mathbf{r}_2^B = \dot\theta W\cos\theta\,\mathbf{k}_1 - \dot\theta W\sin\theta\,\mathbf{j}_1 - \omega W\cos\theta\,\mathbf{i}_1$$

so $\mathbf{v}_0^B = \mathbf{v}_0^C + \boldsymbol{\omega}_{20}\times\mathbf{r}_2^B$ reproduces the boxed velocity. For the acceleration, $\mathbf{a}_0^B = \mathbf{a}_0^C + \boldsymbol{\alpha}_{20}\times\mathbf{r}_2^B + \boldsymbol{\omega}_{20}\times(\boldsymbol{\omega}_{20}\times\mathbf{r}_2^B)$:

$$\boldsymbol{\alpha}_{20} \times \mathbf{r}_2^B = \left(\ddot{\theta}\,\mathbf{i}_1 + \omega\dot{\theta}\,\mathbf{j}_1\right) \times (W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1) = W\ddot{\theta}\cos\theta\,\mathbf{k}_1 - W\ddot{\theta}\sin\theta\,\mathbf{j}_1 + W\omega\dot{\theta}\sin\theta\,\mathbf{i}_1$$

$$\boldsymbol{\omega}_{20} \times \left(\boldsymbol{\omega}_{20} \times \mathbf{r}_2^B\right) = \left(\dot{\theta}\,\mathbf{i}_1 + \omega\,\mathbf{k}_1\right) \times \left(-\omega W\cos\theta\,\mathbf{i}_1 - \dot{\theta}W\sin\theta\,\mathbf{j}_1 + \dot{\theta}W\cos\theta\,\mathbf{k}_1\right)$$

$$= -\dot{\theta}^2 W\sin\theta\,\mathbf{k}_1 - \dot{\theta}^2 W\cos\theta\,\mathbf{j}_1 - \omega^2 W\cos\theta\,\mathbf{j}_1 + \omega\dot{\theta}W\sin\theta\,\mathbf{i}_1$$

(using $\mathbf{i}_1 \times \mathbf{j}_1 = \mathbf{k}_1, \mathbf{i}_1 \times \mathbf{k}_1 = -\mathbf{j}_1, \mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1, \mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$). Adding $\mathbf{a}_0^C = -H\omega^2\,\mathbf{i}_1$: the $\mathbf{i}_1$ component is $2W\omega\dot{\theta}\sin\theta - H\omega^2$, the $\mathbf{j}_1$ component $-W(\ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta + \omega^2\cos\theta)$, the $\mathbf{k}_1$ component $W(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta)$: the same boxed result. Note that the Coriolis term of route 1, $2W\omega\dot{\theta}\sin\theta\,\mathbf{i}_1$, appears in route 2 split in two equal halves, one from $\boldsymbol{\alpha}_{20}$ (the term $\omega\dot{\theta}\,\mathbf{j}_1$ that makes angular accelerations not add) and one from the double cross product.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Reading of the terms.** $-H\omega^2\mathbf{i}_1 - W\omega^2\cos\theta\,\mathbf{j}_1$ is the centripetal acceleration of the spin; $W\ddot{\theta}$ and $W\dot{\theta}^2$ terms are the tangential and normal accelerations of the panel rotation; $2W\omega\dot{\theta}\sin\theta\,\mathbf{i}_1$ is the Coriolis coupling of the two rotations, present only when the panel moves ($\dot{\theta} \neq 0$) with a velocity component normal to the spin axis.
- **Limits:** $\dot{\theta} = 0$ (panel locked) gives the rigid-body motion $\mathbf{v}_0^B = \boldsymbol{\omega}_{10} \times \mathbf{r}_1^B, \mathbf{a}_0^B = -\omega^2\mathbf{r}_\perp$; $\omega = 0$ (no spin) leaves the circular motion of radius W about the hinge.
- **Dimensions:** every velocity component is of the form $[W\omega] = [H\omega] = [W\dot{\theta}] = \mathrm{m\,s^{-1}}$; every acceleration component $[W\omega\dot{\theta}] = [H\omega^2] = [W\ddot{\theta}] = [W\dot{\theta}^2] = \mathrm{m\,s^{-2}}$.
- **Verification (sympy, `verify_P41.py`):** $[{}_0R_1]$, $[{}_1R_2]$, $[{}_0R_2]$ are proper rotations; the inertial first and second derivatives of $[{}_0R_1]\{\mathbf{r}_1^B\}$ equal the Notes Eq. 3.26 and 3.27 expressions; both routes give exactly the official key.
- **Cross-check with Topic 1.** The Topic 1 solution (Poisson formula on the body basis) gives the same velocity and acceleration; no discrepancy with the official key.

Source: Problems.pdf (PDF page 29); key on PDF page 76

PROBLEM 47

Two Repelling Particles on Concentric Wires in a Rotating Plane

Original statement · Problems.pdf, p. 32

Relative Equilibrium

We consider a system of two heavy particles of equal mass m. The first particle P is constrained to move without friction along a circular wire of radius a. The second particle Q is constrained to move without friction along a circular wire of radius $2a$. Both wires are contained in a plane Ox_1y_1 that rotates at a constant, known angular rate Ω about the axis $Oy_1 \equiv Oy_0$ with respect to a fixed reference frame S_0 with axes $Ox_0y_0z_0$ (not shown in the figure). We also define a moving reference frame S_1 with axes $Ox_1y_1z_1$. The two particles repel each other with a force that is proportional to their distance, with constant of proportionality, k.

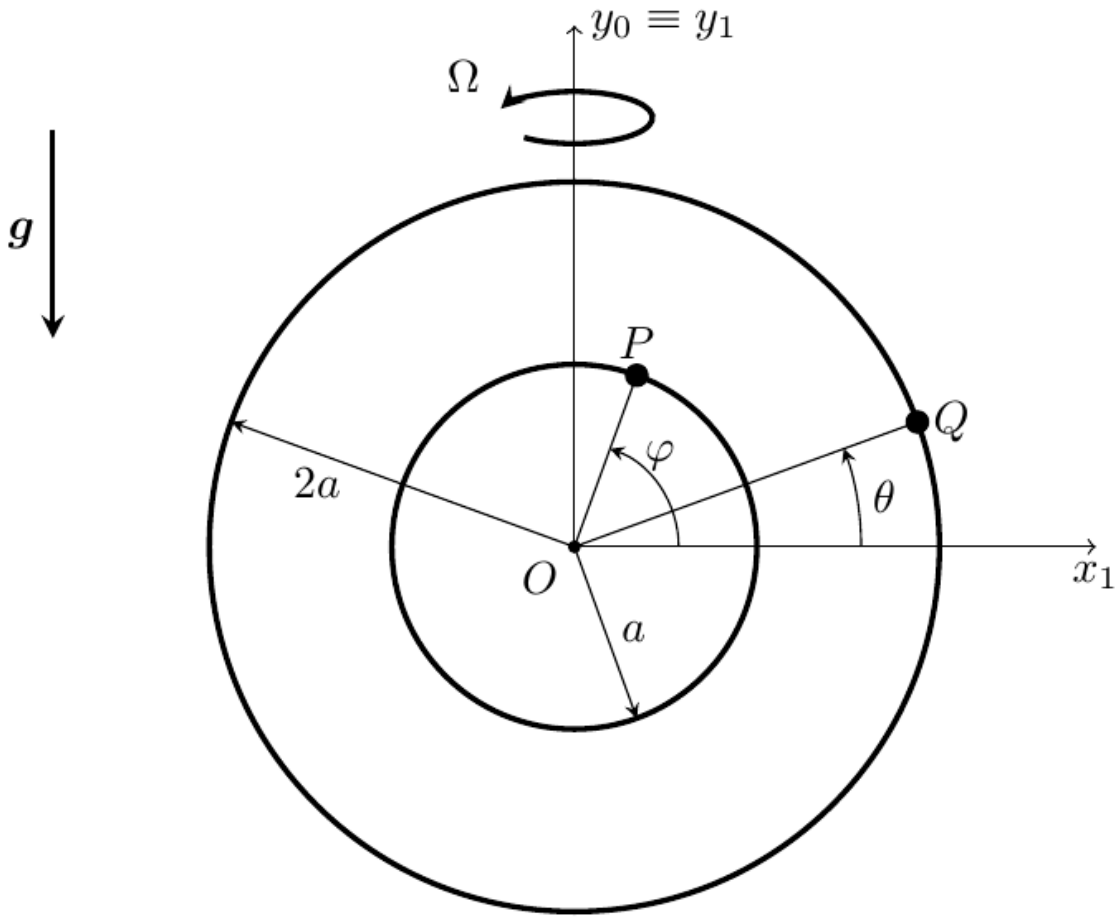


Figure: Concentric circular wires of radii a and 2a centred at O in the plane Ox1y1, particle P at angle varphi and particle Q at angle theta from x1, the plane spins with rate Omega about the vertical axis y0 = y1, gravity g downward (Problems.pdf, p. 32).

(a) How many configuration degrees of freedom has this system?

(b) How many independent components of the reaction forces on the particles need to be considered? Write down the reaction force on each particle.

(c) Write down the relative accelerations of P and Q with respect to S_1 as a function of the coordinates of the problem and their derivatives.

(d) Express the applied forces on the particles as a function of the coordinates of the problem.

(e) Express the fictitious forces of inertia on the particles as seen from S_1, as a function of the coordinates of the problem and their derivatives.

(f) Write down a system of ordinary differential equations to determine the evolution of the coordinates of the problem as a function of time, free of any other unknowns. You do not need to solve this system (since no initial conditions have been provided) but we assume that we know how to solve it, so that from this point we know the coordinates as a function of time.

(g) Determine the values of the reaction forces from the wires as a function of time.

(h) Is the mechanical energy of this system conserved in S_1? If so, write down the corresponding equation.

(i) Determine the values of $\alpha = \dfrac{ka}{mg}$ and $\beta = \dfrac{a\Omega^2}{g}$ so that $\varphi = 30^\circ, \theta = -60^\circ$ is an equilibrium position of the system in the relative motion.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** two heavy particles of mass m; P on a circular wire of radius a and Q on a concentric circular wire of radius $2a$, both centred at O and lying in the plane Ox_1y_1. The plane spins about the vertical axis $Oy_1 \equiv Oy_0$ with constant rate Ω. The particles repel each other with a force k times their distance.
- **Hypotheses:**
  1. $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial; $\mathbf{j}_0$ is vertical upward and gravity is $-g\,\mathbf{j}_0$ (the figure draws $\mathbf{g}$ antiparallel to y_1).
  2. Smooth, bilateral wires: each reaction is perpendicular to the tangent of its wire.
  3. The repulsion acts along the line PQ: on P it is $k\left(\mathbf{r}^P - \mathbf{r}^Q\right)$, on Q it is $k\left(\mathbf{r}^Q - \mathbf{r}^P\right)$ (action and reaction).
  4. φ and θ are measured from Ox_1 in the positive sense about $\mathbf{k}_1$, as in the figure. The sense of Ω only affects the sign of the Coriolis terms; we take $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}_0$ with Ω a signed constant.
- **Why S_1:** both wires are fixed in S_1, so each particle has a single angular coordinate there. The inertia forces (Notes Eq. 4.62) are the price, and with $\boldsymbol{\omega}_{10}$ constant and O fixed only the centrifugal and Coriolis forces appear.

Symbol	Meaning	SI unit
m	mass of each particle	kg
a	radius of the inner wire (outer: $2a$)	m
k	repulsion constant	N/m
Ω	constant rotation rate of the plane	rad/s
g	gravitational acceleration	m/s^2
φ, θ	angular positions of P and Q from Ox_1	rad
$\alpha = ka/(mg), \beta = a\Omega^2/g$	dimensionless parameters	1

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Frames:** $S_0 = \{O; \mathcal{B}_0\}$ and $S_1 = \{O; \mathcal{B}_1\}$ with common origin; S_1 is rotated by the angle Ωt about $\mathbf{j}_0 = \mathbf{j}_1$ (we take $S_1 \equiv S_0$ at $t = 0$). A simple rotation about $\mathbf{j}_0$ gives

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i}_0 - \sin\Omega t\,\mathbf{k}_0, \qquad \mathbf{j}_1 = \mathbf{j}_0, \qquad \mathbf{k}_1 = \sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{k}_0$$

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & 0 & \sin\Omega t \\ 0 & 1 & 0 \\ -\sin\Omega t & 0 & \cos\Omega t \end{pmatrix}$$

- **Checks:** $\det[{}_0R_1] = \cos\Omega t\,(\cos\Omega t) + \sin\Omega t\,(\sin\Omega t) = 1$ (expansion along the second row). Rows: $\cos^2\Omega t + \sin^2\Omega t = 1, 1, \sin^2\Omega t + \cos^2\Omega t = 1$; first and third rows: $-\cos\Omega t\sin\Omega t + \sin\Omega t\cos\Omega t = 0$; so $[{}_0R_1][{}_0R_1]^T = I$. Also $\mathbf{i}_1 \times \mathbf{j}_1 = \sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{k}_0 = \mathbf{k}_1$ (right-handed).
- **Motion of S_1: $\mathbf{a}_0^O = \mathbf{0}$; $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{j}_1$.** Check: $\left.\dfrac{d\mathbf{i}_1}{dt}\right|_0 = -\Omega\,(\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{k}_0) = -\Omega\,\mathbf{k}_1 = \Omega\,\mathbf{j}_1 \times \mathbf{i}_1$. Since Ω and $\mathbf{j}_0$ are constant, $\boldsymbol{\alpha}_{10} = \mathbf{0}$.
- **Polar bases of the two particles** (in the plane Ox_1y_1):

$$\mathbf{e}_r^P = \cos\varphi\,\mathbf{i}_1 + \sin\varphi\,\mathbf{j}_1, \quad \mathbf{e}_\varphi = -\sin\varphi\,\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1; \qquad \mathbf{e}_r^Q = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1, \quad \mathbf{e}_\theta = -\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1$$

Each triad $\{\mathbf{e}_r, \mathbf{e}_\varphi$ or $\mathbf{e}_\theta, \mathbf{k}_1\}$ is orthonormal and right-handed ($\mathbf{e}_r \times \mathbf{e}_\varphi = (\cos^2\varphi + \sin^2\varphi)\,\mathbf{k}_1 = \mathbf{k}_1$). Dot products used below: $\mathbf{e}_r^Q \cdot \mathbf{e}_r^P = \cos(\varphi - \theta)$, $\mathbf{e}_r^Q \cdot \mathbf{e}_\varphi = \sin(\theta - \varphi)$, $\mathbf{e}_r^P \cdot \mathbf{e}_\theta = \sin(\varphi - \theta)$, $\mathbf{j}_1 \cdot \mathbf{e}_\varphi = \cos\varphi$, $\mathbf{j}_1 \cdot \mathbf{e}_r^P = \sin\varphi$, $\mathbf{i}_1 \cdot \mathbf{e}_\varphi = -\sin\varphi$, $\mathbf{i}_1 \cdot \mathbf{e}_r^P = \cos\varphi$ (and the same with θ for Q).

PHASE 3

Step-by-Step Mathematical Deduction

(a) Configuration degrees of freedom

> **Pedagogical Justification:** each particle on a wire (a curve) loses 2 of its 3 coordinates.

P: 3 coordinates minus 2 constraints ($z_1 = 0$, $x_1^2 + y_1^2 = a^2$) = 1. Q: 3 minus 2 ($z_1 = 0$, $x_1^2 + y_1^2 = 4a^2$) = 1. The motion of the plane is prescribed. Hence

$$\text{DOF} = 2, \qquad \text{coordinates } (\varphi, \theta)$$

(b) Reaction forces

> **Pedagogical Justification:** a smooth wire cannot push along its tangent, so its reaction lies in the normal plane of the wire, which is spanned by two independent directions; there is one unknown per removed coordinate.

Two components per particle, four in total:

$$\mathbf{N}^P = N_1\,\mathbf{e}_r^P + N_2\,\mathbf{k}_1, \qquad \mathbf{N}^Q = N_3\,\mathbf{e}_r^Q + N_4\,\mathbf{k}_1$$

(c) Relative accelerations

> **Pedagogical Justification:** $\mathbf{i}_1, \mathbf{j}_1$ are constant in S_1; by the chain rule $\left.\dfrac{d\mathbf{e}_r^P}{dt}\right|_1 = \dfrac{d\mathbf{e}_r^P}{d\varphi}\dot\varphi = \dot\varphi\,\mathbf{e}_\varphi$ and $\left.\dfrac{d\mathbf{e}_\varphi}{dt}\right|_1 = \dot\varphi\,(-\cos\varphi\,\mathbf{i}_1 - \sin\varphi\,\mathbf{j}_1) = -\dot\varphi\,\mathbf{e}_r^P$ (same for θ).

$$\mathbf{r}_1^P = a\,\mathbf{e}_r^P, \qquad \mathbf{v}_1^P = a\dot\varphi\,\mathbf{e}_\varphi, \qquad \mathbf{a}_1^P = a\ddot\varphi\,\mathbf{e}_\varphi + a\dot\varphi\left(-\dot\varphi\,\mathbf{e}_r^P\right) = a\ddot\varphi\,\mathbf{e}_\varphi - a\dot\varphi^2\,\mathbf{e}_r^P$$

$$\mathbf{r}_1^Q = 2a\,\mathbf{e}_r^Q, \qquad \mathbf{v}_1^Q = 2a\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a}_1^Q = 2a\ddot\theta\,\mathbf{e}_\theta - 2a\dot\theta^2\,\mathbf{e}_r^Q$$

(d) Applied forces

> **Pedagogical Justification:** the applied (actual, non-reaction) forces are the weights and the mutual repulsion; their components in each particle's polar basis are what enters the equations.

Weights: $-mg\,\mathbf{j}_1$ on each particle, i.e. $-mg\sin\varphi\,\mathbf{e}_r^P - mg\cos\varphi\,\mathbf{e}_\varphi$ on P and $-mg\sin\theta\,\mathbf{e}_r^Q - mg\cos\theta\,\mathbf{e}_\theta$ on Q.

Repulsion, with $\mathbf{r}^P - \mathbf{r}^Q = a\,\mathbf{e}_r^P - 2a\,\mathbf{e}_r^Q = a\,(\cos\varphi - 2\cos\theta)\,\mathbf{i}_1 + a\,(\sin\varphi - 2\sin\theta)\,\mathbf{j}_1$:

$$\mathbf{F}_k^P = k\left(\mathbf{r}^P - \mathbf{r}^Q\right) = ka\left[1 - 2\cos(\varphi-\theta)\right]\mathbf{e}_r^P + 2ka\sin(\varphi-\theta)\,\mathbf{e}_\varphi$$

$$\mathbf{F}_k^Q = k\left(\mathbf{r}^Q - \mathbf{r}^P\right) = ka\left[2 - \cos(\varphi-\theta)\right]\mathbf{e}_r^Q - ka\sin(\varphi-\theta)\,\mathbf{e}_\theta$$

(For P: $\mathbf{e}_r^Q\cdot\mathbf{e}_r^P = \cos(\varphi-\theta)$ and $-2a\,\mathbf{e}_r^Q\cdot\mathbf{e}_\varphi = -2a\sin(\theta-\varphi) = 2a\sin(\varphi-\theta)$. For Q: $-a\,\mathbf{e}_r^P\cdot\mathbf{e}_\theta = -a\sin(\varphi-\theta)$.)

(e) Inertia forces in S_1

> **Pedagogical Justification:** Notes Eq. 4.62 with $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$: only the centrifugal force $-m\boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1)$ and the Coriolis force $-2m\boldsymbol{\omega}_{10}\times\mathbf{v}_1$ remain (Slide 28).

Cross products with $\mathbf{j}_1\times\mathbf{i}_1 = -\mathbf{k}_1, \mathbf{j}_1\times\mathbf{j}_1 = \mathbf{0}, \mathbf{j}_1\times\mathbf{k}_1 = \mathbf{i}_1$. For P, $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P = \Omega a\left(\cos\varphi\,\mathbf{j}_1\times\mathbf{i}_1 + \sin\varphi\,\mathbf{j}_1\times\mathbf{j}_1\right) = -\Omega a\cos\varphi\,\mathbf{k}_1$ and $\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P\right) = -\Omega^2 a\cos\varphi\,\mathbf{j}_1\times\mathbf{k}_1 = -\Omega^2 a\cos\varphi\,\mathbf{i}_1$. Hence (it is $m\Omega^2\mathbf{r}_\perp$, Slide 30):

$$\mathbf{F}_C^P = m\Omega^2 a\cos\varphi\,\mathbf{i}_1 = m\Omega^2 a\cos\varphi\left(\cos\varphi\,\mathbf{e}_r^P - \sin\varphi\,\mathbf{e}_\varphi\right), \qquad \mathbf{F}_C^Q = 2m\Omega^2 a\cos\theta\left(\cos\theta\,\mathbf{e}_r^Q - \sin\theta\,\mathbf{e}_\theta\right)$$

For the Coriolis force, $\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P = \Omega a\dot\varphi\left(-\sin\varphi\,\mathbf{j}_1\times\mathbf{i}_1 + \cos\varphi\,\mathbf{j}_1\times\mathbf{j}_1\right) = \Omega a\dot\varphi\sin\varphi\,\mathbf{k}_1$, so

$$\mathbf{F}_{COR}^P = -2m\Omega a\dot\varphi\sin\varphi\,\mathbf{k}_1, \qquad \mathbf{F}_{COR}^Q = -4m\Omega a\dot\theta\sin\theta\,\mathbf{k}_1$$

The Coriolis forces are perpendicular to the plane of the wires: only the reactions N_2, N_4 can balance them.

(f) Equations of motion free of unknown reactions

> **Pedagogical Justification:** write $m\mathbf{a}_1 = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63) for each particle and project on its tangent ($\mathbf{e}_\varphi$, $\mathbf{e}_\theta$), which is perpendicular to both reaction components and to the Coriolis force.

P, along $\mathbf{e}_\varphi$: $ma\ddot\varphi = -mg\cos\varphi + 2ka\sin(\varphi-\theta) - m\Omega^2 a\sin\varphi\cos\varphi$.

Q, along $\mathbf{e}_\theta$: $2ma\ddot\theta = -mg\cos\theta - ka\sin(\varphi-\theta) - 2m\Omega^2 a\sin\theta\cos\theta$.

Dividing by ma and $2ma$:

$$\boxed{\ddot{\varphi} = -\frac{g}{a}\cos\varphi + \frac{2k}{m}\sin(\varphi-\theta) - \Omega^2\sin\varphi\cos\varphi, \qquad \ddot{\theta} = -\frac{g}{2a}\cos\theta - \frac{k}{2m}\sin(\varphi-\theta) - \Omega^2\sin\theta\cos\theta}$$

(g) Reactions as functions of time

> **Pedagogical Justification:** the remaining projections ($\mathbf{e}_r$ and $\mathbf{k}_1$) contain one reaction each; with $\varphi(t), \theta(t)$ known from (f), they give the reactions explicitly.

P, along $\mathbf{e}_r^P$: $-ma\dot{\varphi}^2 = N_1 - mg\sin\varphi + ka\left[1 - 2\cos(\varphi-\theta)\right] + m\Omega^2 a\cos^2\varphi$. Along $\mathbf{k}_1$: $0 = N_2 - 2m\Omega a\dot{\varphi}\sin\varphi$.

Q, along $\mathbf{e}_r^Q$: $-2ma\dot{\theta}^2 = N_3 - mg\sin\theta + ka\left[2 - \cos(\varphi-\theta)\right] + 2m\Omega^2 a\cos^2\theta$. Along $\mathbf{k}_1$: $0 = N_4 - 4m\Omega a\dot{\theta}\sin\theta$.

$$\boxed{\begin{aligned} N_1(t) &= -ma\dot{\varphi}^2 + mg\sin\varphi - ka\left[1 - 2\cos(\varphi-\theta)\right] - m\Omega^2 a\cos^2\varphi, \quad N_2(t) = 2m\Omega a\dot{\varphi}\sin\varphi \\ N_3(t) &= -2ma\dot{\theta}^2 + mg\sin\theta - ka\left[2 - \cos(\varphi-\theta)\right] - 2m\Omega^2 a\cos^2\theta, \quad N_4(t) = 4m\Omega a\dot{\theta}\sin\theta \end{aligned}}$$

(h) Mechanical energy in S_1

> **Pedagogical Justification:** yes, it is conserved. In S_1 the wires are fixed and smooth, so $\mathbf{N}\cdot\mathbf{v}_1 = 0$; the Coriolis forces do no work (Slide 31); the Euler force and $-m\mathbf{a}_0^Q$ vanish; the weights, the repulsion (a central force linear in the distance) and the centrifugal forces (constant $\boldsymbol{\omega}_{10}$, Slide 30) all derive from potentials.

Potentials in S_1: weights $mga\sin\varphi + 2mga\sin\theta$ (heights y_1); repulsion $-\frac{1}{2}k\left|\mathbf{r}^P - \mathbf{r}^Q\right|^2$ (minus sign: repulsive, so that $-\nabla_P$ gives $+k(\mathbf{r}^P - \mathbf{r}^Q)$), with

$$\left|\mathbf{r}^P - \mathbf{r}^Q\right|^2 = a^2 + 4a^2 - 4a^2\,\mathbf{e}_r^P\cdot\mathbf{e}_r^Q = 5a^2 - 4a^2\cos(\varphi-\theta)$$

and centrifugal $-\frac{1}{2}m\Omega^2\left(r_\perp^P\right)^2 - \frac{1}{2}m\Omega^2\left(r_\perp^Q\right)^2$ with $r_\perp = |x_1|$ (distance to the axis Oy_1, since $z_1 = 0$): $-\frac{1}{2}m\Omega^2 a^2\cos^2\varphi - 2m\Omega^2 a^2\cos^2\theta$. Kinetic energy: $\frac{1}{2}ma^2\dot{\varphi}^2 + \frac{1}{2}m\,4a^2\dot{\theta}^2$. Dropping the constant $-\frac{5}{2}ka^2$:

$$\boxed{E_1 = \frac{1}{2}ma^2\left(\dot{\varphi}^2 + 4\dot{\theta}^2\right) + mga\left(\sin\varphi + 2\sin\theta\right) + 2ka^2\cos(\varphi-\theta) - \frac{1}{2}m\Omega^2 a^2\left(\cos^2\varphi + 4\cos^2\theta\right) = \text{const}}$$

Check. By the chain rule ($\frac{d}{dt}\cos(\varphi-\theta) = -\sin(\varphi-\theta)(\dot\varphi-\dot\theta)$, $\frac{d}{dt}\cos^2\varphi = -2\cos\varphi\sin\varphi\,\dot\varphi$):

$$\frac{dE_1}{dt} = \dot\varphi\left[ma^2\ddot\varphi + mga\cos\varphi - 2ka^2\sin(\varphi-\theta) + m\Omega^2a^2\sin\varphi\cos\varphi\right] + \dot\theta\left[4ma^2\ddot\theta + 2mga\cos\theta + 2ka^2\sin(\varphi-\theta) + 4m\Omega^2a^2\sin\theta\cos\theta\right]$$

and both brackets vanish by the two equations of (f) (multiplied by ma^2 and $4ma^2$), so $dE_1/dt = 0$.

(i) Parameters for the equilibrium $\varphi = 30^\circ, \theta = -60^\circ$

> **Pedagogical Justification:** a relative equilibrium has $\dot\varphi = \dot\theta = \ddot\varphi = \ddot\theta = 0$, so the right-hand sides of (f) must vanish. Dividing the first by g/a and the second by $g/(2a)$ introduces $\alpha = ka/(mg)$ and $\beta = a\Omega^2/g$.

$$0 = -\cos\varphi + 2\alpha\sin(\varphi-\theta) - \beta\sin\varphi\cos\varphi, \qquad 0 = -\cos\theta - \alpha\sin(\varphi-\theta) - 2\beta\sin\theta\cos\theta$$

With $\varphi-\theta = 90^\circ$ ($\sin = 1$), $\cos 30^\circ = \frac{\sqrt3}{2}$, $\sin 30^\circ = \frac12$, $\cos(-60^\circ) = \frac12$, $\sin(-60^\circ) = -\frac{\sqrt3}{2}$:

$$0 = -\frac{\sqrt3}{2} + 2\alpha - \frac{\sqrt3}{4}\beta, \qquad 0 = -\frac12 - \alpha + \frac{\sqrt3}{2}\beta$$

From the second, $\alpha = \frac{\sqrt3}{2}\beta - \frac12$. Substituting in the first: $-\frac{\sqrt3}{2} + \sqrt3\beta - 1 - \frac{\sqrt3}{4}\beta = 0$, i.e. $\frac{3\sqrt3}{4}\beta = 1 + \frac{\sqrt3}{2} = \frac{2+\sqrt3}{2}$, so

$$\beta = \frac{2\left(2+\sqrt3\right)}{3\sqrt3} = \frac{6+4\sqrt3}{9} \approx 1.4365, \qquad \alpha = \frac{\sqrt3}{2}\cdot\frac{6+4\sqrt3}{9} - \frac12 = \frac{12+6\sqrt3-9}{18} = \frac{1+2\sqrt3}{6} \approx 0.7440$$

$$\boxed{\alpha = \frac{ka}{mg} = \frac{1+2\sqrt3}{6} \approx 0.744, \qquad \beta = \frac{a\Omega^2}{g} = \frac{6+4\sqrt3}{9} \approx 1.4365}$$

Both values are positive, so the equilibrium is physically realisable.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Equilibrium:** at $\varphi - \theta = 90^\circ$ the radii OP and OQ are perpendicular, so the tangential part of the repulsion is as large as it can be ($2ka$ on P, ka on Q), and it balances the tangential components of the weight and of the centrifugal force. P (at 30° above Ox_1) is pushed up its wire by the repulsion from Q, while gravity and the centrifugal force pull it back; Q (at 60° below Ox_1) is pushed down by the repulsion and by gravity, and the centrifugal force (which pulls it towards $\theta = 0$, away from the axis) holds it.
- **Coriolis forces:** they are normal to the rotating plane and only load the wires (N_2, N_4); they never enter the equations of motion or the energy.
- **Limits:** $\Omega \to 0$ removes the centrifugal terms and $N_2 = N_4 = 0$; $k \to 0$ decouples the two particles into two independent rotating-ring pendulums, $\ddot\varphi = -\frac{g}{a}\cos\varphi - \Omega^2 \sin\varphi \cos\varphi$.
- **Dimensions:** $[g/a] = [k/m] = [\Omega^2] = \mathrm{s}^{-2} = [\ddot\varphi]$; $[N_1]$: $[ma\dot\varphi^2] = [mg] = [ka] = [m\Omega^2 a] = \mathrm{N}$; $[N_2] = [m\Omega a\dot\varphi] = \mathrm{N}$; $[E_1] = \mathrm{kg\,m^2/s^2} = [mga] = [ka^2] = \mathrm{J}$; $\alpha = \mathrm{N\,m^{-1}\,m/N}$ and $\beta = \mathrm{m\,s^{-2}/(m\,s^{-2})}$ are dimensionless.
- **Official key** (Problems.pdf, p. 76): $\alpha = 0.744$, $\beta = 1.4365$. Our exact values $\alpha = (1 + 2\sqrt{3})/6 = 0.74402$ and $\beta = (6 + 4\sqrt{3})/9 = 1.43647$ agree.
- **Symbolic check (sympy):** $\mathbf{a}_0$ of each particle obtained by differentiating $[{}_0R_1]\,\mathbf{r}_1$ twice equals the Notes Eq. 3.27 decomposition; the two equations of motion, the four reactions and $dE_1/dt = 0$ are confirmed, and solving the equilibrium conditions gives exactly the values above.

Source: Problems.pdf (PDF page 32); key on PDF page 76

PROBLEM 49

Particle on a Rotating Sphere Attracted to a Point

Original statement · Problems.pdf, p. 34

Relative Equilibria & Bifurcation

A point particle P of mass m slides without friction on a sphere of radius a centered at the origin of the inertial reference frame S_0 as shown in the figure. The particle is attracted by a point A in the equatorial plane, located a distance a from the surface of the sphere as shown. The force of attraction is linearly proportional to the distance between this point and P, with known constant of proportionality $k > 0$. The sphere and the point A rotate rigidly about the vertical with a known angular velocity $\Omega \mathbf{k}_0$. We define the auxiliary reference frame S_1 that rotates with the sphere and point A, with $\mathbf{k}_1 = \mathbf{k}_0$ and $\mathbf{i}_1$ pointing in the direction of A. We introduce the angles ϕ and θ as shown to locate P on the sphere. Note that there is no gravity in this problem.

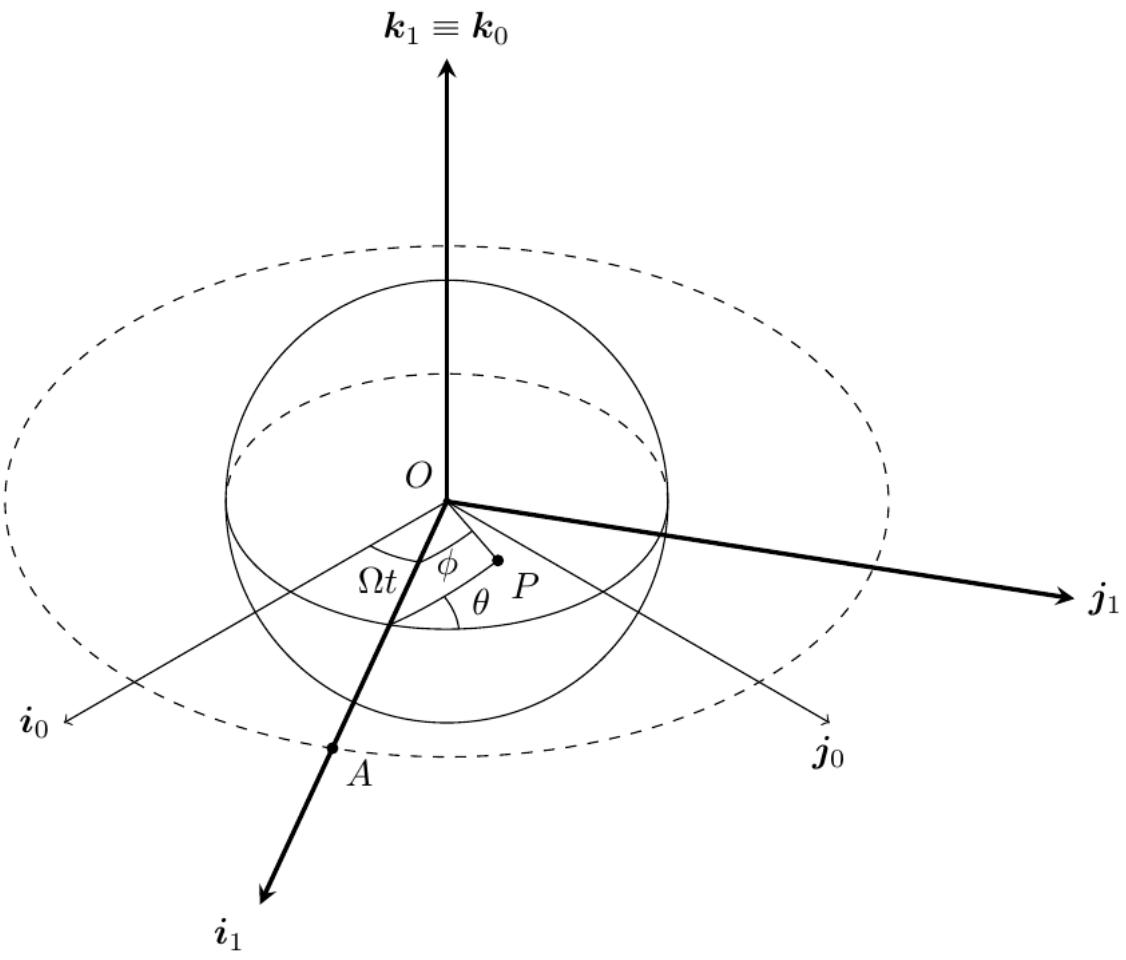


Figure: Sphere of radius a centred at O, point A on the i1 axis outside the sphere in the equatorial plane, particle P located by the angles phi and theta, frame S1 rotated by Omega t about k0 = k1 (Problems.pdf, p. 34).

(a) Express the position, velocity, and acceleration of P with respect to S_1 in terms of ϕ, θ and their derivatives. Remember to define any unit vectors that you want to use in terms of $\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0$ first.

(b) Write down the (vector) equation of motion of P in S_1, and derive from there a set of two scalar second-order differential equations for ϕ, θ that do not depend on any other unknowns. To do so, first identify and write down all the forces needed in your analysis.

(c) Is the mechanical energy of P conserved in S_1? Explain. If so, write down the corresponding conservation law.

(d) Identify all equilibrium configurations of the system, and discuss them as a function of the ratio $m\Omega^2/k$.

(e) Assuming the functions of time $\phi(t)$ and $\theta(t)$ are known, write down the expression of the reaction of the sphere on P.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** a particle P of mass m slides on a smooth sphere of radius a centred at O. A point A of the equatorial plane, at distance a from the surface (so $OA = 2a$), attracts P with the force $k\,\overrightarrow{PA}$. Sphere and A rotate rigidly about the vertical with angular velocity $\Omega\,\mathbf{k}_0$. No gravity.
- **Hypotheses:**
  1. $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial; Ω is constant.
  2. The sphere is smooth and the contact bilateral: the reaction is radial, $\mathbf{N} = N\,\mathbf{e}_r$ with N of either sign.
  3. **Reading of the figure (angles):** ϕ is the angle at O between $\overrightarrow{OA}$ (i.e. $\mathbf{i}_1$) and $\overrightarrow{OP}$; θ is the angle, measured about the axis OA, between the plane containing $\mathbf{i}_1$ and $\overrightarrow{OP}$ and the equatorial plane (the figure draws it at the point where OA crosses the sphere, between the equator and the arc through P). So ϕ is a polar angle about the axis OA and θ the corresponding azimuth, with $\theta = 0$ in the equatorial plane on the side of $\mathbf{j}_1$ and $\theta = \pi/2$ in the upper hemisphere.

- **Why S_1:** in S_1 both the sphere and the attracting point A are fixed, so the attraction depends only on the position of P and the problem has no explicit time dependence; the inertia forces (Notes Eq. 4.62) take that role. With polar axis OA, the distance AP depends on ϕ only, which keeps the attraction simple.

Symbol	Meaning	SI unit
m	mass of P	kg
a	radius of the sphere ($OA = 2a$)	m
k	attraction constant	N/m
Ω	angular rate of sphere and A	rad/s
ϕ, θ	polar angle from OA, azimuth about OA	rad
N	normal reaction	N
$\Lambda = m\Omega^2/k$	ratio of centrifugal to attraction stiffness	1

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Frames:** $S_0 = \{O; \mathcal{B}_0\}$ inertial; $S_1 = \{O; \mathcal{B}_1\}$ rotating with the sphere, $\mathbf{k}_1 = \mathbf{k}_0$, $\mathbf{i}_1$ along $\overrightarrow{OA}$. The figure measures Ωt from $\mathbf{i}_0$ to $\mathbf{i}_1$:

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i}_0 + \sin\Omega t\,\mathbf{j}_0, \qquad \mathbf{j}_1 = -\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0, \qquad \mathbf{k}_1 = \mathbf{k}_0$$

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & -\sin\Omega t & 0 \\ \sin\Omega t & \cos\Omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\Omega t + \sin^2\Omega t = 1$$

The rows are unit vectors and the first two are orthogonal ($\cos\Omega t\sin\Omega t - \sin\Omega t\cos\Omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.

- **Motion of S_1:** $\mathbf{a}_0^O = \mathbf{0}$, $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_1$ (simple rotation, Slide 16), $\boldsymbol{\alpha}_{10} = \mathbf{0}$. The attracting point is $\mathbf{r}_1^A = 2a\,\mathbf{i}_1$.
- **Spherical basis with polar axis $\mathbf{i}_1$.** Let $\mathbf{u}(\theta) = \cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1$ (unit vector of the plane $\mathbf{j}_1\mathbf{k}_1$). Then

$$\mathbf{e}_r = \cos\phi\,\mathbf{i}_1 + \sin\phi\,\mathbf{u} = \cos\phi\,\mathbf{i}_1 + \sin\phi\cos\theta\,\mathbf{j}_1 + \sin\phi\sin\theta\,\mathbf{k}_1$$

$$\mathbf{e}_\phi = \frac{\partial \mathbf{e}_r}{\partial \phi} = -\sin\phi\,\mathbf{i}_1 + \cos\phi\cos\theta\,\mathbf{j}_1 + \cos\phi\sin\theta\,\mathbf{k}_1, \qquad \mathbf{e}_\theta = \frac{d\mathbf{u}}{d\theta} = -\sin\theta\,\mathbf{j}_1 + \cos\theta\,\mathbf{k}_1$$

with $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ given above in terms of $\mathcal{B}_0$. The triad is orthonormal and right-handed: $\mathbf{e}_r \times \mathbf{e}_\phi = \left(\cos^2\phi + \sin^2\phi\right)\mathbf{i}_1 \times \mathbf{u} = \cos\theta\,\mathbf{k}_1 - \sin\theta\,\mathbf{j}_1 = \mathbf{e}_\theta$, hence also $\mathbf{e}_\phi \times \mathbf{e}_\theta = \mathbf{e}_r$, $\mathbf{e}_\theta \times \mathbf{e}_r = \mathbf{e}_\phi$. Inverse relations used below: $\mathbf{u} = \sin\phi\,\mathbf{e}_r + \cos\phi\,\mathbf{e}_\phi$, so

$$\mathbf{i}_1 = \cos\phi\,\mathbf{e}_r - \sin\phi\,\mathbf{e}_\phi, \qquad \mathbf{k}_1 = \sin\theta\,\mathbf{u} + \cos\theta\,\mathbf{e}_\theta = \sin\phi\sin\theta\,\mathbf{e}_r + \cos\phi\sin\theta\,\mathbf{e}_\phi + \cos\theta\,\mathbf{e}_\theta$$

PHASE 3

Step-by-Step Mathematical Deduction

(a) Position, velocity and acceleration in S_1

Pedagogical Justification: $\mathbf{r}_1^P = a\,\mathbf{e}_r$ (Notes Eq. 3.16 with the origin O common to both frames). The relative derivative keeps $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ fixed, so the spherical vectors change only through $\phi(t)$ and $\theta(t)$ (chain rule).

Time derivatives of the basis in S_1:

$$\left.\frac{d\mathbf{e}_r}{dt}\right|_1 = -\sin\phi\,\dot\phi\,\mathbf{i}_1 + \cos\phi\,\dot\phi\,\mathbf{u} + \sin\phi\,\frac{d\mathbf{u}}{d\theta}\dot\theta = \dot\phi\,\mathbf{e}_\phi + \sin\phi\,\dot\theta\,\mathbf{e}_\theta$$

$$\left.\frac{d\mathbf{e}_\phi}{dt}\right|_1 = -\cos\phi\,\dot\phi\,\mathbf{i}_1 - \sin\phi\,\dot\phi\,\mathbf{u} + \cos\phi\,\dot\theta\,\mathbf{e}_\theta = -\dot\phi\,\mathbf{e}_r + \cos\phi\,\dot\theta\,\mathbf{e}_\theta$$

$$\left.\frac{d\mathbf{e}_\theta}{dt}\right|_1 = \dot\theta\,(-\cos\theta\,\mathbf{j}_1 - \sin\theta\,\mathbf{k}_1) = -\dot\theta\,\mathbf{u} = -\sin\phi\,\dot\theta\,\mathbf{e}_r - \cos\phi\,\dot\theta\,\mathbf{e}_\phi$$

Position and velocity:

$$\mathbf{r}_1^P = a\,\mathbf{e}_r, \qquad \mathbf{v}_1^P = a\,\left.\frac{d\mathbf{e}_r}{dt}\right|_1 = a\dot{\phi}\,\mathbf{e}_\phi + a\sin\phi\,\dot{\theta}\,\mathbf{e}_\theta$$

Acceleration, by the product rule on each term, $\frac{d}{dt}(a\dot{\phi}\,\mathbf{e}_\phi) = a\ddot{\phi}\,\mathbf{e}_\phi + a\dot{\phi}\left(-\dot{\phi}\,\mathbf{e}_r + \cos\phi\,\dot{\theta}\,\mathbf{e}_\theta\right)$ and $\frac{d}{dt}(a\sin\phi\,\dot{\theta}\,\mathbf{e}_\theta) = a\left(\cos\phi\,\dot{\phi}\dot{\theta} + \sin\phi\,\ddot{\theta}\right)\mathbf{e}_\theta + a\sin\phi\,\dot{\theta}\left(-\sin\phi\,\dot{\theta}\,\mathbf{e}_r - \cos\phi\,\dot{\theta}\,\mathbf{e}_\phi\right)$; collecting:

$$\mathbf{a}_1^P = -a\left(\dot{\phi}^2 + \sin^2\phi\,\dot{\theta}^2\right)\mathbf{e}_r + a\left(\ddot{\phi} - \sin\phi\cos\phi\,\dot{\theta}^2\right)\mathbf{e}_\phi + a\left(\sin\phi\,\ddot{\theta} + 2\cos\phi\,\dot{\phi}\dot{\theta}\right)\mathbf{e}_\theta$$

(b) Forces, vector equation of motion and the two scalar equations

> **Pedagogical Justification:** in S_1, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63). The actual forces are the attraction and the reaction; the inertia forces (Notes Eq. 4.62) reduce to the centrifugal and Coriolis forces because $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$.

Attraction: $\mathbf{F}_k = k\left(\mathbf{r}_1^A - \mathbf{r}_1^P\right) = k\left(2a\,\mathbf{i}_1 - a\,\mathbf{e}_r\right)$; with $\mathbf{i}_1 = \cos\phi\,\mathbf{e}_r - \sin\phi\,\mathbf{e}_\phi$,

$$\mathbf{F}_k = ka\left(2\cos\phi - 1\right)\mathbf{e}_r - 2ka\sin\phi\,\mathbf{e}_\phi$$

Reaction: $\mathbf{N} = N\,\mathbf{e}_r$.

Centrifugal force. With the identity $\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = (\mathbf{a}\cdot\mathbf{c})\mathbf{b} - (\mathbf{a}\cdot\mathbf{b})\mathbf{c}$ (Slide 30): $\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P\right) = \Omega^2 z_1\,\mathbf{k}_1 - \Omega^2\mathbf{r}_1^P$, where $z_1 = \mathbf{r}_1^P\cdot\mathbf{k}_1 = a\sin\phi\sin\theta$. Hence $\mathbf{F}_C = m\Omega^2\left(\mathbf{r}_1^P - z_1\mathbf{k}_1\right)$, and with $\mathbf{k}_1$ in the spherical basis:

$$\mathbf{F}_C = m\Omega^2 a\left[\left(1 - \sin^2\phi\sin^2\theta\right)\mathbf{e}_r - \sin^2\theta\sin\phi\cos\phi\,\mathbf{e}_\phi - \sin\phi\sin\theta\cos\theta\,\mathbf{e}_\theta\right]$$

Coriolis force. Using $\mathbf{k}_1 = \sin\phi\sin\theta\,\mathbf{e}_r + \cos\phi\sin\theta\,\mathbf{e}_\phi + \cos\theta\,\mathbf{e}_\theta$ and the cyclic products of the right-handed triad:

$$\mathbf{k}_1\times\mathbf{e}_\phi = \sin\phi\sin\theta\,\mathbf{e}_r\times\mathbf{e}_\phi + \cos\theta\,\mathbf{e}_\theta\times\mathbf{e}_\phi = \sin\phi\sin\theta\,\mathbf{e}_\theta - \cos\theta\,\mathbf{e}_r$$

$$\mathbf{k}_1 \times \mathbf{e}_\theta = \sin\phi\sin\theta\,\mathbf{e}_r \times \mathbf{e}_\theta + \cos\phi\sin\theta\,\mathbf{e}_\phi \times \mathbf{e}_\theta = -\sin\phi\sin\theta\,\mathbf{e}_\phi + \cos\phi\sin\theta\,\mathbf{e}_r$$

$$\mathbf{F}_{COR} = -2m\Omega\,\mathbf{k}_1 \times \left(a\dot{\phi}\,\mathbf{e}_\phi + a\sin\phi\,\dot{\theta}\,\mathbf{e}_\theta\right) = 2m\Omega a\left[\left(\cos\theta\,\dot{\phi} - \sin\theta\sin\phi\cos\phi\,\dot{\theta}\right)\mathbf{e}_r + \sin\theta\sin^2\phi\,\dot{\theta}\,\mathbf{e}_\phi - \sin\theta\sin\phi\,\dot{\phi}\,\mathbf{e}_\theta\right]$$

Vector equation of motion in S_1:

$$m\mathbf{a}_1^P = k\left(\mathbf{r}_1^A - \mathbf{r}_1^P\right) + N\,\mathbf{e}_r - m\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) - 2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P$$

Scalar equations free of N: project on $\mathbf{e}_\phi$ and $\mathbf{e}_\theta$, which are perpendicular to $\mathbf{N}$:

$$\boxed{ma\left(\ddot{\phi} - \sin\phi\cos\phi\,\dot{\theta}^2\right) = -2ka\sin\phi - m\Omega^2 a\sin^2\theta\sin\phi\cos\phi + 2m\Omega a\sin\theta\sin^2\phi\,\dot{\theta}}$$

$$\boxed{ma\left(\sin\phi\,\ddot{\theta} + 2\cos\phi\,\dot{\phi}\dot{\theta}\right) = -m\Omega^2 a\sin\phi\sin\theta\cos\theta - 2m\Omega a\sin\theta\sin\phi\,\dot{\phi}}$$

(c) Conservation of mechanical energy in S_1

Pedagogical Justification: yes, E_1^P is conserved. The sphere is fixed in S_1 and smooth, so $\mathbf{N}\cdot\mathbf{v}_1^P = 0$ ($\mathbf{v}_1^P$ has no $\mathbf{e}_r$ component); $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$ (Slide 31); the Euler force and $-m\mathbf{a}_0^O$ are zero; the attraction towards the point A, fixed in S_1, is conservative with $V_k = \frac{1}{2}k\left|\mathbf{r}_1^P - \mathbf{r}_1^A\right|^2$; and the centrifugal force has $V_C = -\frac{1}{2}m\Omega^2 r_\perp^2$ because $\boldsymbol{\omega}_{10}$ is constant (Slide 30).

Potentials on the sphere:

$$\left|\mathbf{r}_1^P - \mathbf{r}_1^A\right|^2 = a^2 - 4a^2\,\mathbf{e}_r\cdot\mathbf{i}_1 + 4a^2 = 5a^2 - 4a^2\cos\phi \;\Rightarrow\; V_k = \frac{5}{2}ka^2 - 2ka^2\cos\phi$$

$$r_\perp^2 = x_1^2 + y_1^2 = a^2 - z_1^2 = a^2 - a^2\sin^2\phi\sin^2\theta \;\Rightarrow\; V_C = -\frac{1}{2}m\Omega^2 a^2 + \frac{1}{2}m\Omega^2 a^2\sin^2\phi\sin^2\theta$$

Dropping the constants, with $T_1^P = \frac{1}{2}m\left|\mathbf{v}_1^P\right|^2 = \frac{1}{2}ma^2\left(\dot\phi^2 + \sin^2\phi\,\dot\theta^2\right)$:

$$\boxed{E_1^P = \frac{1}{2}ma^2\left(\dot\phi^2 + \sin^2\phi\,\dot\theta^2\right) - 2ka^2\cos\phi + \frac{1}{2}m\Omega^2 a^2\sin^2\phi\sin^2\theta = \text{const}}$$

Check. Differentiating with the chain and product rules ($\frac{d}{dt}\sin^2\phi = 2\sin\phi\cos\phi\,\dot\phi$, $\frac{d}{dt}\sin^2\theta = 2\sin\theta\cos\theta\,\dot\theta$) and grouping:

$$\frac{dE_1^P}{dt} = a\dot\phi\left[ma\left(\ddot\phi - \sin\phi\cos\phi\,\dot\theta^2\right) + 2ka\sin\phi + m\Omega^2 a\sin^2\theta\sin\phi\cos\phi\right] + a\sin\phi\,\dot\theta\left[ma\left(\sin\phi\,\ddot\theta + 2\cos\phi\,\dot\phi\dot\theta\right) + m\Omega^2 a\sin\phi\sin\theta\cos\theta\right]$$

By the equations of (b) the brackets equal $2m\Omega a\sin\theta\sin^2\phi\,\dot\theta$ and $-2m\Omega a\sin\theta\sin\phi\,\dot\phi$, so

$$\frac{dE_1^P}{dt} = 2m\Omega a^2\sin\theta\sin^2\phi\,\dot\phi\dot\theta - 2m\Omega a^2\sin\theta\sin^2\phi\,\dot\phi\dot\theta = 0$$

which is exactly $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$. The constant is fixed by initial conditions (not given).

(d) Equilibrium configurations

> **Pedagogical Justification:** a relative equilibrium has $\dot\phi = \dot\theta = \ddot\phi = \ddot\theta = 0$, so the right-hand sides of the equations of (b) must vanish (the Coriolis terms disappear). The points $\phi = 0, \pi$ are the poles of the coordinates (θ undefined there) and must be examined separately; the symbolic check confirms the list below with Cartesian coordinates and a Lagrange multiplier.

With $\Lambda = m\Omega^2/k$, the two conditions are

$$\sin\phi\left(2k + m\Omega^2\sin^2\theta\cos\phi\right) = 0 \iff \sin\phi\left(2 + \Lambda\sin^2\theta\cos\phi\right) = 0, \qquad \sin\phi\sin\theta\cos\theta = 0$$

- $\sin\phi = 0$**:** both are satisfied. **E1:** $\phi = 0$, $\mathbf{r}_1^P = a\,\mathbf{i}_1$, the point of the sphere nearest to A. **E2:** $\phi = \pi$, $\mathbf{r}_1^P = -a\,\mathbf{i}_1$, the point farthest from A. They exist for every Λ: at both, the attraction and the centrifugal force are radial (along $\pm\mathbf{i}_1$) and the reaction balances them.

- $\sin\phi \neq 0$**:** the second condition needs $\sin\theta = 0$ or $\cos\theta = 0$. If $\sin\theta = 0$, the first becomes $2 = 0$: impossible. If $\cos\theta = 0$ ($\theta = \pm\pi/2$, $\sin^2\theta = 1$), the first gives

$$\cos\phi^* = -\frac{2k}{m\Omega^2} = -\frac{2}{\Lambda}$$

which has a solution with $\sin\phi^* \neq 0$ only if $\Lambda > 2$. **E3, E4** ($\theta = \pm\pi/2$):

$$\mathbf{r}_1^P = -\frac{2a}{\Lambda}\,\mathbf{i}_1 \pm a\sqrt{1 - \frac{4}{\Lambda^2}}\,\mathbf{k}_1$$

They lie in the meridian plane $\mathbf{i}_1\mathbf{k}_1$ that contains A, on the hemisphere opposite to A, symmetric above and below the equator.

Discussion in terms of $m\Omega^2/k$:

- $\Lambda = m\Omega^2/k \leq 2$: only E1 and E2.
- $\Lambda > 2$: E1, E2, E3 and E4. As $\Lambda \to 2^+$, $\cos\phi^* \to -1$ and E3, E4 merge into E2 (a pitchfork bifurcation); as $\Lambda \to \infty$, $\cos\phi^* \to 0$ and they tend to the poles $\pm a\,\mathbf{k}_1$.

Extension (stability, beyond what is asked). Near E1, $V_1 \approx \text{const} + k\left(y_1^2 + z_1^2\right) + \frac{1}{2}m\Omega^2 z_1^2$, a strict minimum, so E1 is stable for every Λ (energy argument). Near E2, $V_1 \approx \text{const} - k y_1^2 + \left(\frac{1}{2}m\Omega^2 - k\right) z_1^2$; along $\mathbf{j}_1$ the linearised motion is $m\ddot{y}_1 = 2k y_1$ (no Coriolis coupling there, because $\mathbf{k}_1 \times \mathbf{v}_1^P$ is normal to the sphere at E2), so E2 is always unstable with growth rate $\sqrt{2k/m}$. E3 and E4 are maxima of V_1; the linearisation including the Coriolis terms (computed numerically in the verification script for several $\Lambda > 2$) has eigenvalues with positive real part, so they are unstable as well.

(e) Reaction of the sphere

> **Pedagogical Justification:** the projection of the vector equation on $\mathbf{e}_r$ contains N and only known functions once $\phi(t), \theta(t)$ are known.

$$-ma\left(\dot{\phi}^2 + \sin^2\phi\,\dot{\theta}^2\right) = N + ka\left(2\cos\phi - 1\right) + m\Omega^2 a\left(1 - \sin^2\phi\sin^2\theta\right) + 2m\Omega a\left(\cos\theta\,\dot{\phi} - \sin\theta\sin\phi\cos\phi\,\dot{\theta}\right)$$

$$\boxed{\mathbf{N} = -\left[ma\left(\dot{\phi}^2 + \sin^2\phi\,\dot{\theta}^2\right) + ka\left(2\cos\phi - 1\right) + m\Omega^2 a\left(1 - \sin^2\phi\sin^2\theta\right) + 2m\Omega a\left(\cos\theta\,\dot{\phi} - \sin\theta\sin\phi\cos\phi\,\dot{\theta}\right)\right]\mathbf{e}_r}$$

with $\mathbf{e}_r = \cos\phi\,\mathbf{i}_1 + \sin\phi\cos\theta\,\mathbf{j}_1 + \sin\phi\sin\theta\,\mathbf{k}_1$ and $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in terms of $\mathcal{B}_0$ from Phase 2.

PHASE 4

Physical Interpretation & Dimensional Verification

- **Competition of two effects:** the attraction pulls P towards the point nearest A (E1), and its tangential part has stiffness $\sim k$; the centrifugal force pushes P away from the vertical axis with stiffness $\sim m\Omega^2$. The ratio $\Lambda = m\Omega^2/k$ decides whether, on the far side of the sphere, the centrifugal push towards the equator can be balanced by the attraction: only for $\Lambda > 2$ do the extra equilibria E3, E4 appear.
- **Reaction at E1:** with $\phi = 0$ and no motion, $N = -ka - m\Omega^2 a$: the sphere pulls P inwards, since both the attraction (towards A, outside the sphere) and the centrifugal force push P outwards. The bilateral contact assumed in Phase 1 is needed for that.
- **Limit $\Omega \to 0$:** the inertia forces vanish, $\Lambda \to 0$, and only E1 (stable) and E2 (unstable) remain, as for a fixed sphere with a fixed attracting point.
- **Dimensions:** in the $\mathbf{e}_\phi$ equation, $[ma\ddot{\phi}] = \mathrm{kg\,m\,s^{-2}} = \mathrm{N}$, $[ka] = \mathrm{N}$, $[m\Omega^2 a] = \mathrm{N}$, $[m\Omega a\dot{\theta}] = \mathrm{N}$; $[E_1^P] = [ka^2] = [m\Omega^2 a^2] = \mathrm{J}$; $\Lambda = \mathrm{kg\,s^{-2}/(N/m)}$ is dimensionless.
- **Official key:** none for this problem (Problems.pdf, p. 76 lists no answer for 49).
- **Symbolic check (sympy):** $\mathbf{a}_0^P$ from differentiating $[{}_0R_1]\,\mathbf{r}_1^P$ twice equals the Notes Eq. 3.27 decomposition; the force components, both equations of motion, the reaction and $dE_1^P/dt = 0$ are confirmed; the equilibria found with Cartesian coordinates and a Lagrange multiplier are exactly E1 to E4.

Source: Problems.pdf (PDF page 34); no official key

PROBLEM 51

Particle inside a Hollow Sphere in a Rotating Frame

Original statement · Problems.pdf, p. 36

Non-Inertial Frame + Surface Constraint

A heavy point particle P of mass m is constrained to move inside the surface of a hollow sphere of radius R and center C located at $\mathbf{r}_1^C = R(\mathbf{j}_1 + \mathbf{k}_1)$. The surface of the sphere is smooth. The reference frame S_1 shown in the figure is a non-inertial reference frame, because it rotates with a known angular velocity $\boldsymbol{\omega}_{10} = \Omega(t)\,\mathbf{k}_1$ with respect to inertial space. We introduce the two coordinates θ and ϕ as shown in the figure, in order to characterize the configuration of P.

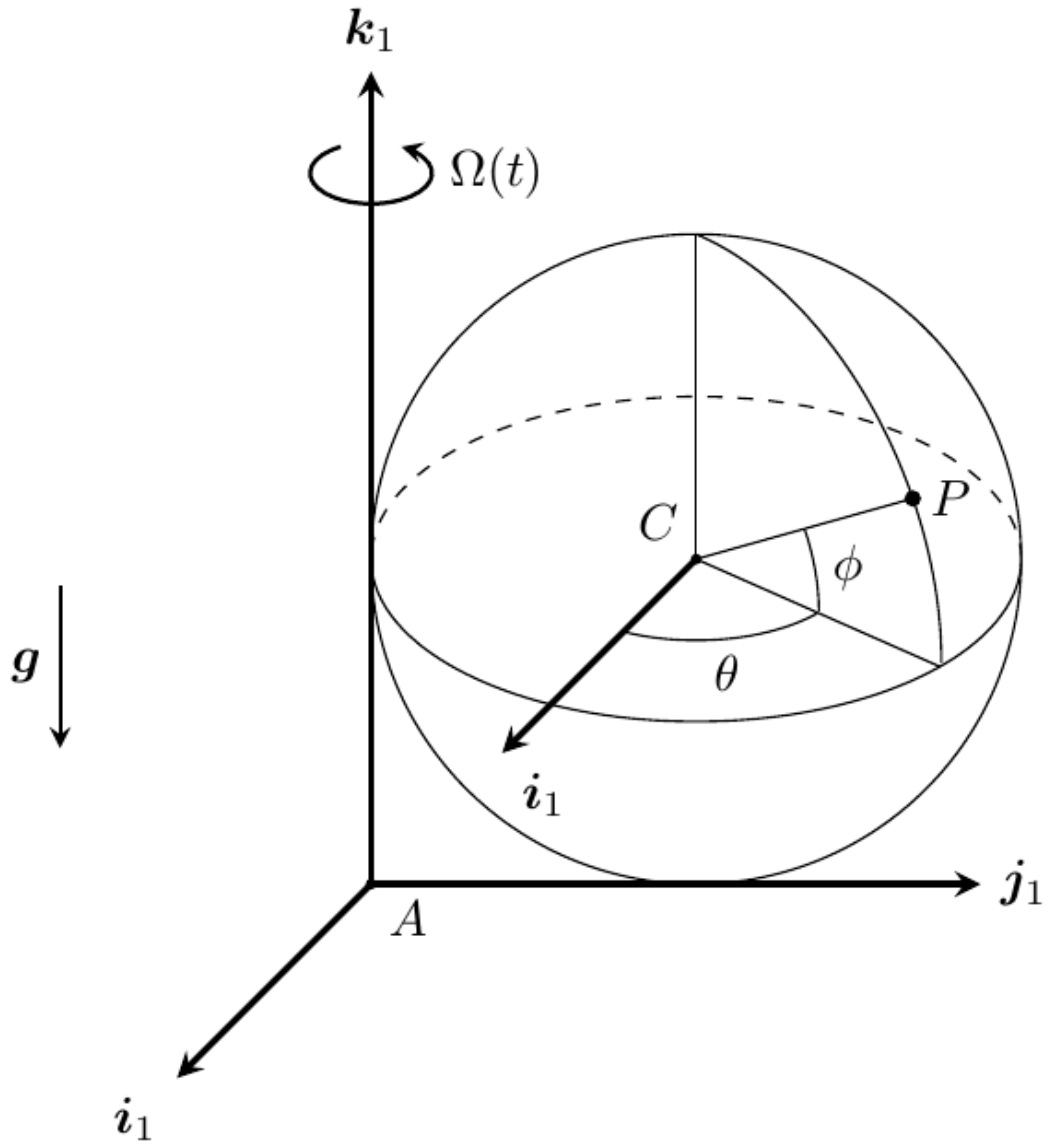


Figure: Rotating frame S1 with axes i1, j1, k1, hollow sphere of centre C at R(j1+k1), particle P located by the angles theta and phi, angular velocity Omega(t) about k1 (Problems.pdf, p. 36).

(a) Give the expressions for the position, velocity, and acceleration vectors of P with respect to S_1, as functions of θ, ϕ and their derivatives. Remember that you may use any unit vectors in your solution, provided that you define them first.

(b) Identify and write down all forces (actual and fictitious) on P as seen from S_1.

(c) Write down two differential equations for θ and ϕ, free of any other unknowns.

(d) Write down an expression for the reaction unknown. This expression may depend on $\theta, \dot{\theta}, \phi, \dot{\phi}$ and the known parameters of the problem. What is the mathematical condition for the particle to separate from the surface?

(e) In the case that $\Omega = \text{const}$, determine and discuss the value of θ, ϕ and the reaction unknown for the equilibrium configurations of P.

Worked Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical system:** a heavy particle P (mass m) slides on the smooth inner surface of a sphere of radius R and centre C, $\mathbf{r}_1^C = R(\mathbf{j}_1 + \mathbf{k}_1)$. The sphere is fixed in the frame $S_1 = \{A; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, which turns about its vertical axis $A\mathbf{k}_1$ with the known rate $\Omega(t)$. The sphere touches the rotation axis at the point $(0, 0, R)$ and the plane $z_1 = 0$ at $(0, R, 0)$.
- **Why S_1.** The constraint (the sphere) is fixed in S_1, so in S_1 the particle simply moves on a fixed sphere described by two angles; all the effect of the rotation is collected in the inertia forces (Notes Eqs. 4.62–4.63). In S_0 the sphere would be a moving constraint.
- **Hypotheses:**
  1. The point A is fixed in an inertial frame S_0 (it lies on the rotation axis), so $\mathbf{v}_0^A = \mathbf{a}_0^A = \mathbf{0}$; $\mathbf{k}_1$ is the ascending vertical and gravity is $-g\,\mathbf{k}_1$ (figure).
  2. The surface is smooth: the reaction is normal to the sphere, along the radius CP. It is unilateral: the particle is inside, so the surface can only push it toward C.
  3. $\Omega(t)$ is a known function; $\dot{\Omega}(t)$ is then known too.

- **Coordinates (from the figure):** θ is the azimuth about the vertical through C, measured from the direction $\mathbf{i}_1$ toward $\mathbf{j}_1$; ϕ is the elevation of CP above the horizontal plane through C. Ranges: $\theta \in (-\pi, \pi]$, $\phi \in [-\pi/2, \pi/2]$ (the poles $\phi = \pm\pi/2$ are coordinate singularities).
- **Degrees of freedom:** 3 coordinates minus 1 constraint $|CP| = R$: **2 DOF**, (θ, ϕ). The reaction adds **1 unknown**, N.

Symbol	Meaning	SI unit
m	mass of P	kg
R	radius of the sphere	m
g	gravitational acceleration	m/s^2
$\Omega(t)$, $\dot{\Omega}(t)$	angular velocity and acceleration of S_1 about $\mathbf{k}_1$	rad/s, rad/s^2
θ, ϕ	azimuth and elevation of P seen from C	rad
N	magnitude of the reaction, $\mathbf{N} = -N\,\mathbf{u}_r$, $N \geq 0$ while in contact	N

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Frames.** $S_0 = \{A; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ inertial, $\mathbf{k}_0 = \mathbf{k}_1$. S_1 turns about $\mathbf{k}_1$ through the angle $\Psi(t)$, with $\dot{\Psi} = \Omega(t)$:

$$[{}_0R_1] = \begin{pmatrix} \cos\Psi & -\sin\Psi & 0 \\ \sin\Psi & \cos\Psi & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\Psi + \sin^2\Psi = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- **Motion of S_1: $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_1$**; since $\mathbf{k}_1$ is fixed in both frames, $\boldsymbol{\alpha}_{10} = \left.\frac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \dot{\Omega}\,\mathbf{k}_1 + \Omega\,\boldsymbol{\omega}_{10} \times \mathbf{k}_1 = \dot{\Omega}\,\mathbf{k}_1$; $\mathbf{r}_0^A = \mathbf{v}_0^A = \mathbf{a}_0^A = \mathbf{0}$.
- **Spherical basis at P (defined in S_1).** With the horizontal unit vector $\mathbf{e}_h = \cos\theta\,\mathbf{i}_1 + \sin\theta\,\mathbf{j}_1$:

$$\mathbf{u}_r = \cos\phi\cos\theta\,\mathbf{i}_1 + \cos\phi\sin\theta\,\mathbf{j}_1 + \sin\phi\,\mathbf{k}_1 = \cos\phi\,\mathbf{e}_h + \sin\phi\,\mathbf{k}_1 \quad (\text{from } C \text{ to } P)$$

$$\mathbf{u}_\theta = -\sin\theta\,\mathbf{i}_1 + \cos\theta\,\mathbf{j}_1 \quad (\text{horizontal, increasing } \theta)$$

$$\mathbf{u}_\phi = -\sin\phi\cos\theta\,\mathbf{i}_1 - \sin\phi\sin\theta\,\mathbf{j}_1 + \cos\phi\,\mathbf{k}_1 = -\sin\phi\,\mathbf{e}_h + \cos\phi\,\mathbf{k}_1 \quad (\text{increasing } \phi)$$

Change-of-basis matrix (columns: $\mathbf{u}_r, \mathbf{u}_\theta, \mathbf{u}_\phi$ in $\mathcal{B}_1$):

$$[{}_1R_u] = \begin{pmatrix} \cos\phi\cos\theta & -\sin\theta & -\sin\phi\cos\theta \\ \cos\phi\sin\theta & \cos\theta & -\sin\phi\sin\theta \\ \sin\phi & 0 & \cos\phi \end{pmatrix}$$

Checks: each column has unit norm ($\cos^2\phi(\cos^2\theta+\sin^2\theta)+\sin^2\phi = 1$, etc.); $\mathbf{u}_r\cdot\mathbf{u}_\theta = -\cos\phi\cos\theta\sin\theta + \cos\phi\sin\theta\cos\theta = 0$, $\mathbf{u}_r\cdot\mathbf{u}_\phi = -\cos\phi\sin\phi + \sin\phi\cos\phi = 0$, $\mathbf{u}_\theta\cdot\mathbf{u}_\phi = \sin\phi\sin\theta\cos\theta - \sin\phi\sin\theta\cos\theta = 0$, so $[{}_1R_u]^T[{}_1R_u] = I$. The determinant is the triple product $\det[{}_1R_u] = \mathbf{u}_r\cdot(\mathbf{u}_\theta\times\mathbf{u}_\phi)$, and $\mathbf{u}_\theta\times\mathbf{u}_\phi = -\sin\phi\,(\mathbf{u}_\theta\times\mathbf{e}_h) + \cos\phi\,(\mathbf{u}_\theta\times\mathbf{k}_1) = \sin\phi\,\mathbf{k}_1 + \cos\phi\,\mathbf{e}_h = \mathbf{u}_r$ (using $\mathbf{u}_\theta\times\mathbf{e}_h = -\mathbf{k}_1$ and $\mathbf{u}_\theta\times\mathbf{k}_1 = \mathbf{e}_h$), hence $\det[{}_1R_u] = \mathbf{u}_r\cdot\mathbf{u}_r = 1$: the triad $\{\mathbf{u}_r, \mathbf{u}_\theta, \mathbf{u}_\phi\}$ is right-handed.

- **Useful inverse relations:** $\mathbf{e}_h = \cos\phi\,\mathbf{u}_r - \sin\phi\,\mathbf{u}_\phi$, $\mathbf{k}_1 = \sin\phi\,\mathbf{u}_r + \cos\phi\,\mathbf{u}_\phi$, $\mathbf{i}_1 = \cos\theta\,\mathbf{e}_h - \sin\theta\,\mathbf{u}_\theta$, $\mathbf{j}_1 = \sin\theta\,\mathbf{e}_h + \cos\theta\,\mathbf{u}_\theta$; and $\mathbf{k}_1\times\mathbf{e}_h = \mathbf{u}_\theta$, $\mathbf{k}_1\times\mathbf{u}_\theta = -\mathbf{e}_h$, $\mathbf{k}_1\times\mathbf{u}_\phi = -\sin\phi\,\mathbf{u}_\theta$.
- **Derivatives in S_1** (chain rule, θ and ϕ depend on t):

$$\left.\frac{d\mathbf{u}_r}{dt}\right|_1 = \frac{\partial\mathbf{u}_r}{\partial\theta}\dot\theta + \frac{\partial\mathbf{u}_r}{\partial\phi}\dot\phi = \dot\theta\cos\phi\,\mathbf{u}_\theta + \dot\phi\,\mathbf{u}_\phi$$

$$\left.\frac{d\mathbf{u}_\theta}{dt}\right|_1 = -\dot\theta\,\mathbf{e}_h = -\dot\theta\,(\cos\phi\,\mathbf{u}_r - \sin\phi\,\mathbf{u}_\phi)\,, \qquad \left.\frac{d\mathbf{u}_\phi}{dt}\right|_1 = -\dot\phi\cos\phi\,\mathbf{e}_h - \sin\phi\,\dot\theta\,\mathbf{u}_\theta - \dot\phi\sin\phi\,\mathbf{k}_1 = -\dot\phi\,\mathbf{u}_r - \dot\theta\sin\phi\,\mathbf{u}_\theta$$

PHASE 3

Step-by-Step Mathematical Deduction

(a) Position, velocity and acceleration of P relative to S_1

Pedagogical Justification: relative quantities are derivatives in S_1; $\mathbf{r}_1^C$ is constant in S_1 and only $\mathbf{u}_r$ changes, through $\theta(t)$ and $\phi(t)$.

$$\boxed{\mathbf{r}_1^P = \mathbf{r}_1^C + \mathbf{CP} = R\,(\mathbf{j}_1 + \mathbf{k}_1) + R\,\mathbf{u}_r}$$

In components of $\mathcal{B}_1$: $x_1 = R\cos\phi\cos\theta$, $y_1 = R(1 + \cos\phi\sin\theta)$, $z_1 = R(1 + \sin\phi)$.

$$\boxed{\mathbf{v}_1^P = R\,\left.\frac{d\mathbf{u}_r}{dt}\right|_1 = R\dot\theta\cos\phi\,\mathbf{u}_\theta + R\dot\phi\,\mathbf{u}_\phi}$$

Differentiating again in S_1 (product rule, $\frac{d}{dt}(\dot\theta\cos\phi) = \ddot\theta\cos\phi - \dot\theta\dot\phi\sin\phi$, and the basis derivatives of Phase 2):

$$\mathbf{a}_1^P = R\left(\ddot\theta\cos\phi - \dot\theta\dot\phi\sin\phi\right)\mathbf{u}_\theta + R\dot\theta\cos\phi\left[-\dot\theta\left(\cos\phi\,\mathbf{u}_r - \sin\phi\,\mathbf{u}_\phi\right)\right] + R\ddot\phi\,\mathbf{u}_\phi + R\dot\phi\left(-\dot\phi\,\mathbf{u}_r - \dot\theta\sin\phi\,\mathbf{u}_\theta\right)$$

Collecting components:

$$\boxed{\mathbf{a}_1^P = -R\left(\dot\phi^2 + \dot\theta^2\cos^2\phi\right)\mathbf{u}_r + R\left(\ddot\theta\cos\phi - 2\dot\theta\dot\phi\sin\phi\right)\mathbf{u}_\theta + R\left(\ddot\phi + \dot\theta^2\sin\phi\cos\phi\right)\mathbf{u}_\phi}$$

(b) Forces on P seen from S_1

> **Pedagogical Justification:** in S_1, Notes Eq. 4.63 reads $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$, where $\mathbf{F}_i$ (Notes Eq. 4.62) has four terms (Slide 28). We evaluate each with $\mathbf{a}_0^A = \mathbf{0}$, $\boldsymbol{\omega}_{10} = \Omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \dot\Omega\mathbf{k}_1$, and project on the spherical basis.

Actual forces.

- Weight: $m\mathbf{g} = -mg\,\mathbf{k}_1 = -mg\sin\phi\,\mathbf{u}_r - mg\cos\phi\,\mathbf{u}_\phi$.
- Reaction (smooth, unilateral, toward C): $\mathbf{N} = -N\,\mathbf{u}_r$, $N \geq 0$ while in contact.

Fictitious forces. Only the part of $\mathbf{r}_1^P$ normal to $\mathbf{k}_1$ enters the cross products with $\mathbf{k}_1$. That part is $\mathbf{r}_\perp = x_1\mathbf{i}_1 + y_1\mathbf{j}_1$; substituting $\mathbf{i}_1 = \cos\theta\,\mathbf{e}_h - \sin\theta\,\mathbf{u}_\theta$ and $\mathbf{j}_1 = \sin\theta\,\mathbf{e}_h + \cos\theta\,\mathbf{u}_\theta$:

$$\mathbf{r}_\perp = R\left[\cos\phi\cos^2\theta + \sin\theta + \cos\phi\sin^2\theta\right]\mathbf{e}_h + R\left[-\cos\phi\cos\theta\sin\theta + \cos\theta + \cos\phi\sin\theta\cos\theta\right]\mathbf{u}_\theta = R\left(\cos\phi + \sin\theta\right)\mathbf{e}_h + R\cos\theta\,\mathbf{u}_\theta$$

- **Origin force:** $\mathbf{F}_A = -m\mathbf{a}_0^A = \mathbf{0}$.
- **Euler force** (Slide 31), with $\mathbf{k}_1 \times \mathbf{e}_h = \mathbf{u}_\theta$, $\mathbf{k}_1 \times \mathbf{u}_\theta = -\mathbf{e}_h$:

$$\mathbf{F}_E = -m\dot{\Omega}\,\mathbf{k}_1 \times \mathbf{r}_\perp = -m\dot{\Omega}R\left[(\cos\phi + \sin\theta)\,\mathbf{u}_\theta - \cos\theta\,\mathbf{e}_h\right]$$

$$\mathbf{F}_E = m\dot{\Omega}R\left[\cos\theta\cos\phi\,\mathbf{u}_r - (\cos\phi + \sin\theta)\,\mathbf{u}_\theta - \cos\theta\sin\phi\,\mathbf{u}_\phi\right]$$

- **Centrifugal force** (Slide 30), $\mathbf{F}_C = m\Omega^2\mathbf{r}_\perp$:

$$\mathbf{F}_C = m\Omega^2 R\left[(\cos\phi + \sin\theta)\cos\phi\,\mathbf{u}_r + \cos\theta\,\mathbf{u}_\theta - (\cos\phi + \sin\theta)\sin\phi\,\mathbf{u}_\phi\right]$$

- **Coriolis force** (Slide 31): $\mathbf{k}_1 \times \mathbf{v}_1^P = R\dot{\theta}\cos\phi\,(-\mathbf{e}_h) + R\dot{\phi}\,(-\sin\phi\,\mathbf{u}_\theta)$, so

$$\mathbf{F}_{COR} = -2m\Omega\,\mathbf{k}_1 \times \mathbf{v}_1^P = 2m\Omega R\left(\dot{\theta}\cos\phi\,\mathbf{e}_h + \dot{\phi}\sin\phi\,\mathbf{u}_\theta\right) = 2m\Omega R\left[\dot{\theta}\cos^2\phi\,\mathbf{u}_r + \dot{\phi}\sin\phi\,\mathbf{u}_\theta - \dot{\theta}\sin\phi\cos\phi\,\mathbf{u}_\phi\right]$$

As required, $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 2m\Omega R^2\left(\dot{\phi}\sin\phi\,\dot{\theta}\cos\phi - \dot{\theta}\sin\phi\cos\phi\,\dot{\phi}\right) = 0$.

(c) Two differential equations for θ and ϕ

> **Pedagogical Justification:** the reaction is along $\mathbf{u}_r$, so the projections of Notes Eq. 4.63 on $\mathbf{u}_\theta$ and $\mathbf{u}_\phi$ contain no unknown force.

Projection on $\mathbf{u}_\theta$ (weight and reaction have no $\mathbf{u}_\theta$ component):

$$mR\left(\ddot{\theta}\cos\phi - 2\dot{\theta}\dot{\phi}\sin\phi\right) = m\Omega^2 R\cos\theta - m\dot{\Omega}R\,(\cos\phi + \sin\theta) + 2m\Omega R\dot{\phi}\sin\phi$$

$$\boxed{\ddot{\theta}\cos\phi - 2\dot{\phi}\sin\phi\left(\dot{\theta} + \Omega\right) = \Omega^2\cos\theta - \dot{\Omega}\,(\cos\phi + \sin\theta)}$$

Projection on $\mathbf{u}_\phi$:

$$mR\left(\ddot{\phi}+\dot{\theta}^2\sin\phi\cos\phi\right)=-mg\cos\phi-m\Omega^2R\left(\cos\phi+\sin\theta\right)\sin\phi-m\dot{\Omega}R\cos\theta\sin\phi-2m\Omega R\dot{\theta}\sin\phi\cos\phi$$

$$\boxed{\ddot{\phi}+\dot{\theta}\left(\dot{\theta}+2\Omega\right)\sin\phi\cos\phi+\frac{g}{R}\cos\phi+\Omega^2\sin\phi\left(\cos\phi+\sin\theta\right)+\dot{\Omega}\cos\theta\sin\phi=0}$$

These two coupled, non-linear ODEs with the known forcing $\Omega(t)$ determine $\theta(t)$, $\phi(t)$ from the initial conditions.

(d) Reaction and separation condition

Pedagogical Justification: the remaining projection, on $\mathbf{u}_r$, contains the only unknown force, N.

$$-mR\left(\dot{\phi}^2+\dot{\theta}^2\cos^2\phi\right)=-N-mg\sin\phi+m\Omega^2R\cos\phi\left(\cos\phi+\sin\theta\right)+m\dot{\Omega}R\cos\theta\cos\phi+2m\Omega R\dot{\theta}\cos^2\phi$$

$$\boxed{N=mR\left(\dot{\phi}^2+\dot{\theta}^2\cos^2\phi\right)-mg\sin\phi+m\Omega^2R\cos\phi\left(\cos\phi+\sin\theta\right)+m\dot{\Omega}R\cos\phi\cos\theta+2m\Omega R\dot{\theta}\cos^2\phi}$$

Grouping $\dot{\theta}^2+2\Omega\dot{\theta}+\Omega^2=(\dot{\theta}+\Omega)^2$ in the $\cos^2\phi$ terms gives the equivalent compact form

$$N=mR\dot{\phi}^2+mR\cos^2\phi\left(\dot{\theta}+\Omega\right)^2+m\Omega^2R\cos\phi\sin\theta+m\dot{\Omega}R\cos\phi\cos\theta-mg\sin\phi$$

It depends only on $\theta,\dot{\theta},\phi,\dot{\phi}$ and the known $m,R,g,\Omega(t),\dot{\Omega}(t)$. **Separation:** the surface can only push ($N\geq 0$). The particle leaves the surface at the first instant at which

$$\boxed{N\left(\theta,\dot{\theta},\phi,\dot{\phi},t\right)=0\ \text{ and would become negative}}$$

i.e. the motion predicted by (c) is valid only while $N\geq 0$.

(e) Equilibrium configurations for $\Omega = \text{const}$

> **Pedagogical Justification:** equilibrium in S_1 means θ, ϕ constant: $\dot\theta = \dot\phi = \ddot\theta = \ddot\phi = 0$, with $\dot\Omega = 0$. Then $\mathbf{F}_E = \mathbf{F}_{COR} = \mathbf{0}$ and the equations of (c) become algebraic. We take $\Omega \neq 0$ (the case $\Omega = 0$ is discussed at the end).

From the θ equation: $\Omega^2 \cos\theta = 0 \Rightarrow \cos\theta = 0$, so $\theta = \pi/2$ or $\theta = -\pi/2$: P must lie in the vertical plane $x_1 = 0$ that contains the rotation axis and C (otherwise the centrifugal force has a component $m\Omega^2 R\cos\theta$ along $\mathbf{u}_\theta$ that nothing balances).

From the ϕ equation, multiplied by R: $g\cos\phi + \Omega^2 R\sin\phi(\cos\phi + \sin\theta) = 0$. The poles $\cos\phi = 0$ do not satisfy it ($\Omega^2 R\sin\phi\sin\theta = \pm\Omega^2 R \neq 0$), so we can divide by $g\cos\phi > 0$. With the dimensionless number $\lambda = \Omega^2 R/g > 0$:

Branch 1, $\theta = \pi/2$ (far side of the sphere from the axis, $y_1 = R(1+\cos\phi) \geq R$):

$$1 + \lambda\,\frac{\sin\phi\,(\cos\phi + 1)}{\cos\phi} = 0 \quad\Longleftrightarrow\quad \boxed{\tan\phi_1 + \sin\phi_1 = -\frac{g}{\Omega^2 R}}$$

The function $f(\phi) = \tan\phi + \sin\phi$ is strictly increasing on $(-\pi/2, \pi/2)$ ($f' = \sec^2\phi + \cos\phi > 0$), goes from $-\infty$ to $+\infty$ and vanishes at 0; so there is **exactly one root,** $\phi_1 \in (-\pi/2, 0)$, below the horizontal plane of C.

Branch 2, $\theta = -\pi/2$ (side of the sphere next to the axis, $y_1 = R(1-\cos\phi)$):

$$1 + \lambda\,\frac{\sin\phi\,(\cos\phi - 1)}{\cos\phi} = 0 \quad\Longleftrightarrow\quad \boxed{\tan\phi_2 - \sin\phi_2 = \frac{g}{\Omega^2 R}}$$

$h(\phi) = \tan\phi - \sin\phi$ has $h' = \sec^2\phi - \cos\phi \geq 0$ (zero only at $\phi = 0$), $h(0) = 0$ and $h \to +\infty$ as $\phi \to \pi/2$, so there is **exactly one root,** $\phi_2 \in (0, \pi/2)$, above the horizontal plane of C.

Reaction at equilibrium. With $\dot\theta = \dot\phi = \dot\Omega = 0$, (d) gives $N = m\Omega^2 R\cos\phi(\cos\phi + \sin\theta) - mg\sin\phi$. The equilibrium relation $\Omega^2 R(\cos\phi + \sin\theta) = -g\cos\phi/\sin\phi$ ($\sin\phi \neq 0$ at both roots) turns it into

$$N = -\frac{mg\cos^2\phi}{\sin\phi} - mg\sin\phi = -\frac{mg\left(\cos^2\phi + \sin^2\phi\right)}{\sin\phi} \quad\Longrightarrow\quad \boxed{N = -\frac{mg}{\sin\phi}}$$

This is simply the vertical balance: the centrifugal force is horizontal, so the vertical component of $\mathbf{N} = -N\mathbf{u}_r$, which is $-N\sin\phi$, must cancel the weight.

Discussion.

- **Branch 1** ($\theta = \pi/2, \phi_1 < 0$): $N_1 = mg/|\sin\phi_1| > 0$. **This is the only physically possible equilibrium.** The particle sits in the lower half of the sphere, on the side away from the axis, where the inward normal $-\mathbf{u}_r$ points up and toward the axis, balancing the weight and the outward centrifugal force. As Ω grows, ϕ_1 rises toward the equator of the sphere: for $\lambda \gg 1$, $\phi_1 \approx -g/(2\Omega^2 R)$ and $N_1 \approx 2m\Omega^2 R$ (the centrifugal force at distance $2R$ from the axis); for $\Omega \to 0$, $\phi_1 \to -\pi/2$ (bottom) and $N_1 \to mg$. Example ($R = 1$ m, $g = 9.81$ m/s^2): $\Omega = 3$ rad/s gives $\phi_1 = -30.3^\circ$ and $N_1 = 1.98\,mg$.
- **Branch 2** ($\theta = -\pi/2, \phi_2 > 0$): $N_2 = -mg/\sin\phi_2 < 0$. It solves the equations of (c), but the reaction would have to **pull** the particle away from C, which the inner surface cannot do: it is **not an admissible equilibrium** for the unilateral contact (it would be one for a particle held on both sides of the surface, or resting on the outside of the sphere). On that side the centrifugal force points toward the inside of the sphere, so it cannot help the surface to hold the particle up.
- **Stability of branch 1** (extension, using the energy in S_1). For constant Ω, A fixed and a smooth sphere fixed in S_1, the energy $E_1^P = T_1^P + V_1$ is conserved (Slides 29–31), with $V_1 = mgz_1 - \frac{1}{2}m\Omega^2(x_1^2 + y_1^2) = mgR(1 + \sin\phi) - \frac{1}{2}m\Omega^2R^2\left(1 + \cos^2\phi + 2\cos\phi\sin\theta\right)$. Its derivatives $\partial V_1/\partial\theta = -m\Omega^2R^2\cos\phi\cos\theta$ and $\partial V_1/\partial\phi = mgR\cos\phi + m\Omega^2R^2\sin\phi(\cos\phi + \sin\theta)$ reproduce the equilibrium equations. At branch 1, $\partial^2V_1/\partial\theta^2 = m\Omega^2R^2\cos\phi_1\sin\theta = m\Omega^2R^2\cos\phi_1 > 0$, $\partial^2V_1/\partial\theta\partial\phi = m\Omega^2R^2\sin\phi_1\cos\theta = 0$ and $\partial^2V_1/\partial\phi^2 = mgR\lambda\cos\phi_1\left(\sec^2\phi_1 + \cos\phi_1\right) > 0$: V_1 has a strict minimum, so the equilibrium is stable (the linearised equations give purely imaginary eigenvalues, see the verification).
- **Case $\Omega = 0$:** the θ equation is satisfied identically and the ϕ equation gives $\cos\phi = 0$: the bottom point $\phi = -\pi/2$ with $N = mg$ (the top point $\phi = \pi/2$ would need $N = -mg < 0$); θ is undefined there (pole of the coordinates).

PHASE 4

Physical Interpretation & Dimensional Verification

- **Role of each inertia force.** The centrifugal force pushes P away from the axis and is responsible for the tilt of the equilibrium; the Euler force (only while Ω changes) drags P against the sense of the angular acceleration, along $-\mathbf{u}_\theta$-like directions; the Coriolis force couples the two angular motions (terms $2\Omega\dot\phi\sin\phi$ and $2\Omega\dot\theta\sin\phi\cos\phi$) and does no work.
- **Absolute reading of N.** In the compact form, $\dot\theta + \Omega$ is the absolute azimuthal rate of the particle about the vertical through C, so $mR\dot\phi^2 + mR\cos^2\phi(\dot\theta + \Omega)^2$ is the centripetal demand of the absolute motion on the sphere.
- **Limit $\Omega \to 0$:** the equations reduce to those of the spherical pendulum, $\ddot\theta\cos\phi - 2\dot\theta\dot\phi\sin\phi = 0$ and $\ddot\phi + \dot\theta^2\sin\phi\cos\phi + (g/R)\cos\phi = 0$, and $N = mR(\dot\phi^2 + \dot\theta^2\cos^2\phi) - mg\sin\phi$.

- **Dimensions:** in (c) every term is s^{-2} ($[\ddot{\theta}] = [\Omega^2] = [\dot{\Omega}] = [\Omega\dot{\phi}] = [g/R] = \mathrm{s}^{-2}$); in (d) every term is $\mathrm{kg\,m\,s}^{-2} = \mathrm{N}$; $\lambda = \Omega^2 R/g$ is dimensionless.
- **Verification (sympy, `verify_P51.py`):** with a general $\Psi(t)$, $\dot{\Psi} = \Omega(t)$, the inertial double derivative of $[{}_0R_1]\{\mathbf{r}_1^P\}$ equals the Notes Eq. 3.27 decomposition; solving $m\mathbf{a}_0^P = -mg\mathbf{k}_1 - N\mathbf{u}_r$ in S_0 returns exactly the two equations of (c) and the N of (d); for $\Omega = 1, 3, 10$ rad/s ($R = 1$ m) both branches satisfy the equations with $N = -mg/\sin\phi$, branch 1 has $N > 0$ and purely imaginary eigenvalues, branch 2 has $N < 0$. No official key exists for this problem.

Source: Problems.pdf (PDF page 36); no official key